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NEET FLIX

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Exercise - I (Conceptual Questions) Build Up Your Understanding

1. The function of tracheal cilia is to

(1) Pass mucus out (2) Pass mucus in

(3) Pass air out (4) Pass air in

BBG001

2. Which of the following prevents collapsing of trachea?

(1) Muscles (2) Diaphragm

(3) Ribs (4) Cartilaginous rings

BBG002

3. Tracheal rings are :–

(1) Complete

(2) Incomplete

(3) Dorsally incomplete

(4) Lateral incomplete

BBG004

4. Inflammation of the lung covering causing severe chest pain is

(1) Emphysema (2) Pleurisy

(3) Asphyxia (4) Hypoxia

BBG008

5. Which of the following is not a part of respiratory tract?

(1) Nasal chamber (2) Oesophagus

(3) Pharynx (4) Trachea

BBG009

6. The epithelium of respiratory bronchioles is:-

(1) Pseudostratified columnar

(2) Simple squamous

(3) Pseudostratified and sensory

(4) Cuboidal and columnar

BBG011

7. Lungs are covered by

(1) Perichondrium (2) Pleural sac

(3) Pericardium (4) Peristomium

BBG014

8. Which one of the following statement is correct?

(1) Chest expands because air enters into the lungs

(2) Air enters into the lungs because chest expands

(3) The muscles of the diaphragm contracts because air enters into the lungs

(4) All of the above statements are correct

BBG015

9. After deep inspiration, capacity of maximum expiration of lung is called :-

(1) Total lung capacity

(2) Functional residual capacity

(3) Vital capacity

(4) Inspiratory capacity

BBG016

10. Which statement is correct ?

(1) Pulmonary ventilation is equal to alveolar ventilation.

(2) Pulmonary ventilation is less than alveolar ventilation.

(3) Alveolar ventilation is more than Pulmonary ventilation.

(4) Alveolar ventilation is less than Pulmonary ventilation.

BBG017

11. About 1500 ml of air left in lungs is called

(1) Tidal volume

(2) Inspiratory reserve volume

(3) Residual volume

(4) Vital capacity

BBG018

12. At high altitude, RBC of human blood will
(1) Increase in number
(2) Decrease in number
(3) Decrease in size
(4) Increase in size
BBG019

13. Which one has the lowest value?
(1) Tidal volume
(2) Vital capacity
(3) Inspiratory reserve volume
(4) Expiratory reserve volume
BBG020

14. Volume of air inspired or expired with each normal breath is known as -
(1) Inspiratory capacity
(2) Total lung capacity
(3) Tidal volume
(4) Residual volume
BBG021

15. Total lung capacity is :-
(1) Total volume of air accommodated in lungs at the end of forced inspiration
(2) RV + ERV + TV + IRV
(3) Vital capacity + Residual volume
(4) All of the above
BBG025

16. Which instrument helps in clinical assessment of pulmonary volumes?
(1) Sphygmomanometer
(2) Stethoscope
(3) Spirometer
(4) Electrocardiograph
BBG026

17. Volume of air remains in the lungs after normal expiration is
(1) ERV + RV (2) IRV + RV
(3) RV + IRV + ERV (4) TV
BBG027

18. Which of the following volume is not included in vital capacity?
(1) ERV (2) TV (3) IRV (4) RV
BBG028

19. The upper region of pharynx in human is called :-
(1) Oropharynx (2) Nasopharynx
(3) Laryngopharynx (4) None of these
BBG029

20. Friction on the lungs surface reduces by
(1) double layered pleura
(2) single layered pleura
(3) ribs covering lungs
(4) mucous membrane surrounding the lungs.
BBG032

21. In lungs, air is separated from venous blood by:
(1) Squamous epithelium + tunica externa of blood vessel
(2) Squamous epithelium + endothelium of blood vessel
(3) Transitional epithelium + tunica media of blood vessel
(4) Columnar epithelium + 3 layered wall of blood vessel.
BBG033

22. In lung, gaseous exchange is done by :-
(1) Simple diffusion
(2) Active transport
(3) Passive transport
(4) Facilitated diffusion
BBG034

23. Which of the following statements is **not true**?
(1) The partial pressure of O_2 in deoxygenated blood is 40 mm Hg.
(2) The partial pressure of O_2 in oxygenated blood is 95 mm Hg.
(3) The partial pressure of O_2 in alveolar air is 104 mm Hg.
(4) The partial pressure of CO_2 in alveolar air is 45 mm Hg.
BBG035

24. Partial pressure of oxygen in alveoli, atmospheric air and tissue will be
(1) 40, 159, 45
(2) 40, 0.3, 45
(3) 104, 159, 40
(4) 104, 0.3, 45

BBG036

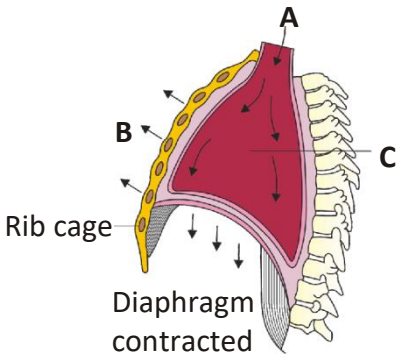
25. What will be the pO_2 and pCO_2 in the atmospheric air compared to those in the alveolar air?
(1) pO_2 lesser, pCO_2 higher
(2) pO_2 higher, pCO_2 lesser
(3) pO_2 higher, pCO_2 higher
(4) pO_2 lesser, pCO_2 lesser

BBG037

26. Partial pressure of CO_2 is higher :-
(1) At alveolar level
(2) At tissue level
(3) In atmosphere
(4) In oxygenated blood

BBG038

27. In the given diagram, identify what is depicted by A, B and C.
Choose the correct option.



(1) A-Air going out from lungs, B-Ribs and sternum relaxed, C-volume of thorax increased
(2) A-Air entering lungs, B-Ribs and sternum relaxed, C-volume of thorax increased
(3) A-Air entering lungs, B-Ribs and sternum raised, C-volume of thorax increased
(4) A -Air going out from lungs, B-Ribs and sternum relaxed, C-volume of thorax decreased

BBG039

28. Almost same pO_2 in humans is found in
(1) alveoli and tissues
(2) oxygenated blood and deoxygenated blood
(3) alveoli and oxygenated blood
(4) alveoli and deoxygenated blood

BBG040

29. Arrange the following in the order if increasing volume
(i) Tidal volume
(ii) Residual volume
(iii) Expiratory reserve volume
(iv) Vital capacity
(1) (i) < (ii) < (iii) < (iv)
(2) (i) < (iii) < (ii) < (iv)
(3) (i) < (iv) < (iii) < (ii)
(4) (i) < (iv) < (ii) < (iii)

BBG042

30. Under normal condition 100 ml blood deliver _____ to tissue.
(1) 4 ml O_2 (2) 10 ml O_2
(3) 5 ml O_2 (4) 25 ml O_2

BBG043

31. Effect of CO_2 concentration on dissociation of oxyhaemoglobin is called:
(1) Bohr's effect
(2) Haldane effect
(3) Hamburger effect
(4) Root effect

BBG046

32. Chloride shift occurs for the transport of:
(1) O_2 (2) CO_2 (3) CO (4) O_3

BBG047

33. Exchange of bicarbonates and chloride ions between RBC and plasma is called:-
(1) Chloride shift.
(2) Bohr's effect.
(3) Haldane's effect.
(4) Intra cellular respiration.
BBG048

34. Determination of oxygen carried by haemoglobin is done by
(1) pH
(2) Partial pressure of oxygen
(3) Partial pressure of carbon dioxide
(4) All the above
BBG049

35. For proper transport of O_2 and CO_2, blood should be
(1) Slightly acidic
(2) Strongly acidic
(3) Strongly alkaline
(4) Slightly alkaline
BBG050

36. What would happen when blood is acidic?
(1) Binding of oxygen with haemoglobin increases
(2) Red blood corpuscles are formed in higher number
(3) Binding of oxygen with haemoglobin decreases
(4) There is no change in oxygen binding nor number of RBC
BBG051

37. The chloride shift is movement of Cl^-
(1) From plasma to RBC
(2) From WBC to plasma
(3) From RBC to plasma
(4) From plasma to WBC
BBG052

38. Chloride shift occurs in respond to :
(1) H^+ (2) K^+ (3) HCO_3^- (4) Na^+
BBG053

39. What happen to the O_2 dissociation curve of Hb if pH is decreased ?
(1) shift to left
(2) shift to right
(3) remain unchanged
(4) will oscillate erratically
BBG054

40. Dissociation curve shifts to the right when:-
(1) CO_2 concentration decreases
(2) CO_2 concentration increases
(3) O_2 concentration increases
(4) H^+ concentration decreases
BBG055

41. Oxyhaemoglobin acts as
(1) Alkali (2) Acid
(3) Neutral (4) Buffer
BBG056

42. The oxygen dissociation curve will show a right shift in case of :-
(1) Less H^+ concentration
(2) High partial pressure of O_2
(3) Low partial pressure of CO_2
(4) High temperature.
BBG057

43. Which of the following conditions are found in the alveoli of lungs :
I. High pO_2 II. Low pCO_2
III. High pCO_2 IV. Low pO_2
V. Low H^+ VI. High H^+
Choose the **correct** option.
(1) I, III and V
(2) III, IV and VI
(3) I, IV and VI
(4) I, II and V
BBG058

44. Binding of CO_2 to oxyhaemoglobin occurs when :
(1) pCO_2 is high and pO_2 is less in tissue
(2) pCO_2 is low and pO_2 is high in tissue
(3) pCO_2 is low and pO_2 is low in tissue
(4) pCO_2 is high and pO_2 is high in tissue

BBG059

45. In Bohr's effect, curve shift to right when :-
(1) pCO_2 decreases and p50 O_2 increases
(2) pCO_2 increases and p50 O_2 decreases
(3) pCO_2 increases and p50 O_2 increases
(4) pCO_2 increases and p50 O_2 decreases and pH increases

BBG060

46. In which three forms CO_2 is transported by blood?
(1) As a solution, Carbamino compounds, bicarbonates
(2) As a solution, Carbamino proteins, $KHCO_3$
(3) As a solution, Carbamino haemoglobin, $NaHCO_3$
(4) As a solution, Carbamino compound, H_2CO_3

BBG061

47. H. Hb is :-
(1) H_2b
(2) Heavy hydrogen bond
(3) Reduced bromide
(4) Haemoglobin acid

BBG062

48. During transport of CO_2, blood does not become acidic due to :
(1) Neutralisation of H_2CO_3 by Na_2CO_3
(2) Absorption by leucocytes
(3) Blood buffers
(4) Non-accumulation

BBG064

49. Rate of respiration is directly affected by
(1) CO_2 concentration
(2) O_2 in trachea
(3) Concentration of O_2
(4) Diaphragm expansion

BBG066

50. CO is more toxic than CO_2 because it :
(1) Damages lungs
(2) It destroys haemoglobin
(3) Affects the nervous system
(4) Reduces the oxygen carrying capacity of haemoglobin

BBG067

51. Carbon monoxide has greater affinity for haemoglobin as compare to oxygen :-
(1) 1000 Times (2) 200 Times
(3) 20 Times (4) 2 Times

BBG068

52. About 97% of oxygen is transported by RBC remaining 3% is :-
(1) Dissolved in plasma & transported
(2) Retained in lungs
(3) Attached to cell membranes
(4) Found inside mitochondria

BBG069

53. Increased asthmatic attacks in certain seasons are related to :-
(1) Hot and humid environment
(2) Eating fruits preserved in tin containers
(3) Inhalation of seasonal pollen
(4) Low temperature

BBG070

54. Disease aggravated by pollution is :-
(1) Haemophilia (2) Rheumatism
(3) Scurvy (4) Bronchitis

BBG071

55. Column-I represents diseases and column II represents their symptoms. Which of the following pairs are correct match for them:-

Column-I	**Column-II**
(P) Asthma	(i) Damaged alveolar walls
(Q) Emphysema	(ii) Accumulation of W.B.C. in alveolus
(R) Pneumonia	(iii) Allergy

(1) P–iii, Q–ii, R–i (2) P–iii, Q–i, R–ii

(3) P–ii, Q–iii, R–i (4) P–ii, Q–i, R–iii

BBG072

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	2	2	2	2	2	3	4	3	1	1	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	4	2	1	2	1	4	3	2	2	3	3	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	1	4	4	3	1	3	2	2	2	4	4	1	3
Question	46	47	48	49	50	51	52	53	54	55					
Answer	3	4	3	1	4	2	1	3	4	2					

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Approximately seventy percent of carbon-dioxide absorbed by the blood will be transported to the lungs:
(1) as bicarbonate ions
(2) in the form of dissolved gas molecules
(3) by binding to R.B.C.
(4) as carbamino - haemoglobin

BBG087

AIPMT 2015

2. When you hold your breath, which of the following gas changes in blood would first lead to the urge to breathe?
(1) rising CO_2 concentration
(2) falling CO_2 concentration
(3) rising CO_2 and falling O_2 concentration
(4) falling O_2 concentration

BBG088

NEET-I 2016

3. Asthma may be attributed to :
(1) bacterial infection of the lungs
(2) allergic reaction of the mast cells in the lungs
(3) inflammation of the trachea
(4) accumulation of fluid in the lungs

BBG089

NEET-II 2016

4. The partial pressure of oxygen in the alveoli of the lungs is :-
(1) Less than that in the blood
(2) Less than that of carbon dioxide
(3) Equal to that in the blood
(4) More than that in the blood

BBG090

5. Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because :-
(1) There is a positive intrapleural pressure
(2) Pressure in the lungs is higher than the atmospheric pressure.
(3) There is a negative pressure in the lungs.
(4) There is a negative intrapleural pressure pulling at the lung walls

BBG091

NEET (UG) 2017

6. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of:
(1) Inspiratory Reserve Volume
(2) Tidal Volume
(3) Expiratory Reserve Volume
(4) Residual Volume

BBG092

NEET(UG) 2018

7. Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively ?
(1) Inflammation of bronchioles; Decreased respiratory surface
(2) Increased number of bronchioles; Increased respiratory surface
(3) Increased respiratory surface; Inflammation of bronchioles
(4) Decreased respiratory surface; Inflammation of bronchioles

BBG093

8. Match the items given Column I with those in Column II and select the ***correct*** option given below :

Column-I	Column-II
a. Tidal volume	i. 2500-3000 mL
b. Inspiratory Reserve volume	ii. 1100-1200 mL
c. Expiratory Reserve volume	iii. 500-550 mL
d. Residual volume	iv. 1000-1100 mL

	a	b	c	d
(1)	iii	ii	i	iv
(2)	iii	i	iv	ii
(3)	iii	iv	ii	iii
(4)	iv	iii	ii	i

BBG094

9. Which of the following is an occupational respiratory disorder ?

(1) Anthracis

(2) Silicosis

(3) Botulism

(4) Emphysema

BBG095

NEET (UG) 2019

10. Tidal Volume and Expiratory Reserve Volume of an athlete is 500 mL and 1000 mL respectively. What will be his Expiratory Capacity if the Residual Volume is 1200 mL?

(1) 1500 mL

(2) 1700 mL

(3) 2200 mL

(4) 2700 mL

BBG096

NEET (UG) 2019 (ODISHA)

11. Select the correct statement.

(1) Expiration occurs due to external intercostal muscles

(2) Intrapulmonary pressure is lower than the atmospheric pressure during inspiration.

(3) Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.

(4) Expiration is initiated due to contraction of diaphragm.

BBG097

12. The maximum volume of air a person can breathe in after a forced expiration is known as:

(1) Expiratory Capacity

(2) Vital Capacity

(3) Inspiratory Capacity

(4) Total lung Capacity

BBG098

NEET (UG) 2020

13. Identify the wrong statement with reference to transport of oxygen.

(1) Low pCO_2 in alveoli favours the formation of oxyhaemoglobin.

(2) Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2.

(3) Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.

(4) Higher H^+ conc. in alveoli favours the formation of oxyhaemoglobin.

BBG099

14. Select the correct events that occur during inspiration.

(a) Contraction of diaphragm

(b) Contraction of external inter-costal muscles

(c) Pulmonary volume decreases

(d) Intra pulmonary pressure increases

(1) only (d) (2) (a) and (b)

(3) (c) and (d) (4) (a), (b) and (d)

BBG100

NEET (UG) 2020 (COVID-19)

15. The Total Lung Capacity (TLC) is the total volume of air accommodated in the lungs at the end of a forced inspiration. This includes:

(1) RV; IC (Inspiratory Capacity); EC (Expiratory Capacity); and ERV

(2) RV; ERV; IC and EC

(3) RV; ERV; VC (Vital Capacity) and FRC (Functional Residual Capacity)

(4) RV (Residual Volume); ERV (Expiratory Reserve Volume); TV (Tidal Volume); and IRV (Inspiratory Reserve Volume)

BBG101

16. Match the following columns and select the correct option :

	Column-I		Column-II
(a)	Pneumotaxic Centre	(i)	Alveoli
(b)	O_2 Dissociation	(ii)	Pons region of brain
(c)	Carbonic Anhydrase	(iii)	Haemoglobin
(d)	Primary site of exchange of gases	(iv)	R.B.C.

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

BBG102

NEET (UG) 2021

17. The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are :

(1) $pO_2 = 104$ and $pCO_2 = 40$

(2) $pO_2 = 40$ and $pCO_2 = 45$

(3) $pO_2 = 95$ and $pCO_2 = 40$

(4) $pO_2 = 159$ and $pCO_2 = 0.3$

BBG103

18. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

(1) High pO_2, low pCO_2, less H^+, lower temperature

(2) Low pO_2, high pCO_2, more H^+, higher temperature

(3) High pO_2, high pCO_2, less H^+, higher temperature

(4) Low pO_2, low pCO_2, more H^+, higher temperature

BBG104

NEET (UG) 2022

19. Which of the following is **not** the function of conducting part of respiratory system ?

(1) Inhaled air is humidified

(2) Temperature of inhaled air is brought to body temperature

(3) Provides surface for diffusion of O_2 and CO_2

(4) It clears inhaled air from foreign particles

BBG105

20. Under normal physiological conditions in human being every 100 ml of oxygenated blood can deliver _____ml of O_2 to the tissues.

(1) 5ml (2) 4 ml (3) 10 ml (4) 2 ml

BBG106

21. Which of the following statements are correct with respect to vital capacity?

(a) It includes ERV, TV and IRV

(b) Total volume of air a person can inspire after a normal expiration

(c) The maximum volume of air a person can breathe in after forced expiration

(d) It includes ERV, RV and IRV.

(e) The maximum volume of air a person can breath out after a forced inspiration.

Choose the **most appropriate answer** from the options given below:

(1) (b), (d) and (e) (2) (a), (c) and (d)

(3) (a), (c) and (e) (4) (a) and (e)

BBG107

NEET (UG) 2023

22. Vital capacity of lung is _____.

(1) IRV + ERV + TV + RV

(2) IRV + ERV + TV - RV

(3) IRV + ERV + TV

(4) IRV + ERV

BBG155

NEET (UG) MANIPUR 2023

23. Select the sequence of steps in Respiration.

(A) Diffusion of gases (O_2 and CO_2) across alveolar membrane.

(B) Diffusion of O_2 and CO_2 between blood and tissues.

(C) Transport of gases by the blood.

(D) Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.

(E) Utilisation of O_2 by the cells for catabolic reactions are resultant release of CO_2.

Choose the **correct** answer from the options given below :-

(1) (D), (A), (C), (B), (E)

(2) (C), (B), (A), (E), (D)

(3) (B), (C), (E), (D), (A)

(4) (A), (C), (B), (E), (D)

BBG156

24. **Assertion (A) :** A person goes to high altitude and experiences 'Altitude Sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) : Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) **(A)** is true but **(R)** is false.

(2) **(A)** is false but **(R)** is true.

(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(4) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

BBG157

NEET (UG) 2024

25. Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

(1) High pO_2 and High pCO_2

(2) High pO_2 and Lesser H^+ concentration

(3) Low pCO_2 and High H^+ concentration

(4) Low pCO_2 and High temperature

BBG158

26. Match **List I** with **List II** :

	List I		List II
A.	Expiratory capacity	I.	Expiratory reserve volume + Tidal Volume + Inspiratory reserve volume
B.	Functional residual capacity	II.	Tidal volume + Expiratory reserve volume
C.	Vital capacity	III.	Tidal volume + Inspiratory reserve volume
D.	Inspiratory capacity	IV.	Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :-

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-II, C-IV, D-I

(3) A-II, B-I, C-IV, D-III

(4) A-I, B-III, C-II, D-IV

BBG159

RE-NEET (UG) 2024

27. Match **List-I** with **List-II.**

	List I		List II
A.	Residual Volume	I.	Maximum volume of air that can be breathed in after forced expiration.
B.	Vital Capacity	II.	Volume of air inspired or expired during normal respiration.
C.	Expiratory Capacity	III.	Volume of air remaining in lungs after forcible expiration.
D.	Tidal Volume	IV.	Total volume of air expired after normal inspiration.

Choose the **correct answer** from the options given below :-

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-IV, C-I, D-III

(3) A-III, B-I, C-IV, D-II

(4) A-I, B-II, C-III, D-IV

BBG160

28. **Assertion A :** During the transportation of gases about 20-25 percent of CO_2 is carried by Haemoglobin as carbamino-haemoglobin.

Reason R : This binding is related to high pCO_2 and low pO_2 in tissues.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **A** is true but **R** is false

(2) **A** is false but **R** is true

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

BBG161

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	2	4	4	4	1	2	2	1	2	2	4	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	2	1	1	3	1	3	3	1	3	2	1	3	3		

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which is a common passage for food and air?
(1) Trachea
(2) Oesophagus
(3) Pharynx
(4) Glottis

BBG108

2. The function of conducting part in respiratory system of human is :-
(1) Clears foreign particles.
(2) Humidifies atmospheric air
(3) Brings the air to body temperature
(4) All of the above

BBG109

3. When there is no air in initial bronchioles, they does not collapse. It is due to :-
(1) Presence of Lecithin
(2) Presence of incomplete cartilaginous rings
(3) Presence of complete cartilaginous rings
(4) Presence of mucous

BBG110

4. Which of the following steps not involved in respiration?
(1) Diffusion of gases across alveolar membrane
(2) Transport of gases by the blood
(3) Provide nutrients to all the living cells of body
(4) Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2.

BBG111

5. Inspiration can occur, when –
(1) Pressure within the lungs is less than the atmospheric pressure.
(2) Pressure within the lungs is more than the atmospheric pressure.
(3) Pressure within the lungs and atmospheric air is same.
(4) No effect of pressure on inspiration.

BBG112

6. By the contraction in diaphragm volume of thoracic chamber increases in the :-
(1) Dorso-ventral axis
(2) Antero-posterior axis
(3) Dorso-posterior axis
(4) Antero-ventral axis

BBG113

7. The primary muscle of inspiration is :-
(1) Diaphragm
(2) Intercostal muscle
(3) Abdominal muscle
(4) Oblique muscle

BBG114

8. Total volume of air a person can expire after normal inspiration is :-
(1) Vital capacity
(2) Functional residual capacity
(3) Inspiratory capacity
(4) Expiratory capacity

BBG115

9. Which of the following factor can affect the rate of diffusion of gases?
(1) Thickness of the membranes involved in diffusion
(2) Solubility of the gases
(3) Pressure of the gases
(4) All of these

BBG116

AIM: FREE EDUCATION TO ALL APUL

10. Which of the following is correct?

	Alveoli	Deoxygenated blood	Tissue
(1)	pO_2=159mmHg	pCO_2=40mmHg	pCO_2=20 mmHg
(2)	pCO_2=40mmHg	pO_2=95mmHg	pO_2=40mmHg
(3)	pO_2=104mmHg	pCO_2=45mmHg	pCO_2=45mmHg
(4)	pO_2=40mmHg	pO_2=40mmHg	pCO_2=45mmHg

BBG117

11. What is true about diffusion capacity?

(1) Diffusion capacity of CO_2 is much higher than O_2

(2) Diffusion capacity of O_2 is much higher than CO_2

(3) Diffusion capacity of O_2 and CO_2 is same

(4) None of the above

BBG118

12. Which of the following statement **is true**?

(1) 20-25 percent CO_2 is transported by RBCs.

(2) 97 percent O_2 is transported by RBCs.

(3) 70 percent CO_2 is carried as bicarbonate.

(4) All of these are true.

BBG119

13. Binding of oxygen with haemoglobin is primarily related to :-

(1) Partial pressure of O_2

(2) Partial pressure of CO_2

(3) H^+ ion concentration

(4) Temperature

BBG120

14. In which the following factors can interfere in binding of O_2 with haemoglobin.

A $\rightarrow$ pCO_2, B $\rightarrow$ H^+ ion concentration

C $\rightarrow$ Temperature

(1) Only A (2) B, C

(3) A and C (4) A, B, C

BBG121

15. The conditions which are favourable for the formation of oxyhaemoglobin :-

(1) $pO_2\uparrow$, $pCO_2\uparrow$, H^+conc.$\downarrow$, Temperature$\downarrow$

(2) $pO_2\downarrow$, $pCO_2\downarrow$, H^+conc.$\uparrow$, Temperature$\uparrow$

(3) $pO_2\uparrow$, $pCO_2\downarrow$, H^+conc.$\downarrow$, Temperature$\downarrow$

(4) $pO_2\uparrow$, $pCO_2\uparrow$, H^+conc.$\downarrow$, Temperature$\uparrow$

BBG122

16. Which of the following statement/s is/are correct?

A. High concentration of carbonic anhydrase is present in RBCs.

B. Minute quantities of carbonic anhydrase is present in plasma.

C. Every 100 ml blood delivers approximately 4 ml of CO_2 to the alveoli.

D. 20-25% CO_2 is carried by haemoglobin as carbaminohaemoglobin.

(1) A, C and D (2) A and D

(3) A, B, C and D (4) Only A

BBG123

17. Which of the following statement is incorrect?

(1) Among vertebrates, fishes use gills whereas amphibians, reptiles, birds and mammals respire through lungs.

(2) Mechanisms of breathing vary among different groups of animals depending mainly on their habitats and levels of organisation.

(3) Lower invertebrates like sponges, coelenterates, flatworms, etc., exchange O_2 with CO_2 by osmosis over their entire body surface.

(4) Earthworms use their moist cuticle and insects have a network of tubes (tracheal tubes) to transport atmospheric air within the body

BBG124

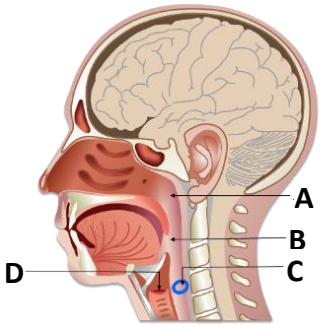
18. A chemo sensitive area is situated adjacent to respiratory rhythm centre which is highly sensitive to _____ and _____ ions.

(1) O_2, H^+ (2) CO_2, OH^-

(3) CO_2, H^+ (4) CO_2, O_2

BBG125

19. Label A, B, C and D in the diagram given below



(1) A–Nasopharynx, B–Oropharynx, C–Glottis, D–Gullet

(2) A–Nasopharynx, B–Oropharynx, C–Gullet, D–Glottis

(3) A–Oropharynx, B–Nasopharynx, C–Gullet, D–Glottis

(4) A–Oropharynx, B–Nasopharynx, C–Glottis, D–Gullet

BBG126

20. Chronic disorder in which alveolar wall are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking :-

(1) Asthma (2) Emphysema

(3) Bronchitis (4) Fibrosis

BBG127

21. What is the correct about alveoli?

(1) Thin, irregular-walled and vascularised bag-like structures

(2) Thick, irregular-walled and vascularised bag-like structures

(3) Thin, regular-walled and vascularised bag-like structures

(4) Thick, regular-walled and vascularised bag-like structures

BBG128

22. Total volume of air accommodate in the lungs at the end of forced inspiration :-

(1) TV + IRV

(2) TV + IRV + ERV

(3) TV + ERV

(4) TV + IRV + ERV + RV

BBG129

23. Mark the true statement among the following with reference to normal breathing

(1) Inspiration is a passive process where as expiration is active

(2) Inspiration is a active process where as expiration is passive

(3) Inspiration and expiration are active processes

(4) Inspiration and expiration are passive processes

BBG130

24. A person breathes in some volume of air by forced inspiration after having a forced expiration. This quantity of air taken in is

(1) Total lung capacity

(2) Tidal volume

(3) Vital capacity

(4) Inspiratory capacity

BBG131

25. Mark the correct pair of muscles involved in the normal breathing in humans :-

(1) External and internal intercostal muscles

(2) Diaphragm and abdominal muscles

(3) Diaphragm and external intercostal muscles

(4) Diaphragm and internal intercostal muscles

BBG132

26. The diffusion membrane is made up of three major layers namely.

(1) The thin squamous epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance in between them.

(2) The thick squamous epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance in between them.

(3) The thin columnar epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance in between them.

(4) The thin columnar and cuboidal epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance in between them.

BBG133

27. Which of the following group of animals, is used lungs for respiration?

(1) Fishes, Frog, Tadpole larva

(2) Fish, Mollusca, aquatic arthropods

(3) Reptiles, Birds, Mammals

(4) Insects, Earthworm, Mollusca

BBG134

(B) ANALYTICAL QUESTIONS

28. Whether a child died after normal birth or died before birth can be confirmed by measuring :

(1) Tidal volume of air

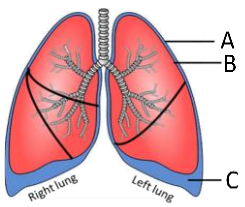
(2) Residual volume of air

(3) The weight of the child

(4) The dead space air

BBG135

29. Find the correct option for the diagram given below :



(1) A–Parietal pleura, B–Visceral pleura, C–Pleural cavity filled with pleural fluid

(2) A–Visceral pleura, B–Parietal pleura, C–Pleural cavity filled with pleural fluid

(3) A–Parietal pericardium, B–Visceral pericardium, C–Pleural cavity filled with pleural fluid

(4) A–Visceral pericardium, B–Parietal pericardium, C–Pleural cavity filled with pleural fluid

BBG136

30. Which of the following statement is correct?

(1) Alveoli are the primary sites of exchange of gases

(2) Exchange of gases occur at tissue level and alveolar surface.

(3) Pressure contributed by an individual gas in a mixture of gases is called partial pressure and is represented as pO_2 for oxygen and pCO_2 for carbon dioxide.

(4) All of the above.

BBG137

31. Identify A, B and C in the given sentence

Inspiration is initiated by ____ A _____ of diaphragm which ____ B ____ the volume of thoracic chamber in _____C _____ axis.

(1) A–relaxation, B–increases, C–antero posterior axis

(2) A–contraction, B–decreases, C–antero posterior axis

(3) A–contraction, B–increases, C–antero posterior axis

(4) A–relaxation, B–contraction, C–antero posterior axis

BBG138

32. In lungs, there is definite exchange of ions between RBC and Plasma. Removal of CO_2 from blood involves :
(1) efflux of Cl^- ions from RBC
(2) influx of Cl^- ions into RBC.
(3) influx of HCO_3^- ions into Plasma.
(4) efflux of HCO_3^- ions from RBC.

BBG139

33. The chloride content of RBC will be higher than that of plasma in :-
(1) Systemic arteries and pulmonary veins.
(2) Systemic veins and pulmonary arteries.
(3) Systemic arteries and pulmonary arteries.
(4) Systemic veins and pulmonary veins.

BBG140

34. Which of the following lung volumes or capacities can be measured by spirometer?
(1) Functional residual capacity [FRC]
(2) Residual volume
(3) Total lung capacity [TLC]
(4) Vital capacity

BBG141

35. Which of the following statement is incorrect?
(1) A healthy man can inspire or expire approximately 6000 to 8000 ml of air per second.
(2) Total volume of air that can accommodated in the lungs at the end of forced inspiration is called total lung capacity.
(3) Total volume of air a person can expire after a normal inspiration is called expiratory capacity.
(4) Amount of air remain in lungs even after deep and forceful expiration is called residual volume of air.

BBG142

36. The respiratory membranes facilitates the exchange of respiratory gases through diffusion. Oxygen enters the deoxygenated blood through diffusion because :-
(1) Partial pressure of oxygen in alveolar air and capillaries is 40 mm Hg and 104 mm Hg respectively.
(2) Partial pressure of oxygen in alveolar air and capillaries is 104 mm Hg and 40 mm Hg respectively.
(3) Partial pressure of oxygen in alveolar air and capillaries is 46 mm Hg and 40mm Hg respectively.
(4) Partial pressure of oxygen in alveolar air and capillaries it 40 mm Hg and 46 mm Hg respectively.

BBG143

37. Blood do not become acidic although it carries CO_2 because :-
(1) CO_2 is continuously diffused through tissues.
(2) CO_2 combines with H_2O to form HCO_3
(3) In CO_2 transport, buffers plays an important role.
(4) CO_2 is absorbed by WBC.

BBG144

38. The amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2. Why?
(1) Because diffusion of gases takes place only at alveoli
(2) Because the solubility of CO_2 is 20-25 times lesser than that of O_2
(3) Because the solubility of CO_2 is 20-25 times higher than that of O_2
(4) Because alveoli are very large in number

BBG145

39. Find out the correct match from the following table:

	Column I	Column II	Column III
(i)	At tissue level	pO_2 = 40 mm Hg	pCO_2 = 45 mm Hg
(ii)	In pulmonary vein	pO_2 = 95 mm Hg	pCO_2 = 50 mm Hg
(iii)	In systemic artery	pO_2 = 40 mm Hg	pCO_2 = 40 mm Hg
(iv)	In alveoli	pO_2 = 104 mm Hg	pCO_2 = 40 mm Hg

(1) Both (i) and (ii) (2) Both (iii) and (iv)

(3) (i), (ii) and (iii) (4) Both (i) and (iv)

BBG146

40. Match the following columns.

Column I		Column II	
A	Inspiratory capacity (IC)	1	Total air, a person can inspire after normal expiration
B	Expiratory capacity (EC)	2	Maximal volume of the air, a person can breath in after a forced expiration
C	Functional residual capacity (FRC)	3	Volume of the air that will remain in lungs after a normal expiration
D	Vital capacity (VC)	4	Total volume of air a person can expire after a normal inspiration

Codes :-

	A	B	C	D
(1)	1	2	3	4
(2)	1	4	3	2
(3)	1	4	2	3
(4)	4	1	3	2

BBG147

41. Which of the following option gives correct information about the coverings of thoracic cage on various surfaces :-

	Anterior surface	Posterior surface	Dorsal surface
(1)	Sternum and Ribs	Ribs	Diaphragm
(2)	Sternum and Ribs	Diaphragm	Clavicle bones and neck
(3)	Sternum and Ribs	Vertebral column	Clavicle bones and neck
(4)	Clavicle bones and neck	Diaphragm	Vertebral column

BBG148

42. Considering the following labels for mentioned structures and select the correct sequence of their arrangement :-

A = Terminal bronchioles

B = Respiratory bronchioles

C = Tertiary bronchi

D = Primary bronchi

E = Secondary bronchi

F = Total pulmonary bronchioles

(1) D $\longrightarrow$ E $\longrightarrow$ C $\longrightarrow$ F $\longrightarrow$ A $\longrightarrow$ B

(2) A $\longrightarrow$ B $\longrightarrow$ D $\longrightarrow$ E $\longrightarrow$ C $\longrightarrow$ F

(3) C $\longrightarrow$ F $\longrightarrow$ A $\longrightarrow$ B $\longrightarrow$ D $\longrightarrow$ E

(4) D $\longrightarrow$ E $\longrightarrow$ F $\longrightarrow$ C $\longrightarrow$ B $\longrightarrow$ A

BBG149

43. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

(a) All the time of inspiration, contraction in diaphragm and external intercostal muscles take place.

(b) Normal breathing is also called as abdominal breathing.

(c) Expiration during sneezing, coughing, yoga is normal expiration during which IICM and abdominal muscles relax.

(d) Inspiration can occur if the pressure within lungs (intra pulmonary pressure) is more then the atmospheric pressure.

(1) Statements (a), (b), (c), (d)

(2) Statements (a) only

(3) Statements (a), (b), (c)

(4) Statements (a) and (b)

BBG150

44. Find out the **correct** match from the following table :-

	Column-I	Column-II	Column-III
(i)	Tidal volume (TV)	(EC - ERV)	500 ml
(ii)	Expiratory reserve volume (ERV)	(VC - IRV - TV)	3000 ml
(iii)	Vital capacity (VC)	IRV + ERV + RV	4500 ml
(iv)	Residual volume (RV)	FRC - ERV	1200 ml

(1) (i) and (ii)

(2) (i), (iii) and (iv)

(3) (i), (ii) and (iii)

(4) (i) and (iv)

BBG151

45. Find out the **correct** match from the following:

(i) **Larynx** – sound box

(ii) **Trachea** – complete cartilaginous rings

(iii) **Epiglottis** – elastic cartilaginous flap

(iv) **Alveoli** – very thin, irregular walled, vascularised bag like structures

(v) **Pleural fluid** – reduces friction on the lung surface

(1) (i), (ii), (iii), (iv), (v)

(2) (i), (iii), (iv), (v)

(3) (i), (iii), (v)

(4) (i) and (v)

BBG152

46. Which of the following statement is **not correct**?

(1) Formation of oxyhaemoglobin is a process of oxidation.

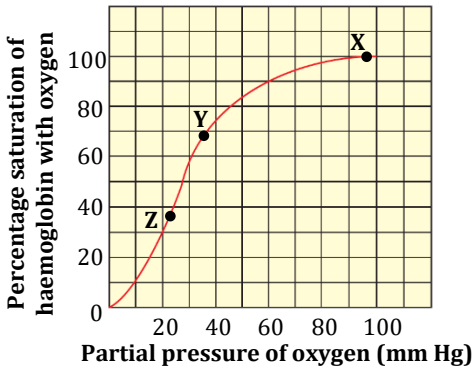
(2) Every 100 ml of oxygenated blood can deliver around 5 ml of O_2 to tissue under physiological condition.

(3) Dissociation curve is curve between percentage saturation of Hb with oxygen and partial pressure of oxygen.

(4) High concentration of CO_2 activates dissociation of oxyhaemoglobin is called Bohr's effect.

BBG153

47. Given below graph shows an oxygen dissociation curve :-



Where in the body will haemoglobin be saturated at the percentage shown at points X, Y and Z.

(1) X-Pulmonary artery, Y-Pulmonary vein, Z-Carotid artery

(2) X-Systemic artery, Y-Pulmonary artery, Z-Pulmonary vein

(3) X-Pulmonary vein, Y-Systemic vein, Z-Systemic vein during exercise

(4) X-Left ventricle, Y-Right ventricle, Z-Systemic artery

BBG154

48. Which one of the following statements is **incorrect** ?

(1) The residual air in lungs slightly decreases the efficiency of respiration in mammals

(2) The presence of non–respiratory air sacs, increases the efficiency of respiration in birds

(3) In insects, circulating body fluids serve to distribute oxygen to tissues

(4) The principle of counter current flow facilitates efficient respiration in gills of fishes

BBG073

49. The majority of carbon dioxide produced by our body cells is transported to the lungs :

(1) Dissolved in the blood

(2) As bicarbonates

(3) As carbonates

(4) Attached to haemoglobin

BBG074

50. What is vital capacity of our lungs ?

(1) Inspiratory reserve volume plus expiratory reserve volume.

(2) Total lung capacity minus Residual volume.

(3) Inspiratory reserve volume plus Tidal volume.

(4) Total lung capacity minus Expiratory reserve volume.

BBG077

51. Listed below are four respiratory capacities (a-d) and four jumbled respiratory volumes of a normal human adult :

	Respiratory capacities	Respiratory volumes
(a)	Residual volume	2500 mL
(b)	Vital capacity	3500 mL
(c)	Inspiratory reserve volume	1200 mL
(d)	Inspiratory capacity	4500 mL

Which one of the following is the **correct** matching of two capacities and volumes?

(1) (a) 4500 mL, (b) 3500 mL

(2) (b) 2500 mL, (c) 4500 mL

(3) (c) 1200 mL, (d) 2500 mL

(4) (d) 3500 mL, (a) 1200 mL

BBG078

52. What is true about RBCs in humans ?

(1) They do not carry CO_2 at all.

(2) They carry about 20-25 per cent of CO_2.

(3) They transport 99.5 per cent of O_2.

(4) They transport about 80 percent oxygen only and the rest 20 per cent of it is transported in dissolved state in blood plasma.

BBG079

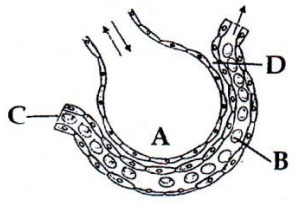
53. Two friends are eating together on a dining table. One of them suddenly starts coughing while swallowing some food. This coughing would have been due to improper movement of-

(1) Epiglottis (2) Diaphragm

(3) Neck (4) Tongue

BBG080

54. The figure given below shows a small part of human lung where exchange of gases takes place. In which one of the options given below, the one part, A, B, C or D is correctly identified along with its function



Options :

(1) C $\rightarrow$ Arterial capillary-passes oxygen to tissues

(2) A $\rightarrow$ Alveolar cavity-main site of exchange of respiratory gases

(3) D $\rightarrow$ Capillary wall-exchange of O_2 and CO_2 takes place here

(4) B $\rightarrow$ Red blood cell-transport of CO_2 mainly

BBG081

55. A large proportion of oxygen is left unused in the human blood even after its uptake by the body tissues. This O_2 :

(1) Acts as a reserve during muscular exercise

(2) Raises the pCO_2 of blood to 75 mm of Hg.

(3) Is enough to keep oxyhaemoglobin saturation at 96%

(4) Helps in releasing more O_2 to the epithelial tissues

BBG082

56. Bulk of carbon dioxide (CO_2) released from body tissues into the blood is present as :

(1) Carbamino-haemoglobin in RBCs

(2) Bicarbonate in blood plasma and RBCs

(3) Free CO_2 in blood plasma

(4) 70% carbamino-haemoglobin and 30% as bicarbonate

BBG083

57. Which one of the following is the correct statement for respiration in humans ?

(1) Workers in grinding and stone-breaking industries may suffer from lung fibrosis

(2) Abut 90% of carbon dioxide (CO_2) is carried by haemoglobin as carbamino haemoglobin

(3) Cigarette smoking may lead to inflammation of bronchi

(4) Neural signals from pneumotaxic centre in pons region of brain can increase the duration of inspiration

BBG084

58. People who have migrated from the planes to an area adjoining Rohtang pass about six months back:

(1) suffer from altitude sickness with symptoms like nausea, fatigue, etc.

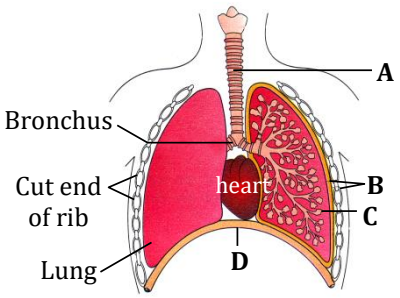
(2) have the usual RBC count but their haemoglobin has very high binding affinity to O_2

(3) have more RBCs and their haemoglobin has a lower binding affinity to O_2

(4) are not physically fit to play games like football.

BBG085

59. The figure shows a diagrammatic view of human respiratory system with labels A, B, C and D. Select the option which gives correct identification and main function and/or characteristic :-



(1) **D** – Lower end of lungs – diaphragm pulls it down during inspiration

(2) **A** – trachea - long tube supported by complete cartilaginous rings for conducting inspired air

(3) **B** – pleural membrane - surround ribs on both sides to provide cushion against rubbing

(4) **C** – Alveoli - thin walled vascular bag like structures for exchange of gases

BBG086

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	2	3	1	2	1	4	4	3	1	4	1	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	2	2	1	4	2	3	3	1	3	2	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	2	4	1	2	3	3	4	2	4	1	4	4	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Answer	1	3	3	2	2	4	2	1	2	1	2	1	3	4	

Exercise - I (Conceptual Questions) Build Up Your Understanding

BLOOD

1. The normal Albumin/Globulin ratio in blood is :-
(1) 2 : 1 (2) 1 : 2
(3) 1 : 4 (4) 1 : 5
BFC001

2. Eosinophilia is caused by :-
(1) Taeniasis (2) Ascariaris
(3) Allergy (4) All of above
BFC002

3. Blood group Antigen are :-
(1) Found in Hb molecule
(2) Found in Plasma protein
(3) Found on RBC
(4) None
BFC003

4. In normal healthy female, the number of RBC/mm^3 of blood is :
(1) 6.5-7.0 million (2) 5.5-6.0 million
(3) 4.5-5.0 million (4) 3.5-4.0 million
BFC007

5. Blood colloidal osmotic pressure mainly maintained by which plasma protein :-
(1) Globulin (2) Albumin
(3) Fibrinogen (4) Prothrombin
BFC008

6. Mammalian RBC are :-
(1) Biconcave, circular, non-nucleated
(2) Biconcave, Nucleated
(3) Oval, Nucleated
(4) None **BFC009**

7. Globulin protein of blood plasma mainly involved in the :-
(1) Clotting
(2) Osmotic balance
(3) Defence mechanism
(4) None
BFC010

8. Which WBCs resist infections and are also associated with allergic reactions?
(1) Lymphocytes (2) Neutrophils
(3) Eosinophils (4) Monocytes
BFC011

9. Persons with _____ and _____ blood group are called universal recipients & universal donors respectively :-
(1) AB^-, O^+ (2) O^+, AB^-
(3) O^-, AB^+ (4) AB^+, O^-
BFC012

10. ABO blood grouping is based on :
(1) Surface antibodies on RBC.
(2) Surface antigen on WBC.
(3) Surface antigen on RBC.
(4) Plasma antigens.
BFC013

11. Which leucocyte has bean shaped nucleus?
(1) Basophil (2) Monocyte
(3) Neutrophil (4) Lymphocyte
BFC014

12. Smallest blood element :-
(1) RBC (2) WBC
(3) Platelets (4) None
BFC015

13. Blood clotting requires :-
(1) $Na^+ + K^+$
(2) Na^+ + Prothrombin
(3) Na^+ + Thromboplastin
(4) Ca^{++} + Thromboplastin
BFC016

14. Lymph differ from blood in possessing :
(1) Only WBC
(2) More RBC & WBC
(3) More RBC & few WBC
(4) More WBC & few RBC
BFC017

15. Blood platelets found in :-
(1) Pisces (2) Reptiles
(3) Birds (4) Mammals
BFC018

16. Diapedesis means :-
(1) Formation of WBC
(2) Formation of RBC
(3) Process by which certain WBCs squeeze through thin capillary wall
(4) Movement of food in gut

BFC019

17. Process by which blood cells are formed in bone marrow :-
(1) Haemopoiesis (2) Haemolysis
(3) Thrombopoiesis (4) Erythroblastosis

BFC021

18. Largest leucocytes :-
(1) Neutrophil (2) Basophil
(3) Monocyte (4) Lymphocyte

BFC022

19. Content of haemoglobin / 100 ml of Blood -
(1) 15 gm (2) 20 gm (3) 10 gm (4) 5 gm

BFC023

20. In comparison to WBC, RBC have :-
(1) Antigen (Agglutinogen) surface on RBC
(2) Carbonic anhydrase
(3) Donnan's membrane
(4) All of above

BFC027

21. One is more in lymph than blood :
(1) RBC (2) Nutrients
(3) Lipids (4) Oxygen

BFC028

22. Serum is :-
(1) Blood – Blood cells
(2) Plasma – Fibrinogen
(3) Blood – Plasma
(4) Blood – RBC

BFC029

23. Blood bank of body is :-
(1) Liver (2) Spleen
(3) Heart (4) Bone marrow

BFC030

24. Ist site of haemopoesis :-
(1) Bone marrow (2) Spleen
(3) Liver (4) Yolk sac

BFC033

25. Blood cells are produced by Bone marrow in
(1) All bones (2) Some bones
(3) Most of the bones (4) None

BFC035

26. Blood :–
(1) Contains plasma
(2) Contains corpuscles
(3) Contains proteins
(4) All of the above

BFC037

27. A reduction in platelets number causes
(1) Clotting disorder
(2) Immune disorder
(3) Digestive disorder
(4) Respiratory disorder

BFC038

28. Which WBC increase in Allergy?
(1) Acidophil (2) Basophil
(3) Lymphocyte (4) Neutrophil

BFC040

29. How many polypeptide chains are present in single molecule of Haemoglobin protein:-
(1) 1 (2) 3 (3) 4 (4) 2

BFC041

30. Thromboplastin is secreted by :–
(1) Kidney (2) Platelets
(3) Leucocyte (4) Erythrocyte

BFC042

31. Antibody are absent in which blood group:
(1) A (2) B (3) AB (4) O

BFC044

32. Blood group 'A' can receive blood from which group :-
(1) A, AB, O (2) A, O
(3) O (4) B, AB

BFC047

33. Which is not a plasma protein :–
(1) Heparin (2) Albumin
(3) Prothrombin (4) Fibrinogen

BFC048

34. Megakaryocyte cell is?
(1) RBC producer
(2) Thrombocyte producer
(3) WBC producer
(4) Protein producer
BFC049

35. Person having 'B' blood group have antibody:
(1) Anti A (2) Anti B
(3) Both (4) None
BFC050

36. In which pair erythroblastosis foetalis can occur :-
(1) Rh^+ male & Rh^- female
(2) Rh^- male & Rh^- female
(3) Rh^+ male & Rh^+ female
(4) Rh^- male & Rh^+ female
BFC053

37. Blood of AB blood group can be donated to:
(1) A (2) B (3) AB (4) O
BFC054

38. The Rh antibodies from the mother (Rh – ve) can leak into the blood of the foetus (Rh + ve) and destroy the:-
(1) Foetal RBCs (2) Mother RBCs
(3) Foetal WBCs (4) Both (1) and (2)
BFC055

39. Agranulocytes are :-
(1) Eosinophils and neutrophils
(2) Monocytes and lymphocytes
(3) Eosinophils and lymphocytes
(4) Lymphocytes and basophils.
BFC056

40. Platelets are a source of :
(1) Fibrinogen (2) Calcium
(3) Thromboplastin (4) Haemoglobin
BFC057

41. Which is unrelated to blood coagulation ?
(1) Fibrinogen (2) Fibrin
(3) Bilirubin (4) Calcium
BFC058

42. Which of the following are involved in body defence :-
(1) Neutrophils (2) Lymphocytes
(3) Macrophages (4) All the above.
BFC062

43. Prothrombin, albumin and fibrinogen are synthesised by :-
(1) Pancreas (2) Bone marrow
(3) Spleen (4) Liver
BFC064

44. Which one is a factor for maturation of erythrocytes?
(1) Vitamin B_{12} (2) Vitamin A
(3) Vitamin D (4) Vitamin C.
BFC065

45. In which state iron is present in haemoglobin:
(1) Unionic (2) Fe^{2+}
(3) Fe^{3+} (4) None of the above
BFC066

46. Immature RBCs of mammals have :
(1) No nucleus
(2) Single beaded nucleus
(3) Many nuclei
(4) Single nucleus
BFC067

47. Megakaryocytes :-
(1) Produce leucocytes.
(2) Forms blood platelets
(3) Are carriers of oxygen.
(4) Are carriers of carbon dioxide.
BFC068

48. During blood clotting, fibrin is produced by
(1) Thrombin (2) Prothrombin
(3) Liver (4) Proteolysis
BFC069

49. RBCs are nucleated in :
(1) Man (2) Rabbit
(3) Rat (4) Frog
BFC072

50. An anticoagulant is :
(1) Heparin (2) Hirudin
(3) EDTA (4) All the above
BFC073

51. Blood has a pH of :-
(1) 7.4 (2) 7.8 (3) 6.9 (4) 6.3
BFC075

52. The RBCs in human are :-
(1) Oval
(2) Circular, biconcave and nucleated
(3) Circular, biconcave and nonnucleated
(4) Oval, nonnucleated, Circular
BFC076

53. Bilirubin and biliverdin are derived from :-
(1) Globin (2) Haem
(3) Iron (4) Fat.
BFC077

54. Protein required for coagulation of blood is
(1) Haemoglobin (2) Globulin
(3) Fibrinogen (4) Albumin
BFC078

55. Globulin is :-
(1) Plasma protein
(2) Antigen
(3) Serum
(4) Found in lymphatic tissue.
BFC079

56. To prevent clotting, donor's blood is treated with-
(1) Sodium glycocholate
(2) Sodium hydroxide
(3) Heparin
(4) Sodium taurocholate.
BFC080

57. Continuous bleeding from an injured part of body is due to deficiency of :-
(1) Vitamin–A (2) Vitamin–B
(3) Vitamin–K (4) Vitamin–E
BFC081

58. Abnormal increase in number of RBC in blood is called
(1) Anaemia (2) Polycythemia
(3) Leukemia (4) Sarcoma
BFC082

59. Liquid which remain after clotting of blood is called as :-
(1) Serum (2) Plasma
(3) Lymph (4) Blood
BFC083

60. Which of the following substances, if introduced into the blood stream, would cause coagulation of blood at the site of its introduction?
(1) Thromboplastin (2) Fibrinogen
(3) Heparin (4) Prothrombin
BFC084

TYPES OF CIRCULATION

61. Closed circulatory system occurs in
(1) Cockroach (2) Tadpole/Fish
(3) Mosquito (4) Housefly
BFC085

62. Systemic heart refers to :-
(1) The heart that contracts under stimulation from nervous system
(2) Left auricle and left ventricle in higher vertebrates
(3) Entire heart in lower vertebrates
(4) The two ventricles together in humans
BFC086

STRUCTURE OF HEART, HEART BEAT, CONDUCTING SYSTEM

63. Where is the pace maker situated :-
(1) In left auricle near opening of pulmonary vein
(2) In right auricle near eustachian valve
(3) On inter - auricular septum
(4) On inter-ventricular septum
BFC087

64. Papillary muscles are found in :-
(1) Haemocoel of cockroach
(2) Auricles of heart
(3) Ventricles of heart
(4) Arm
BFC088

65. To reach the left side of heart the blood must pass through :-
(1) Sinus venosus (2) Kidneys
(3) Liver (4) Lungs
BFC089

66. Characteristics of cardiac muscles are that they :-
(1) Contract quickly and get fatigued
(2) Contract quickly and do not get fatigue
(3) Contract slowly and get fatigued
(4) Contract slowly and do not get fatigue
BFC090

67. In heart of Human, bicuspid valve is situated between :-
(1) Right auricle and pulmonary aorta
(2) Post caval and auricle
(3) Left auricle and left ventricle
(4) Right auricle and right ventricle
BFC091

68. When the right ventricle contracts the blood is pump into :-
(1) Superior vena cava
(2) Dorsal aorta
(3) Pulmonary aorta
(4) Pulmonary veins
BFC092

69. Coronary artery supplies blood to :-
(1) Mammary glands (2) Rib muscles
(3) Skin (4) Heart muscles
BFC096

70. In children, heart rate is :-
(1) More than adult (2) Less than adult
(3) Equal to adult (4) None of these
BFC097

71. The wall of Human heart is thick due to presence of :-
(1) Inner layer endocardium
(2) Middle layer myocardium
(3) Outer most layer pericardium
(4) Outer layer epicardium
BFC098

72. The pulmonary aorta arise from :-
(1) Left ventricle (2) Right ventricle
(3) Left auricle (4) Right auricle
BFC099

73. When right ventricle of human heart contract then blood pumped into :-
(1) All parts of body (2) Lungs
(3) Pulmonary veins (4) Systemic aorta
BFC100

74. Bundle of His originates from :-
(1) Sinu-auricular node
(2) Auriculo-ventricular node
(3) Pulmonary aorta
(4) Systemic aorta
BFC101

75. The papillary muscles are helpful in :-
(1) Movement of eye balls
(2) Movement of eye lids
(3) Closing & opening the valves of heart
(4) Movement of pinnae
BFC104

76. The valves of the heart are attached to papillary muscles by :-
(1) Columnae carnae
(2) Chordae tendinae
(3) Tendinae
(4) Pectinate muscles
BFC106

77. Which has the thickest walls :-
(1) Right auricle (2) Left auricle
(3) Right ventricles (4) Left ventricle
BFC107

78. Blood supply to heart musculature is via :-
(1) Cardiac artery (2) Coronary artery
(3) Aorta (4) Pulmonary vein
BFC108

79. The mitral valve is supported by :-
(1) Bundle of HIS (2) Ductus Arteriosus
(3) Foramen ovale (4) Chordae tendinae
BFC112

80. Normal Heart rate in a two months old infant is
(1) <72/min. (2) 60 to 72/min.
(3) >72/min. (4) 16/min.
BFC113

81. The largest and the thickest heart chamber is
(1) Left ventricle (2) Left atrium
(3) Righ atrium (4) Right ventricle
BFC114

82. Pace maker is :-
(1) Instrument for measuring heart beat
(2) Instrument for measuring pulse rate
(3) Auriculo-ventricular node that provides impulse for heart beat
(4) Sino-auricular node that provides impulse for heart beat
BFC115

83. Tricupsid valve is found in between :
(1) Sinus venosus and right auricle
(2) Right auricle and right ventricle
(3) Left ventricle and left auricle
(4) Ventricle and aorta
BFC116

84. Origin of heart beat and its conduction is represented by :-
(1) AV node → Bundle of His → SA node → Purkinje fibres
(2) SA node → Purkinje fibres → AV node → Bundle of His
(3) Purkinje fibres → AV node → AV node → Bundle of His
(4) SA node → AV node → Bundle of His → Purkinje fibres
BFC117

85. The hormone that stimulates heart beat is:
(1) Insulin (2) Adrenaline
(3) Glucagon (4) Gastrin
BFC118

86. Heart beat is accelerated by :-
(1) Sympathetic nerves and noradrenaline
(2) Cranial nerves and adrenaline
(3) Cranial nerves and acetylcholine
(4) Sympathetic nerves and acetylcholine
BFC119

87. Neurogenic heart is characteristic of :-
(1) Humans (2) Invertebrates
(3) Rat (4) Rabbit
BFC120

88. In circulatory system, valves occur in
(1) Heart and blood vessels of both vertebrates and invertebrates as well as vertebrate lymphatics
(2) Both vertebrate and invertebrate hearts
(3) Vertebrate heart only
(4) Both vertebrate and invertebrate hearts and their blood vessels.
BFC121

89. Pericardial fluid is secreted by :-
(1) Myocardium
(2) Parietal peritoneum
(3) Visceral peritoneum
(4) Pericardium
BFC122

90. Which one generates heart beat?
(1) Purkinje fibres
(2) Cardiac branch of vagus nerve
(3) SA node
(4) AV node
BFC123

91. Match the columns :-

Column I	Column II
a. Superior Vena Cava/SVC	p. Carries deoxygenated blood to lungs
b. Inferior Vena blood Cava/IVC	q. Carries oxygenated from lungs
c. Pulmonary Artery	r. Brings deoxygenated blood from lower parts of body to right atrium
d. Pulmonary Vein	s. Brings deoxygenated blood from upper parts of body into right atrium

(1) a—q, b—s, c—r, d—p
(2) a—s, b—p, c—q, d—r
(3) a—s, b—r, c—p, d—q
(4) a—s, b—p, c—r, d—q
BFC125

92. Blood vessel which brings oxygenated blood to left auricle is :-
(1) Precaval vein/SVC
(2) Post caval vein/IVC
(3) Pulmonary vein
(4) Pulmonary artery
BFC126

93. Bundle of His is a network of :-
(1) Muscle fibres distributed throughout the heart walls
(2) Muscle fibres found only in the inter ventricular septum
(3) Nerve fibres distributed in ventricles
(4) Nerve fibres found throughout the heart
BFC128

94. The cardiac impulse that results into the heart beat is delayed at :-
(1) Internodal tract
(2) AV node
(3) Bundle of His
(4) Purkinje fibres
BFC130

95. Bicuspid (mitral) valve guards the opening in mammals between :-
(1) Left atrium and left ventricle
(2) Pulmonary vein and left auricle
(3) Stomach and intestine
(4) Right atrium and right ventricle
BFC131

96. "Bundle of His" are :-
(1) nervous tissue supplied to ventricles
(2) nervous tissue supplied to heart
(3) muscular tissue supplied to ventricles
(4) muscular tissue supplied to heart
BFC132

97. The heart beat of which animal is myogenic in nature :-
(1) Cockroach (2) Leech
(3) Elephant (4) All of these
BFC134

REGULATION OF HEART BEAT, CARDIAC CYCLE AND HEART SOUNDS

98. Blood pressure and heart beat is influenced by:
(1) Insulin
(2) Adrenaline
(3) Optic nerve
(4) Growth hormone
BFC135

99. The heart sound "DUP" is produced when :-
(1) Mitral valve opens
(2) Mitral valve closes
(3) Semilunar valve at the base of aorta closes
(4) Tricuspid valve opens
BFC137

100. When heart beat decreases than normal is called -
(1) Bradycardia (2) Tachycardia
(3) Hypocardia (4) Nicardia
BFC138

101. The 'Lubb' and "Dup" heart sound are due to:-
(1) Opening of heart valves
(2) Action of papillary muscles
(3) Closing of heart valves
(4) Activity of pace maker
BFC139

102. Normal Cardiac output is :-
(1) 15 Litres/min. (2) 5 Litres × 72/min.
(3) 5 Litres/min. (4) 5/72 Litres/min.
BFC140

103. Acetylcholine causes :-
(1) Bradycardia (2) Tachycardia
(3) Both (4) None
BFC141

104. Ist Heart sound is :-
(1) 'LUBB' at end of systole
(2) 'DUBB' at end of systole
(3) 'LUBB' at beginning of Ventricular systole
(4) 'DUP' at beginning of Ventricular systole
BFC142

105. Heart beat becomes faster on stimulation by
(1) Sympathetic nerves and adrenaline
(2) Sympathetic and parasympathetic nerves
(3) Parasympathetic nerves and epinephrine
(4) Parasympathetic nerves and acetylcholine

BFC143

106. The sound of lubb is produced during closure of :
(1) Bicuspid valve (2) Tricuspid valve
(3) Semilunar valves (4) Both (1) and (2)

BFC144

107. In diastole, the Heart is filled with the blood. This blood is :-
(1) Deoxygenated
(2) Venous blood
(3) Oxygenated blood
(4) Partial oxygenated blood

BFC146

108. In human oxygenated blood flows from :–
(1) Left auricle to left ventricle during auricular systole
(2) Right auricle to right ventricle during ventricular systole
(3) Right ventricle to aorta during ventricular systole
(4) Pulmonary vein to left auricle during auricular systole

BFC147

109. A heart "murmur" disorder indicates a defect of :-
(1) Bundle of His
(2) Heart valves
(3) Sinuauricular node
(4) Atrioventricular node

BFC149

110. Blood enters into the heart because muscles of :
(1) Atria relax (2) Ventricle contract
(3) Ventricle relax (4) Atria contract

BFC150

BLOOD PRESSURE, BLOOD VESSELS, PORTAL SYSTEM AND LYMPHATIC SYSTEM AND OTHERS

111. Blood Capillaries are made of :-
(1) Endothelium and thin coat of connective tissue
(2) Endothelium and thin coat of muscle fibres
(3) Endothelium and thin coat of connective tissue and muscle fibres.
(4) Only endothelium

BFC151

112. Cardiac center is present in :
(1) Cerebrum
(2) Medulla oblongata
(3) Pons
(4) Epithalamus

BFC152

113. Pulmonary veins are those which :-
(1) Carry deoxygenated blood from lungs to heart
(2) Carrying oxygenated blood From lungs to heart
(3) Carry deoxygenated blood from heart to lung
(4) Carry oxygenated blood from heart to lungs

BFC153

114. Oxygenated blood is carried by :-
(1) Pulmonary artery
(2) Pulmonary vein
(3) Renal vein
(4) Hepatic portal vein

BFC154

115. Lymph can be defined as :-
(1) Blood minus corpuscles
(2) Blood minus Plasma
(3) Blood minus WBC
(4) Blood minus RBC & Platelets

BFC155

116. Which has no muscular walls :-
(1) Capillary (2) Arteriole
(3) Veins (4) Artery
BFC157

117. Pulse beat is measured in :-
(1) Veins (2) Artery (Radial)
(4) Nerve (4) Capillary
BFC158

118. In which of the following character a vein differs from an artery :-
(1) Having valves to control flow of blood
(2) Having narrow lumen
(3) Having muscular wall
(4) Having pigmented wall to give dark look
BFC160

119. Systolic pressure is higher than diastolic pressure due to :-
(1) Volume of blood in the heart is greater during systole
(2) Arteries contract during systole
(3) Blood vessels offer resistance to flowing blood during systole
(4) Blood is forced into arteries during systole.
BFC161

120. The hepatic portal system is :-
(1) Between digestive tract and hypothalamus
(2) Between digestive tract and liver
(3) Between liver and hypothalamus
(4) Between liver and Pituitary
BFC162

121. Which artery supplies blood to the diaphragm :-
(1) Phrenic (2) Splenic
(3) Renal (4) Caudal
BFC163

122. Which one of the following organ can be called a sort of "blood bank"?
(1) Heart (2) Liver
(3) Spleen (4) Lungs
BFC164

123. All arteries carry oxygenated blood except:
(1) Systemic (2) Hepatic
(3) Pulmonary (4) Cardiac
BFC166

124. Coagulation of lymph is :-
(1) Faster than blood
(2) Not possible
(3) Slower than blood
(4) A passive process
BFC168

125. An artery can be distinguished from a vein in having
(1) Thicker wall
(2) Lesser lumen
(3) No valves
(4) All of the above
BFC169

126. Removal of which organ will have least effect in an adult Human :-
(1) Spleen (2) Liver
(3) Pancreas (4) Pituitary
BFC171

127. Which one of the following is the main graveyard of RBC :-
(1) Bone marrow
(2) Spleen
(3) Liver
(4) Kidney
BFC172

128. A portal system is that in which :-
(1) A vein begins from an organ and ends in heart
(2) An artery breaks up in an organ & restarts by the union of its capillaries
(3) The blood from gut is brought in to kidneys before it is poured in to heart
(4) A vein breaks up in an organ into capillaries & restarts by their union as a new vein in the same organ
BFC174

129. Indicate correct statement for Human :-
(1) Arteries always carry oxygenated blood while veins always carry deoxygenated blood
(2) Venous blood is returned to left auricle
(3) Arteries are provided with valves while veins are devoid of valves
(4) Arteries always carry blood away from the heart, while veins always carry blood towards the heart

BFC175

130. Blood circulation that starts in capillaries and ends in capillaries is called :-
(1) Portal circulation
(2) Hepatic circulation
(3) Cardiac arrest
(4) Coronary circulation

BFC176

131. What is true about vein
(1) All veins carry deoxygenated blood
(2) All veins carry oxygenated blood
(3) They carry blood from organs towards heart
(4) They carry blood from heart towards organs

BFC177

132. In mammals the role of spleen is :-
(1) Graveyard of RBC
(2) Reservoir of blood
(3) Haemopoietic organ
(4) All of the above

BFC178

133. Which of the following is valve less?
(1) Arteries
(2) Veins
(3) Lymphatics
(4) Chambers in Heart

BFC179

134. Which of the following carries only deoxygenated blood?
(1) Carotid artery
(2) Pulmonary artery
(3) Pulmonary vein
(4) Aorta

BFC180

135. Pericardium in made up of -
(1) Simple cuboidal epithelium
(2) Simple squamous epithelium
(3) Simple columnar epithelium
(4) Stratified squamous epithelium

BFC181

136. Lymph :
(1) Transports O_2 to brain
(2) Transports CO_2 to lungs
(3) Returns interstitial fluid to blood
(4) Returns RBCs and WBCs to lymph nodes

BFC182

137. Glucose is carried from digestive tract to liver by-
(1) Hepatic artery
(2) Hepatic portal vein
(3) Pulmonary vein
(4) Pulmonary artery

BFC183

138. Pulmonary artery differs from pulmonary vein in having
(1) Thick wall (2) Thin wall
(3) Valves (4) Both (2) and (3)

BFC184

139. All veins have deoxygenated blood except
(1) Renal vein
(2) Hepatic vein
(3) Hepatic portal vein
(4) Pulmonary veins

BFC186

140. Normal pulse pressure is
(1) 80 mm Hg (2) 120 mm Hg
(3) 40 mm Hg (4) 320 mm Hg

BFC187

141. Fully digested food reaches to liver by :-
(1) Hepatic portal vein
(2) Hepatic artery
(3) Hepatic vein
(4) All the above

BFC188

142. Which of the following statement is true for Lymph?
(1) WBC and serum
(2) All components of blood except RBCs, Platelets and some proteins
(3) RBCs, WBCs and Plasma
(4) RBCs, Proteins and Platelets

BFC189

143. Lymph vessels pour their materials in :-
(1) Sub clavian vein
(2) Pulmonary artery
(3) Artery which enters in legs
(4) Right ventricle

BFC190

144. Hepatic portal system starts from :-
(1) Digestive system to liver
(2) Kidney to liver
(3) Liver to heart
(4) Liver to kidney

BFC191

145. Blood leaving liver and moving to heart will have more concentration of :-
(1) Bile
(2) Urea
(3) Glycogen
(4) Amino acid

BFC192

146. In the veins, the blood is under low pressure and blood flow is ____ due to this pressure.
(1) Slow
(2) Rapid
(3) Frequently
(4) Remains constant

BFC193

147. The structure of which of the following consist of a layer of single cell thickness ?
(1) Blood capillary (2) Artery
(3) Aorta (4) arteriole

BFC194

148. Coronary artery disease is due to :
(1) *Streptococci* bacteria
(2) Inflammation of pericardium
(3) Weakening of the heart valves
(4) Insufficient blood supply to the heart muscles

BFC195

149. An artery is a vessel that carries blood :
(1) Away from the heart
(2) Towards the heart
(3) Which is deoxygenated without any exception
(4) None of these

BFC196

150. Which one indicates the hypertension ?
(1) 90/60
(2) 120/85
(3) 110/70
(4) 140/100

BFC197

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	3	2	1	3	3	4	3	2	3	4	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	3	1	4	3	2	2	4	2	4	1	1	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	1	2	1	1	3	1	2	3	3	4	4	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	2	1	4	4	1	3	2	3	1	3	3	2	1	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	2	2	3	4	2	3	3	4	1	2	2	2	2	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	4	2	4	3	1	4	2	4	2	1	2	1	4	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	3	2	2	1	3	3	2	3	1	3	3	1	3	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	4	2	1	2	1	4	2	2	2	4	1	2	1	4	2
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	1	3	3	3	4	1	2	4	4	1	3	4	1	2	2
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	3	2	1	4	3	1	2	1	1	2	1	1	4	1	4

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. How do parasympathetic neural signals affect the working of the heart ?
(1) Reduce both heart rate and cardiac output.
(2) Heart rate is increased without affecting the cardiac output.
(3) Both heart rate and cardiac output increase.
(4) Heart rate decreases but cardiac output increases.

BFC208

AIPMT 2015

2. Blood pressure in the mammalian aorta is maximum during :
(1) Diastole of the right ventricle
(2) Systole of the left ventricle
(3) Diastole of the right atrium
(4) Systole of the left atrium

BFC209

NEET-I 2016

3. Blood pressure in the pulmonary artery is :-
(1) same as that in the aorta.
(2) more than that in the carotid.
(3) more than that in the pulmonary vein.
(4) less than that in the venae cavae.

BFC210

NEET-II 2016

4. Name the blood cells, whose reduction in number can cause clotting disorder, leading to excessive loss of blood from the body.
(1) Neutrophils (2) Thrombocytes
(3) Erythrocytes (4) Leucocytes

BFC211

5. Serum differs from blood in :-
(1) Lacking clotting factors
(2) Lacking antibodies
(3) Lacking globulins
(4) Lacking albumins

BFC212

NEET (UG) 2017

6. Adult human RBCs are enucleated. Which of the following statement(s) is/are **most appropriate** explanation for this feature ?
(a) They do not need to reproduce
(b) They are somatic cells
(c) They do not metabolize
(d) All their internal space is available for oxygen transport
(1) only (a) (2) (a), (c) and (d)
(3) (b) and (c) (4) only (d)

BFC213

7. The hepatic portal vein drains blood to liver from :
(1) Stomach (2) Kidneys
(3) Intestine (4) Heart

BFC214

NEET (UG) 2018

8. Match the items given in Column I with those in Column II and select the ***correct*** option given below:

Column I	Column II
a. Tricuspid valve	i. Between left atrium and left ventricle
b. Bicuspid valve	ii. Between right ventricle and pulmonary artery
c. Semilunar valve	iii. Between right atrium and right ventricle

(1) a-iii, b-i, c-ii

(2) a-i, b-iii, c-ii

(3) a-i, b-ii, c-iii

(4) a-ii, b-i, c-iii

BFC216

9. Match the items given in Column I with those in Column II and select the ***correct*** option given below :-

	Column I		**Column II**
a.	Fibrinogen	i.	Osmotic balance
b.	Globulin	ii.	Blood clotting
c.	Albumin	iii.	Defence mechanism

(1) a-iii, b-ii, c-i (2) a-i, b-ii, c-iii

(3) a-i, b-iii, c-ii (4) a-ii, b-iii, c-i

BFC217

NEET(UG) 2019

10. Match the **Column - I** with **Column -II**

	Column - I		**Column - II**
(a)	P-wave	(i)	Depolarisation of ventricles
(b)	QRS complex	(ii)	Repolarisation of ventricles
(c)	T-wave	(iii)	Coronary ischemia
(d)	Reduction in the size of T-wave	(iv)	Depolarisation of atria
		(v)	Repolarisation of atria

Select the **correct** option :-

(1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(v)

(3) (a)-(ii), (b)-(i), (c)-(v), (d)-(iii)

(4) (a)-(ii), (b)-(iii), (c)-(v), (d)-(iv)

BFC218

NEET (UG) 2019 (ODISHA)

11. All the components of the nodal tissue are auto excitable. Why does the SA node act as the normal pacemakar?

(1) SA node has the lowest rate of depolarisation.

(2) SA node is the only component to generate the threshold potential.

(3) Only SA node can convey the action potential to the other components.

(4) SA node has the highest rate of depolarisation.

BFC219

12. A specialised nodal tissue embedded in the lower corner of the right atrium, close to Atrio-ventricular septum, delays the spreading of impulses to heart apex for about 0.1 sec. The delay allows.

(1) blood to enter aorta.

(2) the ventricles to empty completely.

(3) blood to enter pulmonary arteries.

(4) the atria to empty completely.

BFC220

NEET(UG) 2020

13. Match the following columns and select the correct option.

Column-I		**Column-II**	
A.	Eosinophils	I.	Immune response
B.	Basophils	II.	Phagocytosis
C.	Neutrophils	III.	Release histaminase, destructive enzymes
D.	Lymphocytes	IV.	Release granules containing histamine

(1) A-(II), B-(I), C-(III), D-(IV)
(2) A-(III), B-(IV), C-(II), D-(I)
(3) A-(IV), B-(I), C-(II), D-(III)
(4) A-(I), B-(II), C-(IV), D-(III)

BFC221

NEET (UG) 2020 (COVID-19)

14. Which of the following conditions cause erythroblastosis foetalis ?

(1) Mother Rh^{+ve} and foetus Rh^{-ve}
(2) Mother Rh^{-ve} and foetus Rh^{+ve}
(3) Both mother and foetus Rh^{-ve}
(4) Both mother and foetus Rh^{+ve}

BFC222

NEET (UG) 2021

15. Persons with 'AB' blood group are called as 'Universal recipients'. This is due to :-

(1) Absence of antigens A and B on the surface of RBCs
(2) Absence of antigens A and B in plasma
(3) Presence of antibodies, anti-A and anti-B, on RBCs
(4) Absence of antibodies, anti-A and anti-B, in plasma

BFC223

16. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins?

(1) Thrombin (2) Renin
(3) Epinephrine (4) Thrombokinase

BFC224

NEET (UG) 2022

17. Which one of the following statements is **correct** ?

(1) The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria
(2) Blood moves freely from atrium to the ventricle during joint diastole.
(3) Increased ventricular pressure causes closing of the semilunar valves.
(4) The atrio-ventricular node (AVN) generates an action potential to stimulate atrial contraction

BFC225

18. Given below are two statements:

Statement I: The coagulum is formed of network of threads called thrombins.

Statement II: Spleen is the graveyard of erythrocytes.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

BFC226

RE-NEET (UG) 2022

19. A unique vascular connection between the digestive tract and liver is called _____.

(1) Hepato–pancreatic system
(2) Hepatic portal system
(3) Renal portal system
(4) Hepato–cystic system

BFC227

20. Arrange the following formed elements in the decreasing order of their abundance in blood in humans :

(a) Platelets (b) Neutrophils
(c) Erythrocytes (d) Eosinophils
(e) Monocytes

Choose **the most appropriate answer** from the options given below :-

(1) (c), (a), (b), (e), (d)
(2) (c), (b), (a), (e), (d)
(3) (d), (e), (b), (a), (c)
(4) (a), (c), (b), (d), (e)

BFC228

NEET (UG) 2023

21. Match **List I** with **List II**.

	List I		List II
A.	P-wave	I.	Beginning of systole
B.	Q-wave	II.	Repolarisation of ventricles
C.	QRS complex	III.	Depolarisation of atria
D.	T-wave	IV.	Depolarisation of ventricles

Choose the **correct** answer from the options given below :-
(1) A-IV, B-III, C-II, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-I, C-IV, D-II

BFC292

22. Which of the following statements are correct?
A. Basophils are most abundant cells of the total WBCs
B. Basophils secrete histamine, serotonin and heparin
C. Basophils are involved in inflammatory response
D. Basophils have kidney shaped nucleus
E. Basophils are agranulocytes

Choose the **correct** answer from the options given below:
(1) C and E only (2) B and C only
(3) A and B only (4) D and E only

BFC293

NEET (UG) MANIPUR 2023

23. Match **List-I** with **List-II**.

	List-I (ECG)		List-II (Electrical activity of heart)
(A)	P-wave	(I)	Depolarisation of ventricles
(B)	QRS complex	(II)	End of systole
(C)	T wave	(III)	Depolarisation of atria
(D)	End of T wave	(IV)	Repolarisation of ventricles

Choose the **correct** answer from the options given below :
(1) A-(IV), B-(I), C-(III), D-(II)
(2) A-(I), B-(IV), C-(III), D-(II)
(3) A-(IV), B-(III), C-(I), D-(II)
(4) A-(III), B-(I), C-(IV), D-(II)

BFC294

24. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Eosinophils	(I)	6 – 8%
(B)	Lymphocytes	(II)	2 – 3%
(C)	Neutrophils	(III)	20 – 25%
(D)	Monocytes	(IV)	60 – 65 %

Choose the **correct** answer from the options given below :-
(1) A-(IV), B-(I), C-(II), D-(III)
(2) A-(IV), B-(I), C-(III), D-(II)
(3) A-(II), B-(III), C-(IV), D-(I)
(4) A-(II), B-(III), C-(I), D-(IV)

BFC295

NEET (UG) 2024

25. Following are the stages of pathway for conduction of an action potential through the heart :-

A. AV bundle B. Purkinje fibres
C. AV node D. Bundle branches
E. SA node

Choose the **correct** sequence of pathway from the options given below :-

(1) E-C-A-D-B (2) A-E-C-B-D
(3) B-D-E-C-A (4) E-A-D-B-C

BFC297

26. Match **List I** with **List II**.

List I		List II	
A.	P wave	I.	Heart muscles are electrically silent.
B.	QRS complex	II.	Depolarisation of ventricles.
C.	T wave	III.	Depolarisation of atria.
D.	T-P gap	IV.	Repolarisation of ventricles.

Choose the **correct** answer from the options given below :-

(1) A-I, B-III, C-IV, D-II
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-IV, B-II, C-I, D-III

BFC298

RE-NEET (UG) 2024

27. A person with blood group A Rh^- can receive the blood transfusion from which of the following types?

A. B Rh^- B. AB Rh^-
C. O Rh^- D. A Rh^-
E. A Rh^+

Choose the **correct** answer from the options given below :-

(1) D and E only (2) D only
(3) A and B only (4) C and D only

BFC299

28. 'Lub' sound of Heart is caused by the _____.

(1) closure of the semilunar valves.
(2) opening of tricuspid and bicuspid valves.
(3) opening of the semilunar valves.
(4) closure of the tricuspid and bicuspid valves.

BFC300

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	3	2	1	4	3	1	4	1	4	4	2	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	1	2	3	2	1	4	2	4	3	1	2	4	4		

Exercise - III

Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which of the following WBCs are phagocytic cells.

(a) Monocytes (b) Neutrophils

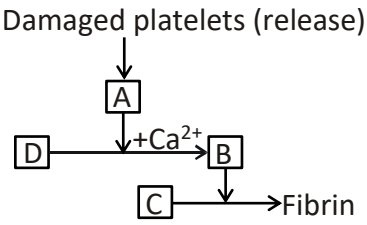
(c) Basophils (d) Eosinophils

(1) Only (a) (2) (a) and (b)

(3) (a) and (c) (4) (c) and (d)

BFC229

2. Identify A, B and C in the given below blood clotting process.



Options :-

	A	B	C
(1)	Thromoplastin	Prothrombin	Fibrinogen
(2)	Thrombin	Fibrinogen	Thrombo-Kinase
(3)	Thromboplastin	Thrombin	Fibrinogen
(4)	Prothrombin	Thrombin	Fibrinogen

BFC230

3. Platelets are cells fragments produced from:-

(1) Thrombocytes (2) Megakaryocytes

(3) Monocytes (4) Mast cell

BFC231

4. Read the following statements (A-D) :-

A. RBCs are the most abundant of all the cells in blood

B. A healthy adult man has on an average 5 billions to 5.5 billions of RBCs mm^{-3} of blood

C. RBCs are formed in liver in the adults

D. RBCs are devoid of nucleus in most of the mammals and are biconcave in shape

How many of the above statements are incorrect

(1) Four (2) Three

(3) Two (4) One

BFC232

5. Erythroblastosis foetalis can be avoided by administering.....to the mother immediately after the delivery of first child:-

(1) Vitamins

(2) Antibiotics

(3) Anti-Rh antibodies

(4) Rh-antigen

BFC233

6. Read the following (A - D) Statements :-

(A) Plasma is a straw coloured, viscous fluid constituting 55 percent of the blood.

(B) 90-92 percent of plasma is water and proteins contribute 6-8 percent of it

(C) Globulins are needed for clotting or coagulation of Blood

(D) Fibrinogens are primarily involved in defence mechanism of the body.

How many of the above statements are correct:

(1) Four (2) Three

(3) Two (4) One

BFC234

7. Which of the following is incorrect match of W.B.Cs with its functions :-

(1) Neutrophils – Phagocytic cells

(2) Eosinophils – Resist infections and are also associated with allergic reactions

(3) Basophils – Secrete histamine serotonin and Heparin

(4) T-Lymphocytes – Produce antibodies

BFC235

8. The state of heart when it is not pumping blood effectively enough to meet the needs of the body, is :-

(1) Heart attack (2) Cardiac arrest

(3) Angina pectoris (4) Heart Failure

BFC236

9. The opening between the right atrium and the right ventricle is guarded by :-

(1) Tricuspid valve (2) Bicuspid value

(3) Semilunar valve (4) Eustachian valve

BFC237

10. Incomplete double circulation is found in :-

(1) Reptiles and Birds

(2) Amphibian and Reptiles

(3) Birds and Mammals

(4) Fish and Amphibia

BFC238

11. The first heart sound is associated with :-

(1) Closure of the semilunar valves

(2) Closure of the tricuspid and bicuspid valves

(3) Opening of the semilunar valves

(4) Opening of the eustachian valve

BFC239

12. Correctly match column-I with column-II

Column-I	**Column-II**
A-Cardiac arrest	(i) Heart not pumping blood effectively
B-Heart Failure	(ii) Heart muscle is suddenly damaged
C-Heart attack	(iii) Acute chest pain
D-Angina	(iv) Heart stops beating

(1) A→(i), B→(ii), C→(iii), D-(iv)

(2) A→(iv), B→(ii), C→(i), D-(iii)

(3) A→(iv), B→(i), C→(ii), D-(iii)

(4) A→(ii), B→(iii), C→(i), D-(iv)

BFC240

13. Match the Column-I with Column-II.

Column-I	**Column-II**
(A) Fish	(i) 3-chambered heart
(B) Amphibia	(ii) Incomplete double circulation
(C) Birds	(iii) 4-chambered heart
	(iv) Single circulation
	(v) 2-chambered heart
	(vi) Double circulation

(1) A→(i), (ii) B→(iii), (vi) C→(iv), (v)

(2) A→ (i), (iv) B→(v), (ii) C→(iii), (vi)

(3) A→(v), (iv) B→(i), (ii) C→(iii), (vi)

(4) A→(iii), (ii) B→(i), (iv) C→(v), (vi)

BFC241

14. In which the following can increase the rate of heart beat?

(A) Sympathetic neural signals.

(B) Parasympathetic neural signals.

(C) Adrenal medullary hormones.

(D) Vagus nerve.

(E) Thyroxine hormone

(F) Acetylcholine

(1) A, C, E (2) D, E, F

(3) A, C, D (4) B, D, F

BFC242

15. During joint diastole :-
(1) Tricuspid and bicuspid valves are open
(2) Semilunar valves are closed
(3) All the four chambers of heart are in a relaxed state.
(4) All of the above

BFC243

16. Which is responsible for initiating and maintaining the rhythmic contractile activity of the heart?
(1) Sino-atrial node (SAN)
(2) Atrio-ventricular node (AVN)
(3) Purkinje fibres
(4) Bundle of his

BFC244

17. In which the following has the ability to generate action potentials without any external stimuli?
(1) Sino-atrial node (SAN)
(2) Atrio-ventricular node (AVN)
(3) Purkinje fibres
(4) All of the above

BFC245

18. Which information is **incorrect** about cardiac output?
(1) It's average value is 5000 ml
(2) The stroke volume multiplied by the heart rate, gives the cardiac output.
(3) It is the volume of blood pumped out by each ventricle per minute.
(4) The body has no ability to alter the cardiac output.

BFC246

19. Birds and Mammals have :-
(1) Single and closed type circulation
(2) Double and open type circulation
(3) Double and closed type circulation
(4) Single and open type circulation

BFC247

20. Open circulatory system is present in :-
(1) Annelids and Chordates
(2) Annelids and Arthropods
(3) Arthropods and Chordates
(4) Arthropods and Mollusca

BFC248

21. Identify the correct sequence of events in a cardiac cycle :-
(1) Joint diastole $\rightarrow$ Atrial systole $\rightarrow$ Ventricular systole
(2) Joint diastole $\rightarrow$ Atrial diastole $\rightarrow$ Ventricular systole
(3) Ventricular systole $\rightarrow$ Atrial systole $\rightarrow$ Joint diastole
(4) Atrial systole $\rightarrow$ Joint diastole $\rightarrow$ Ventricular systole

BFC249

22. During ventricular systole :-
(1) Semilunar valves are closed
(2) About 30 percent blood is pumped into aorta from ventricles.
(3) Tricuspid and Bicuspid valves are closed
(4) Ventricular pressure declines

BFC250

23. The cardiac impulse is initiated and conducted further upto ventricle. The correct sequence of conduction of impulse is :-

(1)	S A Node	A V Node	Purkinje fiber	A V Bundle
(2)	S A Node	Purkinje fiber	A V Node	A V Bundle
(3)	S A Node	A V Node	A V Bundle	Purkinje fiber
(4)	S A Node	Purkinje fiber	A V Bundle	A V Node

BFC251

24. The second heart sound (dup) is associated with the closure of
(1) Tricuspid valve
(2) Semilunar valves
(3) Bicuspid valve
(4) Tricuspid and bicuspid valves.

BFC252

25. Match the terms given under Column 'I' with their functions given under Column 'II' and select the answer from the options given below:

Column-I	Column-II
A. Lymphatic System	i. Carries oxygenated blood
B. Pulmonary vein	ii. Immune Response
C. Thrombocytes	iii. To drain back the tissue fluid to the circulatory system
D. Lymphocytes	iv. Coagulation of blood

(1) A – (i), B – (ii), C – (iii), D – (iv)
(2) A – (iii), B – (i), C – (iv), D – (ii)
(3) A – (iii), B – (i), C – (ii), D – (iv)
(4) A – (ii), B – (i), C – (iii), D – (iv)

BFC253

26. Cardiac activity could be moderated by the autonomous neural system. Tick the correct answer :-
(1) The parasympathetic system stimulates heart rate and stroke volume
(2) The sympathetic system stimulates heart rate and stroke volume
(3) The parasympathetic system decreases the heart rate but increase stroke volume
(4) The sympathetic system decreases the heart rate but increase stroke volume

BFC254

27. Which among the following is correct during each cardiac cycle ?
(1) The volume of blood pumped out by the Rt and Lt ventricles is same.
(2) The volume of blood pumped out by the Rt and Lt ventricles is different
(3) The volume of blood received by each atrium is different
(4) The volume of blood received by the aorta and pulmonary artery is different

BFC255

(B) ANALYTICAL QUESTIONS

28. Blood platelets are found only in the blood of :-
(1) Birds (2) Reptiles
(3) Mammals (4) Amphibians

BFC256

29. What is the main difference in human and frog RBC?
(1) Human RBC are non-nucleated
(2) Haemoglobin is found only in human RBC
(3) Human RBC have nucleus
(4) Human RBC are multinucleated

BFC257

30. Prothrombin is found in :-
(1) Intestine and helps in cellulose digestion
(2) Liver and helps in production of bile
(3) Blood and gives red colour
(4) Blood and helps in blood clotting

BFC258

31. Which type of WBCs are most abundant in blood of rabbit and other vertebrates?
(1) Acidophils (2) Basophils
(3) Lymphocytes (4) Neutrophils

BFC259

32. The basophils of white blood cells liberates or secretes :-

(1) Heparin (2) Serotonin

(3) Histamine (4) All of these

BFC260

33. Circular, biconcave and non-nucleated RBC's are found in

(1) Rat (2) Rabbit

(3) Man (4) All of the above

BFC261

34. In the ABO blood system, you normally type blood group of following types :-

(1) A (2) B, AB

(3) ABO (4) A, B, AB or O

BFC262

35. Read the following statements and choose the correct option -

(a) Plasma consists of nearly 45% part of blood

(b) Fibrinogen, albumin & globulin are major proteins of plasma

(c) Albumin helps in osmotic balance & retention of water in plasma

(d) Plasma without the clotting factors is called serum

(1) only b and c correct

(2) b, c & d correct

(3) a, b & c correct

(4) only c & d correct

BFC263

36. Which of the following should be avoided in biological marriages?

(1) A^+ boy and A^+ girl (2) A^+ boy and A^- girl

(3) O^+ boy and O^+ girl (4) O^- boy and O^+ girl

BFC264

37. After examining the blood group of husband and wife , the doctor advised them not to have more than one child. The blood groups of the couple are likely to be :

(1) Male Rh^- and female Rh^+

(2) Female Rh^- and male Rh^+

(3) Male Rh^+ and female Rh^+

(4) Male Rh^- and female Rh^-

BFC265

38. Mammalian RBC is :-

(1) Biconcave, circular, non-Nucleated

(2) Biconcave, Nucleated

(3) Oval Nucleated

(4) Biconvex and nucleated

BFC266

39. Which one of the following is agranulocyte?

(1) Neutrophil (2) Eosinophil

(3) Basophil (4) Monocyte

BFC267

40. Which among of the following is incorrect about platelets?

(1) Also known as Thrombocytes.

(2) Normal platelets count is 1.5 – 3.5 lakh/mm^3 of blood.

(3) They are nucleated cell fragments.

(4) Platelets are only found in mammals.

BFC268

41. The valve that checks the back flow of blood from left ventricle to left atrium and is having two cusps :-

(1) Tricuspid valve

(2) Mitral valve

(3) Pulmonary semilunar valve

(4) Aortic semilunar valve **BFC269**

42. Which of the following receives oxygenated blood from the lungs through pulmonary veins?
(1) Left atrium
(2) Right atrium
(3) Both left and right atrium
(4) Statement is wrong because left atrium receive deoxygenated blood
BFC270

43. Which of the following, does not help in clotting of blood?
(1) Heparin (2) Prothrombin
(3) Ca^{2+} (4) Exposure to O_2
BFC271

44. The SA node is located at :-
(1) In the lower left ventricle wall
(2) In the septum between the atria and ventricles
(3) In the septum between the left and right ventricles
(4) In the upper right atrial wall
BFC272

45. Which one of the following couple were suggested by Doctors to not have more than one child
(1) Rh^+ male and Rh^- female
(2) Rh^- male and Rh^+ female
(3) Rh^+ male and Rh^+ female
(4) Rh^- male and Rh^- female
BFC273

46. The pH of blood is :
(1) Between 7–8 (2) Between 2–4
(3) Between 12–14 (4) Between 2–5
BFC274

47. Universal blood recipient is :
(1) Blood group–O (2) Blood group–AB
(3) Blood group–A (4) Blood group–B
BFC275

48. Life span of RBC is :
(1) 50 days (2) 70 days
(3) 120 days (4) 220 days
BFC276

49. During blood clotting which of the following is used
(1) Co (2) Ca^{++} (3) Na^+ (4) Cl^-
BFC277

50. The fibers that conduct the impulse to the ventricles are known as the :-
(1) SA node
(2) AV bundle
(3) Right bundle branch
(4) Purkinje fibers
BFC278

51. Which of the following does not play a role in blood coagulation ?
(1) Vitamin K (2) Vitamin D
(3) Calcium ions (4) Fibrinogen
BFC279

52. The cardiac output of a person is 4.2 lit/minute and pulse rate is 60/minute. Calculate the stroke volume :
(1) 70 ml (2) 90 ml (3) 3.80 ml (4) 100 ml
BFC280

53. Read the following statements and choose which are not correct -
(a) Plasma has 90 - 92% water & 6 - 8% proteins.
(b) Erythrocytes are called as formed elements & constitute 55% of blood.
(c) Granulocytes include basophil, neutrophil & monocyte.
(d) Reduction in number of platelets may cause excessive blood loss.
Choose the correct answer from options given below :
(1) a and b (2) b and c
(3) c and d (4) b, c and d
BFC281

54. Cardiac output :-
(1) Stroke volume x heart rate
(2) Stroke volume x end systolic volume
(3) Heart rate x end systolic volume
(4) End diastolic volume x End systolic volume

BFC282

55. Which of the following waves of ECG represent Repolarization?
(1) P wave (2) QRS complex
(3) RS wave (4) T wave

BFC283

56. Choose the option which has all correct statements related to ECG -
(a) QRS complex represents polarisation of ventricles
(b) ECG is graphical representation of electrical activity of heart
(c) T-wave represent end of diastole & repolarisation
(d) Number of QRS complexes is equal to number of heart beat of an individual.
(1) a, b and d (2) b, c and d
(3) b and c (4) b and d

BFC284

57. The hepatic portal system is :-
(1) Between digestive tract and hypothalamus
(2) Between digestive tract and liver
(3) Between liver and hypothalamus
(4) Between liver and Pituitary

BFC285

58. The hormonal regulation of Cardiac output is:
(1) Mediated by adrenal cortex
(2) Mediated by adrenal medulla
(3) Mediated by thyroid
(4) Mediated by pineal

BFC286

59. Given below are two statements:
Statement-I : Cardiac output is blood volume pumped by each ventricle per minute.
Statement-II : Body has the ability to alter heart rate but not stroke volume.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II both are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BFC287

60. Given below are two statements:
Statement-I : Sympathetic nervous system can increase the strength of ventricular contraction & cardiac output also.
Statement-II : Adrenal medullary hormones can decrease the cardiac output.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II both are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BFC288

61. Match the columns

Column-I	Column-II
a. Bicuspid valve	p. Brain
b. Nephron	q. Liver
c. Alveoli	r. Heart
d. Cerebrum	s. Kidney
	t. Lungs

(1) a–s, b–r, c–p, d–t
(2) a–r, b–t, c–s, d–p
(3) a–r, b–s, c–t, d–p
(4) a–s, b–q, c–p, d–t

BFC289

62. Vasoconstriction causes
(1) Increase in heart beat
(2) Decrease in heart beat
(3) Increase in blood pressure
(4) Decrease in blood pressure

BFC290

63. Given below are two statements:

Statement-I : Lymph is derived from blood vascular system & again drain back to the blood.

Statement-II : Lymph is important carrier for nutrients, hormones & fatty acid etc.

In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II both are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BFC291

64. Which one of the following mammalian cells is not capable of metabolising glucose to carbon-dioxide aerobically ?
(1) Red blood cells
(2) White blood cells
(3) Unstriated muscle cells
(4) Liver cells

BFC199

65. A drop of each of the following, is placed separately on four slides. Which of them will not coagulate?
(1) Whole blood from pulmonary vein
(2) Blood plasma
(3) Blood serum
(4) Sample from the thoracic duct of lymphatic system

BFC200

66. Which type of white blood cells are concerned with the release of histamine and the natural anticoagulant heparin ?
(1) Eosinophils (2) Monocytes
(3) Neutrophils (4) Basophils

BFC201

67. In humans, blood passes from the post caval to the diastolic right atrium of heart due to :-
(1) stimulation of the sino auricular node
(2) pressure difference between the post caval and atrium
(3) pushing open of the venous valves
(4) suction pull

BFC202

68. The most active phagocytic white blood cells are:-
(1) Eosinophils and lymphocytes
(2) Neutrophils and monocytes
(3) Neutrophils and eosinophils
(4) Lymphocytes and macrophages

BFC203

69. The most popularly known blood grouping is the ABO grouping. It is named ABO and not ABC, because "O" in it refers to having:-
(1) No antigens A and B on RBCs
(2) Other antigens besides A and B on RBCs
(3) Over dominance of this type on the genes for A and B types
(4) One antibody only - either anti–A or anti–B on the RBCs

BFC204

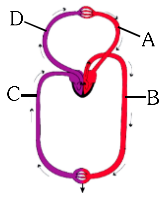
70. The letter T in T-lymphocyte refers to :-
(1) Thymus (2) Thyroid
(3) Thalamus (4) Tonsil

BFC205

71. Compared to blood our lymph has :-
(1) More RBCs and less WBCs
(2) No plasma
(3) Plasma without proteins
(4) More WBCs and no RBCs

BFC206

72. Figure shows schematic plan of blood circulation in humans with labels A to D, Identify the label and give its function/s.



(1) D-Dorsal aorta-takes blood from heart to body parts, PO_2 = 95 mm Hg
(2) A-Pulmonary vein-takes impure blood from body parts, PO_2 = 60 mm Hg
(3) B-Pulmonary artery-takes blood from heart to lungs, PO_2 = 90 mm Hg
(4) C-Vena Cava-takes blood from body parts to the right auricle, PCO_2 = 45mm Hg

BFC207

73. Match Column I with Column II :

Column I	Column II
(a) Eosinophils	(i) Coagulation
(b) RBC	(ii) Universal Recipient
(c) AB Group	(iii) Resist Infections
(d) Platelets	(iv) Contraction of Heart
(e) Systole	(v) Gas transport

(1) a-v , b-ii , c-iii , d-i , e-iv
(2) a-ii , b-v , c-iii , d-iv , e-i
(3) a-iii , b-v , c-ii , d-i , e-iv
(4) a-iv , b-ii , c-v , d-iii , e-i

BFC296

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	3	3	3	4	4	1	2	2	3	3	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	4	3	4	1	3	3	2	2	2	1	3	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	4	4	2	2	2	1	4	3	2	1	1	4	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	2	3	2	2	2	1	2	1	4	4	2	2	2	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73		
Answer	3	3	4	1	3	4	2	2	1	1	4	4	3		

Exercise - I (Conceptual Questions) Build Up Your Understanding

AXIAL SKELETON

1. Cranium of human is made up of -
(1) 8 bones (2) 12 bones
(3) 16 bones (4) 14 bones
BLC001

2. The number of floating ribs and true ribs in human body -
(1) 6 pairs, 8 pairs (2) 5 pairs, 4 pairs
(3) 3 pairs, 2 pairs (4) 2 pairs, 7 pairs
BLC002

3. How many bones are present in human skull skeleton?
(1) 22 (2) 29 (3) 12 (4) 42
BLC003

4. Number of cervical and thoracic vertebrae in human are
(1) 7, 12 (2) 7, 10
(3) 7, 11 (4) 7, 14
BLC004

5. Cranial bone related to ear is
(1) Occipital (2) Frontal
(3) Nasal (4) Temporal
BLC005

6. These facial bones are unpaired -
(1) Lacrimal, Vomer
(2) Vomer, Mandible
(3) Nasal, Vomer
(4) Palatine, Mandible
BLC006

7. In human beings the cranium is formed by
(1) Ten bones of which two are paired
(2) Eight bones of which two are paired
(3) Fourteen bones of which six are paired
(4) Twelve bones of which four are paired
BLC007

8. Which of the following is wrong statement about hyoid bone?
(1) one in number
(2) U-shaped
(3) Not attached with any other bone of the body
(4) Present at the top of the buccal cavity
BLC008

9. Largest, Heaviest mammalian vertebra is :-
(1) Thoracic (2) Sacral
(3) Lumbar (4) Cervical
BLC009

10. Which one of the following is the *correct description* of a certain part of a normal human skeleton ?
(1) First vertebra is axis which articulates with the occipital condyles.
(2) The 9th and 10th pairs of ribs are called the floating ribs.
(3) Glenoid cavity is a depression to which the thigh bone articulates.
(4) Parietal bone and the temporal bone of the skull are joined by fibrous joint.
BLC010

11. Which of the following vertebrae are fused?
(1) Cervical, Thoracic (2) Cervical, Sacrum
(3) Sacrum, Coccyx (4) Thoracic, coccyx
BLC011

12. The increase in vertebrae size from the neck to the lower back is due to :-
(1) Supporting more weight in the lower back than in the neck.
(2) Supporting less weight in the lower back than in the neck.
(3) Evolution
(4) Vertebrae don't increase in size.
BLC012

13. Symphysis is made of :-
(1) fibrocartilage (2) synovial fluid
(3) elastic cartilage (4) hyaline cartilage
BLC013

APPENDICULAR SKELETON

14. Part of pectoral girdle is:-
(1) Ileum (2) Glenoid cavity
(3) Acetabulum (4) Sternum
BLC014

15. Scapula and clavicle form
(1) Pelvic girdle (2) Pectoral girdle
(3) Skull (4) Vertebral column
BLC015

16. Pectoral girdle, pelvic girdle and limb bones constitute :
(1) Visceral skeleton
(2) Outer skeleton
(3) Axial skeleton
(4) Appendicular skeleton
BLC016

17. Glenoid cavity is found in :
(1) Pelvic girdle (2) Skull
(3) Pectoral girdle (4) Sternum
BLC017

18. Acromion process is part of :
(1) Vertebral column (2) Pelvic girdle
(3) Femur (4) Pectoral girdle
BLC018

19. Pelvic girdle consists of :
(1) Ilium
(2) Ilium and ischium
(3) Ilium, ischium and pubis
(4) Ischium and pubis
BLC019

20. Which of the following is a sesamoid bone?
(1) Pelvic (2) Patella
(3) Axis (4) Pectoral girdle
BLC020

21. Number of bones in thumb of right forelimb of man are
(1) 5 (2) 4 (3) 3 (4) 2
BLC021

22. The clavicle articulates with _____ of scapula
(1) Acromion process
(2) Glenoid cavity
(3) Acetabulum cavity
(4) Ball and socket joint
BLC022

23. Which one is a flat bone ?
(1) Scapula (2) Carpal
(3) Patella (4) Tarsals
BLC023

24. Digital formula of foot of Human is :-
(1) 03332 (2) 03333
(3) 03322 (4) 23333
BLC024

25. Presence of obturator foramen is feature of:-
(1) Pectoral girdle
(2) Vertebrae
(3) Pelvic girdle
(4) Occipital bone
BLC025

26. The other name for knee cap is
(1) Clavicle (2) Patella
(3) Radius (4) Femur
BLC026

27. An acromian process is characteristically found in the:-
(1) Pectoral girdle of mammals
(2) Sperm of mammals
(3) Pelvic girdle of mammals
(4) Skull of frog
BLC027

JOINTS AND DISORDERS

28. Study of joints known as :-
(1) Osteology
(2) Arthrology
(3) Craniology
(4) Kinesiology

BLC028

29. Saddle joint occurs between :-
(1) Carpal and first metacarpal of thumb
(2) Femur and pelvic girdle
(3) All the vertebrae
(4) Phalanges

BLC029

30. Joints between phalanges are :-
(1) Ball and socket joints
(2) Gliding joints
(3) Hinge joints
(4) Pivot joints

BLC030

31. It is an outcome of irregularities in metabolism of nitrogenous waste :-
(1) Osteoporosis
(2) Osteo-arthritis
(3) Gouty arthritis
(4) Rheumatoid arthritis

BLC031

32. Which of the following is correct for the given assertion (A) and reason (R)?
Assertion (A) : Knee joint is hinge joint type joint.
Reason (R) : Femur , patella and fibula are associated with knee joint.
(1) Both (A) and (R) are true but (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true statement but (R) is false.
(4) Both (A) and (R) are false

BLC032

33. Synovial fluid is secreted by :-
(1) Blood
(2) Cartilage
(3) Bone
(4) Synovial membrane

BLC033

34. A joint between humerus and radius ulna is:
(1) Ball & socket (2) Pivot
(3) Gliding (4) Hinge

BLC034

35. The number of saddle joints in human beings is :-
(1) 4 (2) 8 (3) 2 (4) 10

BLC035

Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	1	4	2	2	4	3	4	3	1	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	4	3	2	4	1	1	4	3	2	1	2	1	3
Question	31	32	33	34	35										
Answer	3	3	4	4	3										

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Select the correct matching of the type of the joint with the example in human skeletal system :

	Type of Joint	Example
(1)	Cartilaginous joint	Between frontal and parietal
(2)	Pivot joint	Between third and fourth cervical vertebrae
(3)	Hinge joint	Between humerus and pectoral girdle
(4)	Gliding joint	Between carpals

BLC036

AIPMT 2015

2. Glenoid cavity articulates :-
(1) Scapula with acromion
(2) Clavicle with scapula
(3) Humerus with scapula
(4) Clavicle with acromion

BLC037

Re-AIPMT 2015

3. Which of the following is not a function of the skeletal system?
(1) Locomotion
(2) Production of erythrocytes
(3) Storage of minerals
(4) Production of body heat

BLC038

4. Which of the following joints would allow no movement?
(1) Ball and Socket joint
(2) Fibrous joint
(3) Cartilaginous joint
(4) Synovial joint

BLC039

NEET-II 2016

5. Osteoporosis, an age–related disease of skeletal system, may occur due to :-
(1) Decreased level of estrogen
(2) Accumulation of uric acid leading to inflammation of joints.
(3) Immune disorder affecting neuro-muscular junction leading to fatigue.
(4) High concentration of Ca^{++} and Na^{+}.

BLC040

NEET-UG 2017

6. Out of 'X' pairs of ribs in humans only 'Y' pairs are true ribs. Select the option that correctly represents values of X and Y and provides their explanation :-
(1) X = 12, Y = 5 True ribs are attached dorsally to vertebral column and sternum on the two ends.
(2) X = 24, Y = 7 True ribs are dorsally attached to vertebral column but are free on ventral side.
(3) X = 24, Y = 12 True ribs are dorsally attached to vertebral column but are free on ventral side.
(4) X = 12, Y = 7 True ribs are attached dorsally to vertebral column and ventrally to the sternum.

BLC041

7. The pivot joint between atlas and axis is a type of :-
(1) Cartilaginous joint (2) Synovial joint
(3) Saddle joint (4) Fibrous joint

BLC042

NEET-UG 2019

8. Select the correct option .
(1) 8th, 9th and 10th pairs of ribs articulate directly with the sternum.
(2) 11th and 12th pairs of ribs are connected to the sternum with the help of hyaline cartilage.
(3) Each rib is a flat thin bone and all the ribs are connected dorsally to the thoracic vertebrae and ventrally to the sternum.
(4) There are seven pairs of vertebrosternal, three pairs of vertebrochondral and two pairs of vertebral ribs.

BLC043

NEET-UG 2019 (Odisha)

9. Match the following joints with the bones involved:-

Column - I		Column - II
(a) Gliding joint	(i)	Between carpal and metacarpal of thumb
(b) Hinge joint	(ii)	Between Atlas and Axis
(c) Pivot joint	(iii)	Between the carpals
(d) Saddle joint	(iv)	Between Humerus and Ulna.

Select the correct option from the following

(1) (a)-(iii), (b)-(iv), (c)-(ii), d-(i)

(2) (a)-(iv), (b)-(i), (c)-(ii), d-(iii)

(3) (a)-(iv), (b)-(ii), (c)-(iii), d-(i)

(4) (a)-(i), (b)-(iii), (c)-(ii), d-(iv)

BLC044

NEET-UG 2020

10. Match the following columns and select the correct option

Column - I		Column - II
(a) Gout	(i)	Decreased levels of estrogen.
(b) Osteoporosis	(ii)	Low Ca^{++} ions in the blood.
(c) Tetany	(iii)	Accumulation of uric acid crystals.
(d) Muscular dystrophy disorder.	(iv)	Auto immune dystrophy
	(v)	Genetic disorder

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(v)

(3) (a)-(iv), (b)-(v), (c)-(i), (d)-(ii)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

BLC045

11. Match the following columns and select the correct option

Column - I	Column - II
(a) Floating Ribs	(i) Located between second and seventh ribs
(b) Acromion	(ii) Head of the Humerus
(c) Scapula	(iii) Clavicle
(d) Glenoid cavity	(iv) Do not connect with the sternum

	(a)	(b)	(c)	(d)
(1)	(iv)	(iii)	(i)	(ii)
(2)	(ii)	(iv)	(i)	(iii)
(3)	(i)	(iii)	(ii)	(iv)
(4)	(iii)	(ii)	(iv)	(i)

BLC046

NEET-UG 2021

12. Match list - I with list – II

List I		List II	
(a)	Scapula	(i)	Cartilaginous bone
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous bone
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the correct answer from the options given below :-

(1) a-i, b-iii, c-ii, d-iv (2) a-ii, b-iii, c-iv, d-i

(3) a-iv, b-ii, c-iii, d-i (4) a-iv, b-iii, c-ii, d-i

BLC047

NEET-UG 2022

13. Given below are two statements: one is labelled as : Assertion (A) and the other is labelled as Reason (R).

Assertion : Osteoporosis is characterised by decreased bone mass and increased chances of fractures.

Reason : Common cause of osteoporosis is increased levels of estrogen.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(2) Assertion is correct but Reason is not correct.

(3) Assertion is not correct but Reason is correct.

(4) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

BLC048

NEET-UG 2023

14. Match **List I** with **List II**.

	List I (Type of Joint)		List II (Found between)
A.	Cartilaginous	I.	Between flat skull bones Joint
B.	Ball and Socket Joint	II.	Between adjacent vertebrae in vertebral column
C.	Fibrous Joint	III.	Between carpal and metacarpal of thumb
D.	Saddle Joint	IV.	Between Humerus and Pectoral girdle

Choose the **correct** answer from the options given below :-

(1) A-II, B-IV, C-I, D-III (2) A-I, B-IV, C-III, D-II

(3) A-II, B-IV, C-III, D-I (4) A-III, B-I, C-II, D-IV

BLC049

NEET-UG 2023 (Manipur)

15. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Palm bones	(I)	Phalanges
(B)	Wrist bones	(II)	Metacarpals
(C)	Ankle bones	(III)	Carpals
(D)	Digit bones	(IV)	Tarsals

Choose the **correct** answer from the options given below :-

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(3) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

BLC050

NEET (UG) 2024

16. Match **List-I** with **List-II**.

	List I		List II
A.	Fibrous joints	I.	Adjacent vertebrae, limited movement
B.	Cartilaginous joints	II.	Humerus and Pectoral girdle, rotational
C.	Hinge joints	III.	Skull, don't allow any movement
D.	Ball and socket joints	IV.	Knee, help in locomotion

Choose the correct answer from the options given below :-

(1) A-IV, B-II, C-III, D-I

(2) A-I, B-III, C-II, D-IV

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-I, C-IV, D-II

BLC051

RE-NEET (UG) 2024

17. Match **List-I** with **List-II**.

	List-I Location of Joint		List-II Type of Joint
A.	Joint between humerus and pectoral girdle	I.	Gliding joint
B.	Knee joint	II.	Ball and Socket joint
C.	Joint between atlas and axis	III.	Hinge Joint
D.	Joint between carpals	IV.	Pivot joint

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-IV, D-I (2) A-III, B-II, C-I, D -IV

(3) A-I, B-IV, C-III, D-II (4) A-II, B-I, C-III, D-IV

BLC052

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	4	2	1	4	2	4	1	2	1	4	2	1	4
Question	16	17													
Answer	4	1													

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Match the columns and pick out the correct answer :-

	Column-A	Column-B
(i)	Atlas/axis	(a) Saddle joint
(ii)	Carpal/metacarpal of thumb	(b) Hinge joint
(iii)	Between phalanges	(c) Fibrous joint
(iv)	Femur / acetabulum	(d) Ball and socket joint
(v)	Between cranial bones	(e) Pivot joint

	(i)	(ii)	(iii)	(iv)	(v)
(1)	e	a	b	c	d
(2)	e	a	b	d	c
(3)	e	a	d	b	c
(4)	e	a	d	c	b

BLC053

2. Joint between pubic bones in the pelvic girdle is-

(1) Cartilagenous joint

(2) Fibrous joint

(3) Hinge joint

(4) Ball and socket joint

BLC054

3. 11th and 12th pairs of ribs are called

(1) True ribs

(2) False ribs

(3) Vertebrochondral ribs

(4) Floating ribs

BLC055

4. The number of phalanges in both fore limbs of human is :-

(1) 14 (2) 28

(3) 12 (4) 24

BLC056

5. Identify the correct and wrong statements:-

(a) Human skeleton has 206 bones

(b) There are 12 pairs of ribs in human

(c) Sternum is present on the dorsal side of the body.

(d) All mammals (except of few) have 8 cervical vertebrae.

(e) Ribs are bicephalic

	correct	wrong
(1)	a, b, c	d, e
(2)	a, b, d	c, e
(3)	a, b, e	c, d
(4)	a, c, d	b, e

BLC057

6. Which of the following is **wrong** match for each limb?

(1) Wrist bones $\rightarrow$ 8 in number

(2) Palm bones $\rightarrow$ 5 in number

(3) Ankle bones $\rightarrow$ 7 in number

(4) Digits $\rightarrow$ 12 in number

BLC058

7. Each coxal bone is formed by fusion of three bones?

(1) Ilium, ischium, pubis

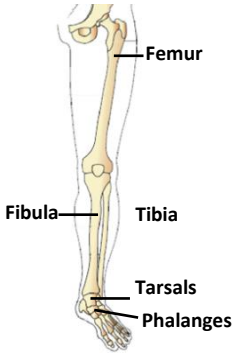
(2) Ilium, sacrum, pubis

(3) Ilium, sacrum, coccyx

(4) Ilium sacrum, ischium

BLC059

8. Given below is a diagram of the bones of the left human hind limb as seen from front. It has certain mistakes in labelling.



Two of the wrongly labelled bones are :-
(1) Tibia and tarsals
(2) Femur and fibula
(3) Fibula and phalanges
(4) Tarsals and femur

BLC060

9. This bone is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh ribs. Identify this bone :-
(1) Femur (2) Hip bone
(3) Scapula (4) Sacrum

BLC061

10. _____ are characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones :-
(1) Fibrous joints
(2) Synovial joints
(3) Cartilagenous joints
(4) Hinge joints

BLC062

11. Inflammation of joints due to accumulation of uric acid crystals is called :-
(1) Osteoporosis
(2) Gout
(3) Arthritis
(4) Tetany

BLC063

12. ____ and _____ constitute our skeletal system.
(1) Bones and blood
(2) Blood and lymph
(3) Bones and cartilages
(4) Bones and nervous

BLC064

13. Which of the following type of joint allows considerable movements and therefore, play a significant role in locomotion?
(1) Synovial joints
(2) Fibrous joints
(3) Cartilagenous joints
(4) All of the above

BLC065

14. Match the columns

Column-A	**Column-B**
(a) Ribs	(i) 26
(b) Vertebrae	(ii) 14
(c) Phalanges	(iii) 1
(d) Sternum	(iv) 24

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(i)	(iii)	(ii)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(iv)	(iii)	(i)	(ii)

BLC066

15. Match the List-1 with List-2 correct and find your answer from the code given below :-

List-1	**List-2**
(i) Palm bones	(A) 14 in number
(ii) Wrist bones	(B) 5 in number
(iii) Ankle bones	(C) 8 in number
(iv) Phalanges	(D) 7 in number

Code :

(1)	(i)-A	(ii)-B	(iii)-C	(iv)-D
(2)	(i)-B	(ii)-C	(iii)-D	(iv)-A
(3)	(i)-A	(ii)-C	(iii)-D	(iv)-B
(4)	(i)-C	(ii)-D	(iii)-B	(iv)-A

BLC067

16. Between humerus and pectoral girdle which type of joint is present :-
(1) Pivot joint (2) Hinge joint
(3) Ball & socket joints (4) Gliding joints

BLC068

17. Find out incorrect statements :-
(1) Brain box is formed of 8 bones
(2) In skull the frontal bone is single
(3) A single U-shaped bone present at the base of the buccal cavity is quadrate
(4) Hyoid bone is the part of skull

BLC069

(B) ANALYTICAL QUESTIONS

18. Pelvic girdle of man consists of :-
(1) Ilium, ischium and pubis
(2) Ilium, ischium and coracoid
(3) Coracoid, scapula and clavicle
(4) Ilium, coracoid and scapula

BLC070

19. Hinge joint is present between :-
(1) Femur and ulna
(2) Humerus and ulna
(3) Femur and pectoral girdle
(4) Femur and pelvic girdle

BLC071

20. Thoracic cage in Human is made up of :-
(1) Ribs, vertebral column and diaphragm
(2) Ribs, diaphragm
(3) Vertebral column, diaphragm and sternum
(4) Ribs, vertebral column and sternum

BLC072

21. In which one of the following bones acromian process is found ?
(1) Femur (2) Coxal bones
(3) Atlas (4) Scapula

BLC073

22. The number of floating ribs in human body
(1) 6 pairs (2) 5 pairs
(3) 3 pairs (4) 2 pairs

BLC074

23. Axial skeletal is made up of
(1) Vertebral column complete
(2) Skull
(3) Sternum
(4) All of the above

BLC075

24. The shoulder and hip joints are
(1) Pivot joint
(2) Hinge joint
(3) Ellipsoid joint
(4) Ball and socket joint

BLC076

25. Bones become fragile in -
(1) Gout (2) Osteoporosis
(3) Arthritis (4) None of these

BLC077

26. Part of mammalian body has single pairs of bones -
(1) Pelvic girdle
(2) External ear
(3) Lower jaw
(4) All of the above

BLC078

27. Which one of the following items gives its *correct total* number?
(1) Types of diabetes – 3
(2) Cervical vertebrae in humans – 8
(3) Floating ribs in humans – 4
(4) Amino acids found in proteins – 16

BLC079

28. Elbow joint is an example of :-
(1) Ball and socket joint
(2) Pivot joint
(3) Hinge joint
(4) Gliding joint

BLC080

29. Three of the following pairs of human skeletal parts are correctly matched with their respective inclusive skeletal category and one pair is not matched. Identify the nonmatching pair :-

Pair of skeletal parts	**Category**
(1) Malleus and stapes	Ear ossicles
(2) Sternum and ribs	Axial skeleton
(3) Clavicle and glenoid cavity	Pelvic girdle
(4) Humerus and ulna	Appendicular Skeleton

BLC081

30. Select the correct statement regarding the specific disorder of muscular or skeletal system-

(1) Myasthenia gravis – Auto immune disorder which inhibits sliding of myosin filaments.

(2) Gout – inflammation of joints due to extra deposition of calcium.

(3) Muscular dystrophy – age related shortening of muscles.

(4) Osteoporosis-decrease in bone mass and higher chances of fractures with advancing age.

BLC082

31. Select the correct statement with respect to locomotion in humans :-

(1) The joint between adjacent vertebrae is a fibrous joint.

(2) A decreased level of progesterone causes osteoporosis in old people.

(3) Accumulation of uric acid crystals in joints causes their inflammation.

(4) The vertebral column has 10 thoracic vertebrae.

BLC083

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	4	2	3	4	1	3	3	2	2	3	1	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	1	2	4	4	4	4	4	2	1	3	3	3	4
Question	31														
Answer	3														

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION, STRUCTURE OF MUSCLE

1. Epimysium of muscles are made up of :-
(1) White fibrous connective tissue
(2) Adipose connective tissue
(3) Reticular connective tissue
(4) Areolar connective tissue

BLC135

2. Muscles of Iris & Ciliary body originate :-
(1) Ectoderm
(2) Mesoderm
(3) Endoderm
(4) All of above

BLC136

3. Striated muscle fibres found in :-
(1) Trachea (2) Lung
(3) Leg (4) Gall bladder

BLC137

4. The type of muscles present in our :
(1) Upper arm are smooth muscle fibres fusiform in shape.
(2) Heart are involuntary and unstriated smooth muscles.
(3) Intestine are striated and involuntary.
(4) Thigh are striated and voluntary.

BLC138

5. Which one of the following is *correct* pairing of a body part and the kind of muscle tissue that moves it ?
(1) Iris – Involuntary smooth muscle
(2) Heart wall–Involuntary unstriated muscle
(3) Biceps of upper arm–Smooth muscle fibres
(4) Abdominal wall–Skeletal muscle

BLC139

6. One myosin filament in the myofibril of skeletal muscle fibre is surrounded by how many actin filaments ?
(1) Two (2) Four (3) Six (4) Three

BLC140

7. Which character is not associated with red muscle fibres?
(1) High quantity of myoglobin is present
(2) Amount of mitochondria are more
(3) Amount of sacroplasmic reticulum is high
(4) Contraction is slow and for long duration

BLC141

8. Ciliary movement occurs in most of our internal tubular organs which are lined by ___A___ epithelium. The coordinated movement of cilia in the ___B___ help us in removing dust particles.
Identify A and B complete the given statement.
(1) A–squamous, B–trachea
(2) A–cuboidal, B–trachea
(3) A–ciliated, B–trachea
(4) A–stratified, B–trachea

BLC142

9. Skeletal muscles are closely associated with the ...A... components of the body. They have ...B... appearance under the microscope and hence are called ...C... muscles.
Choose the correct options to fill A, B and C, so as to complete the given statement.
(1) A–muscular, B–stripped, C–striated
(2) A–visceral, B–stripped, C–striated
(3) A–skeletal, B–stripped, C–striated
(4) A–microfibrillar, B–stripped, C–striated

BLC143

MUSCLE PROTEIN, MUSCLE CONTRACTION

10. During contraction of muscles :-
(1) Actin Filament slide over actin
(2) Myosin filament slide over myosin
(3) Actin filament slide over myosin
(4) Myosin filament slide over actin

BLC144

11. Contractile unit of muscle fibres :-
(1) Muscle fibre (2) Sarcomere
(3) Myofibril (4) Sacosome

BLC145

12. ATP-ase activity found in :-
(1) Myosin filament
(2) Actin filament
(3) Both
(4) Troponin

BLC146

13. Chemical Ions responsible for muscles contraction :-
(1) Ca^{++} & K^{+} (2) Na^{+} & K^{+}
(3) Na^{+} & Ca^{++} (4) Ca^{++} & Mg^{++} Ions

BLC147

14. Immediate source of energy for muscle contraction is:-
(1) Glucose
(2) GTP
(3) ATP
(4) Creatine phosphate

BLC148

15. The muscle fatigue occurs due to accumulation of:-
(1) CO_2
(2) Lactic acid
(3) Creatine phosphate
(4) Myosin ATPase

BLC149

16. ___A___ band contains actin and is called ___B___ band, whereas the ___C___ band called ___D___ band contains myosin.
Choose the correct options for A, B, C and D to complete the given statement :-
(1) A–Light, B–I, C–dark, D–A
(2) A–Dark, B–I, C–light, D–A
(3) A–Dark, B–A, C–light, D–I
(4) A–Light, B–A, C–dark, D–I

BLC150

17. In a ...A... state, the edge of thin filaments on either side of thick filaments ...B... overlap the free ends of ...C... filaments leaving the central part of thick filaments. This central part of thick filament, not overlapped by thin filaments is called ...D... zone.
Choose the correct options to fill the gaps A, B, C and D, so as to complete the given statement
(1) A–resting, B–partially, C–thick, D–H
(2) A–resting, B–partially, C–thick, D–A
(3) A–resting, B–partially, C–thick, D–I
(4) A–resting, B–partially, C–thick, D–M

BLC151

INVOLUTARY, CARDIAC MUSCLES, DISORDERS

18. Functionally, cardiac muscles are similar to-
(1) Unstriped muscles
(2) Striped muscles
(3) Striped & Unstriped muscles
(4) None

BLC152

19. Mitochondria in cardiac muscles :-
(1) More than other muscles fibres
(2) Less than other muscles fibres
(3) Equal than other muscles fibres
(4) None

BLC153

20. Unstriped muscle are also known as :-
(1) Visceral
(2) Smooth
(3) Involuntary
(4) All

BLC154

21. Cardiac muscles Fibres :-
(1) Involuntary
(2) Non-fatigue
(3) Striated like
(4) All

BLC155

22. Smooth muscles fibres are :-
(1) Spindle shaped
(2) Unbranched & Involuntary
(3) Uninucleated
(4) All of above

BLC156

23. Which filamentous proteins are found in muscle?
(1) Leucine & myosin protein
(2) Actin & myosin protein
(3) Actin & leucine protein
(4) Actin protein only

BLC157

24. I. Sarcoplasmic reticulum are abundant.
II. Myoglobin content is high.
III. Sarcoplasmic reticulum are moderate.
IV. Aerobic muscles.
V. Depends on anaerobic respiration for energy.
VI. Less myoglobin content.
Select the option with correct statements for red muscle.
(1) I, II and IV (2) I, II and III
(3) II, III and IV (4) II and IV

BLC158

Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	4	1	3	3	3	3	3	2	1	4	3	2
Question	16	17	18	19	20	21	22	23	24						
Answer	1	1	1	1	4	4	4	2	3						

AIM: FREE EDUCATION TO ALL APUL

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Stimulation of a muscle fiber by a motor neuron occurs at :
(1) the neuromuscular junction
(2) the transverse tubules
(3) the myofibril
(4) the sacroplasmic reticulum

BLC159

AIPMT 2015

2. Sliding filament theory can be best explained as :-
(1) Actin and Myosin filaments shorten and slide pass each other
(2) Actin and Myosin filaments do not shorten but rather slide pass each other
(3) When myofilaments slide pass each other, Myosin filaments shorten while Actin filaments do not shorten
(4) When myofilaments slide pass each other Actin filaments shorten while Myosin filament do not shorten

BLC160

NEET-I 2016

3. Lack of relaxation between successive stimuli in sustained muscle contraction is known as:-
(1) Spasm (2) Fatigue
(3) Tetany (4) Tonus

BLC161

NEET-II 2016

4. Smooth muscles are :-
(1) Involuntary, cylindrical, striated
(2) Voluntary, spindle–shaped, uninucleate
(3) Involuntary, fusiform, non-striated
(4) Voluntary, multinucleate, cylindrical

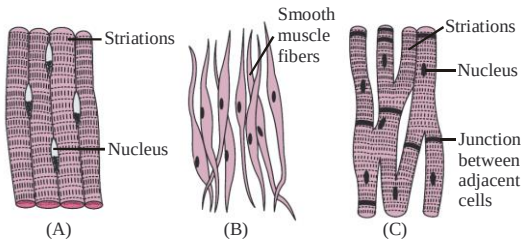
BLC162

5. Name the ion responsible for unmasking of active sites for myosin for cross–bridge activity during muscle contraction.
(1) Sodium (2) Potassium
(3) Calcium (4) Magnesium

BLC163

NEET-UG 2017

6. Select the **correct** match :-



(1) A-Skeletal muscles → biceps, striated muscles, involuntary
(2) B-Smooth muscles → walls of stomach, voluntary
(3) C-Cardiac muscles → heart, dorsal wall of aorta, voluntary
(4) A-Skeletal muscle → biceps, striated, voluntary

BLC164

NEET-UG 2018

7. Calcium is important in skeletal muscle contraction because it :-
(1) binds to troponin to remove the masking of active sites on actin for myosin.
(2) activates the myosin ATPase by binding to it.
(3) detaches the myosin head from the actin filament.
(4) prevents the formation of bonds between the myosin cross bridges and the actin filament.

BLC165

NEET-UG 2019 (Odisha)

8. Which of the following diseases is an autoimmune disorder?

(1) Myasthenia gravis

(2) Arthritis

(3) Osteoporosis

(4) Gout

BLC166

NEET-UG 2019

9. Which of the following muscular disorders is inherited?

(1) Tetany

(2) Muscular dystrophy

(3) Myasthenia gravis

(4) Botulism

BLC167

10. Chronic autoimmune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :-

(1) Arthritis

(2) Muscular dystrophy

(3) Myasthenia gravis

(4) Gout

BLC168

11. Match the following columns and select the correct option

Column - I	**Column - II**
(a) Gout	(i) Decreased levels of estrogen
(b) Osteoporosis	(ii) Low Ca^{++} ions in the blood
(c) Tetany	(iii) Accumulation of uric acid crystals
(d) Muscular dystrophy	(iv) Auto immune disorder
	(v) Genetic disorder

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(v)

(3) (a)-(iv), (b)-(v), (c)-(i), (d)-(ii)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

BLC169

NEET-UG 2021

12. Which of the following statements wrongly represents the nature of smooth muscle?

(1) These muscle have no striations

(2) They are involuntary muscles

(3) Communication among the cells is performed by intercalated discs

(4) These muscles are present in the wall of blood vessels

BLC170

13. During muscular contraction which of the following events occur ?

(a) 'H' zone disappears

(b) 'A' band widens

(c) 'I' band reduces in width

(d) Myosin hydrolyzes ATP, releasing the ADP and Pi

(e) Z-lines attached to actins are pulled inwards

Choose the **correct** answer from the options given below.

(1) (a), (c), (d), (e) only

(2) (a), (b), (c), (d) only

(3) (b), (c), (d), (e) only

(4) (b), (d), (e), (a) only

BLC171

NEET-UG 2022

14. Which of the following is a **correct** match for disease and its symptoms ?

(1) Tetany - high Ca^{2+} level causing rapid spasms.

(2) Myasthenia gravis - Genetic disorder resulting in weakening and paralysis of skeletal muscle

(3) Muscular dystrophy - An auto immune disorder causing progressive degeneration of skeletal muscle

(4) Arthritis - Inflammed joints

BLC172

Re-NEET-UG 2022

15. According to the sliding filament theory:

(1) Actin and myosin filaments slide over each other to increase the length of the sarcomere.

(2) Length of A–band does not change.

(3) I–band increases in length

(4) The actin filaments slide away from A–band resulting in shortening of sarcomere.

BLC173

NEET-UG 2023

16. Which of the following statements are correct regarding skeletal muscle ?

(A) Muscle bundles are held together by collagenous connective tissue layer called fascicle.

(B) Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.

(C) Striated appearance of skeletal muscle fibre is due to distribution pattern of actin and myosin proteins.

(D) M line is considered as functional unit of contraction called sarcomere.

Choose the ***most appropriate*** answer from the options given below:

(1) B and C only

(2) A, C and D only

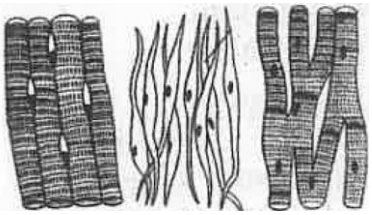
(3) C and D only

(4) A, B and C only

BLC174

NEET (UG) 2024

17. Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body :-



(a) (b) (c)

Name of muscle/location

(1) (a) Smooth - Toes
(b) Skeletal - Legs
(c) Cardiac - Heart

(2) (a) Skeletal - Triceps
(b) Smooth - Stomach
(c) Cardiac - Heart

(3) (a) Skeletal - Biceps
(b) Involuntary - Intestine
(c) Smooth - Heart

(4) (a) Involuntary - Nose tip
(b) Skeletal - Bone
(c) Cardiac - Heart

BLC175

18. Which of the following are Autoimmune disorders?

(A) Myasthenia gravis.

(B) Rheumatoid arthritis.

(C) Gout

(D) Muscular dystrophy.

(E) Systemic Lupus Erythematosus (SLE).

Choose the most appropriate answer from the options given below :-

(1) A, B & D only (2) A, B & E only

(3) B, C & E only (4) C, D & E only

BLC176

RE-NEET (UG) 2024

19. Select the **correct** statements regarding mechanism of muscle contraction.

(A) It is initiated by a signal sent by CNS via sensory neuron.

(B) Neurotransmitter generates action potential in the sarcolemma.

(C) Increased Ca^{++} level leads to the binding of calcium with troponin on actin filaments.

(D) Masking of active site for actin is activated.

(E) Utilising the energy from ATP hydrolysis to form cross bridge.

Choose the **most appropriate** answer from the options given below:

(1) B, C and E only

(2) C, D and E only

(3) A and D only

(4) B, D and E only

BLC177

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	3	3	3	4	1	1	2	3	2	3	1	4	2
Question	16	17	18	19											
Answer	1	2	2	1											

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. In a muscle fibre Ca^{++} is stored in :-
(1) Sarcoplasmic reticulum
(2) Sarcosome
(3) Sarcomere
(4) Sarcolemma
BLC178

2. Cardiac muscles are :-
(1) Striated, voluntary
(2) Striated, involuntary
(3) Unstriated, voluntary
(4) Unstriated, involuntary
BLC179

3. Ciliary movement is found in:
(1) Macrophages and leucocytes
(2) Fallopian tube and vasa deferentia
(3) Fallopian tube and Trachea
(4) Tongue and Limbs
BLC180

4. Amoeboid movement is found in :-
(1) Limbs, Jaw
(2) Macrophages, Leucocytes
(3) Trachea, Fallopian tube
(4) Tongue, Leucocytes
BLC181

5. In the center of I-Band, is an elastic fiber, which bisect it
(1) 'H'-zone (2) 'M'-line
(3) 'Z' line (4) A-band
BLC182

6. Motor neuron along with muscle fibers are connected & constitute a :-
(1) Neuro muscular junction
(2) Motor end plate
(3) Motor unit
(4) Both (1) & (2)
BLC183

7. Myosin head contains :-
(1) ATPase
(2) ATP binding sites
(3) Active sites for actin
(4) All of these
BLC184

8. Which points are correct for cardiac muscles?
(A) They are striated
(B) They are unbranched
(C) They are involuntary
(D) Intercalated disc are present
(1) A, B, C (2) B, C, D
(3) A, C, D (4) A, B, D
BLC185

9. In *Paramoecium*, the movement of food through cytopharynx take place by :-
(1) Cilia (2) Flagella
(3) Microvilli (4) Myonemes
BLC186

10. The striated appearance of the myofibril is due to the distribution pattern of :-
(1) Actin (2) Myosin
(3) Calmodulin (4) Both (1) and (2)
BLC187

11. Repeated activation of the muscles can lead to the accumulation of :-
(1) Acetic acid (2) Acetyl choline
(3) Lactic acid (4) Lactose
BLC188

12. Muscles contains a red coloured oxygen storing pigment called :-
(1) Haemoglobin (2) Sarcosome
(3) Myoglobin (4) Hemerythrins
BLC189

(B) ANALYTICAL QUESTIONS

13. In a resting state, the central part of thick filament which is not overlapped by thin filament is called :-
(1) A-band (2) M-line
(3) H-zone (4) I-band
BLC190

14. During contraction of muscle :
(1) Thin filaments slides over the thick filaments.
(2) Thick filament slides over the thin filaments
(3) Both filaments slide over each other.
(4) No filament slides during contraction.
BLC191

15. Which points are correct with the regarding of skeletal muscles?
(A) Their activities are under the voluntary control of the nervous system.
(B) They are located in the inner walls of visceral organs.
(C) They are primarily involved in locomotary actions and changes of body postures.
(D) They have no striped appearance
(1) B, D (2) B, C (3) A, D (4) A, C

BLC192

16. Which points are correct about visceral muscles?
(1) These are located in the inner walls of hollow visceral organs of the body.
(2) They do not exhibit any striations.
(3) Their activities are not under the voluntary control of the nervous system.
(4) All of the above

BLC193

17. Correctly match column-I with column-II.

Column-I	Column-II
(A) Tetany	(i) Auto immune disorder.
(B) Osteoporosis	(ii) Progressive degeneration of skeletal muscle.
(C) Muscular dystrophy	(iii) Inflammation of joints.
(D) Arthritis	(iv) Rapid spasms in muscle.
(E) Myasthenia Gravis	(v) Bone mass decreased.

(1) A→(i), B→(ii), C→(iii), D→(iv), E→(v)
(2) A→(iv), B→(v), C→(i), D→(iii), E→(ii)
(3) A→(iv), B→(v), C→(ii), D→(iii), E→(i)
(4) A→(ii), B→(iii), C→(i), D→(v), E→(iv)

BLC194

18. The functional unit of contraction is :-
(1) Muscle fibre (2) Sarcomere
(3) Sarosomes (4) Sarcoplasm

BLC195

19. The correct sequence of events during relaxation of muscle is :-
(i) Inactivation of the actin
(ii) Ca^{++} returned to SR
(iii) Cross bridges are broken
(1) (i), (ii), (iii) (2) (ii), (i), (iii)
(3) (ii), (iii), (i) (4) (iii), (ii), (i)

BLC196

20. Striated muscles are primarily involved in
(A) Locomotary actions
(B) Changes in body postures
(C) Transportation of food through digestive tract
(D) Transportation of gametes through genital tract
(1) A, B (2) B, C (3) C, D (4) A, D

BLC197

21. Which character is/are associated with locomotion?
(A) All locomotion are movements
(B) All movements are not locomotions
(C) Methods of locomotion vary with animals habitats
(D) Locomotion is also for escape from predators
(1) A and B (2) B, C and D
(3) A, B and C (4) A, B, C and D

BLC198

22. The thick filaments in 'A' band are held together in the middle of this band by a thin membrane called
(1) Z-line (2) M-line
(3) H-line (4) O-line

BLC199

23. A neural signal reaching neuromuscular junction releases a neurotransmitter namely
(1) Dopamine (2) Acetylcholine
(3) GABA (4) Glycine

BLC200

24. **Statement I :** All locomotion are movement but all movements are not locomotion.
Statement II : Walking, running, climbing and swimming such types of voluntary movements are called locomotion.
In the light of above statements, choose the correct answer from the option given below.
(1) Both statement-I and II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and II are correct.

BLC201

25. **Statement I :** Each myofibril has alternate dark and light band on it.
Statement II : Myosin filaments are thinner as compared to the actin filaments.
In the light of the above statements, choose the correct answer from the option given below.
(1) Both statement-I and II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and II are correct.

BLC202

26. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Increase Ca^{+2} level leads to the binding of calcium with a subunit of troponin on actin filaments.
Reason (R) : Action potential spreads through the muscle fibre and causes the release of calcium ion into the sarcoplasm.
In the light of the above statements, choose the correct answer from the option given below.
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BLC203

27. The type of muscles present in our :
(1) upper arm are smooth muscle fibres fusiform in shape.
(2) heart are involuntary and unstriated smooth muscles.
(3) intestine are striated and involuntary.
(4) thigh are striated and voluntary.

BLC204

28. Which one of the following pairs of chemical substances, is correctly categorised ?
(1) Troponin and myosin – Complex proteins in striated muscles
(2) Secretin and rhodopsin – Polypeptide hormones
(3) Calcitonin and thymosin – Thyroid hormones
(4) Pepsin and prolactin – Two digestive enzymes secreted in stomach

BLC205

29. The H-zone in the skeletal muscle fibre is due to:
(1) extension of myosin filaments in the central portion of the A-band
(2) the absence of myofibrils in the central portion of A-band
(3) the central gap between myosin filaments in the A-band
(4) the central gap between actin filaments extending through myosin filaments in the A-band

BLC206

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	3	2	3	3	4	3	1	4	3	3	3	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	4	3	2	2	1	4	2	2	4	2	4	4	1	4	

Exercise - I (Conceptual Questions) Build Up Your Understanding

NEURAL SYSTEM AND NEURAL TISSUE

1. Nissl's bodies found in neurons are :-
(1) Made of DNA
(2) Masses of RER
(3) Help in formation of neurofibrils
(4) Masses of mitochondria

BNC001

2. "Nodes of Ranviers" are found in :-
(1) Brain (2) Heart
(3) Axon (4) Eye

BNC002

3. Afferent nerve fiber conducts impulse from
(1) C.N.S. to effector
(2) Receptor to C.N.S.
(3) Receptor to effector
(4) Effector to receptor

BNC003

4. Nissl granules occur in which part and what is their function ?
(1) Neurons and help in protein synthesis
(2) Blood and help in nutrition and excretion
(3) Sarcoplasm and help in contraction
(4) Mucous cell and secrete mucous

BNC004

5. The parts of the neurons that perform basic cellular functions such as protein synthesis etc. :–
(1) Axons (2) Axon hillock
(3) Synaptic knobs (4) Soma

BNC005

6. The nerves leading to the central nervous system are called :-
(1) Afferent (2) Efferent
(3) Motor (4) None

BNC006

7. Unit of nervous system :-
(1) Neuron (2) Neuroglia
(3) Axon (4) Cyton

BNC007

8. Integrative system in the body are :-
(1) Endocrine system
(2) Nervous system
(2) Blood vascular system
(4) Both (1) & (2)

BNC008

9. The Schwann sheath is :-
(1) A non-myelinated nerve fibres
(2) Associated with myelin sheath
(3) A connective tissue cell
(4) Associated with myelinated & non-myelinated nerve fibre

BNC009

10. Rapid integration of the functional activities in human is achieved by :-
(1) Nervous system
(2) Endocrine system
(3) Blood
(4) Muscular system

BNC010

11. Which one of the following types of neurons are most numerous in the body ?
(1) Unipolar (2) Multipolar
(3) Bipolar (4) Pseudounipolar

BNC011

12. Sheath of schwann occurs on :-
(1) Neurons (2) Axons
(3) Dendrons (4) Neuroglia

BNC012

13. Which cell-organellae absent in neurons ?
(1) Mitochondria (2) Ribosome
(3) Centriole (4) Nucleus
BNC013

14. Nerve fibres are surrounded by an insulating fatty layer called :-
(1) Adipose sheath (2) Myelin sheath
(3) Hyaline sheath (4) Peritoneum
BNC014

15. Myelinogenesis (Myelin formation) process occur in C.N.S. (central nervous system)
(1) By schwann cells
(2) By oligodendrocytes
(3) By Axolemma
(4) By neurolemma
BNC015

16. Which of the following statement is correct for node of Ranvier of nerve ?
(1) Neurilemma is discontinuous
(2) Myelin sheath is discontinuous
(3) Both neurilemma & Myelin sheath are discontinuous
(4) Covered by myelin sheath
BNC016

17. The "Nissle's granules" of nerve cell's are made up of :-
(1) Ribosome (2) Protein
(3) DNA (4) Mitochondria
BNC017

18. Non Myelinated axons differ from myelinated in that they :-
(1) Are more excitable
(2) Lack nodes of Ranvier
(3) Are not capable of regeneration
(4) Are not associated with Schwann cells
BNC018

19. If myelin sheath is continuous in myelinated nerve fibre than what will happen in neuronal conduction?
(1) Velocity is increased
(2) Conduction is slow
(3) Conduction is stopped
(4) No effect
BNC019

20. Nerve cells do not possess -
(1) Neurilemma (2) Sarcolemma
(3) Dendrites (4) Axon
BNC020

21. Myelin sheath covers which of the following?
(1) Muscle fibre (2) Nerve fibre
(3) Collagen fiber (4) Tendons
BNC021

22. Dendrites are associated with which system
(1) Nervous system
(2) Digestive system
(3) Muscular system
(4) Blood vascular system
BNC022

23. The gray matter differs from white matter in the:-
(1) Absence of axons
(2) Absence of myelin sheath
(3) Presence of myelin sheath
(4) Absence of nurilemma
BNC023

24. Nerve cells possess
(a) Dendrites (b) Axon
(c) Sarcolemma (d) Neurilemma
(1) a, b (2) a, b, c
(3) a, b, d (4) a, b, c, d
BNC024

25. The nervous system is derived from :-
(1) Ectoderm
(2) Endoderm
(3) Mesoderm
(4) Ecto and Mesoderm

BNC025

26. The nerve cell can be distinguished from other cells of the body by the presence of :-
(1) Neuroplasm (2) Neurolemma
(3) Mitochondria (4) Neurites

BNC026

27. Function of nervous tissue is :-
(1) Irritability or Excitability
(2) Sensitivity
(3) Elasticity
(4) Contraction

BNC027

28. Which of the following processes occur only in animals ?
(1) Hormonal control
(2) Respiration
(3) Nervous control
(4) Nutrition

BNC028

29. Which cell in our body is more than a feet long?
(1) Nerve cell (2) Muscle cell
(3) Bone cell (4) Gland cell

BNC029

30. Which cell stop dividing after birth ?
(1) Epithelium (2) Neuron
(3) Glial cells (4) Liver

BNC030

31. In which animal, nerve cell is present but brain is absent ?
(1) Sponge (2) Earthworm
(3) Cockroach (4) Hydra

BNC031

32. Intercellular communication in multicellular organism occurs through -
(1) Nervous system only
(2) Digestive system only
(3) Respiratory system only
(4) Both nervous and endocrine system

BNC032

33. Myelin sheath is derived from :
(1) Astrocytes cells (2) Schwann cells
(3) Nerve cells (4) All of these

BNC033

34. Synaptic vesicles are found in –
(1) Presynaptic neuron
(2) Post synaptic neuron
(3) Synaptic cleft
(4) None of these

BNC034

35. Nissl's granules are absent in :
(1) Axon (2) Cyton
(3) Dendron (4) Dendrites

BNC035

36. In a myelinated neuron, two adjacent myelin sheaths are separated by gaps called
(1) Nodes of Ranvier (2) Synaptic cleft
(3) Synaptic knob (4) Neural plate

BNC036

37. Nissl's granules are found in :
(1) Liver cells (2) Nerve cells
(3) Kidney (4) Heart

BNC037

NERVE IMPULSE CONDUCTION

38. Chemical transmission of nerve impulses from one neuron to another at a synapse is by:-
(1) Cholesterol (2) Acetylcholine
(3) Cholecystokinin (4) ATP

BNC038

39. Saltatory conduction occurs in :-
(1) Non-myelinated fibers
(2) Myelinated fibers
(3) Both of them
(4) None of them

BNC039

40. When a nerve fibers is stimulated the inside of the membrane becomes :-
(1) Filled with acetyl choline
(2) Negatively charged
(3) Positively charged
(4) Neutral

BNC040

41. "Jumping of the action potential" at the nodes of ranvier is known as :-
(1) Saltatory conduction
(2) Neuro transmission
(3) Recovery phase
(4) Active phase

BNC041

42. Nerve impulses are generated by nerve fibers only when the membrane shall become more permeable to :-
(1) Adrenaline (2) Phosphorus
(3) Sodium ions (4) Potassium ions

BNC042

43. The functional connection between two neurons is called :-
(1) Synapse (2) Synapsis
(3) Chiasma (4) Chiasmata

BNC044

44. Conduction of nerve impulse is :-
(1) Faster in non-myelinated fibres
(2) Faster in myelinated fibres
(3) No difference in the rate of conduction in myelinated & non-myelinated fibres
(4) None of the above

BNC045

45. When the axons membrane is positively charged outside and negatively charged inside, then the condition is known as :-
(1) Action potential
(2) Resting potential
(3) Active potential
(4) Differential potential

BNC047

46. The main function of acetylcholine is to :-
(1) Increase heart beat
(2) Help in synaptic transmission of nerve impulse
(3) Help in conduction of nerve impulse through axon
(4) Control reflex action

BNC048

47. Depolarization of axolemma during nerve conduction takes place because of-
(1) Equal amount of Na^+ & K^+ move out across axolema
(2) Na^+ move inside
(3) More Na^+ outside
(4) None

BNC049

48. In the resting state of the neural membrane, diffusion due to concentration gradients, if allowed, would drive :-
(1) K^+ and Na^+ out of the cell
(2) Na^+ into the cell
(3) Na^+ out of the cell
(4) K^+ into the cell

BNC050

49. Unidirectional transmission of a nerve impulse through synapse fibre is due to :–
(1) Nerve fibre is insulated by a medullary sheath.
(2) Sodium pump starts operating only at the cyton and then continues into the nerve fibre.
(3) Neurotransmitters are released by dendrites and not by axon endings.
(4) Neurotransmitters are released by the axon endings and not by dendrites.

BNC051

50. Repolarisation of Neuron is occured due to:-
(1) Influx of Na^+ (2) Influx of K^+
(3) Efflux of Na^+ (4) Efflux of K^+

BNC052

51. If GABA is released at synapse area then what will happens :-
(1) Depolarization of neuron
(2) Repolarization of neuron
(3) Hyperpolarization of neuron
(4) No effect

BNC053

52. Nerve impulse travel through synapse with the help of
(1) Acetylcholine and sympathetin
(2) Choline and acetylcholine
(3) Adrenaline and noradrenaline
(4) None of the above

BNC054

53. A typical value of resting membrane potential is:
(1) –40 mv (2) –60 mv
(3) –70 mv (4) –80 mv

BNC055

54. Pre synaptic membrane is part of :
(1) Dendron (2) Axon hillock
(3) Telodendria (4) Soma

BNC056

55. Acetylcholinesterase enzyme splits acetylcholine into :
(1) Acetone and choline
(2) Acetic acid and choline
(3) Amino acid and choline
(4) Aspartic acid and acetylcholine

BNC057

56. During nerve impulse conduction permeability of a membrane is greater for :
(1) Na^+
(2) K^+
(3) Equal for both (1) and (2)
(4) Ca^{+2}

BNC058

57. Acetylcholinesterase help in breaking :
(1) Synapse (2) Acetylcholine
(3) Dendrites (4) Axon

BNC059

58. Action potential of a nerve cell is generated by :
(1) Na^+ (2) K^+ (3) Ca^{++} (4) Cl^-

BNC060

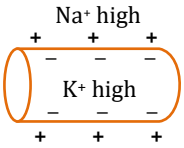
59. Connection between axon and dendrite is :
(1) Synapse (2) Synapsis
(3) Desmosome (4) Tight junction

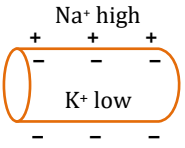
BNC061

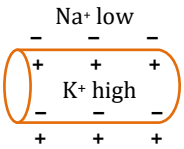
60. During repolarisation of nerve :
(1) K^+ gates closes and Na^+ gates opens
(2) Na^+ channels are closed and K^+ channels are open
(3) Both gates remain open
(4) Both K^+ and Na^+ gates are closed

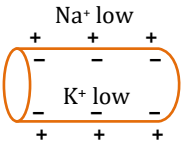
BNC062

61. Which of the following diagram illustrates the distribution of Na^+ and K^+ ions in a section of non-myelinated axon which is at resting potential?

(1) 

(2) 

(3) 

(2) 

BNC063

62. The one way or unidirectional transmission of nerve cells is due to :

(1) Synapses (2) Myelin sheath

(3) Membrane polarity (4) Interneurons

BNC064

63. The forebrain develops into-

(1) Diencephalon and Cerebrum

(2) Diencephalon and Cerebellum

(3) Diencephalon and Medulla

(4) Diencephalon and Pons

BNC065

64. Nerve impulse initiates with the movement of:

(1) K^+ (2) Na^+ (3) Ca^{++} (4) Mg^{+2}

BNC066

65. Na^+/K^+ pump in a cell is an example of –

(1) Osmosis (2) Diffusion

(3) Passive transport (4) Active transport

BNC067

STRUCTURE AND FUNCTION OF BRAIN

66. Corpus callosum connects :-

(1) Two cerebral hemisphere

(2) Two optic lobes

(3) Two olfactory lobes

(4) Optic chiasma

BNC068

67. Outer most covering of brain is called :-

(1) Choroid (2) Duramater

(3) Piamater (4) Arachnoid

BNC069

68. The membrane which cover the brain and the spinal cord is :-

(1) White matter (2) Grey matter

(3) Peritonium (4) Menix

BNC070

69. Cerebellum is concerned with :-

(1) Co-ordination of muscular movement

(2) Memory

(3) Vision

(4) Reflex action

BNC071

70. Crura cerebri is located in :-

(1) Fore brain (2) Hind brain

(3) Mid brain (4) None

BNC072

71. Piamater is :-

(1) Inner most meninge (2) Middle meninge

(3) Outer meninge (4) None

BNC073

72. The box like bony structure which encloses the brain is called :-

(1) Cranium (2) Pericardium

(3) Peritoneum (4) Periosteum

BNC074

73. Which one of the following menix is present only in mammalian brain :-

(1) Duramater (2) Arachnoid

(3) Piamater (4) None of them

BNC076

74. Small, solid and four optic lobes or colliculus called corpora quadrigemina are found in :-

(1) Mammals (2) Amphibians

(3) Aves (4) Reptiles

BNC077

75. Hypothalamus is situated on the :-
(1) Upper lateral surface of diencephalon
(2) Lower lateral surface of diencephalon
(3) Ventral side of optic lobes
(4) Dorsal side of optic lobes

BNC078

76. Epithalamus is situated on the :-
(1) Roof of diencephalon
(2) Lateral wall of diencephalon
(3) Dorsal side of optic lobes
(4) Floor of diencephalon

BNC079

77. Which of the following is a richly vascular layer with lots of blood capillaries :-
(1) Duramater (2) Piamater
(3) Epidermis of skin (4) Both (1) & (2)

BNC080

78. Which of the following is not a part of hind brain :–
(1) Medulla oblongata (2) Thalamus
(3) Cerebellum (4) Pons

BNC081

79. Which is correct about pons varolii ?
(1) Situated between midbrain & M.O.
(2) Pons contains pneumotaxic centre
(3) Inner gray, outer white matter
(4) All of the above

BNC082

80. If the corpus callosum is removed in mammalian brain then what will be affected:-
(1) Coordination of Cerebrum
(2) Involuntary activity of brain
(3) Coordination of Cerebellum
(4) Behaviour and emotional disturbances

BNC083

81. The name of nervous band connecting both the cerebral hemispheres in your brain -
(1) Corpus albicans
(2) Corpus callosum
(3) Corpus striatum
(4) Corpus spongiosum

BNC084

82. Arbor vitae is a part of -
(1) Cerebrum (2) Cerebellum
(3) Midbrain (4) Forebrain

BNC085

83. Human brain is covered by a hard layer called
(1) White matter
(2) Duramater
(3) Piamater
(4) Gray matter

BNC086

84. Which of the following is the part of mid brain?
(1) Cerebrum
(2) Diencephalon
(3) Corpora quadrigemina
(4) None of these

BNC087

85. The function of cerebrospinal fluid does not include:
(1) Protection of brain and spinal cord by containing antibody
(2) Protection of delicate brain and spinal cord from shock
(3) As a medium for excretion of waste product
(4) Buoyancy to brain

BNC088

86. Cerebral hemispheres of mammals consist of-
(1) Outer gray matter and central white matter
(2) Outer white matter and central gray matter
(3) Gray matter and white matter inter mingled
(4) Gray matter only
BNC089

87. Which of the following forms the cerebro-spinal fluid:-
(1) Choroid plexus
(2) Duramater
(3) Arachnoid mater
(4) Cerebrum and spinal cord
BNC090

88. If Broca's area is completely injured then what happen firstly:–
(1) Concerning speech muscle are paralysed
(2) Speech stuttered & not clear
(3) Unable to speak
(4) Only able to speak written word
BNC091

89. In man the osmotic centres are situated in -
(1) Cerebrum
(2) Hypothalamus
(3) Pituitary gland
(4) Medulla oblongata
BNC092

90. Hypothalamus does not control -
(1) Libido
(2) Osmoregulation
(3) Creative thinking and consciousness
(4) Thermoregulation
BNC093

91. Choroid plexus is a network of -
(1) Nerves (2) Muscle fibres
(3) Capillaries (4) Lymph vessels
BNC094

92. Voluntary activites of body are controlled by-
(1) Diencephalon (2) Cerebrum
(3) Crura cerebri (4) Cerebellum
BNC095

93. Thermo regulatory center in human brain is :-
(1) Pituitary (2) Diencephalon
(3) Hypothalamus (4) None
BNC096

94. Respiratory control in brain occurs in :-
(1) Medulla oblongata (2) Cerebellum
(3) Hypothalamus (4) Pericardium
BNC097

95. Drinking of alcohol affects mostly:-
(1) Cerebrum (2) Cerebellum
(3) Medulla oblongata (4) Diencephalon
BNC098

96. Which part of the brain regulates the body temperature, hunger and water balance :-
(1) Hypothalamus (2) Infundibulum
(3) Medulla oblongata (4) Pons veroli
BNC099

97. Most of the involuntary action are controlled by :-
(1) Medulla oblongata (2) Cerebrum
(3) Cerebellum (4) Diencephalon
BNC100

98. When the medulla oblongata (M.O.) is damaged, then what happen ?
(1) Immediately die
(2) Die after few hrs.
(3) Live at 1 hrs & after it may die
(4) No affect
BNC101

99. Cerebellar hemisphere is the centre of :-
(1) Taste (2) Smell
(3) Balance (4) Thinking
BNC102

100. Which part of brain is supposed to be damaged if in an accident, a person lost control of water balance, hunger and body temp. :-
(1) Cerebellum
(2) Hypothalamus
(3) Medula oblongata
(4) Corpora quadrigemina

BNC103

101. Which part of brain controls emotions like love, anger and pleasure –
(1) M. O. (2) Hypothalamus
(3) Mid brain (4) Cerebellum

BNC104

102. Hippocampal lobes are the parts of :-
(1) Olfactory lobes
(2) Cerebrum
(3) Cerebellum
(4) Medulla Oblongata

BNC107

103. Which part of the brain is more developed in human?
(1) Medulla (2) Cerebellum
(3) Cerebrum (4) Optic lobes

BNC108

104. Which of the following is not correctly matched :-
(1) Cerebrum – Olfaction
(2) Hypothalamus – Pituitary
(3) Cerebellum – Balance
(4) Mid brain – Temperature regulation

BNC109

105. The correct sequence of meninges of brain from outside to inside is :
(1) Duramater → Arachnoid → Piamater
(2) Arachnoid → Duramater → Piamater
(3) Piamater → Duramater → Arachnoid
(4) Duramater → Piamater → Arachnoid

BNC110

106. The thermoregulatory centre in the body of man is found in :
(1) Diencephalon
(2) Hypothalamus
(3) Pituitary
(4) Skin

BNC111

107. Part of brain concerned with the muscular movement is :
(1) Cerebellum
(2) Thalamus
(3) Hippocampus
(4) Temporal lobe of cerebrum

BNC112

108. In human corpus callosum connects :
(1) The two optic lobes
(2) Bone and muscle
(3) The two cerebral hemispheres
(4) Two lobes of pituitary gland

BNC113

109. Broca's area of speech is present in :
(1) Parietal lobe
(2) Temporal lobes
(3) Temporal and occipital lobe
(4) Frontal lobe

BNC114

110. In human body muscular co-ordination is controlled by:
(1) Spinal cord
(2) Cortex
(3) Cerebellum
(4) Cerebral hemisphere

BNC115

111. The optic lobe in humans are represented by the corpora –
(1) Bigemina
(2) Arenacea
(3) Striata
(4) Quadrigemina

BNC116

SPINAL CORD, PNS, ANS

112. Parasympathetic system increase activity of:
(1) Lacrimal gland, sweat gland, arrector pili
(2) Heart, lacrimal gland, pancreas
(3) Heart, adrenal gland and sweat gland
(4) Gut, urinary bladder and digestive gland
BNC128

113. Number of spinal nerves in human :-
(1) 31 pairs (2) 32 pairs
(3) 12 pairs (4) 37 pairs
BNC132

114. In human, autonomic nervous system is composed of :-
(1) Sympathetic and parasympathetic nerves
(2) Cranial and spinal nerves
(3) Brain and spinal nerves
(4) Medullated and non-medullated nerves
BNC134

115. Autonomic nervous system controls :-
(1) Conditioned reflexes
(2) Functioning of spinal cord
(3) Functioning of visceral organs
(4) Reflex actions
BNC139

116. Stimulation of sympathetic nervous system causes :-
(1) Constriction of blood vessels and high blood pressure
(2) Dilation of bronchi & pupil
(3) Erection of hair
(4) All of the above
BNC140

117. The cranial and spinal nerves are included under-
(1) Autonomic nervous system
(2) Peripheral nervous system
(3) Central nervous system
(4) Cutaneous nervous system
BNC144

118. Conservation of energy take place by :-
(1) Sympathetic nervous system
(2) Parasympathetic nervous system
(3) Reflex action
(4) None
BNC145

119. If parasympathetic nerve is cut, then heart beat–
(1) Unaffected (2) Decreases
(3) Increases (4) Stop
BNC146

120. Norepinephrine leads to increase in :–
(1) Blood pressure
(2) Urine production
(3) Cellular respiration
(4) Release of epinephrine
BNC147

121. In emergency condition, all changes occur in our body except –
(1) Heart beat increases
(2) Dilates blood vessels of brain, lungs, heart and striated muscle
(3) Brochodilation
(4) Micturition is done
BNC150

122. Which of the following is released by parasympathtic nervous system ?
(1) Serotonin (2) Acetylcholine
(3) Epinephrine (4) Norepinephrine
BNC155

123. Number of cranial nerve in human is -
(1) Ten only (2) Ten pairs
(3) Twenty pairs (4) Twelve pairs
BNC157

124. Which of the following pair is mismatched–
(1) Cerebrum – voluntary activities
(2) Cerebellum – body balance
(3) M. O. – Pneumotaxic centre
(4) Spinal cord – reflex action
BNC158

125. After sympathetic stimulation, which type of activities are not present in human being :
(1) Tachycardia
(2) Bronchodilation
(3) Micturition
(4) Semen Ejaculation

BNC159

126. You are watching a horror movie and you notice your heart is beating fast and mouth is dry. It is because of -
(1) Fight and flight response
(2) Autonomic nervous system
(3) Sympathetic nervous system
(4) Both (1) and (3)

BNC161

127. ANS effects on :-
(1) Reflex actions
(2) Sensory organs
(3) Internal organs
(4) None of these

BNC164

128. Which one of the following is not an effect of sympathetic nervous system –
(1) Dilation of pupil
(2) Inhibition of peristalsis
(3) Elevation of blood pressure
(4) Stimulation for saliva secretion

BNC167

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	1	4	1	1	4	4	1	2	2	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	2	3	2	2	1	2	3	1	4	1	3	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	2	1	1	1	2	2	2	3	1	3	1	2	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	2	4	4	3	1	3	3	2	1	2	1	1	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	1	1	2	4	1	2	4	1	3	1	1	2	1	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	2	2	4	1	2	2	2	3	1	1	1	3	2	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	2	3	1	2	1	1	1	3	2	2	2	3	4	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	1	3	4	3	4	4	1	1	3	4	2	2	3	1
Question	121	122	123	124	125	126	127	128							
Answer	4	2	4	3	3	4	3	4							

AIM: FREE EDUCATION TO ALL APUL

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Injury localized to the hypothalamus would most likely to disrupt :
(1) short - term memory.
(2) co-ordination during locomotion.
(3) executive functions, such as decision making.
(4) regulation of body temperature.

BNC182

AIPMT 2015

2. Which of the following regions of the brain is incorrectly paired with its function?
(1) Cerebellum- language comprehension
(2) Corpus callosum-communication between the left and right cerebral cortex
(3) Cerebrum- calculation and contemplation
(4) Medulla oblongata - homeostatic control

BNC183

NEET-UG 2017

3. Myelin sheath is produced by:
(1) Astrocytes and Schwann cells
(2) Oligodendrocytes and Osteoclasts
(3) Osteoclasts and Astrocytes
(4) Schwann cells and Oligodendrocytes

BNC184

4. Receptor sites for neurotransmitters are present on:
(1) Pre-synaptic membrane
(2) Tips of axons
(3) Post-synaptic membrane
(4) Membrane of synaptic vesicles

BNC185

NEET-UG 2018

5. Which of the following structures or regions is ***incorrectly*** paired with its function?

(1)	Medulla oblongata	controls respiration and cardiovascular reflexes.
(2)	Limbic system	consists of fibre tracts that interconnect different regions of brain; controls movement.
(3)	Hypothalamus	production of releasing hormones and regulation of temperature hunger and thirst.
(4)	Corpus callosum	band of fibres connecting left and right cerebral hemispheres.

BNC186

NEET-UG 2019

6. Which part of the brain is responsible for thermoregulation?
(1) Cerebrum
(2) Hypothalamus
(3) Corpus callosum
(4) Medulla oblongata

BNC187

NEET-UG 2019 (Odisha)

7. Which of the following statements is **not** correct?
(1) An action potential in an axon does not move backward because the segment behind is in a refractory phase
(2) Depolarisation of hair cells of cochlea results in the opening of the mechanically gated potassium -ion channels.

(3) Rods are very sensitive and contribute to daylight vision.

(4) In the knee-jerk reflex, stimulus is the stretching of muscle and response is its contraction.

BNC188

NEET-UG 2022

8. Select the incorrect statement regarding synapses.

(1) Electrical current can flow directly from one neuron into the other across the electrical synapse.

(2) Chemical synapses use neurotransmitters

(3) Impulse transmission across a chemical synapse is always faster than that across an electrical synapse.

(4) The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.

BNC189

Re-NEET-UG 2022

9. Match **List-I** with **List-II.**

	List-I		List-II
(a)	Multipolar neuron	(i)	Somatic neural system
(b)	Bipolar neuron	(ii)	Cerebral cortex
(c)	Myelinated nerve fibre	(iii)	Retina of Eye
(d)	Unmyelinated nerve fibre	(iv)	Spinal nerves

Choose the **correct answer** from the options given below :-

(1) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

(2) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i)

(3) (a)–(ii), (b)–(iii), (c)–(i). (d)–(iv)

(4) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)

BNC190

NEET-UG 2023

10. The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc. are

(1) Corpora quadrigemina & hippocampus

(2) Brain stem & epithalamus

(3) Corpus callosum and thalamus

(4) Limbic system & hypothalamus

BNC225

NEET-UG 2023 (MANIPUR)

11. Brain stem of human brain consists of :-

(1) Mid-brain, Pons and Medulla Oblongata

(2) Forebrain, Cerebellum and Pons

(3) Thalamus, Hypothalamus and Corpora quadrigemina

(4) Amygdala, Hippocampus and Corpus Callosum

BNC226

NEET (UG) 2024

12. Match **List I** with **List II** :-

List I		List II	
A.	Pons	I.	Provides additional space for Neurons, regulates posture and balance.
B.	Hypothalamus	II.	Controls respiration and gastric secretions
C.	Medulla	III.	Connects different regions of the brain
D.	Cerebellum	IV.	Neuro secretory cells

Choose the correct answer from the options given below :-

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-IV, C-II, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-II, B-I, C-III, D-IV

BNC227

13. Given below are two statements :-

Statement-I : The cerebral hemispheres are connected by nerve tract known as corpus callosum.

Statement-II : The brain stem consists of the medulla oblongata, pons and cerebrum.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

BNC228

RE-NEET (UG) 2024

14. Match **List-I** with **List-II.**

	List-I		List-II
(a)	Schwann cells	(i)	Neurotransmitter
(b)	Synaptic knob	(ii)	Cerebral cortex
(c)	Bipolar neurons	(iii)	Myelin sheath
(d)	Multipolar neurons	(iv)	Retina

Choose the **correct answer** from the options given below :-

(1) A-III, B-I, C-IV, D-II
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-III, C-I, D-IV

BNC229

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	4	1	4	3	2	2	3	3	4	4	1	2	3	1	

Exercise - III

Master Your Understanding

(A) NCERT Based QUESTIONS

1. Which system provides an organised network of point to point connection :-

(1) Integrated system

(2) Neuro-endocrine system

(3) Endocrine system

(4) Nervous system

BNC191

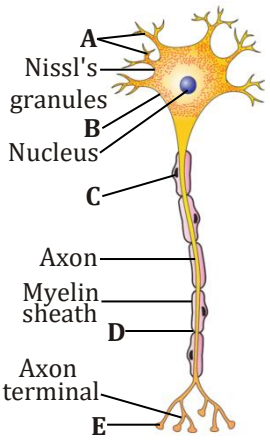
2. Which role of neuron regarding different kinds of stimuli is absent :-

(1) detect (2) receive

(3) transmit (4) protect

BNC192

3. The accompanied diagram shows the structure of neuron. Identify A to E.



	A	B	C	D	E
(1)	Nerve fibre	Cyton or cell body	Schwann cell	Node of Ranvier	Synaptic knob
(2)	Dendrites	Cyton or cell body	Schwann cell	Node of Ranvier	Synaptic knob
(3)	Dendrites	Nerve cell	Schwann cell	Node of Ranvier	Synaptic knob
(4)	Dendrites	Cyton or cell body	Nerve cell	Node of Ranvier	Synaptic knob

BNC193

4. When a neuron is not conducting any impulse i.e. resting, the axonal membrane is-

(1) Comparatively more permeable to K^+ and impermeable (nearly impermeable) to Na^+

(2) Impermeable to negatively charged proteins present in the axoplasm

(3) (1) & (2) Both

(4) More permeable to Na^+ ions than K^+ ion

BNC194

5. Pick out the incorrect statement?

(1) Myelinated nerve fibres are found in spinal and cranial nerve.

(2) Unmyelinated nerve fibre is enclosed by a schwann cell.

(3) In resting stage the axonal membrane is comparatively more permeable to potassium ion and nearly impermeable to sodium ions.

(4) Axolemma is more permeable to negatively charged protein present in the axoplasm.

BNC195

6. During depolarization of the neuronal membrane-

(1) Na^+ ions rapidly move inside the cell

(2) Na^+ ions rapidly move outside the cell

(3) K^+ ions rapidly move outside the cell

(4) K^+ ions rapidly move inside the cell

BNC196

7. Identify the correct set of statements :-

(a) Limbic lobe consists of amygdala and pons.

(b) Cerebral aqueduct passes through the midbrain.

(c) The medulla contains centre which control respiration, cardiovascular reflexes and gastric secretions.

(d) Brain stem forms the connections between the cerebrum and midbrain.

(e) Knee jerk reflex is polysynaptic reflex.

Choose the correct answer from options given below :

(1) (a) and (d) only

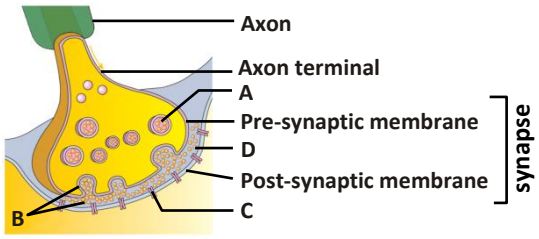
(2) (b), (c), (d) and (e) only

(3) (a), (b), (d) and (e) only

(4) (b) and (c) only

BNC197

8. Study the diagram of synapse-



I. Which alphabet indicate the location of the receptor molecules ?

II. Which alphabet points to a synaptic vesicles

III. Which alphabet points to neurotransmitter

IV. Which alphabet points to synaptic cleft

	I	II	III	IV
(1)	C	A	B	D
(2)	B	A	C	D
(3)	C	A	D	B
(4)	C	D	A	B

BNC198

9. Which of the following statements is false about the electrical synapse ?

I. At electrical synapses, the membranes of pre and post synaptic neurons are in very close proximity.

II. Electrical current can flow directly from one neuron into the other across the synapses.

III. Transmission of an impulse across electrical synapses is very similar to impulse conduction along single axon.

IV. Electrical synapses pass electrical signal between cells with the use of Ach

V. Electrical synapses are fast.

VI. Electrical synapses are rare in our system.

(1) I and II (2) Only II

(3) Only IV (4) Only V

BNC199

10. Identify the correct set of statements :

(a) Myelinated nerve fibres are found in autonomic and the somatic neural systems.

(b) The gaps between two adjacent myelin sheaths are called nodes of Ranvier.

(c) The axonal membrane is impermeable to negatively charged proteins present in the axoplasm.

(d) Electrical synapses are rare in our system.

(e) Synaptic cleft is present in chemical synapse.

Choose the correct answer from options given below :

(1) (a) and (d) only

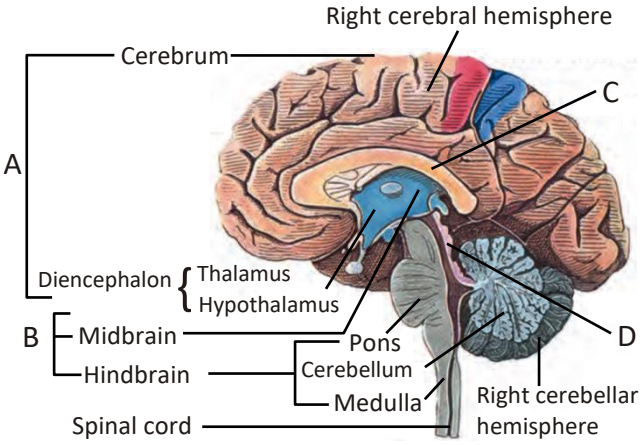
(2) (b), (c), (d) and (e) only

(3) (a), (b), (d) and (e) only

(4) (b) and (c) only

BNC200

11. Identify A, B, C and D in given diagram –



(1) A–Forebrain, B–Brainstem, C–Corpus callosum, D–Cerebral aqueduct

(2) A–Forebrain, B–Brainstem, C–Cerebral aqueduct, D–Corpus callosum

(3) A–Brainstem, B–Forebrain, C–Corpus callosum, D–Cerebral aqueduct

(4) A–Brainstem, B–Forebrain, C–Cerebral aqueduct, D–Corpus luteum

BNC201

12. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Multipolar neurons are found in the cerebral cortex.

Reason (R) : They consists of one axon and two or more dendrites.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BNC202

13. The name of nervous band connecting the cerebral hemispheres is -

(1) Corpus albicans

(2) Corpus callosum

(3) Corpus striatum

(4) Corpus spongiosum

BNC203

14. Which of the following statements is incorrect about cortex of cerebrum ?

(1) It consists of grey matter

(2) It consists of white matter

(3) It shows prominent folds

(4) It contains motor areas, sensory areas and association areas.

BNC204

15. Association areas in cerebral cortex are -

(1) Sensory areas

(2) Motor areas

(3) Responsible for intersensory associations, memory and communication

(4) None of the above is correct

BNC205

16. The cerebrum wraps around a structure called thalamus, which is-

(1) A major coordinating centre for sensory signal only

(2) A major centre for motor signaling

(3) A major coordinating centre for sensory and motor signaling

(4) Not a nervous part of a brain

BNC206

17. Hypothalamus does not control -

(1) Thermoregulation

(2) Urge for eating and drinking

(3) Produces hormones that regulate the synthesis and secretion of pituitary hormone

(4) Creative thinking and consciousness

BNC207

(B) ANALYTICAL QUESTIONS

18. The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc; form a complex structure called-

(1) Reticular system

(2) Corpora quadrigemina

(3) Limbic lobe/limbic system

(4) Arbor vitae

BNC208

19. Which of the following statements or structures is not correct about the midbrain?

(1) Located between the thalamus / hypothalamus and pons

(2) Has arbor vitae

(3) Has a canal (Cerebral aqueduct)

(4) Its dorsal part consists of 4 lobes (corpora quadrigemina)

BNC209

20. Read the following statements and choose the correct option –

I. Cerebellum has very convoluted surface in order to provide the additional space for more neurons.

II. The medulla is connected to the spinal cord

III. Medulla contains controlling centres for respiration, cardiovascular reflexes and gastric secretion.

(1) All are correct

(2) Only I is correct

(3) Only I and III are correct

(4) Only II is correct.

BNC210

21. Where A stands for axon, D for dendrite, S for synapse, and CB for cell body, a typical sequence of structures between a receptor and an efferent is?

(1) D – CB – A – S – D – CB – A

(2) A – D – CB – S – A – D – CB

(3) D – CB – A – S – A – CB – D

(4) D – A – S – CB – D– A – CB

BNC212

22. Read the following statements :-

(A) Nervous system provides an organised network of point to point connection.

(B) The endocrine system provides chemical integration through hormone.

(C) The neural organisation is very complex in lower invertebrates.

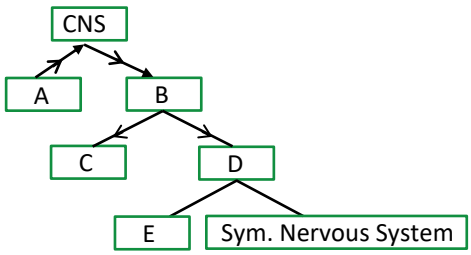
(D) Neuron can detect & receive stimuli but can't transmit.

How many of above statements are false.

(1) 4 (2) 3 (3) 2 (4) 1

BNC214

23. Which of the following answer shows the correct arrangement of nerve fibre.



(1) (A)–Centrifugal, (B)–Efferent, (C)–PNS, (D–ANS, (E)–Parasympathetic

(2) (A)–Centripetal, (B)–Afferent, (C)–PNS, (D)–ANS, (E) Parasympathetic

(3) (A)–Centrifugal, (B)–Afferent, (C)–SNS, (D)–ANS, (E) Parasympathetic

(4) (A)–Centripetal, (B)–Efferent, (C)–SNS, (D) ANS, (E)–Parasympathetic

BNC216

24. Which statement is correct regarding neuron.

(1) Neuron is composed of 4 parts containing cell body dendrite, axon & telodendria

(2) Nissl's granules are found in both cell body & axon

(3) Impulse are divided into apolar, bipolar & multipolar

(4) Dendrites transmit impulse toward the body while axon transmit impulse away from the body

BNC217

25. Which of the nervous system transmit impulse from CNS to involuntary organs & smooth muscle ?

(1) Sympathetic nervous system

(2) Parasympathetic nervous system

(3) Autonomic nervous system

(4) Somatic nervous system

BNC218

26. Read the following statement :-

(A) Peripheral nervous system divide's in somatic nervous system & autonomic nervous system.

(B) Central nervous system includes brain & vertebral column.

(C) Afferent nerve fibre transmit impulse from CNS and vise versa is also correct

How many of above statements is/are false

(1) A & B (2) B & C

(3) C & A (4) All

BNC219

27. Which of the following statement is correct.

(A) The electrical potential difference across the polarised plasma membrane is called as action potential in resting stage.

(B) Na^+ ion is responsible for generating an action potential.

(C) The impulse is action potential

(1) A & B (2) B & C

(3) C & A (4) All

BNC220

28. Which of the statement is false regarding synapse?

(1) Synapse is formed by 2 membrane first presynaptic membrane of synaptic knob & second post synaptic membrane of dendrite.

(2) Synaptic membrane always be separated by a gap called synaptic cleft.

(3) Electrical synapse in very similar to impulse conduction along a single axon.

(4) In chemical synapse, neurotransmitter is released and either excitatory or inhibitory potential is generated on post synaptic membrane.

BNC221

29. $Na^+ – K^+$ pump –

(A) Transports $3Na^+$ inwards & $2K^+$ outwards.

(B) Maintain ionic gradients by active transport.

(C) Works against a concentration gradient.

How many of above statements are false?

(1) All are correct (2) 3

(3) 2 (4) 1

BNC222

30. Which of the following are controlled by limbic system :-

(A) Emotional reaction

(B) Sexual behaviour

(C) Respiration

(D) Olfaction

(E) Body balance

Choose the correct option –

(1) A, B, C, D, E (2) A, B, D

(3) A, B, C, D (4) A, B, D, E

BNC223

31. The prime area of brain where different type of information are integrated –

(1) CNS (2) ANS

(3) PNS (4) SNS

BNC224

32. Which one of the following statements is correct?

(1) Neither hormones control neural activity nor the neuron control endocrine activity

(2) Endocrine glands regulate neural activity, but not vice versa

(3) Neurons regulate endocrine activity, but not vice versa

(4) Endocrine glands regulate neural activity, and nervous system regulates endocrine glands

BNC173

33. Which one of the following does not act as a neurotransmitter?

(1) Norepinephrine

(2) Cortisone

(3) Acetylcholine

(4) Epinephrine

BNC174

34. During the transmission of nerve impulse through a nerve fibre, the potential on the inner side of the plasma membrane has which type of electric charge?

(1) First positive, then negative and continue to be negative

(2) First negative, then positive and continue to be positive

(3) First positive, then negative and again back to positive

(4) First negative, then positive and again back to negative

BNC175

35. Which one of the following pairs of structures distinguishes a nerve cell from other types of cell?

(1) Vacuoles and fibres

(2) Flagellum and medullary sheath

(3) Nucleus and mitochondria

(4) Perikaryon and dendrites

BNC176

36. During the propagation of a nerve impulse, the action potential results from the movement of :

(1) K^+ from intracellular fluid to extra-cellular fluid

(2) Na^+ from extracellular fluid to intra-cellular fluid

(3) K^+ from extracellular fluid to intra-cellular fluid

(4) Na^+ from intracellular fluid to extra-cellular fluid

BNC177

37. The nerve centres which control the body temperature and the urge for eating are contained in:

(1) Cerebellum

(2) Thalamus

(3) Hypothalamus

(4) Pons

BNC178

38. A person entering an empty room suddenly finds a snake right in front on opening the door. Which one of the following is likely to happen in his neuro-hormonal control system?

(1) Hypothalamus activates the parasympathetic division of brain

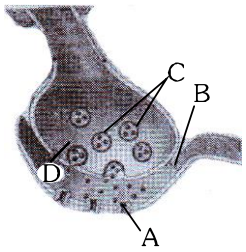
(2) Sympathetic nervous system is activated releasing epinephrine and norepinephrine from adrenal cortex

(3) Sympathetic nervous system is activated releasing epinephrine and norepinephrine from adrenal medulla

(4) Neurotransmitters diffuses rapidly across the synaptic cleft and transmit a nerve impulse

BNC179

39. A diagram showing axon terminal and synapse is given below. Identify correctly at least two of A-D.



(1) C-Neurotransmitter, D-Ca^{++}

(2) A-Receptor, C-Synaptic vesicles

(3) B-Synaptic connection, D-K^{+}

(4) A-Neurotransmitter, B-Synaptic cleft

BNC180

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	2	3	4	1	4	1	3	2	1	1	2	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	3	2	1	1	3	4	4	3	2	2	2	4	2
Question	31	32	33	34	35	36	37	38	39						
Answer	1	4	2	4	4	2	3	3	2						

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION AND MECHANISM OF HORMONE ACTION

1. A hormone is :-
(1) An enzyme
(2) Chemical messenger
(3) Primary messenger
(4) Both (2) and (3)

BCI001

2. Endocrine glands can be defined as those glands which pour their secretion :-
(1) Directly into blood
(2) Into blood or ducts
(3) When they are cut
(4) into particular organ

BCI003

3. The receptor for protein hormones are present on :-
(1) Nucleus
(2) Endoplasmic reticulum
(3) Cytoplasm
(4) Cell-surface

BCI004

4. Hormones are :-
(1) Internal secretion mostly discharged in the blood by endocrine glands
(2) Secretion of exocrine glands
(3) Chemical substances secreted into the gut
(4) Inorganic catalysts

BCI005

5. Hormones are :-
(1) Produced in low amount
(2) Easily diffusible
(3) Non - antigenic
(4) All

BCI006

6. Statement not correct for hormones is that, these:-
(1) Are not all protein
(2) Are secreted in small amount
(3) Affect metabolism
(4) Acts as catalyst

BCI008

7. "Secondary messenger" is :-
(1) Cyclic AMP (2) ATP
(3) ADP (4) DNA

BCI009

8. Hormones are chemically :-
(1) Amino acid (2) Protein
(3) Steroid (4) All

BCI010

9. First discovered hormone :-
(1) Thyroxine (2) Adrenaline
(3) Secretin (4) Insulin

BCI011

10. Which of the following is not a steroid hormone?
(1) Androgen (2) Aldosterone
(3) Estrogen (4) Relaxin

BCI012

11. Which of the following is not an endocrine gland?
(1) Pancreas (2) Adrenal gland
(3) Thyroid gland (4) Salivary gland

BCI013

12. Which of the following hormones is not proteinaceous is nature ?
(1) TSH (2) Aldosterone
(3) LH (4) FSH

BCI014

13. Which of the hormone is polypeptide ?

(1) LH (2) FSH

(3) Insulin (4) Thyroxine

BCI015

14. Steroid hormones transmit their information by :

(1) Stimulating the receptors present on cell membrane.

(2) Entering into the cell and modifying cellular contents.

(3) Entering into the cell and modifying nuclear organisation.

(4) The help of an intracellular second messenger.

BCI016

15. Which is not a peptide hormone-

(1) Insulin (2) Cortisol

(3) Glucagon (4) All of these

BCI018

16. If receptor molecule is removed from target organ for hormone action, the target organ will :

(1) Continue to respond but require higher concentration of hormone.

(2) Continue to respond but in opposite way.

(3) Continue to respond without any difference.

(4) Not respond to hormone.

BCI019

17. Prostaglandins are –

(1) Amino acid (2) Steroid

(3) Fatty acid (4) Carbohydrate

BCI020

PITUITARY GLAND AND HYPOTHALAMUS

18. Pituitary gland does not control the secretory activity of :-

(1) Thyroid (2) Adrenal cortex

(3) Adrenal medulla (4) Testes

BCI021

19. Which of the following controls spermatogenesis :-

(1) FSH (2) LTH

(3) LH (4) Vasopressin

BCI022

20. Which is called "Master gland" of the body :-

(1) Thyroid (2) Pituitary

(3) Thymus (4) Adrenal

BCI023

21. Hormone secreted by pituitary gland are chemically –

(1) All protein

(2) All steroid

(3) Complex compounds of proteins and carbohydrates

(4) Some steroid and some protein

BCI026

22. Gonadotrophic hormone is produced by :-

(1) Interstitial cells of testis

(2) Adrenal cortex

(3) Adenohypophysis

(4) Posterior part of thyroid

BCI028

23. The hormones FSH and LH are together called:

(1) Emergency hormone

(2) Neuro hormone

(3) Gonadotrophic hormone

(4) Antistress hormone

BCI029

24. Gigantism and acromegaly are due to :-
(1) Hyperpituitrism (2) Hypopituitrism
(3) Hypothyroidism (4) Hyperthyroidism
BCI030

25. If amount of ADH decrease in blood, micturition :-
(1) Remains unchanged
(2) Decreases
(3) Increases
(4) Does not occur
BCI031

26. The follicle stimulating hormone is secreted from:-
(1) Posterior lobe of pituitary gland
(2) Reproductive gland
(3) Thyroid gland
(4) Anterior lobe of pituitary gland
BCI033

27. Pituitary gland is under control of :-
(1) Hypothalamus (2) Adrenal gland
(3) Pineal gland (4) Thyroid gland
BCI034

28. Vasopressin is responsible for :-
(1) Controlling Oogenesis
(2) Regulating blood pressure and act on the nephron tubules.
(3) Regulating formation of pigment.
(4) Controlling spermatogenesis.
BCI036

29. The main function of prolactin hormone is to:
(1) Influence the activity of thyroid gland
(2) Control development of Graffian follicles
(3) Initiate and maintain secretion of milk by mammary gland
(4) Cause ejection of milk
BCI037

30. I.C.S.H. in male acts on :-
(1) Cells of leydig (2) Sertoli cells
(3) Spermatids (4) Spermatogonia
BCI039

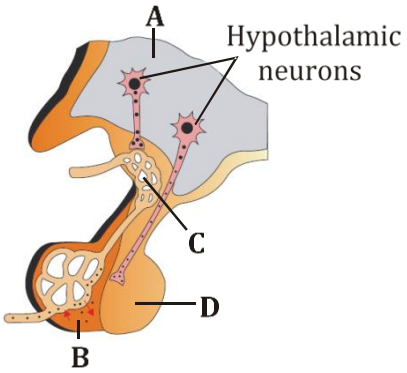
31. Hypophysis cerebri is the other name of :-
(1) Adenohypophysis
(2) Islets of langerhans
(3) Neurohypophysis
(4) Pituitary
BCI040

32. Which of the following hormone helps in facultative water reabsorption by nephrons :-
(1) MSH (2) FSH
(3) ADH (4) ACTH
BCI041

33. Hormone of hypothalamus are called :-
(1) Regulatory hormones
(2) Growth hormones
(3) Tropic hormones
(4) Both (1) and (3)
BCI042

34. Diabetes insipidus disease is caused due to the deficiency of hormone produced by :-
(1) Pituitary (2) Adrenal
(3) Pancreas (4) Thyroid
BCI043

35.



Which of the following option in given table is correct identification of the structures labelled as A,B,C and D and their corresponding function in the above figure:-

(1)	(A)	Hypothalamus	Produces Prolactin hormone
(2)	(B)	Posterior pituitary	Release & FSH and LH
(3)	(C)	Portal circulation	Supply blood from hypothalamus to posterior pituitary
(4)	(D)	Posterior pituitary	Release oxytocin and vasopressin

BCI044

36. MSH produced by the pars intermedia of pituitary causes in lower vertebrates :-
(1) Darkening of skin
(2) Light colouration of skin
(3) Both
(4) Body growth

BCI045

37. LTH is also known as :-
(1) Lactogenic Hormone
(2) Prolactin
(3) Mammotropic Hormone
(4) All

BCI046

38. Growth hormone of pituitary is more effective in:-
(1) Presence of thyroxine
(2) Absence of thyroxine
(3) Absence of Insulin
(4) Presence of adrenaline

BCI048

39. Gonadotropic hormone are :-
(1) FSH (2) LH
(3) LTH (4) FSH, LH

BCI049

40. MSH is secreted in man by which part of pituitary?
(1) Anterior Pituitary
(2) Middle lobe of pituitary
(3) Posterior lobe of pituitary
(4) Infundibulum

BCI050

41. Oxytocin is used in :-
(1) Milk ejection
(2) Parturition
(3) Milk let down process
(4) All of the above

BCI051

42. Hyper secretion of STH leads to :-
(1) Dwarfism & Acromegaly
(2) Goitre, Sterility
(3) Cretinism, Myxoedema
(4) Gigantism & Acromegaly

BCI052

43. Oxytocin mainly helps in :-
(1) Milk production (2) Child birth
(3) Diuresis (4) Gametogenesis

BCI053

44. Which hormone is concerned with the concentration of urine ?
(1) Oxytocin (2) Vasopressin
(3) Prolactin (4) Cortisol

BCI054

45. Acromegaly is caused by :-
(1) Excess of S.T.H.
(2) Excess of Thyroxine
(3) Deficiency of Thyroxine
(4) Excess of Adrenaline

BCI055

46. Oxytocin is released from :-
(1) Adenohypophysis (Anterior lobe)
(2) Adenohypophysis (Posterior lobe)
(3) Hypothalamus
(4) Neurohypophysis

BCI056

47. FSH is :-
(1) Glycoprotein (2) Metalloprotein
(3) Glycolipid (4) Phospholipid

BCI057

48. Which one hormone of the pituitary of the human controls the protein metabolism and growth of skeleton ?
(1) Iodo thyroxine
(2) Leutotrophic hormone
(3) Somatotrophic hormone
(4) Oxytocin

BCI059

49. Ovulation in mammals occurs mainly under the influence of :-
(1) TSH and ACTH (2) FSH and LH
(3) TSH and STH (4) MTH and ACTH

BCI060

50. Secretion of estrogen is controlled by :-
(1) HCG (2) Progesterone
(3) LH (4) F.S.H.

BCI061

51. Immediate cause of induction of ovulation in human female is plasma surge of :-
(1) Progesterone (2) LH
(3) FSH (4) Estradiol

BCI062

52. Stimulation of uterine contraction during child birth is brought about by :-
(1) Adrenaline (2) Progesterone
(3) Oxytocin (4) Prolactin

BCI063

53. Which gland secretion is under nervous control ?
(1) Adrenal cortex (2) Anterior pituitary
(3) Posterior pituitary (4) Pineal body

BCI064

54. Which of the following is correct ?
(A) Pars distalis produces GH, PRL, TSH. ACTH, LH, FSH
(B) Pars intermedia secretes only one hormone called melatonin
(C) Posterior lobe of pituitary is also called as neurohypophysis or pars nervosa
(D) Posterior pituitary, stores and releases two hormones called oxytocin and vasopressin
(1) A, B & C (2) B, C & D
(3) A, C & D (4) B and C

BCI065

55. ADH responsible for reabsorption of water and reduction of urine secretion is synthesize by :
(1) Posterior pituitary gland
(2) Juxtaglomerular apparatus
(3) Anterior pituitary gland
(4) Hypothalamus

BCI066

56. The hormones that initiates ejection of milk, stimulates milk production and growth of ovarian follicles are respectively known as :
(1) PRL, OT and LH
(2) OT, PRL and FSH
(3) LH, PRL and FSH
(4) PRH, OT and LH

BCI067

57. Match the hormone in column I with their function in column II :

Column I		Column II	
a	FSH	i	Prepare endometrium for implantation
b	LH	ii	Develop female secondary sexual characters
C	Progesterone	iii	Contraction of uterine wall
D	Estrogen	iv	Development of corpus luteum
		v	Maturation of Graafian follicle

(1) a-v, b-iv, c-i, d-ii
(2) a-iii, b-iv, c-i, d-ii
(3) a-iv, b-iii, c-ii, d-i
(4) a-i, b-ii, c-iii, d-iv

BCI068

58. Which of the following is gonadotrophic hormone?
(1) Collip's hormone
(2) Prolactin
(3) Oxytocin
(4) Luteinizing hormone

BCI069

59. FSH is produced by :
(1) Adrenal cortex
(2) Anterior lobe of pituitary gland
(3) Middle lobe of pituitary gland
(4) Posterior lobe of pituitary gland

BCI070

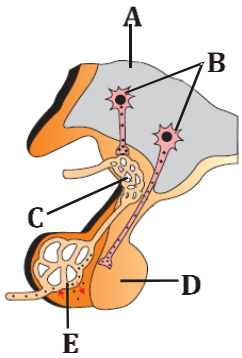
60. Hormone that promotes cell division, protein synthesis and bone growth is –
(1) ACTH (2) ADH (3) PTH (4) GH

BCI071

61. An adenohypophysis hormone which is regulated by feedback mechanism is –
(1) Oxytocin (2) TSH
(3) Vaspressin (4) Cortisone

BCI072

62. On the basis of below diagram choose the correct option which match the physiological function of hormones :-



(1) A–Only releases, D–Release and synthesis, E–Only synthesis
(2) A–Only release, D–Only synthesis, E–Synthesis and release
(3) A–Synthesis and release, D–Only releases, E–Synthesis and releasing
(4) A–Synthesis and release, D–Only synthesis, E–Only release

BCI073

63. Vasopressin influences :
(1) Electrolyte efflux
(2) Nerve excitability
(3) Water reabsorption
(4) All of these

BCI074

64. If ADH level of blood is less :
(1) Volume of urine increases
(2) Volume of urine decreases
(3) Volume of urine is normal
(4) Volume of urine is unaffected

BCI075

65. Which of the following hormones helps in the contraction of uterus during child birth:
(1) ADH (2) Androgen
(3) Oxytocin (4) Glucocorticoid

BCI078

66. Which of the following hormones stimulates the secretion of milk from female ?
(1) LH (2) Prolactin
(3) Oxytocin (4) Progesterone

BCI079

67. The formation of egg and sperm is affected by:
(1) LH (2) MSH (3) TSH (4) FSH

BCI080

68. Mammalian prolactin is secreted by –
(1) Adenohypophysis
(2) Neurohypophysis
(3) Adrenal cortex
(4) Adrenal medulla

BCI081

69. Hypersecretion of growth hormone in the period of growth lead to :
(1) Acromegaly (2) Cushing syndrome
(3) Midgets (4) Gigantism
BCI082

70. Acromegaly is a disease caused by :
(1) Over secretion of growth hormone in childhood
(2) Over secretion of growth hormone in adulthood
(3) Under secretion of growth hormone in adulthood
(4) Deficiency of calcium and phosphorous in the diet.
BCI083

71. In absence of ADH, the disease caused is –
(1) Diabetes mellitus (2) Diabetes insipidus
(3) Oligouria (4) Acromegaly
BCI084

72. Gonadotropic hormones are :
(1) Estrogen and progesterone
(2) Luteinizing hormone and follicle stimulating hormone
(3) Testosterone and Androsterone
(4) Prolactin and Luteotropin
BCI086

73. Which hormone is responsible for milk ejection after the birth of the baby ?
(1) Oxytocin (2) Progesterone
(3) Prolactin (4) Estrogen
BCI087

THYROID, PARATHYROID AND ADRENAL GLANDS

74. How many statements are correct regarding parathyroid gland ?
(a) Four parathyroid gland present on front side of thyroid gland
(b) It secretes parathromone which is steroidal in nature
(c) It increase blood Ca^{++} level
(d) It acts on bone and stimulate bone resorption
(1) One (2) Two
(3) Three (4) Four
BCI089

75. Injection of which of the following increases metabolic rate ?
(1) STH (2) Insulin
(3) Thyroxine (4) Testosterone
BCI092

76. Hypothyroidism in adults causes :-
(1) Addison's disease
(2) Myxoedema
(3) Sterility
(4) Cretinism
BCI093

77. Parathormone deficiency in man causes :-
(1) Hyper calcemia (2) Hypocalcaemia
(3) Goitre (4) All
BCI097

78. Cretinism is due to abnormal secretion of :-
(1) ACTH (2) Thyroxine
(3) Calcitonin (4) Parathormone
BCI098

79. Exopthalmic goitre is caused due to hypersecretion of :-
(1) Adrenal (2) Thyroid
(3) Parathyroid (4) Thymus
BCI100

80. The main function of thyroid gland is to control :-
(1) Growth
(2) Reproduction
(3) Secondary sexual characters
(4) Basal metabolic rate
BCI101

81. The two lobes of thyroid gland are joined by a horizontal connection called :-
(1) Inter thyroidal connective
(2) Inter thyroidal commissure
(3) Intermediary lobe
(4) Isthmus

BCI102

82. Hyper secretion of Parathyroid hormone result in
(1) Stronger bones due to increased incorporation of calcium in them.
(2) Deposition of calcium in various skeletal structure
(3) No effect on the constitution of bones
(4) Weaker bones due to increased removal of calcium from them

BCI106

83. Hormone that decrease calcium level in blood:
(1) Thyroxine (2) Parathormone
(3) Thyrocalcitonin (4) Cortisol

BCI108

84. BMR is increased due to :-
(1) ADH
(2) Adrenaline
(3) Parasympathetic nervous system
(4) Thyroxine

BCI109

85. Goitre is a pathological condition associated with:
(1) Glucagon (2) Thyroxine
(3) Progesterone (4) Testosterone

BCI110

86. Effects of thyroxine on metabolic rate is:-
(1) Decreases (2) No effect
(3) Increases (4) Uncertain

BCI111

87. Deficiency of which of the following may cause bone deformation :-
(1) Oxytocin (2) Vitamin D
(3) STH (4) Thyroxine

BCI112

88. Function of thyrocalcitonin :-
(1) To reduce the calcium level in blood
(2) To increase the calcium level in blood
(3) Oppose the action of thyroxine
(4) Maturation of gonads

BCI113

89. Parathormone deficiency leads to :-
(1) Decrease of Ca^{+2} level in blood
(2) Increase of Ca^{+2} level in blood
(3) Osteoporosis
(4) Hypercalcemia

BCI114

90. Parathyroid hormone –
(1) is produced by the thyroid gland
(2) is released when blood calcium levels fall
(3) stimulates osteoblasts to lay down new bone
(4) stimulates calcitonin release.

BCI116

91. Hyposecretion of aldosterone causes :-
(1) Gull's disease (2) Grave's disease
(3) Cushing's disease (4)Addison's disease

BCI119

92. A tumour in the adrenal zona glomerulosa can cause hyper secretion of hormones produced in that region. Which of the following might you expect to find in a patient with such a tumour ?
(1) Increased blood sodium levels
(2) Increased blood glucose levels
(3) Decreased blood calcium levels
(4) Increased dehydration

BCI121

93. Epinephrine and norepinephrine together known as :-
(1) Steroid (2) Protein
(3) Catecholamines (4) Glycoprotein
BCI123

94. Retention of sodium in body depends up on hormone from :-
(1) Adrenal cortex (2) Adrenal medulla
(3) Parathyroid (4) Thyroid
BCI125

95. Adrenal cortex also controls the carbohydrate metabolism through :-
(1) Adrenaline
(2) Noradrenaline
(3) Glucocorticoids
(4) Mineralocorticoids
BCI126

96. "4s gland" is :-
(1) Pancreas (2) Liver
(3) Thyroid (4) Adrenal
BCI127

97. Adrenaline increases :-
(1) Heart beat (2) Blood pressure
(3) Both (4) Salivation
BCI129

98. Which hormone controls activity of zona glomerulosa of adrenal gland ?
(1) Renin (2) Thyroxine
(3) ADH (4) FSH
BCI130

99. All of the following are functions of adrenaline except:-
(1) Increases blood supply in skeletal muscle
(2) Hyperglycaemia
(3) Uterine relaxation
(4) Tachycardia
BCI131

100. When the primary sexual organ does not develop, puberty may appear due to :-
(1) Stimulation of adrenal cortex
(2) Stimulation of adrenal medulla
(3) Excessive secretion from gonads
(4) Reduced secretion from gonads
BCI132

101. Norepinephrine hormone is secreted from :-
(1) Zona glomerulosa
(2) Zona fasiculata
(3) Zona reticularis
(4) Medulla of adrenal
BCI133

102. Which gland is concerned with salt equilibrium in body :–
(1) Anterior pituitary (2) Pancreas
(3) Adrenal (4) Thyroid
BCI134

103. Norepinephrine leads to increase in :–
(1) Blood pressure
(2) Urine production
(3) Cellular respiration
(4) Release of epinephrine
BCI135

104. Largest amount of iodine is found in :-
(1) Adrenals (2) Liver
(3) Thyroid (4) Testes
BCI136

105. Which gland prepares you for flight, fear and fight during adverse conditions :-
(1) Thyroid (2) Parathyroid
(3) Pituitary (4) Adrenal
BCI137

106. Blood pressure is controlled by :-
(1) Adrenal gland (2) Thyroid gland
(3) Pituitary gland (4) Pancreas
BCI138

107. Life saving hormone are secreted by :-
(1) Pituitary (2) Pineal
(3) Adrenal (4) Thyroid
BCI139

108. Hypocalcemia causes -
(1) Tetany (2) Osteoporosis
(3) Myxoedema (4) Cretinism
BCI140

109. Temperature of body is controlled by which endocrine gland:-
(1) Pituitary (2) Thyroid
(3) Adrenal (4) Pancreas
BCI141

110. During emergency which of the following hormone is secreted ?
(1) Aldosterone (2) Thyroxine
(3) Adrenaline (4) Calcitonin
BCI142

111. Corticosteroids are secreted by :
(1) Adrenal gland (2) Pineal gland
(3) Pituitary gland (4) Thyroid gland
BCI143

112. Aldosterone is secreted by :
(1) Zona glomerulosa (2) Zona fasciculata
(3) Zona reticularis (4) Zona pellucida
BCI144

113. Which gland stores hormone before its secretion and then release it ?
(1) Thyroid (2) Pancreas
(3) Pineal (4) Pituitary
BCI145

114. Which of the following disease is not related to thyroid gland ?
(1) Goitre (2) Cretinism
(3) Myxoedema (4) Acromegaly
BCI146

115. Grave's disease is due to :
(1) Hyperactivity of thyroid gland
(2) Hypoactivity of adrenal cortex
(3) Hyperactivity of adrenal medulla
(4) Hypoactivity of islets of Langerhans
BCI147

116. Hypothyroidism causes in adult :
(1) Obesity (2) Diabetes
(3) Cretinism (4) Myxoedema
BCI148

117. The hormone that controls the level of calcium and phosphorus in the blood is secreted by :
(1) Thyroid (2) Parathyroid
(3) Pituitary (4) Thymus
BCI149

118. Obesity of face, hyperglycemia and virilism in females is characteristic of –
(1) Grave's disease
(2) Addison's disease
(3) Conn's disease
(4) Cushing's syndrome
BCI150

119. Muscular tetany can be caused by deficiency of:
(1) Oxytocin (2) STH
(3) ADH (4) Parathyroid hormone
BCI151

120. Addison's disease is caused due to :
(1) hypersecretion of adrenal cortical hormones
(2) hypersecretion of growth hormone
(3) hypersecretion of thymus
(4) none of the above
BCI152

121. Para-thyroid hormone is a :
(1) Peptide (2) Carbohydrate
(3) Lipid (4) Steroid
BCI153

122. Parathormone is secreted during :
(1) Increased blood calcium level
(2) Decreased blood calcium level
(3) Increased blood sugar level
(4) Decreased blood sugar level
BCI155

123. Chronic disturbance in hormone secretion of thyroid gland causes :
(1) Goitre (2) Diabetes
(3) Addison's disease (4) Colour blindness
BCI156

124. In man removal of Parathyroid gland leads to:
(1) Acromegaly (2) Tetany
(3) Polyuria (4) Diabetes insipidus
BCI160

125. Parathormone induces :
(1) Increase in blood sugar level
(2) Decrease in serum calcium level
(3) Increase in serum calcium level
(4) Decrease in blood sugar level
BCI161

126. Which disease is caused by under secretion of adrenal cortex ?
(1) Cretinism (2) Dwarfism
(3) Sterility (4)Addison's disease
BCI162

127. Mammals born without a thymus gland fail to manufacture :-
(1) B - Lymphocytes (2) T - Lymphocytes
(3) Plasma cells (4) Basophils
BCI165

128. If thymectomy is done during adult hood then what possibility is there ?
(1) Immunosuppressant
(2) Die immediately
(3) No adverse reaction
(4) Myasthenia gravis
BCI166

129. Thymosin stimulates :-
(1) Milk secretion (2) Erythrocytes
(3) T-lymphocytes (4) Melanocytes
BCI167

130. Glucagon is secreted by :-
(1) Leydig cells
(2) Islets of Langerhans
(3) Corpus luteum
(4) Glisson's capsule
BCI168

131. Oversecretion of glucagon can cause :-
(1) Tetany (2) Diabetes insipidus
(3) Acromegaly (4) Glycosuria
BCI171

132. Which of the following is not function of insulin ?
(1) Increase glycogenesis
(2) Increase glycogenolysis
(3) Increase up take of amino acid by liver and muscle
(4) Promote oxidation of glucose
BCI173

133. Injection of Insulin to human leads to increased :-
(1) Glucose level of blood
(2) Glucose level of urine
(3) Glucose level of cells
(4) Calcium level of blood
BCI174

134. In old age, immune system becomes weak due to gradual degeneration of :-

(1) Pineal gland (2) Parathyroid gland

(3) Thymus gland (4) Adrenal gland

BCI176

135. T–cells mature in :

(1) Peyer's patch (2) Lymph node

(3) Thymus (4) Brusa of Fabricius

BCI178

136. The diabetes mellitus is caused by :-

(1) Hyper secretion of Insulin

(2) Hyposecretion of Insulin

(3) Hyposecretion of ADH

(4) Hyper secretion of ADH

BCI179

137. In diabetes mellitus disease, the urine contains :-

(1) Salt (2) Fat

(3) Protein (4) Sugar

BCI180

138. A patient of diabetes mellitus drink more water and he eliminates extra amount of which substance from blood :-

(1) Protein (2) Sugar

(3) Fat (4) Hormone

BCI181

139. Which gland decreases in size with increasing age?

(1) Thyroid (2) Adrenal

(3) Thymus (4) Pituitary

BCI182

140. Insulin is secreted by :

(1) α–cell of islets of Langerhans

(2) δ–cell of islets of Langerhans

(3) β–cell of islets of Langerhans

(4) Pancreatic acinar cell

BCI183

141. Which one of the following endocrine gland functions as a biological clock and neurosecretory transducer ?

(1) Adrenal gland (2) Thyroid gland

(3) Pineal gland (4) Thymus gland

BCI184

142. Mammalian thymus is mainly concerned with:

(1) Regulation of body temperature

(2) Regulation of body growth

(3) Immunological functions

(4) Secretion of thyrotropin

BCI185

143. The islets of Langerhans are found in :

(1) Pancreas (2) Stomach

(3) Liver (4) Alimentary canal

BCI186

144. Melatonin is secreted by :

(1) Pineal gland (2) Parathyroid gland

(3) Pituitary gland (4) Thyroid gland

BCI187

145. Insulin is related with :

(1) Diabetes (2) Migrain

(3) Jaundice (4) All of the above

BCI188

146. A patient of diabetes mellitus excretes glucose in urine even when he is kept on a carbohydrate free diet. It is because :

(1) Fats are catabolised to form glucose

(2) Amino acids are catabolised in liver

(3) Amino acids are discharged in blood stream from liver

(4) Glycogen from muscles are discharged in blood stream from liver

BCI189

147. The modern idea about ageing is that our body slowly loses the power of defence against the invasion of germs and pathogens. This process starts by the disappearance of which organ ?
(1) Spleen
(2) Thymus gland
(3) Pituitary gland
(4) Parathyroid gland
BCI192

148. Which of the following hormones secreted by pancreas ?
(1) Insulin and glucagon
(2) Epinephrine and nor-epinephrine
(3) Thyroxin and melanin
(4) Prolactin and oxytocin
BCI193

149. The function of glucagon hormone is :
(1) To increase glycogenesis
(2) To decrease blood sugar level
(3) To release glucose from liver cells and glycogenolysis promotion
(4) To increase the absorption of glucose and fatty acids through cell
BCI194

GONADS AND OTHER NON-ORGANISED GLANDS

150. Leydig cells are meant for :-
(1) Formation of sperm
(2) To produce progesterone
(3) To produce testosterone
(4) Nutrition of sperm
BCI197

151. The "Estrogen" secretion is controlled by :-
(1) FSH (2) LH
(3) Progesterone (4) GH
BCI201

152. Which of the following hormone is not secreted by gastro-intestinal tract ?
(1) Gastrin (2) Secretin
(3) Cholecystokinin (4) Erythropoietin
BCI202

153. Which one is a female sex hormone ?
(1) Estrogen (2) Progesterone
(3) Estradiol (4) All of these
BCI203

154. Atrial wall of the heart muscle secretes a peptide hormone to reduce the blood pressure is:
(1) Cholycystokinin
(2) Erythropoetin
(3) Atrial natriuretic factor
(4) Epinephrine
BCI204

155. Which temporary endocrine gland forms in ovary after ovulation :-
(1) Corpus callosum (2) Corpus albicans
(3) Corpus luteum (4) Corpus striata
BCI206

156. Secretin stimulates the activity of :-
(1) Salivary glands (2) Gastric gland
(3) Pancreas (4) Gall-bladder
BCI207

157. Placenta produces which hormone
(1) GH (2) Gastrin
(3) ACTH (4) Progesterone
BCI209

158. Hormone which is responsible for maintenance of pregnancy is :
(1) Estrogen (2) Aldosterone
(3) Progesterone (4) Testosterone
BCI210

159. Which of the following steroid sex hormone influence secondary sex organs?

(1) Progesterone

(2) Oestrogen

(3) LH

(4) LTH

BCI211

160. Progesterone is secreted from :

(1) Testes

(2) Adrenal gland

(3) Pituitary gland

(4) corpus luteum

BCI212

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	4	1	4	4	1	4	3	4	4	2	3	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	3	1	2	1	3	3	1	3	4	1	2	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	4	1	4	1	4	1	4	1	4	4	2	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	1	3	2	4	2	3	3	3	4	2	1	4	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	3	3	1	3	2	4	1	4	2	2	2	1	2	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	2	2	2	4	4	4	3	4	2	3	2	1	1	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	4	1	3	1	3	4	3	1	3	1	4	3	1	3	4
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	1	3	1	2	3	1	1	1	4	1	4	2	4	4	4
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	1	2	1	2	3	4	2	3	3	2	4	2	3	3	3
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	2	4	2	3	3	3	3	1	1	1	1	2	1	3	3
Question	151	152	153	154	155	156	157	158	159	160					
Answer	1	4	4	3	3	3	4	3	2	4					

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Identify the hormone with its **correct** matching of source and function :

(1) Oxytocin - posterior pituitary, growth and maintenance of mammary glands.

(2) Melatonin - pineal gland, regulates the normal rhythm of sleepwake cycle.

(3) Progesterone - corpus-luteum, stimulation of growth and activities of female secondary sex organs.

(4) Atrial natriuretic factor - ventricular wall increases the blood pressure.

BCI228

2. Fight-or-flight reactions cause activation of

(1) the parathyroid glands, leading to increased metabolic rate.

(2) the kidney, leading to suppression of renin-angiotensin-aldosterone pathway.

(3) the adrenal medulla, leading to increased secretion of epinephrine and norepinephrene.

(4) the pancreas leading to a reduction in the blood sugar levels.

BCI229

Re-AIPMT 2015

3. Which one of the following hormones is **not** involved in sugar metabolism ?

(1) Glucagon

(2) Cortisone

(3) Aldosterone

(4) Insulin

BCI230

4. Which one of the following hormones though synthesised elsewhere, is stored and released by the master gland ?

(1) Melanocyte stimulating hormone

(2) Antidiuretic hormone

(3) Luteinizing hormone

(4) Prolactin

BCI231

NEET-I 2016

5. Which of the following pairs of hormones are **not** antagonistic (having opposite effects) to each other?

(1) Parathormone – Calcitonin

(2) Insulin – Glucagon

(3) Aldosterone – Atrial Natriuretic Factor

(4) Relaxin – Inhibin

BCI232

NEET-II 2016

6. Graves' disease is caused due to :-

(1) Hyposecretion of adrenal gland

(2) Hypersecretion of adrenal gland

(3) Hyposecretion of thyroid gland

(4) Hypersecretion of thyroid gland

BCI233

7. Name a peptide hormone which acts mainly on hepatocytes, adipocytes and enhances cellular glucose uptake and utilization.

(1) Secretin (2) Gastrin

(3) Insulin (4) Glucagon

BCI234

AIM: FREE EDUCATION TO ALL APUL

8. Osteoporosis, an age–related disease of skeletal system, may occur due to :-
(1) Decreased level of estrogen
(2) Accumulation of uric acid leading to inflammation of joints.
(3) Immune disorder affecting neuro–muscular junction leading to fatigue.
(4) High concentration of Ca^{++} and Na^{+}.

BCI235

9. The posterior pituitary gland is **not** a 'true' endocrine gland because :-
(1) It is under the regulation of hypothalamus
(2) It secretes enzymes
(3) It is provided with a duct
(4) It only stores and releases hormones

BCI236

NEET(UG) 2017

10. A temporary endocrine gland in the human body is :-
(1) Corpus cardiacum
(2) corpus luteum
(3) Corpus allatum
(4) Pineal gland

BCI237

11. GnRH, a hypothalamic hormone, needed in reproduction, acts on:
(1) anterior pituitary gland and stimulates secretion of LH and FSH.
(2) posterior pituitary gland and stimulates secretion of oxytocin and FSH.
(3) posterior pituitary gland and stimulates secretion of LH and relaxin.
(4) anterior pituitary gland and stimulates secretion of LH and oxytocin.

BCI238

12. Hypersecretion of Growth Hormone in adults does not cause further increase in height, because:
(1) Epiphyseal plates close after adolescence.
(2) Bones loose their sensitivity to Growth Hormone in adults.
(3) Muscle fibres do not grow in size after birth.
(4) Growth Hormone becomes inactive in adults.

BCI239

NEET(UG) 2018

13. Which of the following is an amino acid derived hormone ?
(1) Epinephrine (2) Ecdysone
(3) Estradiol (4) Estriol

BCI240

14. Which of the following hormones can play a significant role in osteoporosis ?
(1) Aldosterone and Prolactin
(2) Progesterone and Aldosterone
(3) Estrogen and Parathyroid hormone
(4) Parathyroid hormone and Prolactin

BCI241

NEET(UG) 2019

15. How does steroid hormone influence the cellular activities?
(1) Changing the permeability of the cell membrane.
(2) Binding to DNA and forming a gene-hormone complex.
(3) Activating cyclic AMP located on the cell membrane.
(4) Using aquaporin channels as second messenger.

BCI242

16. Match the following hormones with the respective disease :-

(a) Insulin	(i) Addison's disease
(b) Thyroxin	(ii) Diabetes insipidus
(c) Corticoids	(iii) Arcomegaly
(d) Growth Hormone	(iv) Goitre
	(v) Diabetes mellitus

Select the **correct** option.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(v)	(i)	(ii)	(iii)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(v)	(iv)	(i)	(iii)
(4)	(ii)	(iv)	(i)	(iii)

BCI243

NEET (UG) 2019 (Odisha)

17. Which of the following hormones is responsible for both the milk ejection reflex and the foetal ejection reflex ?

(1) Estrogen (2) Prolactin

(3) Oxytocin (4) Relaxin

BCI244

18. Which of the following conditions will stimulate parathyroid gland to release parathyroid hormone?

(1) Fall in active Vitamin D levels

(2) Fall in blood Ca^{+2} levels

(3) Fall in bone Ca^{+2} levels

(4) Rise in blood Ca^{+2} levels

BCI245

19. Artificial light, extended work-time and reduced sleep–time disrupt the activity of

(1) Thymus gland

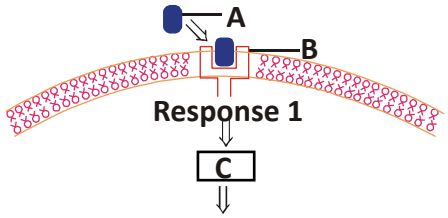
(2) Pineal gland

(3) Adrenal gland

(4) Posterior pituitary gland

BCI246

20. Identify A, B and C in the diagramatic representation of the mechanism of hormone action.



Physiological Responses

Select the correct option from the following

(1) A-Steroid Hormone; B-Hormone-receptor Complex, C-Protein

(2) A-Protein Hormone, B-Receptor; C-Cyclic AMP

(3) A-Steroid Hormone; B-Receptor, C - Second Messenger

(4) A-Protein Hormone; B-Cyclic AMP, C-Hormone-receptor Complex

BCI247

NEET (UG) 2020

21. Select the correct statement.

(1) Insulin is associated with hyperglycemia

(2) Glucocorticoids stimulate gluconeogenesis

(3) Glucagon is associated with hypoglycemia.

(4) Insulin acts on pancreatic cells and adipocytes.

BCI248

22. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus ?

(1) Renal calculi and Hyperglycaemia

(2) Uremia and Ketonuria

(3) Uremia and Renal Calculi

(4) Ketonuria and Glycosuria

BCI249

23. Match the following columns and select the **correct** option :

Column-I	Column-II
(a) Pituitary gland	(i) Grave's disease
(b) Thyroid gland	(ii) Diabetes mellitus
(c) Adrenal gland	(iii) Diabetes insipidus
(d) Pancreas	(iv) Addison's disease

	(a)	(b)	(c)	(d)
(1)	(ii)	(i)	(iv)	(iii)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(iii)	(i)	(iv)	(ii)

BCI250

NEET (UG) 2020 (Covid-19)

24. Match the following columns and select the correct option :-

Column-I	Column-II
(a) Pituitary hormone	(i) Steroid
(b) Epinephrine	(ii) Neuropeptides
(c) Endorphins	(iii) Peptides, proteins
(d) Cortisol	(iv) Biogenic amines

(1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

BCI251

25. Hormones stored and released from neurohypophysis are :-
(1) Thyroid stimulating hormone and Oxytocin
(2) Oxytocin and Vasopressin
(3) Follicle stimulating hormone and Leutinizing hormone
(4) Prolactin and Vasopressin

BCI252

NEET (UG) 2021

26. Erythropoietin hormone which stimulates R.B.C. formation is produced by :-
(1) Alpha cells of pancreas.
(2) The cells of rostral adenohypophysis.
(3) The cells of bone marrow.
(4) Juxtaglomerular cells of the kidney.

BCI253

NEET (UG) 2022

27. Which of the following are not the effects of Parathyroid hormone ?
(a) Stimulates the process of bone resorption
(b) Decreases Ca^{2+} level in blood
(c) Reabsorption of Ca^{2+} by renal tubules
(d) Decreases the absorption of Ca^{2+} from digested food
(e) Increases metabolism of carbohydrates
Choose the **most appropriate** answer from the options given below:
(1) (b), (d) and (e) only
(2) (a) and (e) only
(3) (b) and (c) only
(4) (a) and (c) only

BCI280

NEET (UG) 2023

28. Match **List I** with **List II.**

List I	List II
(A) CCK	(I) Kidney
(B) GIP	(II) Heart
(C) ANF	(III) Gastric gland
(D) ADH	(IV) Pancreas

Choose the **correct** answer from the options given below :-
(1) A-III, B-II, C-IV, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-IV, B-II, C-III, D-I
(4) A-IV, B-III, C-II, D-I

BCI281

29. Which of the following are NOT under the control of thyroid hormone?

(A) Maintenance of water and electrolyte balance

(B) Regulation of basal metabolic rate

(C) Normal rhythm of sleep-wake cycle

(D) Development of immune system

(E) Support the process of R.B.Cs formation

Choose the **correct** answer from the options given below:

(1) B and C only (2) C and D only

(3) D and E only (4) A and D only

BCI282

NEET (UG) Manipur 2023

30. Which of the following statements are **correct** with respect to the hormone and its function ?

(A) Thyrocalcitonin (TCT) regulates the blood calcium level.

(B) In males, FSH and androgens regulate spermatogenesis.

(C) Hyperthyroidism can lead to goitre.

(D) Glucocorticoids are secreted in Adrenal Medulla.

(E) Parathyroid hormone is regulated by circulating levels of sodium ions.

Choose the **most appropriate** answer from the options given below :-

(1) (C) and (E) only

(2) (A) and (B) only

(3) (B) and (C) only

(4) (A) and (D) only

BCI283

NEET (UG) 2024

31. Match **List-I** with **List-II**.

List-I		List-II	
A.	Exophthalmic goitre	I.	Excess secretion of cortisol, moon face & hyperglycemia.
B.	Acromegaly	II.	Hypo-secretion of thyroid hormone and stunted growth.
C.	Cushing's syndrome	III.	Hyper secretion of thyroid hormone & protruding eye balls.
D.	Cretinism	IV.	Excessive secretion of growth hormone.

Choose the correct answer from the options given below :-

(1) A-I, B-III, C-II, D-IV

(2) A-IV, B-II, C-I, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-IV, C-I, D-II

BCI284

RE-NEET (UG) 2024

32. Match **List-I** with **List-II**.

List-I		List-II	
A.	Epinephrine	I.	Hyperglycemia
B.	Thyroxine	II.	Smooth muscle contraction
C.	Oxytocin	III.	Basal metabolic rate
D.	Glucagon	IV.	Emergency hormone

Choose the **correct** answer from the options given below :-

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-II, C-I, D -IV

(3) A-IV, B-III, C-II, D-I

(4) A-I, B-IV, C-III, D-II

BCI285

33. Diuresis is prevented by :-

(1) Renin form JG cells via switching off the osmoreceptors

(2) ANF from atria of the heart

(3) Aldosterone from adrenal medulla

(4) Vasopressin from Neurohypophysis

BCI286

34. Identify the wrong statements :-

(A) Erythropoietin is produced by juxtaglomerular cells of the kidney

(B) Leydig cells produce Androgens

(C) Atrial Natriuretic factor, a peptide hormone is secreted by the seminiferous tubules of the testes.

(D) Cholecystokinin is produced by gastro-intestinal tract

(E) Gastrin acts on intestinal wall and helps in the production of pepsinogen

Choose the **most appropriate** answer from the options given below :-

(1) D and E only

(2) A and B only

(3) C and E only

(4) A and C only

BCI287

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	2	4	4	3	1	4	2	1	1	1	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	2	2	2	2	4	4	2	2	4	1	4	2	2
Question	31	32	33	34											
Answer	4	3	4	3											

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Steroid hormones –
(1) Have only cell surface receptors
(2) Are lipophobic
(3) Have receptors within the nucleus
(4) Are produced by only adrenal cortex.

BCI254

2. Both adrenaline and cortisol are secreted in response to stress. Which of the following statements is also true for both of these hormones?
(1) They act to increase blood glucose
(2) They are secreted by the adrenal cortex
(3) Their secretion is stimulated by adrenocorticotropin
(4) They are secreted into the blood within seconds of the onset of stress.

BCI255

3. Which one is incorrect for hypothalamus ?
(1) Basal part of diencephalon
(2) Regulate narrow spectrum of body functions
(3) Neural control of posterior pituitary
(4) Release somatostatin for GH Inhibition

BCI256

4. Pineal gland is not related with :-
(1) Body temperature
(2) Defence capability
(3) Metabolism
(4) Kidney functions

BCI257

5. Vigrous contraction of uterus muscles is stimulated by:-
(1) ADH (2) MSH
(3) GH (4) Oxytocin

BCI258

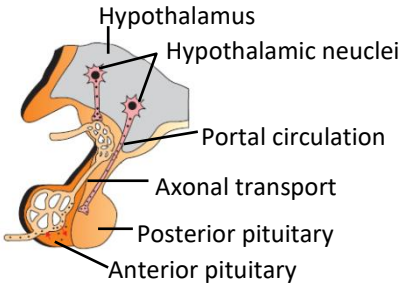
6. Find out correctly matched :-
(A) Thymus – AMI
(B) PTH – Ca^{+2} absorption
(C) Adrenal – glucocorticoids
(D) Thyroid – Anti-inflammatory response
(1) A, B, D (2) A, B, C
(3) B, C, D (4) A, C, D

BCI259

7. Which one is incorrect statement ?
(1) Hypothalamus regulate a wide spectrum of body functions.
(2) Pituitary, pineal, testes, heart and kidney are organised endocrine gland of body.
(3) Hormones are non-nutrient chemicals and intercellular messenger.
(4) LH helps in maintenance of corpus luteum after ovulation.

BCI260

8. Find out incorrect labelling in following diagram



(1) Axonal transport and Anterior pituitary
(2) Portal circulation and Posterior pituitary
(3) Anterior pituitary and Posterior pituitary
(4) Portal circulation and Axonal transport

BCI261

9. Find out incorrect match of hormone with respective function :-

	Hormone	Function
(1)	**Melatonin**	Sleep wake cycle and body temperature
(2)	**FSH**	Growth of ovarian follicles and stimulate gonadial activity.
(3)	**Adrenaline**	Increase concentration of glucose in blood.
(4)	**Progesterone**	Stimulate growth and activities of female secondary sex organs

BCI262

10. Hyperglycemic and hypoglycemic hormones are

(1) Insulin and glucagon

(2) Adrenalin and glucagon

(3) Adrenalin and Insulin

(4) Glucagon and growth hormone

BCI263

11. Group of hormones which is related with cytoplasmic bounded receptors ?

(1) Hypothalamic hormones and epinephrine

(2) Thyroid hormone and estradiol

(3) Insulin and glucagon

(4) GH and MSH

BCI264

12. Two hormones .....(a)..... and .....(b)..... synthesize in hypothalamus and transported in pituitary gland through .....(c)..... and .....(d)..... respectively.

(1) a = oxytocin ⇒ c = portal circulation
b = ADH ⇒ d = direct release

(2) a = ADH ⇒ c = axonal transport
b = TSHRF ⇒ d= portal circulation

(3) a = ACTH ⇒ c = axonal transport
b = MSH ⇒ d= portal circulation

(4) a = TSHRF ⇒ c = axonal transport
b = ADH ⇒ d= portal circulation

BCI265

(B) ANALYTICAL QUESTIONS

13. Which one is correctly matched ?

(a) Pineal gland – Metabolism, Mental retardation

(b) Thymus – Mysthenia gravis

(c) Thyroid – Anti inflammatory reaction

(d) Pancreas – Prolonged hyperglycemia

(1) a, c (2) b, c (3) c, d (4) b, d

BCI266

14. After ovulation, ruptured follicle secrete hormone that helps in :-

(1) Libido

(2) Growth of facial hair

(3) High pitch voice

(4) Pregnancy support

BCI268

15. Diabetic patients are successfully treated with:-

(1) GH therapy

(2) More amount of thyroxine

(3) Insulin therapy

(4) Both (2) and (3)

BCI269

16. Find out suitable match for the following hormones and related organ :-

(1) ANF – Heart, Calcitonin – parathyroid

(2) Renin – Kidney, Relaxin – Placenta

(3) Calcitonin – Kidney, HCG → Ovary

(4) Oestrogen – Testes, Progesterone → Graafian follicle

BCI270

17. Parathormone causes :-
(1) Hypercalcemia
(2) Hyperglycemia
(3) Hyperkalemia
(4) Hypocalcemia and hypoglycemia both

BCI271

18. In mammals the female secondary sexual characters are developed mainly by the hormone?
(1) HCG (2) Progesterone
(3) Estrogens (4) All of these

BCI272

19. Which of the following is an accumulation and release centre of Neurohormones?
(1) Anterior Pituitary lobe
(2) Neurohypophysis
(3) Pars intermedia
(4) Hypothalamus

BCI273

20. BMR and Temperature of body is controlled by which endocrine gland?
(1) Adrenal cortex (2) Thymus
(3) Thyroid (4) Pituitary

BCI274

21. Reabsorption of Na^+ along with excretion of K^+ is controlled by which one of the following hormones?
(1) Prostaglandins (2) Aldosterone
(3) Estrogen (4) Cortisol

BCI275

22. Location and secretion of Leydig cells are?
(1) Pancreas – Glucagon
(2) Ovary – Estrogen
(3) Ovary – Progesterone
(4) Testis – Testosterone

BCI276

23. Urinary excretion of Na^+ is regulated by–
(1) Anterior Pituitary
(2) Adrenal cortex
(3) Neurohyphophysis
(4) Pars intermedia

BCI277

24. Thyrocalcitonin is secreted during –
(1) Increased blood calcium level
(2) Decreased blood calcium level
(3) Both (1) and (2)
(4) Increased blood sugar level.

BCI278

25. Which of the following hormones increase alertness, piloerection and sweating ?
(1) TCT (2) Catecholamines
(3) Cortisol (4) Thymosins

BCI279

26. Which hormone causes dilation of blood vessels, increased oxygen consumption and glucogenesis?
(1) Adrenalin (2) Glucagon
(3) ACTH (4) Insulin

BCI213

27. A steroid hormone which regulates glucose metabolism is –
(1) 11-deoxycorticosterone
(2) Cortisone
(3) Cortisol
(4) Corticosterone

BCI215

28. Injury to adrenal cortex is not likely to affect the secretion of which one of the following
(1) Adrenaline
(2) Cortisol
(3) Aldosterone
(4) Both Androstenedione and Dehydroepiandrosterone

BCI219

29. Select the *correct* matching of a hormone, its source and function.

	Hormone	Source	Function
(1)	Norepine phrine	Adrenal medulla	Increases heart beat, rate of respiration and alertness
(2)	Glucagon	Beta-cells of langerhans	Stimulates glycogenolysis
(3)	Prolactin	Posterior pituitary	Regulates growth of mammary glands and milk formation in females
(4)	Vasopressin	Posterior pituitary	Increases loss of water through urine

BCI222

30. Which one of the following pairs of hormones are the examples of those that can easily pass through the cell membrane of the target cell and bind to a receptor inside it (mostly in the nucleus) :-

(1) Somatostatin, oxytocin

(2) Cortisol, testosterone

(3) Insulin, glucagon

(4) Thyroxin, Insulin

BCI224

31. A pregnant female delivers a baby who suffers from stunted growth, mental retardation, low intelligence quotient and abnormal skin.

This is the result of :

(1) Over secretion of pars distalis

(2) Deficiency of iodine in diet

(3) Low secretion of growth hormone

(4) Cancer of the thyroid gland

BCI225

32. Which of the following statements is **correct** in relation to the endocrine system?

(1) Releasing and inhibitory hormones are produced by the pituitary gland.

(2) Adenohypophysis is under direct neural regulation of the hypothalamus.

(3) Organs in the body like gastrointestinal tract, heart, kidney and liver do not produce any hormones.

(4) Non-nutrient chemicals produced by the body in trace amount that act as intercellular messenger are known as hormones.

BCI226

Exercise - III — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	2	4	4	2	2	4	4	3	2	2	4	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	3	2	3	2	4	2	1	2	1	3	1	1	2
Question	31	32													
Answer	2	4													

Exercise - I (Conceptual Questions) Build Up Your Understanding

1. Epithelial tissue with thin flat cells appearing like packed tiles occurs on :-
(1) Inner lining of intestine
(2) Inner lining of stomach
(3) Inner lining of fallopian tubes
(4) Outer surface of Intestine

BSA001

2. Germinal Epithelium of ovary is formed of:-
(1) Columnar Epithelium
(2) Squamous Epithelium
(3) Cuboidal Epithelium
(4) Stratified Epithelium

BSA002

3. Epidermis of skin of vertebrates comprises:
(1) Simple Epithelium
(2) Stratified Epithelium
(3) Transitional Epithelium
(4) Columnar Epithelium

BSA003

4. Inner lining of Blood vessels and heart is tesselleted Epithelium. Which is :-
(1) Simple squamous due to wavy appearance
(2) Simple squamous due to tile like appearance
(3) Simple cuboidal due to wavy appearance
(4) Simple columnar Epithelium

BSA004

5. Which of the following statement is incorrect about the simple squamous epithelial tissue?
(1) The squamous epithelium is made of a single thin layer of flattened cells with regular boundaries.
(2) Also called tesselated epithelium due to its wavy appearance in blood vessel and coelom.
(3) All the cells of this epithelium rest upon basement membrane
(4) This epithelium is also called pavement epithelium

BSA005

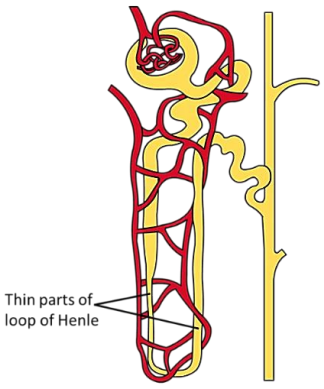
6. This epithelium is made up of a single thin layer of flattened cells and is involved in diffusion, it is found in :-
(1) Walls of blood vessels
(2) Ducts of glands
(3) Tubular parts of nephrons
(4) All of these

BSA006

7. Mucus cells (Goblet cells) :-
(1) Unicellular gland
(2) Multicellular glands
(3) Endocrine glands
(4) Parietal cells of gastric glands

BSA007

8. Which of the following option is correct about the labelled part of diagram given below?



(1) Lined by single layer of cells
(2) Epithelium in this part helps in forming a diffusion boundary
(3) Epithelium in this part is made up of single thin layer of flattened cells
(4) All of the above

BSA008

9. Trachea consist of _____ epithelium :-
(1) Simple columnar (2) PSCCGE
(3) Simple cuboidal (4) Stratified cuboidal

BSA009

10. Lining of blood vessel and air sacs of lungs are made up of :-
(1) Squamous Epithelium
(2) Columnar Epithelium
(3) Cuboidal Epithelium
(4) Pseudo stratified Epithelium

BSA010

11. Brush border Epithelium (Microvilli containing) found in:-
(1) PCT (2) Loop of Henle
(3) Collecting duct (4) Bowman's capsule

BSA011

12. Characteristic of simple epithelium is :-
(1) They are arranged in discriminately
(2) They make a definite layer
(3) Continue to divide and help in organ function
(4) Never divide

BSA012

13. Basement membrane of epithelium tissue is composed of:-
(1) Lipoprotein
(2) Polypeptide
(3) Mucopolysaccharide
(4) Mucopolysaccharide and glycoprotein

BSA013

14. Secretory part of sweat gland is composed of:-
(1) Cuboidal epithelium
(2) Columnar epithelium
(3) Pseudostratified epithelium
(4) Squamous epithelium

BSA014

15. Glands of vertebrates are originated from :-
(1) Ectodermal (2) Endodermal
(3) Mesodermal (4) All the above

BSA015

16. Select the false statement with respect to epithelial tissue.
(1) It has a free surface
(2) It faces body fluids sometimes
(3) It faces the external environment sometimes
(4) It sometimes forms middle structure part of organs

BSA016

17. Inner lining of stomach, rectum and colon is made of :-
(1) Simple squamous epithelium
(2) Simple cuboidal epithelium
(3) Simple columnar epithelium
(4) Pseudostratified epithelium

BSA017

18. Wall of Bowman's capsule in nephron is made up of :-
(1) Cuboidal epithelium
(2) Columnar epithelium
(3) Squamous epithelium
(4) Glandular epithelium

BSA018

19. Which of the following is made up of cube like cells?
(1) Epithelium of fallopian tubes
(2) Epithelium of PCT
(3) Epithelium of stomach
(4) Epithelium of alveoli

BSA019

20. Sweat glands of human being originate from
(1) Ectoderm (2) Mesoderm
(3) Endoderm (4) All the above

BSA020

21. Vesicles of thyroid gland is composed of:-
(1) Simple squamous epithelium
(2) Simple cuboidal epithelium
(3) Stratified squamous epithelium
(4) Stratified cuboidal epithelium

BSA021

22. Which of the following statement is incorrect?
(1) A cell or a group of cells which secretes chemical substances are called glands.
(2) All glands are composed of connective tissue.
(3) Sweat glands are ectodermal in origin.
(4) On the basis of number of cells, glands are of two types unicellular and multicellular.

BSA022

23. Compound squamous epithelium occurs in
(1) Stomach (2) Pharynx
(3) Intestine (4) Trachea

BSA023

24. Epithelial tissue is
(1) Protective covering
(2) Reproductive structure
(3) Nerve cells
(4) Corpuscles.

BSA024

25. Regeneration after injury is absent in
(1) Nervous tissue
(2) Skin epidermis
(3) Tendon
(4) Smooth muscles

BSA025

26. Brush border epithelium occurs in
(1) Trachea (2) Stomach
(3) Small intestine (4) Oesophagus

BSA026

27. Which of the following is not an endocrine gland ?
(1) Thyroid (2) Thymus
(3) Adrenal (4) Salivary

BSA027

28. Simple epithelium is made of
(1) Noncellular layer of hyaluronic acid
(2) Actively dividing cells
(3) Loosely arranged cells
(4) Compactly packed single layer of cells.

BSA028

29. Sebaceous glands are
(1) endocrine glands
(2) exocrine glands
(3) unicellular glands
(4) all of the above

BSA029

30. Characteristic of epithelial tissues is
(1) Never produce glands
(2) Cells can undergo rapid divisions
(3) Abundant vascularisation
(4) Large intercellular spaces.

BSA030

31. Our heart consists of :-
(i) Epithelial tissue
(ii)Connective tissue
(iii) Muscular tissue (iv) Nervous tissue
(1) Only ii (2) i & iii only
(3) ii, iii & iv only (4) All of these

BSA031

32. Supportive connective tissue means :-
(1) Tendon (2) Cartilage & Bone
(3) Ligaments (4) Blood & Lymph

BSA032

33. Fibroblast secretes :-
(1) Matrix (2) Fibres
(3) Both (1) and (2) (4) Cells

BSA033

34. Which of the following statement is incorrect about fibroblasts?
(1) Smallest cells of connective tissue proper
(2) Maximum in number
(3) Irregular in shape due to long cytoplasmic processes
(4) Cytoplasm is rich in rough ER

BSA034

35. All the following cells produces collagen except :-
(1) Osteoblast (2) Chondroblast
(3) Fibroblast (4) Mast cells

BSA035

36. Which of the following is not correct about areolar connective tissue?
(1) Also known as spongy connective tissue.
(2) It is most widely distributed tissue in the body.
(3) In this tissue minimum intercellular space and matrix is present.
(4) It serves as a support framework for epithelium.

BSA036

37. Matrix of cartilage produced by :-
(1) Chondrocytes (2) Chondroclasts
(3) Osteocytes (4) Histiocytes

BSA037

38. Which of following is lack of blood supply :-
(1) Bone (2) Cartilage
(3) Connective tissue (4) All

BSA038

39. Sprain of body is due to pulling of :-
(1) Muscles (2) Ligaments
(3) Tendon (4) Nerves

BSA039

40. Skeletal tissue originated from which embryonic germinal layer :-
(1) Ectoderm (2) Mesoderm
(3) Endoderm (4) All the above

BSA040

41. Which protein is present in largest amount in human body :-
(1) Collagen (2) Elastin
(3) Albumin (4) Keratin

BSA041

42. Yellow fibers are present in :-
(1) Arrange in bundles
(2) Singly and branched
(3) Singly and unbranched
(4) Unbranched & arrange in bundles

BSA042

43. Which of the following are specialised connective tissue :-
(1) Cartilage (2) Bone
(3) Blood (4) All

BSA043

44. Cartilage is present in :-
(1) Between adjacent bones of vertebral column
(2) In middle of the long bone
(3) Both (1) and (2)
(4) None

BSA044

45. Which cartilage present on the end of long bone:-
(1) Hyaline cartilage (2) Fibrous cartilage
(3) Calcified cartilage (4) Elastic cartilage

BSA045

46. Which of the following have hard and non-pliable ground substance :-
(1) Cartilages (2) Bones
(3) Both (4) Areolar tissues

BSA046

47. Mammalian pinna is supported by -
(1) Hyaline cartilage
(2) Calcified cartilage
(3) Elastic cartilage
(4) White fibrous connective tissue
BSA047

48. Fibres present in the form of bundles are -
(1) Elastic
(2) Reticular
(3) Elastic and reticular
(4) Collagen
BSA048

49. Bones are 'connected' by :-
(1) Cartilage (2) Muscles
(3) Ligaments (4) Tendons
BSA049

50. Mast cells occur in :-
(1) Connective tissue (2) Epithelial tissue
(3) Skeletal tissue (4) Nervous tissue
BSA050

51. White fibrous tissue is :-
(1) Nervous (2) Muscular
(3) Ligaments (4) Tendons.
BSA051

52. Loose connective tissue is :-
(1) Areolar (2) Bone
(3) Blood (4) Cartilage.
BSA052

53. Ligament is :-
(1) Modified white fibrous tissue
(2) Solid white fibrous tissue
(3) Modified elastic connective tissue
(4) Cartilage
BSA053

54. Collagen and elastin are formed by :-
(1) Macrophages (2) Fibroblasts
(3) Mast cells (4) Chondrocytes
BSA054

55. Which is not a component of areolar tissue
(1) Macrophage (2) Plasma cell
(3) Schwann cell (4) Adipose cell
BSA055

56. Which of the following is not true?
(1) Limb bones are long bones.
(2) The function of long bones is weight bearing and bringing out movement.
(3) Bone marrows in all such long bones are site of blood cells production.
(4) All of these
BSA056

57. Nasal septum gets damaged. Its recovery requires cartilage called :-
(1) Fibrous cartilage
(2) Elastic cartilage
(3) Hyaline cartilage
(4) Calcified cartilage
BSA057

58. What will happen if ligaments are cut or broken :-
(1) Bones will move freely at joints
(2) No movement at joint
(3) Bone will become unfix
(4) Bone will become fixed
BSA058

59. Which one of the following contains the largest quantity of extracellular material :-
(1) Striated muscle
(2) Areolar tissue
(3) Stratified epithelium
(4) Myelinated nerve fibres
BSA059

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Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	1	1	1	1	4	2	1	1	2	4	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	3	2	1	2	2	2	1	1	3	4	4	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	3	1	4	3	1	2	2	2	1	2	4	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Answer	2	3	4	3	1	4	1	3	2	3	3	3	3	2	

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Choose the correctly matched pair :-
(1) Tendon – Specialized connective tissue
(2) Adipose tissue – Dense connective tissue
(3) Areolar tissue – Loose connective tissue
(4) Cartilage – Loose connective tissue

BSA060

Re-AIPMT 2015

2. The function of the gap junction is to :
(1) stop substance from leaking across a tissue
(2) performing cementing to keep neighbouring cells together
(3) Facilitate communication between adjoining cells by connecting the cytoplasm for rapid transfer of ions, small molecules and some large molecules
(4) separate two cells from each other.

BSA061

NEET-I 2016

3. Which type of tissue correctly matches with its location ?

	Tissue	Location
(1)	Smooth muscle	Wall of intestine
(2)	Areolar tissue	Tendons
(3)	Transitional epithelium	Tip nose
(4)	Cuboidal epithelium	Lining of stomach

BSA062

4. Connective tissues are derived from embryonic
(1) Ectoderm (2) Endoderm
(3) Endo mesoderm (4) Mesoderm

BSA063

NEET-UG 2019

5. The ciliated epithelial cells are required to move particles or mucus in a specific direction. In humans, these cells are mainly present in :
(1) Bile duct and Bronchioles
(2) Fallopian tubes and Pancreatic duct
(3) Eustachian tube and Salivary duct
(4) Bronchioles and Fallopian tubes

BSA064

NEET-UG 2019 (Odisha)

6. Match the following cell structure with its characteristic feature:

(a)	Tight junctions	(i)	Cement neighbouring cells together to form sheet
(b)	Adhering Junctions	(ii)	Transmit information through chemical to another cells
(c)	Gap junctions	(iii)	Establish a barrier to prevent leakage of fluid across epithelial cells
(d)	Synaptic junctions	(iv)	Cytoplasmic channels to facilitate communication between adjacent cells

Select correct option from the following :-
(1) (a)-(ii), (b)-(iv), (c)-(i), d-(iii)
(2) (a)-(iv), (b)-(ii), (c)-(i), d-(iii)
(3) (a)-(iii), (b)-(i), (c)-(iv), d-(ii)
(4) (a)-(iv), (b)-(iii), (c)-(i), d-(ii)

BSA065

NEET-UG 2020 (COVID-19)

7. Select the incorrectly matched pair from following:

(1) Chondrocytes - Smooth muscle cells

(2) Neurons - Nerve cells

(3) Fibroblast - Areolar tissue

(4) Osteocytes - Bone cells

BSA066

NEET-UG 2020

8. Goblet cells of alimentary canal are modified from :

(1) Compound epithelial cells

(2) Squamous epithelial cells

(3) Columnar epithelial cells

(4) Chondrocytes

BSA067

9. Cuboidal epithelium with brush border of microvilli is found in :

(1) eustachian tube

(2) lining of intestine

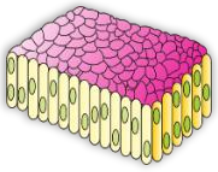
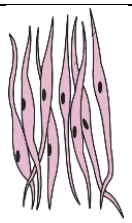
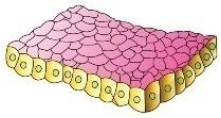
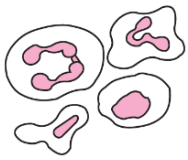
(3) ducts of salivary glands

(4) proximal convoluted tubule of nephron

BSA068

NEET-UG 2021 (Paper-2)

10. Match the following and choose the correct combination.

Column-I (Figure)		Column-II (Character)	
a.		i.	Component of connective tissue
b.		ii.	Found in duct of glands
c.		iii.	Secretion and absorption
d.		iv.	Functioning cannot be directly controlled

(1) a–ii, b–iii, c–i, d–iv

(2) a–iii, b–iv, c–ii, d–i

(3) a–iv, b–iii, c–ii, d–i

(4) a–i, b–ii, c–iv, d–iii

BSA069

NEET-UG 2022

11. Which of the following is present between the adjacent bones of the vertebral column?

(1) Cartilage

(2) Areolar tissue

(3) Smooth muscle

(4) Intercalated discs

BSA070

12. Which of the following is **not** a connective tissue?

(1) Adipose tissue (2) Cartilage

(3) Neuroglia (4) Blood

BSA071

Re-NEET-UG 2022

13. Which of the following types of epithelium is present in the bronchioles and Fallopian tubes?

(1) Simple squamous epithelium

(2) Simple columnar epithelium

(3) Ciliated epithelium

(4) Stratified squamous epithelium

BSA072

NEET-UG 2023

14. **Statement I :** Ligaments are dense irregular tissue.

Statement II : Cartilage is dense regular tissue.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) Both Statement-I and II are false.

(2) Statement-I is true but Statement-II is false.

(3) Statement-I is false but Statement-II is true.

(4) Both Statement-I and II are true.

BSA073

15. Match **List I** with **List II.**

List-I		List-II	
A.	Mast cells	I.	Ciliated epithelium
B.	Inner surface of bronchiole	II.	Areolar connective tissue
C.	Blood	III.	Cuboidal epithelium
D.	Tubular parts of nephron	IV.	specialised connective tissue

Choose the **correct** answer from the options give below :-

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-I, B-II, C-IV, D-III

BSA074

NEET-UG 2023 (Manipur)

16. **Statement I :** Goblet cells are unicellular glands.

Statement II : Earwax is the secretion of exocrine gland.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement-I is true but Statement-II is false

(2) Statement-I is false but Statement-II is true

(3) Both Statement-I and II are true

(4) Both Statement-I and II are false.

BSA075

17. Match **List-I** with **List-II**.

List-I		List-I	
(A)	Columnar epithelium	(I)	Ducts of glands
(B)	Ciliated epithelium	(II)	Inner lining of stomach and intestine
(C)	Squamous epithelium	(III)	Inner lining of bronchioles
(D)	Cuboidal epithelium	(IV)	Endothelium

Choose the **correct** answer from the options given below :-

(1) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(2) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

BSA076

NEET (UG) 2024

18. Match **List I** with **List II**.

	List I		List II
A.	Unicellular glandular epithelium	I.	Salivary glands
B.	Compound epithelium	II.	Pancreas
C.	Multicellular glandular epithelium	III.	Goblet cells of alimentary canal

D.	Endocrine glandular epithelium	IV.	Moist surface of buccal cavity

Choose the **correct** answer from the options given below :-

(1) A-II, B-I, C-III, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

BSA077

RE-NEET (UG) 2024

19. Which of the following statements is **correct** about the type of junction and their role in our body ?

(1) Adhering junctions facilitate the cells to communicate with each other.

(2) Tight junctions help to stop substances from leaking across a tissue.

(3) Tight junctions help to perform cementing to keep neighbouring cells together.

(4) Gap junctions help to create gap between the cells and tissues.

BSA078

20. In which of the following connective tissues, the cells secrete fibres of collagen or elastin?

A. Cartilage B. Bone

C. Adipose tissue D. Blood

E. Areolar tissue

Choose the **most appropriate** answer from the options given below :-

(1) B, C, D and E only

(2) A, B, C and E only

(3) B, C and D only

(4) A, C and D only

BSA079

21. Match **List-I** with **List-II**.

	List-I		List-II
A.	Squamous Epithelium	I.	Goblet cells of alimentary canal
B.	Ciliated Epithelium	II.	Inner lining of pancreatic ducts
C.	Glandular Epithelium	III.	Walls of blood vessels
D.	Compound Epithelium	IV.	Inner surface of Fallopian tubes

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-IV, C-III, D-I

(3) A-III, B-I, C-II, D-IV

(4) A-III, B-IV, C-I, D-II

BSA080

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	1	4	4	3	1	3	4	2	1	3	3	1	2
Question	16	17	18	19	20	21									
Answer	3	4	3	2	2	4									

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. How many of the following junctions are found in epithelium tissue?

Tight junction, Gap and Adhering junction.

(1) Four (2) Three

(3) Two (4) One

BSA081

2. Read the following (A-D) statements

A. It is made of more than one layer (multi-layered) of cells.

B. It has a limited role in secretion and absorption.

C. Their main function is to provide protection against chemical and mechanical stress

D. It covers the dry surface of the skin and the moist surface of buccal cavity.

How many of the above statements are correct for compound epithelium?

(1) Four (2) Three

(3) Two (4) One

BSA082

3. How many of the following substances are secreted by exocrine glands?

mucus, thyroxine, saliva, earwax, insulin, oil, milk, digestive enzymes, melatonin and adrenalin:-

(1) Four (2) Five

(3) Six (4) Seven

BSA083

4. Which connective tissues are present beneath the skin?

A. Areolar tissue

B. Adipose tissue

C. Dense irregular connective tissue

(1) Only B

(2) Only A

(3) Only A and B

(4) All A, B and C

BSA084

5. Read the following (A-D) statements :-

A. Connective tissue are most abundant and widely distributed in the body of complex animals

B. They are named connective tissues because of their special function of linking and supporting other tissues/organs of the body

C. They range from soft connective tissues to specialised types, which include cartilage, bone, adipose and blood

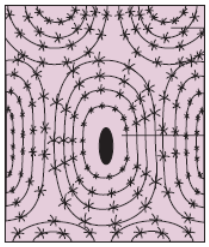
D. The cells of connective tissue secrete modified polysaccharides, which accumulate between cells and fibres and act as matrix

How many of the following statements are correct?

(1) Four (2) Three (3) Two (4) One

BSA085

6. Identify the given below tissue with its type and select the right option for the two together



Option :

Tissue	Type
(1) Cartilage	Specialised connective tissue
(2) Tendon	Dense irregular connective tissue
(3) Ligament	Dense regular connective tissue
(4) Bone	Specialised connective tissue

BSA086

7. Which of the following is incorrect statement for the simple columnar epithelium?

(1) It is composed of a single layer of tall and slender cells

(2) Their nuclei are located at the base

(3) Free surface may have microvilli

(4) They are found in the walls of blood vessels and air sac of lungs

BSA087

8. Connective tissues includes :-

(a) Cartilage (b) Bone

(c) Adipose tissue (d) Blood

(1) a, b, and d

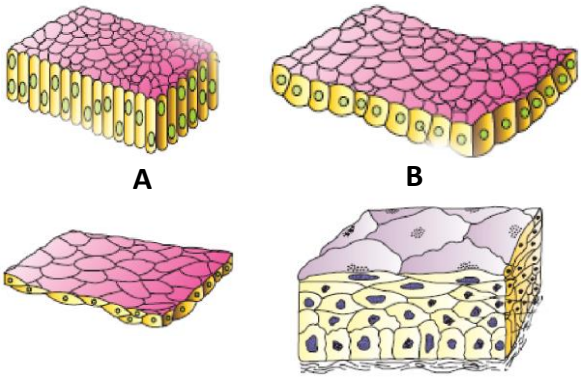
(2) a, b and c

(3) b and d

(4) All a, b, c and d

BSA088

9. The four sketches (A, B, C and D) given below represent four different types of animal tissues. Which one of these is correctly identified in the option given along with its correct location and function?



A B C D

		Tissue	Location	Function
1.	B	Simple squamous epithelium	Fallopian tube	Transport of gamete
2.	C	Simple cuboidal epithelium	Wall of blood vessels and air sac of lungs	Diffusion boundary
3.	D	Compound epithelium	Skin	Protection
4.	A	Simple columnar epithelium	Tubular part of nephron	Secretion

BSA089

10. Read the following (A - D) Statements :-

(A) Tight junctions help to stop substances from leaking across a tissue

(B) Adhering junctions perform cementing to keep neighbouring cells together.

(C) The simple epithelium consists of two or more cell layers and has protective function.

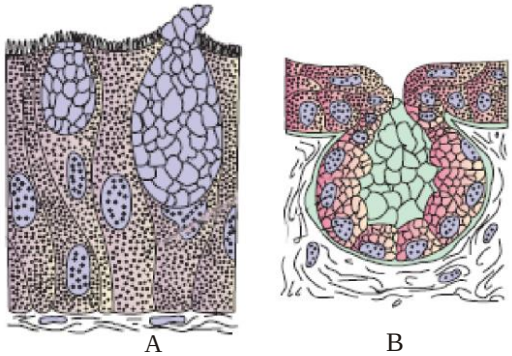
(D) The columnar epithelium is made of a single layer of flattened cells with irregular boundaries

How many of the above statements are correct?

(1) Four (2) Three (3) Two (4) One

BSA090

11. Identify the glands (A) and (B) shown below and select the right option for location and function:-



A B

		Gland	Location	Function
1.	A	Unicellular Gland	Alveoli	Secrete Saliva
2.	B	Multicellular gland	Oesophagus	Secrete enzyme
3.	A	Multicellular gland	Alimentary Canal	Secrete Mucous
4.	B	Multicellular gland	Buccal Cavity	Secrete Saliva

BSA091

12. Which of the following is correct match of epithelial tissue ?

(1)	Squamous epithelium	Bronchioles and fallopian tube
(2)	Columnar eqithelium	Ducts of glands and tubular part of nephron
(3)	Cuboidial epithelium	Walls of blood vessels and air sacs of lungs
(4)	Compound epithelium	Buccal cavity and pharynx

BSA092

13. Read the following (A - D) statements :-

(A) Areolar tissue present beneath the skin

(B) Adipose tissue is a type of dense connective tissue

(C) Tendons attach one bone to another

(D) Ligaments attach skeletal muscles to bones

How many of the above statements are incorrect?

(1) Four (2) Three

(3) Two (4) One

BSA093

14. In multicellular animals a group of similar cells along with intercellular substances perform a specific function such an organisation is called :-

(1) Tissue

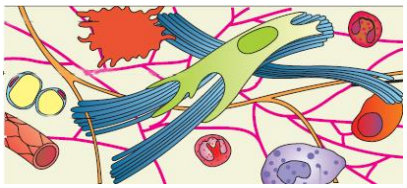
(2) Organ

(3) Body

(4) Organ system

BSA094

15. Identify the figure of animal tissue given below, along with it's correct location :-



	Tissue	Location
(1)	Dense regular Connective tissue	Heart
(2)	Dense irregular Connective tissue	At bone joints
(3)	Adipose tissue	Beneath skin
(4)	Areolar connective tissue	Beneath skin

BSA095

(B) ANALYTICAL QUESTIONS

16. Ligament connects :

(1) Bone to bone (2) Bone to muscle

(3) Muscle to muscle (4) Both (2) and (3)

BSA096

17. Which of the following pair is example of dense regular connective tissue ?

(1) Ligament and tendon

(2) Perichondrium and pericardium

(3) Ligament and pericardium

(4) Perichondrium and tendon

BSA097

18. Match the following columns and choose the correct answer from the options given below

	Column-A		Column-B
(a)	Epithelial tissue	(i)	Helps in contraction and relaxation
(b)	Connective tissue	(ii)	Form the lining and covering of body organs
(c)	Muscular tissue	(iii)	Conduction and transmission
(d)	Nervous tissue	(iv)	Connect the various parts of the body

(1) (a) – (i) , (b) – (ii) , (c) – (iii) , (d) – (iv)

(2) (a) – (ii) , (b) – (iv) , (c) – (i) , (d) – (iii)

(3) (a) – (ii) , (b) – (iii) , (c) – (i) , (d) – (iv)

(4) (a) – (iv) , (b) – (ii) , (c) – (iii) , (d) – (i)

BSA098

19. Which of the following statement is incorrect about the epithelial tissue?

(1) Simple epithelium composed of single layer of cells

(2) Functions of simple epithelium is to form lining for body cavities, ducts and tubes

(3) Compound epithelium is also called stretchable epithelium

(4) Compound epithelium has protective functions as it does in our skin

BSA099

20. Collagen fibres are secreted by :

(1) Mast cells

(2) Macrophage

(3) Histiocytes

(4) Fibroblasts

BSA100

21. Tendons and ligaments are specialized types of-

(1) Nervous tissue

(2) Epithelial tissue

(3) Muscular tissue

(4) Fibrous connective tissue

BSA101

22. Choose the statements which are related with simple and stratified epithelium?

(a) Composed of single layer of cells

(b) Consists of two or more layers of cells

(c) Functions as lining for body cavities, ducts and tubes.

(d) It has protective function as it does in our skin

	a	b	c	d
(1)	Simple	Simple	Stratified	Stratified
(2)	Simple	Stratified	Simple	Stratified
(3)	Simple	Stratified	Stratified	Simple
(4)	Stratified	Simple	Simple	Stratified

BSA102

23. Which of the following statement is incorrect about the diagram given below ?



(1) This epithelium has free surface

(2) Made up of single layer of flat cells

(3) Also called tessellated epithelium

(4) This epithelium helps in absorption and secretion

BSA103

24. Ciliated epithelium is present in -

(1) Trachea (2) Ureter

(3) Intestine (4) Stomach

BSA104

25. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion(A) : Compound epithelium has a limited role in secretion and absorption.

Reason (R) : Compound epithelium is multilayered and it's main function is to provide protection against chemical and mechanical stress.

In the light of the above statements, choose the correct answer from the option given below.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BSA105

26. Identify the correct set of statements.

(a) Loose connective tissue has cells and fibres compactly arranged in a semi-fluid ground substance.

(b) Tendons attach skeletal muscles to bones.

(c) Chondrocytes are enclosed in small cavities within the matrix.

(d) All cells in epithelium are held together with the little intercellular material.

Choose the correct answer from the options given below.

(1) (a) and (b) only

(2) (a) and (d) only

(3) (a), (b) and (c) only

(4) (b), (c) and (d) only

BSA106

27. Which of the following part is not lined by modification of simple columnar epithelium?

(1) Small intestine

(2) Blood vessels of small intestine and alveoli

(3) Liver and bile duct

(4) Fallopian tubes

BSA107

28. Choose the statements which are related with simple and stratified epithelium?

(a) Composed of single layer of cells

(b) Consists of two or more layers of cells

(c) Functions as lining for body cavities, ducts and tubes.

(d) It has protective function as it does in our skin

	a	b	c	d
(1)	Simple	Simple	Stratified	Stratified
(2)	Simple	Stratified	Simple	Stratified
(3)	Simple	Stratified	Stratified	Simple
(4)	Stratified	Simple	Simple	Stratified

BSA108

29. Which of the following is lined by compound epithelium?

A - Moist surface of buccal cavity

B - Alveoli of lungs

C - Dry surface of skin

D - Inner lining of ducts of salivary glands

E - Endothelium

(1) A, C and D

(2) A, B and D

(3) A, D and E

(4) A, B, C and D

BSA109

30. Find the mismatch :-

(1) Tendon — Dense regular white fibrous connective tissue.

(2) Ligament — Dense regular yellow fibrous connective tissue.

(3) Wall of alveoli of lungs — Dense irregular yellow fibrous connective tissue.

(4) Periosteum — Dense regular white fibrous connective tissue

BSA110

31. The main function of ligament is :

(1) Joining of two bones

(2) Joining of muscles

(3) Joining of muscle to bone

(4) Joining of muscle to nerves

BSA111

32. Identify the diagram given below and its examples in our body?

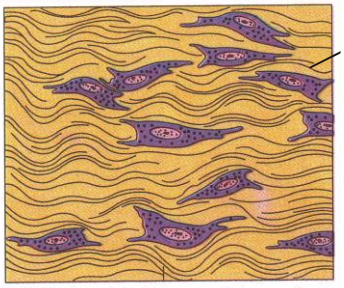


	Diagram	Examples
(1)	Dense regular CT	Ligament
(2)	Dense irregular CT	Ligament, Tendon
(3)	Dense regular CT	Ligament, Tendon
(4)	Dense irregular CT	Tendon, Perichondrium

BSA112

33. How many statements are incorrect for bones?

(a) Bones have hard and pliable ground substance

(b) Matrix is rich in collagen fibres

(c) Bone is a specialized connective tissue

(d) They serve weight-bearing function

(e) Osteoblasts are present in lacunae

(1) three (2) four

(3) one (4) two

BSA113

34. Identify correct set of statements.

(a) In nearly all animal tissues specialised junction provide both structural and functional links between its individual cells.

(b) Gap junctions help to stop substances from leaking across a tissue.

(c) Tight junctions perform connecting the cytoplasm to keep neighbouring cells together.

(d) Epithelium tissue are most abundant and widely distributed in the body of complex animals.

Choose the correct answer from the options given below.

(1) (a) and (d) only

(2) (b) and (c) only

(3) (a) only

(4) (a),(c) and (d) only

BSA114

35. Mucus cells (Goblet cells) :-

(1) Unicellular gland

(2) Multicellular glands

(3) Endocrine glands

(4) Parietal cells of gastric glands

BSA115

36. Given below are two statements –

Statement I : Bones have a solid and pliable ground substances rich in calcium salts and collagen fibre.

Statement II : Most of cartilages in vertebrate embryos are replaced by bone in adults.

In the light of the above statements, choose the correct answer from the option given below.

(1) Both statement–I and statement–II are incorrect.

(2) Statement–I is correct but statement–II is incorrect.

(3) Statement–I is incorrect but statement – II is correct.

(4) Both statement–I and statement–II are correct.

BSA116

37. Epithelial tissues are arise from :-

(1) Ectoderm (2) Endoderm

(3) Mesoderm (4) All the above

BSA117

38. Given below are two statements –

Statement I : Neuroglia make up more than one half the volume of neural tissue in our body.

Statement II : The bone marrow in all bones is the site of production of blood cells.

In the light of the above statements, choose the correct answer from the option given below.

(1) Both statement – I and statement–II are incorrect.

(2) Statement – I is correct but statement-II is incorrect.

(3) Statement–I is incorrect but statement–II is correct.

(4) Both statement–I and statement–II are correct.

BSA118

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39. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion(A) : Some of the columnar or cuboidal cells get specialised for secretion.

Reason (R) : Columnar epithelium is commonly found in duct of glands and tubular part of nephron in kidneys.

In the light of the above statements, choose the correct answer from the option given below.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

BSA119

40. Areolar connective tissue joins –

(1) Fat body with muscles

(2) Integument with muscles

(3) Bones with muscles

(4) Bones with bones

BSA120

41. The cell junctions called tight, adhering and gap junctions are found in :-

(1) Neural tissue (2) Muscular tissue

(3) Connective tissue (4) Epithelial tissue

BSA121

42. The kind of tissue that forms the supportive structure in our pinna (external ears) is also found in :-

(1) Tip of the nose (2) Vertebrae

(3) Nails (4) Ear ossicles

BSA122

43. The epithelial tissue present on the inner surface of bronchioles and fallopian tubes is

(1) Squamous

(2) Cuboidal

(3) Glandular

(4) Ciliated

BSA123

44. The kind of epithelium which forms the inner walls of blood vessels is :

(1) squamous epithelium

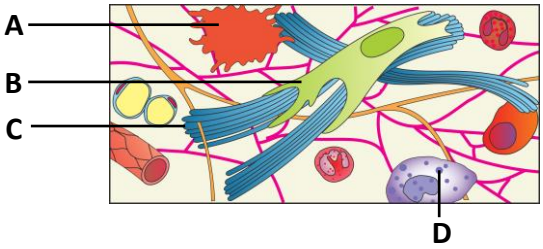
(2) cuboidal epithelium

(3) columnar epithelium

(4) ciliated columnar epithelium

BSA124

45. Given below is the diagrammatic sketch of a certain type of connective tissue. Identify the parts labelled A, B, C and D, and select the right option about them.



Options :

(1) A–Macrophage, B–Collagen fibres, C–Fibroblast, D–Mast cell

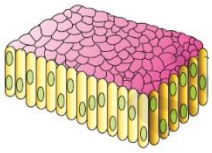
(2) A–Mast cell, B–Collagen fibres, C–Fibroblast, D–Macrophage

(3) A–Macrophage, B–Fibroblast, C–Collagen fibres, D–Mast cell

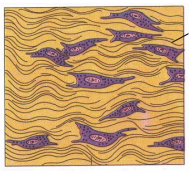
(4) A–Mast cell, B–Macrophage, C–Fibroblast, D–Collagen fibres

BSA125

46. The four sketches (A, B, C and D) given below, represent four different types of animal tissues. Which one of these is correctly identified in the options given, along with its correct location and function?

(A) 

(B) 

(C) 

(D) 

		Tissue	Location	Function
(1)	(D)	Smooth muscle tissue	Heart	Heart contraction
(2)	(A)	Columnar epithelium	Nephron	Secretion and absorption
(3)	(B)	Glandular epithelium	Intestine	Secretion
(4)	(C)	Collagen fibres	cartilage	Attach skeletal muscles to bones

BSA126

47. The supportive skeletal structures in the human external ears and in the nose tip are examples of :-

(1) bone (2) cartilage

(3) ligament (4) areolar tissue

BSA127

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	3	3	1	4	4	4	3	3	4	4	2	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	2	3	4	4	2	4	1	4	4	2	2	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	4	3	1	3	4	2	2	2	4	1	4	1	3
Question	46	47													
Answer	3	2													

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION AND MORPHOLOGY

1. Zoological name of cockroach is :
(1) *Glossina palpalis*
(2) *Periplaneta americana*
(3) *Musca nebulo*
(4) *Apis indica*
BSA170

2. Diagonistic feature of insects is :
(1) Segmented body
(2) Chitin in body wall
(3) Three pair of legs
(4) One pair compound eyes
BSA171

3. Characteristic of group insecta is :
(1) Joined appendages
(2) 3 pair of jointed legs
(3) Chitinous exoskeleton
(4) Compound eyes
BSA172

4. Cockroaches are brown or black bodied animals that are included in class _____ of phylum _____.
(1) reptilia; annelida
(2) insecta; arthropoda
(3) insecta; annelida
(4) reptilia; arthropoda
BSA173

5. Cockroaches are serious pests and vectors of several diseases like :-
(1) malaria and cholera
(2) cholera and dysentery
(3) giardiasis and filariasis
(4) malaria and cancer
BSA174

6. The dorsal plate of skeleton found on the abdomen of cockroach is called :
(1) Pleuron (2) Sternum
(3) Tergum (4) Vertex
BSA175

7. Mandibles of cockroach are :
(1) Short with grinding teeth
(2) Long and pointed
(3) short without teeth
(4) long and coiled
BSA176

8. Head of cockroach according to its position is known as :
(1) Hypopharynx (2) Hypocyrebral
(3) Hypognathus (4) Supragnathus
BSA177

9. Tongue like structure in *Periplaneta* is :
(1) Maxilla (2) Hypopharynx
(3) Labium (4) Labrum
BSA178

10. Which structure is known as lower lip of cockroach ?
(1) Labrum (2) Labium
(3) Mantum (4) Submentum
BSA179

11. Wings of cockroach are mainly helpful in :
(1) Egg laying (2) Preying
(3) Finding the mate (4) Protection
BSA180

12. Pseudopodia of *Amoeba* are similar to :
(1) Legs of cockroach
(2) Teeth in rabbit
(3) Spicules of *Leucosolenia*
(4) Suckers of *Taenia*
BSA181

13. Main character for the distinction between male and female cockroach :
(1) Antennae (2) Mandibles
(3) Anal cerci (4) Anal style
BSA182

14. Anal styles are found in :
(1) Housefly
(2) Female cockroach
(3) Male cockroach
(4) Both male & female cockroach
BSA183

15. Body segments are definite and well marked in:
(1) *Hydra* (2) *Taenia*
(3) *Ascaris* (4) Cockroach
BSA184

16. In cockroach the audioreceptors are situated on :
(1) On legs (2) Antenna
(3) Sensory receptor (4) Anal cerci
BSA185

17. Exoskeleton of cockroach is made up of :
(1) Cartilage (2) Cuticle
(3) Chitin (4) Amino acids
BSA186

BODY CAVITY, BODY WALL, DIGESTION, RESPIRATION, CIRCULATION AND EXCRETION

18. The body cavity of cockroach is called :
(1) Pseudocoel (2) Coelom
(3) Hydrocoel (4) Haemocoel
BSA187

19. Hepatic caecae in cockroach are derived from:
(1) Crop (2) Gizzard
(3) Midgut (4) Proctodaeum
BSA188

20. Saliva of cockroach contains enzyme :
(1) Lipase (2) Amylase
(3) Pepsin (4) Trypsin
BSA189

21. Salivary glands of cockroach open on :
(1) Maxilla (2) Hypopharynx
(3) Labium (4) Labrum
BSA190

22. Mouth part of cockroach are suited for :
(1) Piercing (2) Absorbing
(3) Biting & chewing (4) Drinking
BSA191

23. Cockroach is :
(1) Omnivorous (2) Herbivorous
(3) Carnivorous (4) Sanguivorous
BSA192

24. In cockroach food is crushed in which part :
(1) Crop (2) Gizzard
(3) Mesenteron (4) Oesophagus
BSA193

25. Complete digestion takes place in which part of cockroach ?
(1) In crop (2) In Gizzard
(3) In mesenteron (4) In oesophagous
BSA194

26. Oxygen is carried to the tissues of cockroach by which organ :
(1) Skin
(2) Trachea
(3) Plasma
(4) Respiratory pigment
BSA195

27. Alary muscles in cockroach are associated or connected with :
(1) Trachea (2) Heart
(3) Legs (4) Alimentary canal
BSA196

28. Number of pairs of spiracles in cockroach are:
(1) 4 (2) 6
(3) 8 (4) 10
BSA197

29. In cockroach O_2 is carried to tissue by :
(1) Haemoglobin and haemocyanin
(2) Blood plasma
(3) Tracheal tubes
(4) Diffusion through integument
BSA198

30. Blood of cockroach does not contain haemoglobin because :
(1) It respires through atmosphere
(2) Respires through book lungs
(3) It does not respire
(4) It has some other means to carry oxygen direct into the tissues
BSA199

31. Give the name of blood vessel, which arises from first chamber of heart in cockroach :
(1) Nephrocyte (2) Fenestrae
(3) Ostia (4) Anterior aorta
BSA200

32. Number of chambers in the heart of cockroach :
(1) 5 (2) 9 (3) 13 (4) 16
BSA201

33. Blood circulation in insects :
(1) Flows in arteries and veins
(2) With red blood corpuscles
(3) Open type
(4) Absent
BSA202

34. Heart of cockroach is :
(1) Four chamberd
(2) Vertral to gut
(3) Elongated muscular tube
(4) Three chambered
BSA203

35. The colour of haemolyph of cockroach is :
(1) Yellow (2) Red
(3) Green (4) Colourless
BSA204

36. Physiologically the heart of cockroach is :
(1) Neurogenic (2) myogenic
(3) Epigenic (4) Agenic
BSA205

37. Main excretory product of cockroach is :
(1) Urea (2) Ammonia
(3) Uric acid (4) Amino acid
BSA206

38. Function of Malpighian tubules of cockroach:
(1) Digestion (2) Respiration
(3) Excretion (4) Reproduction
BSA207

39. Exeretory organs of cockroach are :
(1) Trachea (2) Kidney
(3) Nephridia (4) Malpighian tubules
BSA208

NERVOUS SYSTEM, COMPOUND EYE, REPRODUCTION AND DEVELOPEMENT

40. The nerve cord of cockroach is :
(1) Double, ventral and solid
(2) Double dorsal and hollow
(3) Single, dorsal and solid
(4) Single, ventral and hollow
BSA209

41. In cockroach which is helpful in sexual attraction?
(1) Hormone ecdysone
(2) Pheromone
(3) Juvenile hormone
(4) Anal cerci
BSA210

42. Ommatidia are found in :
(1) Eyes of birds (2) Eye of frog
(3) Eye of insects (4) Eye of rabbit
BSA211

43. Structural units found in the compound eye of cockroach are called :
(1) Rhabdom (2) Cone cells
(3) Ommatidia (4) Simple eye
BSA212

44. What type of vision is found in cockroach ?
(1) Mosaic (2) Super position
(3) Binocular (4) None of them
BSA213

45. Ommatidia are units in the eyes of :
(1) Amphibians (2) Mammals
(3) Insects (4) Fishes
BSA214

46. Ootheca is formed in cockroach by :
(1) Phallic Gland (2)Conglobate gland
(3) Utricular gland (4) Collaterial Gland
BSA215

47. Ootheca of Cockroach has fertilized eggs, this numbers is :
(1) 6 (2) 8 (3) 16 (4) 24
BSA216

48. From egg laying to the development of adult cockroach. How many moluting occur?
(1) 13 times (2) 9 times
(3) 6 times (4) 3-4 times
BSA217

49. Conglobate gland is found in the reproductive organs of :
(1) Earth worm (2) Housefly
(3) Male cockroach (4) Female cockroach
BSA218

50. Number of moultings in cockroach after hatching and development of complete animal are :
(1) Less than 7 (2) Upto 8
(3) Upto 9 (4) more than 10
BSA219

51. How many ovarioles are found in each ovary of cockroach ?
(1) 3 (2) 6 (3) 8 (4) 16
BSA220

52. Collaterial glands of cockroach help in :
(1) Fertilization
(2) Formation of ootheca
(3) Copulation
(4) Formation of oothecal chamber
BSA221

53. Juvenile of cockroach is known as :
(1) Tadepole (2) Amocyte
(3) Nymph (4) Naid
BSA222

54. In which of the following moulting is not found ?
(1) *Hydra* (2) Mosquito
(3) Housefly (4) Cockroach
BSA223

55. Life history of cockroach represents :
(1) Ametaboly (2) Holometaboly
(3) Paurometaboly (4) Hypermetaboly
BSA224

56. Which has/have chitinous exo-skeleton ?
(1) Cockroach
(2) Mosquito
(3) Butterfly
(4) All of the above
BSA225

57. Cockroach is :
(1) Nocturnal and omnivorous
(2) Diurinal and omnivorous
(3) Nocturnal and carnivorous
(4) None
BSA226

Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	2	2	3	1	3	2	2	4	1	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	4	3	2	2	3	1	2	3	2	2	4	3	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	3	3	4	1	3	3	4	1	2	3	3	1	3
Question	46	47	48	49	50	51	52	53	54	55	56	57			
Answer	4	3	1	3	4	3	2	3	1	3	4	1			

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. The terga, sterna and pleura of cockroach body are joined by :-
(1) Muscular tissue
(2) Arthrodial membrane
(3) Cartilage
(4) Cementing glue

BSA227

Re-AIPMT 2015

2. The body cells in cockroach discharge their nitrogenous waste in the haemolymph mainly in the form of :-
(1) Calcium carbonate (2) Ammonia
(3) Potassium urate (4) Urea

BSA228

NEET-I 2016

3. Which of the following features is not present in Periplaneta americana ?
(1) Schizocoelom as body cavity
(2) Indeterminate and radial cleavage during embryonic development
(3) Exoskeleton composed of N-acetylglucosamine
(4) Metamerically segmented body

BSA229

NEET-II 2016

4. In male cockroaches, sperms are stored in which part of the reproductive system?
(1) Testes
(2) Vas deferens
(3) Seminal vesicles
(4) Mushroom glands

BSA230

NEET - 2017

5. Which of the following structures does not open into the genital chamber of female cockroaches?
(1) A single median oviduct
(2) Spermatheca
(3) A pair of anal cerci
(4) A pair of collateral glands

BSA231

6. Select the incorrect option with respect to features present in three animals :-

	Characters	Cockroach	Earthworm	Frog
(1)	Blood vascular body	Open	Closed	Closed
(2)	Body surface	Dry	Moist	Moist
(3)	Eyes	Compound	Absent	Simple
(4)	Development	Direct	Indirect	Indirect

BSA232

NEET (UG) 2018

7. Which of the following features is used to identify a male cockroach from a female cockroach?
(1) Presence of a boat shaped sternum on the 9th abdominal segment
(2) Presence of caudal styles or anal styles
(3) Forewings with darker tegmina
(4) Presence of anal cerci

BSA233

NEET (UG) 2019

8. Select the correct sequence of organs in the alimentary canal of cockroach starting from mouth :-
(1) Pharynx $\rightarrow$ Oesophagus $\rightarrow$ Crop $\rightarrow$ Gizzard $\rightarrow$ Ileum $\rightarrow$ Colon $\rightarrow$ Rectum

(2) Pharynx → Oesophagus → Gizzard → Crop → Ileum → Colon → Rectum

(3) Pharynx → Oesophagus → Gizzard → Ileum → Crop → Colon → Rectum

(4) Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum

BSA234

NEET (UG) - 2020

9. If the head of cockroach is removed, it may live for few days because: -

(1) the supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.

(2) the cockroach does not have nervous system.

(3) the head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.

(4) the head holds a 1/3rd of a nervous system while the rest is situated along the dorsal part of its body.

BSA235

NEET (UG) (COVID-19) - 2020

10. In cockroach, identify the parts of the foregut in correct sequence: -

(1) Mouth → Oesophagus → Pharynx → Crop → Gizzard

(2) Mouth → Crop → Pharynx → Oesophagus → Gizzard

(3) Mouth → Gizzard → Crop → Pharynx → Oesophagus

(4) Mouth → Pharynx → Oesophagus → Crop → Gizzard

BSA236

11. Match the following columns with reference to cockroach and select the correct option :

Column - I	Column - II
(a) Grinding of the food particles	(i) Hepatic caecal
(b) Secrete gastric juice	(ii) 10th segment
(c) 10 pairs	(iii) Proventriculus
(d) Anal cerci	(iv) Spiracles
	(v) Alary muscles

(1) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

(2) (a)–(iv), (b)–(iii), (c)–(v), (d)–(ii)

(3) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii)

(4) (a)–(ii), (b)–(iii), (c)–(i), (d)–(iv)

BSA237

NEET (UG) - 2021

12. Which of the following characteristics is incorrect with respect to cockroach?

(1) A ring of gastric caeca is present at the junction of midgut and hind gut.

(2) Hypopharynx lies within the cavity enclosed by the mouth parts.

(3) In females, 7th - 9th sterna together form a genital pouch.

(4) 10th abdominal segment in both sexes, bears a pair of anal cerci.

BSA238

NEET (UG) - 2022

13. Tegmina in cockroach, arises from:

(1) Mesothorax

(2) Metathorax

(3) Prothorax and Mesothorax

(4) Prothorax

BSA239

Re-NEET (UG) - 2022

14. Match List–I with List-II regarding the organs of Cockroach:

List–I	List–II
(a) Crop	(i) grinding the food particles
(b) Proventriculus	(ii) secretion of digestive juice
(c) Hepatic caecae	(iii) removal of nitrogenous waste
(d) Malpighian tubules	(iv) storage of food

Choose the correct answer from the options given below:

(1) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)

(2) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)

(3) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)

(4) (a) – (i), (b) – (iv), (c) – (iii), (d) – (ii)

BSA240

NEET (UG) - 2023

15. Which of the following is characteristic feature of cockroach regarding sexual dimorphism?

(1) Presence of anal styles

(2) Presence of sclerites

(3) Presence of anal cerci

(4) Dark brown body colour and anal cerci

BSA241

NEET (UG) – 2023 (Manipur)

16. The Cockroach is :

(1) Ammonotelic only

(2) Uricotelic only

(3) Ureotelic only

(4) Ureotelic and Uricotelic

BSA242

17. Given below are two statements :

Statement I : In cockroach, the forewings are transparent and prothoracic in origin.

Statement II : In cockroach, the hind wings are opaque, leathery and mesothoracic in origin.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is correct but Statement II is false.

(2) Statement I is incorrect but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

BSA243

NEET (UG) 2024

18. In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on :

(1) 5th segment

(2) 10th segment

(3) 8th and 9th segment

(4) 11th segment

BSA244

19. Match **List-I** with **List-II** related to digestive system of cockroach.

	List I		List II
A.	The structures used for storing of food.	I.	Gizzard
B.	Ring of 6-8 blind tubules at junction of foregut and midgut.	II.	Gastric Caeca

C.	Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	III.	Malpighian tubules
D.	The structures used for grinding the food.	IV	Crop

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-III, B-II, C-IV, D-I

BSA245

RE-NEET (UG) 2024

20. Which of the following is/are present in female Cockroach ?

A. Collateral gland B. Mushroom gland

C. Spermatheca D. Anal style

E. Phallic gland

Choose the most **appropriate answer** from the options given below :

(1) B and D only (2) B and E only

(3) A only (4) A and C only

BSA246

Exercise-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	3	3	4	2	1	3	4	1	1	1	1	1
Question	16	17	18	19	20										
Answer	2	4	2	1	4										

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which of the following incorrect statement about cockroach

(1) Nymph grows by moulting about 13 times to reach the adult form.

(2) Next to last nymphal stage has wing pad

(3) Many species of cockroaches are wild and are no economic importance.

(4) Have pair of compound eyes and two pair of antenna.

BSA247

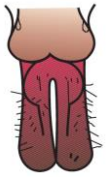
2. How many filamentous malphigian tubules are in cockroach?

(1) 100 – 150 (2) 50 – 100

(3) 150 – 200 (4) 200 – 250

BSA248

3. Identify the A, B, C and D in the given diagram and choose the correct option :

(A) 

(B) 

(C) 

(D) 

	A	B	C	D
1	Hypopharynx	Mandible	Labrum	Maxilla
2	Labrum	Maxilla	Mandible	Hypopharynx
3	Hypopharynx	Labrum	Maxilla	Mandible
4	Mandible	Hypopharynx	Labrum	Maxilla

BSA249

4. Choose the correct statement regarding cockroach.

(1) Sperms are store in the ootheca

(2) fertilization occurs in seminal vessicles

(3) Anal style found in female cockroach

(4) Pair of spermatheca present in 6th segment

BSA250

5. A pair of testis in cockroach lie in -

(1) 2 – 6 abdominal segment

(2) 4 – 7 abdominal segment

(3) 4 – 6 abdominal segment

(4) 6 – 8 abdominal segment

BSA251

6. Exo-skeleton plates of cockroach are connected by thin flexible membrane that is known as:

(1) Arthrodial membrane

(2) Peritoneal membrane

(3) Pleural membrane

(4) Perichordial membrane

BSA252

7. Match the column I and II correctly.

	Column-I		Column-II
(A)	Ovary	(i)	10th segment
(B)	Gizzard	(ii)	Excretary organ
(C)	Malpighian tubules	(iii)	Grinding of food
(D)	Anal cerci	(iv)	2-6th abdominal segment

(1) A-i, B-ii, C-iii, D-iv

(2) A-ii, B-iii, C-iv, D-i

(3) A-iv, B-iii, C-ii, D-i

(4) A-iii, B-iv, C-i, D-ii

BSA253

8. How many oothecae are produced by female cockroach?

(1) 14 – 16 (2) 9 – 10

(3) 12 – 13 (4) 1 – 4

BSA254

(B) ANALYTICAL QUESTIONS

9. Antennae in cockroaches have sensory receptors that help in :-

(1) Thinking and reasoning

(2) Monitoring the environment

(3) Fighting and running

(4) Tasting and watching

BSA255

10. **Assertion (A) :** If the head of cockroach is cut off, it will remain alive for as long as one week.

Reason (R): The head of cockroach hold only a bit of nervous system.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (B) are true, but (R) is not the correct explanation of A

(3) (A) is true, but (R) is false

(4) (A) is false, but (R) is true

BSA256

11. Select the incorrect statements about the hind wings of cockroach :-

I. These are broad.

II. They are not used in flying

III. They are also known as mesothoracic wings

IV. They are transparent and membranous

(1) Only I (2) II and III

(3) I and IV (4) I, II, III and IV

BSA257

12. Which of the following is called brain in cockroach?

(1) Supraoesophageal ganglion

(2) Suboesophageal ganglion

(3) Ventral ganglion

(4) Lateral ganglion

BSA258

13. Male cockroach and female cockroach can be differentiated by :

(1) Ocelli (2) Anal cerci

(3) Anal styles (4) Compound Eye

BSA259

14. Read the following statements about cockroach :

Statement-I : In male cockroach, a characteristic mushroom shaped gland is present in the 6th-7th abdominal segments. Which functions as an accessory reproductive gland.

Statement-II : Blood from sinuses enter heart through ostia and is pumped anteriorly to sinuses again.

Choose the correct answer from the options given below :

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and II are correct.

BSA260

15. Uric acid is the chief nitrogenous component of the excretory products of:-

(1) Frog

(2) Man

(3) Earthworm

(4) Cockroach

BSA261

16. Which of the following correctly state as it happens in the common cockroach?

(1) Malpighian tubules are excretory organs projecting out from the colon.

(2) Oxygen is transported by haemoglobin in blood.

(3) Nitrogenous excretory product is urea.

(4) The food is grinded by mandibles and gizzard

BSA262

Exercise-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	1	4	3	1	3	2	2	1	2	1	3	4	4
Question	16														
Answer	4														

Exercise - I (Conceptual Questions) Build Up Your Understanding

1. Bidder's canal in frog is found in :
(1) Kidney (2) Testes
(3) Urinary bladder (4) Cloaca
BSA302

2. Which of the following is the function of kidney in frog ?
(1) Excretion (2) Osmoregulation
(3) Haemopoiesis (4) Both (1) & (2)
BSA303

3. Excretory product of frog is :-
(1) Urea (2) Ammonia
(3) Uric acid (4) Amino acids
BSA304

4. The venous system of frog differs from that of rabbit in the presence of :-
(1) Hepatic portal system
(2) Renal portal system
(3) Hepatic vein
(4) Three vena cava
BSA305

5. Three chambered heart is found in:-
(1) Frog (2) Butterfly
(3) Crocodile (4) Fish
BSA306

6. Dorsal part of frog is..... and ventral part is .....
(1) Olivegreen, uniformly pale yellow
(2) Olivegreen, uniformly dark yellow
(3) Olivegreen, nonuniformly pale yellow
(4) Olivegreen, nonuniformly dark yellow
BSA307

7. Common Indian bull frog is :-
(1) *Rana tigrina* (2) *Rana esculenta*
(3) *Rana silvatica* (4) *Rana goliath*
BSA308

8. Summer sleep of frog is called :-
(1) Hibernation (2) Aestivation
(3) Both (4) Salivation
BSA309

9. Select the true statement regarding frog (*Rana tigrina*):-
(1) Frog is a homeothermal animal
(2) Frog drinks about 5 litres H_2O per day
(3) poisonous glands are not found on the body of frog.
(4) A very long tail is present in posterior part of frog's body.
BSA310

10. Vocal sacs are found in :-
(1) male frog (2) female frog
(3) both (4) none
BSA311

11. The glands present in the skin of toad are :-
(1) mucous and poisonous
(2) sweat and mammary
(3) sweat and sebaceous
(4) mucous and sweat
BSA312

12. A hibernating frog respires with :-
(1) Lung
(2) Diaphragm
(3) Buccal epithelium
(4) Skin
BSA313

13. Frog belongs to amphibian because :-
(1) tail is not found in it
(2) lungs are found in it
(3) its tadpoles are aquatic in nautre
(4) it lives in water as well as on land
BSA314

14. Skull of frog is :-
(1) Tricondylic (2) Monocondylic
(3) Dicondylic (4) Polyconylic
BSA315

15. Frog reproduces :-
(1) in water
(2) on trees
(3) in sand
(4) on lotus flowers in water
BSA316

16. Number of cranial nerves found in frog :-
(1) 10 (2) 12 (3) 20 (4) 24
BSA317

17. The third eyelid in frog is called :-
(1) Pineal eye
(2) Lower eyelid
(3) Nictitating membrane
(4) Upper eyelid
BSA318

18. Alimentary canal in frog terminates into
(1) Cloaca (2) Stomach
(3) Ileum (4) Duodenum
BSA319

19. The liquified semidigested acidic food is known as :-
(1) Bile (2) Bolus
(3) Chyme (4) Faeces
BSA320

20. Skin helps in exchange of gases because :-
(1) It is kept moist by the mucus and water
(2) It is thin
(3) It is richly supplied with blood
(4) All of these
BSA321

21. Air enters into the lungs through the
(1) Skin (2) Buccal cavity
(3) Mouth (4) Cloaca
BSA322

22. Pericardium is a
(1) Thin, transparent and two layered sac.
(2) Thick, cuticular and two layered sac.
(3) Thick, transparent and three-layered sac.
(4) Thin, opaque and four layered sac.
BSA323

23. In frogs, urine is temporarily stored in
(1) Kidneys (2) Urinary bladder
(3) Ureters (4) Cloaca
BSA324

24. Vasa-efferentia in male frog opens into
(1) Vas deferens
(2) Bidder's canal
(3) Cloaca
(4) Urinogenital duct
BSA325

Exercise-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	1	2	1	1	1	2	3	1	1	4	4	3	1
Question	16	17	18	19	20	21	22	23	24						
Answer	3	3	1	3	4	2	1	2	2						

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT (Mains) - 2011

1. Frogs differ from the humans in possessing:

(1) Thyroid as well as parathyroid

(2) Paired cerebral hemispheres

(3) Hepatic portal system

(4) Nucleated red blood cells

BSA326

AIPMT - 2017

2. Select the correct route for the passage of sperms in male frog :-

(1) Testes → Vasa efferentia → Kidney → Seminal vesicle → Urinogenital duct → Cloaca

(2) Testes → Vasa efferentia → Bidder's canal → Ureter → Cloaca

(3) Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca

(4) Testes → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca

BSA327

3. Frog's heart when taken out of the body continuous to beat for sometimes.

Select the best option from the following statements

(a) Frog is a poikilotherm

(b) Frog does not have any coronary circulation

(c) Heart is "Myogenic" in nature

(d) Heart is autoexcitable

Option :-

(1) Only d

(2) a and b

(3) c and d

(4) Only c

BSA328

Exercise-II (Previous Year Questions) ANSWER KEY

Question	1	2	3
Answer	4	3	3

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Body of frog is divisible into :
(1) Head and trunk (2) Head and tail
(3) Head and neck (4) Head, trunk & tail

BSA329

2. Read the following four statement about sense organs of *Rana tigrina* :–
(1) Frog has simple eyes possessing only one unit
(2) External ear is present in frog
(3) External ear is absent in frog and only tympanum can be seen externally
(4) Eyes and internal ear are well organised structures and the rest are cellular aggregation
How many of the above statements are correct?
(1) 4 (2) 1 (3) 2 (4) 3

BSA330

3. Female frogs do not have :
(1) Tympanum
(2) Nictitating membrane
(3) Vocal sacs and copulatory pads
(4) Hibernation and aestivation

BSA331

4. Frogs are :
(1) Carnivorous (2) Herbivorous
(3) Omnivorous (4) Sanguivores

BSA332

5. Find out correct match for respiration in frog :-

	In Water	On land
(1)	Lungs	Skin
(2)	Cutaneous	Pulmonary
(3)	Buccal-cavity	Lungs
(4)	Pulmonary	Buccal-cavity

BSA333

6. In the heart of frog sinus venosus joins to the
(1) Right atrium (2) Left atrium
(3) Right ventricle (4) Left ventricle

BSA334

7. Venous connection between kidney and lower parts of body in Indian bull frog is known as :
(1) Hepatic portal system
(2) Hypophyseal portal system
(3) Renal portal system
(4) Coronary circulation

BSA335

8. Find out the incorrect option for frog :
(1) Enucleated RBC
(2) WBC present
(3) Nucleated RBC
(4) Platelets present

BSA336

9. The mid brain of frog is characterised by :
(1) Two pairs of optic lobe
(2) One optic lobe
(3) Three optic lobes
(4) A pair of optic lobes

BSA337

(B) ANALYTICAL QUESTIONS

10. Excretory waste of frog is :
(1) Ammonia (2) Urea
(3) TMAO (4) Uric acid

BSA338

11. In frog partially digested food called chyme is passed from stomach to :
(1) Duodenum (2) Ileum
(3) Jejunum (4) Rectum

BSA339

12. Which one of the following is common in both male and female frog?
(1) Bidder's canal
(2) Vocal sac
(3) Cloaca
(4) Copulatory pad

BSA340

13. The larva of *Rana tigrina* is
(1) Tornaria
(2) Tadpole
(3) Trochophore
(4) Ascidian

BSA341

14. Which of the following feature is not found in Indian bull frog?
(1) External fertilisation
(2) Indirect development
(3) Metamorphosis
(4) Internal fertilisation

BSA342

15. The system for control and coordination is highly developed in the frog, it includes:
(1) Both neural system and endocrine glands
(2) Only neural system
(3) Only endocrine glands
(4) None

BSA343

Exercise-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	1	2	1	3	1	4	2	1	3	2	4	1

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

NON-CHORDATES

1. If a food particle is placed near the osculum of a sponge then it will -
(1) Be thrown away
(2) Be ingested
(3) Rotate around osculum
(4) Be ingested first and then thrown out through ostia

BAD005

2. Bath sponge is common name of :
(1) *Spongilla*
(2) *Euspongia*
(3) *Euplectella*
(4) *Leucosolenia*

BAD006

3. Which one of the following is Coelenterate?
(1) Sea urchin (2) Sea cucumber
(3) Sea fan (4) Sea horse

BAD007

4. Which animal do not possess larval stage in its life cycle?
(1) *Ascaris* (2) Frog
(3) *Taenia* (4) *Pheretima*

BAD008

5. Which of the following is an Annelid?
(1) sea horse (2) sea mouse
(3) sea hare (4) sea pen

BAD009

6. Which of the following characteristic is shared by all arthropods?
(1) Complete metamorphosis
(2) Wings
(3) Jointed appendages
(4) Tracheal system

BAD011

7. Members of phylum Arthropoda lack one of the following feature :
(1) External skeleton made of chitin
(2) Compound eyes
(3) Excretion by malphigian tubules
(4) Usually a close type of blood vascular system

BAD012

8. Cockroach, house fly and mosquitoes are insect because they have :-
(1) Chitinous exoskeleton and body divided into head and cephalothorax
(2) Six legs, compound eyes and body divided into head, thorax and abdomen
(3) Segmented body with jointed feet and chitinous exoskeleton
(4) Three pairs of legs, one pair of antennae and flame cells

BAD013

9. Which of the following animal is considered as "Living fossil"?
(1) *Limulus* (2) Spider
(3) *Sycon* (4) Scorpion

BAD014

10. Select the economically important insect :
(1) *Locusta* (2) *Culex*
(3) *Aedes* (4) *Apis*

BAD015

11. Which animal has visceral hump ?
(1) *Culex* (2) *Apis*
(3) *Nereis* (4) *Pila*

BAD016

12. Which mollusca is called tusk shell?
(1) *Neopilina* (2) *Pila*
(3) *Dentalium* (4) *Doris*

BAD017

13. Secretion of shell in molluscs take place by:
(1) Foot (2) Ink gland
(3) Radula (4) Mantle
BAD018

14. Which of the following characteristic is not of phylum mollusca?
(1) Soft body covered usually by a shell
(2) Segmented body
(3) Triploblastic
(4) Presence of coelom
BAD019

15. The most characteristic feature of molluscs is the presence of :
(1) Hard - body (2) Shell
(3) Mantle (4) Foot
BAD020

16. Open type of blood vascular system and presence of haemocyanin pigment are the characters of :
(1) Mollusca (2) Annelida
(3) Echinodermata (4) Hemichordata
BAD021

17. Member of Echinodermata has a specific system, which is not found in other phylum, it is :
(1) Canal system
(2) Water vascular system
(3) Respiratory system
(4) Jointed appendages
BAD022

18. Segmentation of body is not represented in:
(1) Cray – fish (2) *Sepia*
(3) Grass hopper (4) Silver fish
BAD023

19. Which of the following is not related with Echinodermata ?
(1) Madreporite (2) Parapodia
(3) Marine habitat (4) Autotomy
BAD025

20. Locomotory organs of echinoderms are :-
(1) Tube feet (2) Setae
(3) Parapodia (4) Pseudopodia
BAD026

21. Absence of excretory organs, great power of regeneration and exclusively marine animals belongs to the phylum :-
(1) Mollusca (2) Arthropoda
(3) Echinodermata (4) Annelida
BAD027

22. Stomochord (Buccal diverticulum) is present in:
(1) Amphioxus
(2) *Herdmania*
(3) *Balanoglossus*
(4) *Asterias*
BAD028

23. Which of the following character does not belong to Arthropoda?
(1) Closed circulatory system
(2) Movable jaws
(3) Segmented body
(4) Jointed legs
BAD029

24. Chiton is included in phylum :
(1) Annelida (2) Arthropoda
(3) Mollusca (4) Echinodermata
BAD030

25. Coxal glands are the excretory organ of :
(1) Cray fish (2) Devil fish
(3) Jelly - fish (4) Scorpion
BAD031

26. Most of the adult insects have six legs and:
(1) Two antennae and four wings
(2) Four antennae and four wings
(3) Two antennae and two or four wings
(4) Four wings and four eyes
BAD032

27. In *Hydra*, waste material of food digestion and nitrogenous waste material removed respectively from :
(1) Mouth and mouth
(2) Body wall and body wall
(3) Mouth and body wall
(4) Mouth and tentacles

BAD035

28. Given below are four matchings of an animal and its kind of respiratory organ :
A. Silver fish – Trachea
B. Scorpion – Book lung
C. Sea squirt – Pharyngeal gills
D. Dolphin – Skin
The correct matchings are :-
(1) A and D (2) A, B and C
(3) B and D (4) C and D

BAD036

29. In contrast to Annelids, the Platyhelminthes show :
(1) Absence of body cavity
(2) Presence of pseudocoel
(3) Radial symmetry
(4) Bilateral symmetry

BAD037

30. The canal system is characteristic feature of:
(1) Sponges (2) Helminthes
(3) Echinoderms (4) Coelenterates

BAD039

31. Which of the following is pseudoceolomate?
(1) *Hydra* (2) *Periplaneta*
(3) *Ascaris* (4) *Pheretima*

BAD041

32. *Sycon* belongs to a group of animals, which are best described as :
(1) Unicellular or acellular
(2) Multicellular without any tissue organization
(3) Multicellular with a gastrovascular system
(4) Multicellular having tissue organization, but no body cavity

BAD044

33. Which one of the following phyla is correctly matched with its general characteristics ?
(1) Porifera - Cellular level of organisation and external fertilisation
(2) Coelenterata - Diploblastic and mostly Asymmetric
(3) Aschelminthes - Pseudocoelomates and dioecious
(4) Hemichordata - Coelomates and closed circulatory system

BAD045

34. Which one of the following is a matching pair of a body feature and the animal possessing it?
(1) Canal system – *Asterias*
(2) Metagenesis – *Obelia*
(3) Dorsal nerve cord – *Pheretima*
(4) Muscular pharynx – *Taenia*

BAD047

35. One of the following animals is not hermaphrodite :
(1) Earthworm (2) Leech
(3) *Taenia* (4) *Ascaris*

BAD048

36. In Annelids formation of larva is usually absent but when present the larva is called:
(1) Tadpole (2) Planula
(3) Trochophore (4) Ephyra

BAD049

37. Excretion in most Annelids take place by :
(1) Kidneys (2) Flame cells
(3) Nephridia (4) Ciliated funnel

BAD050

38. Sea-anemone belongs to phylum :
(1) Protozoa (2) Porifera
(3) Coelenterata (4) Echinodermata

BAD054

39. Skeleton is made up of _____ in porifera.
(1) Spicules (2) Spongin
(3) Both 1 and 2 (3) Chitin
BAD060

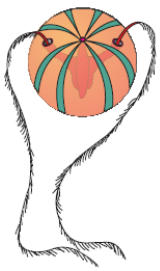
40. Find the odd
(1) Sea fan
(2) sea horse
(3) sea cucumber
(4) sea lily
BAD061

41. Which animal has only cylindrical form?
(1) *Physalia* (2) *Adamsia*
(3) *Hydra* (4) Both (2) and (3)
BAD062

42. Which animal is umbrella-shaped and free swiming?
(1) *Aurelia* (2) Jelly-fish
(3) *Hydra* (4) (1) and (2)
BAD063

43. Corals have a skeleton composed of _____.
(1) $CaCO_3$ (2) $CaPO_4$
(3) $CaCl_2$ (4) $CaSiO_2$
BAD064

44. Identify the animal shown in diagram



(1) Tape worm
(2) *Pleurobrachia*
(3) *Nereis*
(4) Octopus
BAD065

CHORDATES

45. Choose the correct combination of the given option?
(1) *Calotes* - garden lizard
(2) *Chameleon* - krait
(3) *Naja* - viper
(4) *Crocodilus* - tortoise
BAD066

46. Which one of the following mammalia live in water?
(1) Bat (2) Platypus
(3) *Macropus* (4) Blue whale
BAD067

47. The distinguishing feature of all chordates is:
(1) A ventrally placed nerve cord
(2) A water vascular system
(3) A chitinous exoskeleton
(4) An elastic rod (Notochord)
BAD068

48. Chordates possess :
(1) Dorsal nerve cord placed below gut
(2) Single hollow nerve cord placed dorsal to gut
(3) Double ventral nerve cord
(4) Single, solid and ventral nerve cord
BAD069

49. In vertebrates, the notochord :
(1) Persists throughout the life
(2) Is partially replaced by vertebral column
(3) Is completely replaced by vertebral column
(4) Is partially or fully replaced by vertebral column
BAD070

50. Which one is the distinguishing feature of all vertebrates?
(1) An internal bony or cartilagenous skeleton
(2) A hairy body covering
(3) Presence of mammary glands
(4) Open type of circulatory system
BAD071

51. Which one of the following amphibia is common toad?
(1) *Ichthyophis* (2) *Bufo*
(3) Frog (4) *Salamandra*
BAD076

52. Which of the following group of animals have a constant body temperature?
(1) Reptiles, aves & Mammalia
(2) Aves & cyclostomata
(3) Pisces & amphibia
(4) Aves & Mammalia

BAD079

53. Which is poikilothermic animal?
(1) Tortoise (2) Pigeon
(3) Penguin (4) Whale

BAD084

54. Which of the following is not the feature of snakes ?
(1) Eggs (2) Limbs
(3) Scales (4) Kidney

BAD085

55. Which of the following groups includes Endothermic animals (Homothermal) :
(1) Crocodile, alligator, turtle
(2) Whale, pigeon, bat
(3) Sea-horse, dog fish, cat fish
(4) Toad, frog, salamander

BAD086

56. Number of cranial nerves in a crocodile is :
(1) 8-pairs (2) 10-pairs
(3) 12-pairs (4) 14-pairs

BAD088

57. Mostly birds are :
(1) Uricotelic and oviparous
(2) Uricotelic and viviparous
(3) Ammonotelic and oviparous
(4) Uricotelic and ovoviviparous

BAD089

58. The primitive characters of Duck billed platypus suggest that it :
(1) Is a degenerated mammals
(2) Shows the stage between aves and mammals
(3) Shows the stage between reptiles and birds
(4) Shows the stage between reptiles and mammals

BAD090

59. Find out correct statement :
(1) All birds can fly
(2) All snakes are poisonous
(3) All amphibians lay eggs in water
(4) All mammals are viviparous

BAD091

60. Oviparous animal is :
(1) *Scoliodon* (2) Viper
(3) Scorpion (4) Cobra

BAD092

61. Characteristic of *Corvus* is :
(1) Unisexual and sexual dimorphism absent
(2) Bisexual and sexual dimorphism absent
(3) Unisexual and sexual dimorphism present
(4) Bisexual and sexual dimorphism present

BAD093

62. Which of the following is a "Lizard- bird"?
(1) *Archaeopteryx*
(2) *Struthio*
(3) *Aptenodytes*
(4) *Humming bird*

BAD094

63. Uropygial or preen glands occur in :
(1) Reptiles (2) Birds
(3) Amphibians (4) Mammals

BAD095

64. The character found only in birds is :
(1) Toothless beak
(2) Bipedal locomotion
(3) Wings
(4) Feathers

BAD098

65. Which of the following characters is not related to Aves?
(1) Homiothermy and presence of oil glands in skin
(2) Female genital organs of both sides functional
(3) Female genital organs of one sides functional
(4) Exoskeleton of feathers

BAD099

66. Muscular diaphragm is found in :
(1) Amphibian (2) Mammal
(3) Reptilian (4) Fish

BAD101

67. Whales are included in the same taxonomic class as :
(1) Sharks (2) Crocodile
(3) Sea horse (4) Gorilla

BAD102

68. Which of the following group of vertebrates have dicondylic skull?
(1) Amphibia & mammals
(2) Reptilia & Aves
(3) Aves & mammals
(4) Reptilia & mammals

BAD103

69. Hair are not found in adult whale, its body temperature is regulated by :
(1) Sweat glands (2) Blubber
(3) Blood circulation (4) Water

BAD104

70. Two chambered heart is the feature of :
(1) Mammals (2) Aves
(3) Amphibia (4) Pisces

BAD105

71. Which of the following animals have character of both reptiles and mammals?
(1) *Ornithorhynchus* (2) *Pteropus*
(3) *Macropus* (4) *Canis*

BAD106

72. An egg laying mammal is :
(1) Kangaroo (2) Platypus
(3) Koala (4) Whale

BAD107

73. Presence of gills in the tadpole of frog indicates that :
(1) Fishes evolved from frog like ancestors
(2) Frogs will have gills in future
(3) Frogs evolved from gilled ancestors
(4) Fishes were amphibious in the past

BAD108

74. Hollow air filled bones (pneumatic bones) occur in :
(1) Mammals (2) Reptiles
(3) Urodela (4) Aves

BAD109

75. Which of the following character is not found in all the chordates?
(1) Diaphragm
(2) Coelom
(3) Pharyngeal gill slits
(4) Dorsal nerve cord

BAD110

76. In which of the following animal post anal tail is found :
(1) Earthworm
(2) Lower invertebrate
(3) Scorpion
(4) Snake

BAD111

77. In which of the following notochord is present in embryonic stage :
(1) All chordates (2) Some chordates
(3) Vertebrates (4) Non-chordates

BAD112

78. Axolotl larva is the name of larva of :
(1) *Amphioxus* (2) Silkworm
(3) *Salamandra* (4) Round worm

BAD113

79. Most favourable land adaptation in reptiles is:
(1) Lungs
(2) Scales
(3) Moist skin
(4) Pentadactylous limbs

BAD115

80. *Ichthyophis* belongs to :
(1) Amphibia (2) Mollusca
(3) Annelida (4) Reptilia

BAD117

81. Air sacs connected to lungs supplement respiration in :
(1) Lung fishes (2) Birds
(3) Reptilia (4) Mammals

BAD120

82. Which of the following reptile is non-poisonous ?
(1) Cobra (2) Krait
(3) Viper (4) Python

BAD121

83. Tailed amphibian is :
(1) *Icthyophis* (2) Frog
(3) *Salamandra* (4) Toad

BAD122

84. Monocondylar, Poikilothermal animals are:
(1) Birds
(2) Amphibia
(3) Reptiles
(4) Mammals

BAD123

85. Which one of the following is a true fish ?
(1) Cray fish (2) Jelly fish
(3) Silver fish (4) Cat fish

BAD124

86. Corpus callosum is found in the brain of :
(1) Elephant (2) Pigeon
(3) Crocodile (4) Frog

BAD125

87. One of the following is a very unique feature of the mammalian body :
(1) Presence of muscular diaphragm
(2) Four chambered heart
(3) Rib cage
(4) Homeothermy

BAD126

88. Which is the smallest taxonomic group having cranium, vetebral column, ventral heart, pulmonary respiration and two pairs of legs ?
(1) Chordata (2) Gnathostomata
(3) Vertebrata (4) Tetrapoda

BAD127

89. Which of the following has no teeth in its mouth?
(1) Tortoise (2) Crocodile
(3) *Hemidactylus* (4) *Draco*

BAD130

90. Feathers of birds are modified :
(1) Epidermal scales (2) Bony scales
(3) Plates (4) Scutes

BAD131

91. Identify the animal -



(1) *Ascidia* (2) *Salpa*
(3) *Amphioxus* (4) *Doliolum*

BAD132

92. Notochord is present only in larval tail of :
(1) Urochordata
(2) Cephalochordata
(3) Vertebrata
(4) Protochordates

BAD133

93. Vertebrates have ventral muscular heart with _____ chambers.
(1) 2 (2) 3
(3) 4 (4) All above

BAD134

94. Larve of _____ after metamorphosis return to the ocean.
(1) *Scoliodon* (2) Shark
(3) Lamprey (4) Flying fish
BAD135

95. Air bladder is absent in _____.
(1) Dog fish (2) *Catla*
(3) Rohu (4) Flying fish
BAD136

96. Sea horse is _____.
(1) a bird (2) a mammal
(3) an amphibian (4) a fish
BAD137

97. Which group of vertebrates have characteristic features the presence of feathers and most of them can fly ?
(1) Mammals (2) Fishes
(3) Reptiles (4) Birds
BAD258

98. Which class name refers to their creeping or crawling mode of locomotion (Latin, repere or reptum, to creep or crawl)?
(1) Aves (2) Mammalia
(3) Amphibia (4) Reptilia
BAD259

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	3	4	2	3	4	2	1	4	4	3	4	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	2	1	3	3	1	3	4	3	3	2	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	3	2	4	3	3	3	3	2	4	4	1	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	2	4	1	2	4	1	2	2	3	1	4	3	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	1	2	4	2	2	4	1	2	4	1	2	3	4	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	1	3	2	1	2	4	3	3	4	1	1	4	1	1
Question	91	92	93	94	95	96	97	98							
Answer	1	1	4	3	1	4	4	4							

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which one of the following living organisms completely *lacks* a cell wall?

(1) Cyanobacteria

(2) Sea – fan *(Gorgonia)*

(3) *Saccharomyces*

(4) Blue–green algae

BAD175

2. Select the Taxon mentioned that represents both marine and fresh water species :

(1) Echinoderms (2) Ctenophora

(3) Cephalochordata (4) Cnidaria

BAD176

3. *Planaria* possess high capacity of :

(1) Metamorphosis

(2) Regeneration

(3) Alternation of generation

(4) Bioluminescence

BAD177

4. A marine cartilaginous fish that can produce electric current is :

(1) *Pristis* (2) *Torpedo*

(3) *Trygon* (4) *Scoliodon*

BAD178

AIPMT 2015

5. Which of the following animals is **not** viviparous?

(1) Elephant (2) Platypus

(3) Whale (4) Flying fox (Bat)

BAD179

6. Which of the following represents the correct combination without any exception?

	Characteristics	Class
(1)	Mouth ventral, gills without operculum; skin with placoid scales; persistent notochord	Chondrichthyes
(2)	Sucking and circular mouth; jaws absent, integument without scales; paired appendages	Cyclostomata
(3)	Body covered with feathers; skin moist and glandular; fore-limbs form wings; lungs with air sacs	Aves
(4)	Mammary gland; hair on body; pinnae; two pairs of Limbs	Mammalia

BAD180

Re-AIPMT 2015

7. Metagenesis refers to :

(1) Presence of a segmented body and parthenogenetic mode of reproduction

(2) Presence of different morphic forms

(3) Alternation of generation between asexual and sexual phases of an organism

(4) Occurrence of a drastic change in form during post-embryonic development

BAD183

8. Which one of the following animals has two separate circulatory pathways ?

(1) Shark (2) Frog

(3) Lizard (4) Whale

BAD184

9. Body having meshwork of cells, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum :

(1) Protozoa (2) Coelenterata

(3) Porifera (4) Mollusca

BAD185

10. A jawless fish, which lays eggs in fresh water and whose ammocoetes larvae after metamorphosis return to the ocean is :

(1) *Petromyzon*

(2) *Eptatretus*

(3) *Myxine*

(4) *Neomyxine*

BAD186

NEET-I 2016

11. Which of the following characteristic features always holds true for the corresponding group of animals?

(1)	Cartilaginous endoskeleton	Chondrichthyes
(2)	Viviparous	Mammalia
(3)	Possess a mouth with an upper and a lower jaw	Chordata
(4)	3 - chambered heart with one incompletely divided ventricle	Reptilia

BAD187

12. Which one of the following characteristics is **not** shared by birds and mammals ?

(1) Ossified endoskeleton

(2) Breathing using lungs

(3) Viviparity

(4) Warm blooded nature **BAD188**

NEET-II 2016

13. Choose the **correct** statement.

(1) All reptiles have a three-chambered heart.

(2) All pisces have gills covered by an operculum.

(3) All mammals are viviparous.

(4) All cyclostomes do not possess jaws and paired fins.

BAD189

NEET (UG) 2017

14. An important characteristic that Hemichordates share with Chordates is :

(1) Ventral tubular nerve cord

(2) Pharynx with gill slits

(3) Pharynx without gill slits

(4) Absence of notochord

BAD190

15. Which among these is the correct combination of aquatic mammals ?

(1) Dolphins, Seals, *Trygon*

(2) Whales, Dolphins, Seals

(3) *Trygon*, Whales, Seals

(4) Seals, Dolphins, Sharks

BAD191

16. In case of poriferans, the spongocoel is lined with flagellated cells called:

(1) oscula

(2) choanocytes

(3) mesenchymal cells

(4) ostia

BAD192

NEET (UG) 2018

17. Which of the following animals does ***not*** undergo metamorphosis ?

(1) Earthworm (2) Tunicate

(3) Moth (4) Starfish

BAD193

18. Identify the vertebrate group of animals characterized by crop and gizzard in its digestive sytstem.

(1) Amphibia (2) Reptilia

(3) Aves (4) Osteichthyes

BAD194

19. Which one of these animals is ***not*** a homeotherm?

(1) *Macropus* (2) *Chelone*

(3) *Camelus* (4) *Psittacula*

BAD195

NEET (UG) 2019

20. Match the following organisms with their respective characteristics :

(a) *Pila* (i) Flame cells

(b) *Bombyx* (ii) Comb plates

(c) *Pleurobrachia* (iii) Radula

(d) *Taenia* (iv) Malpighian tubules

Select the **correct** option from the following:

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(ii)	(iv)	(iii)	(i)
(4)	(iiii)	(ii)	(iv)	(i)

BAD197

21. Consider following features :

(a) Organ system level of organisation

(b) Bilateral symmetry

(c) True coelomates with segmentation of body

Select the **correct** option of animal groups which possess all the above characteristics.

(1) Annelida, Arthropoda and Chordata

(2) Annelida, Arthropoda and Mollusca

(3) Arthropoda, Mollusca and Chordata

(4) Annelida, Mollusca and Chordata

BAD198

NEET (UG) 2019 (Odisha)

22. Match the following genera with their respective phylum:

(a) *Ophiura* (i) Mollusca

(b) *Physalia* (ii) Platyhelminthes

(c) *Pinctada* (iii) Echinodermata

(d) *Planaria* (iv) Coelenterata

Select the correct option:

(1) a–(iv), b–(i), c–(iii), d–(ii)

(2) a–(iii), b–(iv), c–(i), d–(ii)

(3) a–(i), b–(iii), c–(iv), d–(ii)

(4) a–(iii), b–(iv), c–(ii), d–(i)

BAD199

23. Which of the following animals are true coelomates with bilateral symmetry?

(1) Adult Echinoderms

(2) Aschelminthes

(3) Platyhelminthes

(4) Annelids

BAD200

NEET (UG) - 2020

24. Match the following columns and select the **correct** option.

Column -I	**Column -II**
(a) 6 -15 pairs of gill slits	(i) *Trygon*
(b) Heterocercal caudal fin	(ii) Cyclostomes
(c) Air Bladder	(iii) Chondrichthyes
(d) Poison sting	(iv) Osteichthyes

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(iv)	(iii)	(ii)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iv)	(ii)	(iii)	(i)

BAD201

25. Which of the following statements are true for the phylum-Chordata ?

(a) In Urochordata notochord extends from head to tail and it is present throughout their life.

(b) In Vertebrata notochord is present during the embryonic period only.

(c) Central nervous system is dorsal and hollow.

(d) Chordata is divided into 3 subphyla : Hemichordata, Tunicata and Cephalochordata.

(1) (b) and (c) (2) (d) and (c)

(3) (c) and (a) (4) (a) and (b)

BAD202

26. Match the following columns and select the **correct** option.

Column-I	**Column-II**
(a) Gregarious, polyphagous pest	(i) *Asterias*
(b) Adult with radial symmetry and larva with bilateral symmetry	(ii) Scorpion
(c) Book lungs	(iii) *Ctenoplana*
(d) Bioluminescence	(iv) *Locusta*

(1) a–(ii), b–(i), c–(iii), d–(iv)

(2) a–(i), b–(iii), c–(ii), d–(iv)

(3) a–(iv), b–(i), c–(ii), d–(iii)

(4) a–(iii), b–(ii), c–(i), d–(iv)

BAD203

27. Bilaterally symmetrical and acoelomate animals are exemplified by:

(1) Annelida

(2) Ctenophora

(3) Platyhelminthes

(4) Aschelminthes

BAD204

NEET (UG)-2020 (COVID-19)

28. Which of the following options does correctly represent the characteristic features of phylum Annelida ?

(1) Triploblastic, unsegmented body and bilaterally symmetrical.

(2) Triploblastic, segmented body and bilaterally symmetrical.

(3) Triploblastic, flattened body and acoelomate condition.

(4) Diploblastic, mostly marine and radially symmetrical.

BAD205

29. Match the following group of organisms with their respective distinctive characteristics and select the correct option

Organisms	**Characteristics**
(a) Platyhelminthes	(i) Cylindrical body with no segmentation
(b) Echinoderms	(ii) Warm blooded animals with direct development
(c) Hemichordates	(iii) Bilateral symmetry with incomplete digestive system
(d) Aves	(iv) Radial symmetry with Indirect development

(1) a–(iii), b–(iv), c–(i), d–(ii)

(2) a–(ii), b–(iii), c–(iv), d–(i)

(3) a–(iv), b–(i), c–(ii), d–(iii)

(4) a–(i), b–(ii), c–(iii), d–(iv)

BAD206

30. All vertebrates are chordates but all chordates are not vertebrates, why ?

(1) Notochord is replaced by vertebral column in adult of some chordates.

(2) Ventral hollow nerve cord remains throughout life in some chordates.

(3) All chordates possess vertebral column.

(4) All chordates possess notochord throughout their life.

BAD207

31. Match the following columns and select the correct option :

	Column-I		**Column-II**
(a)	*Aptenodytes*	(i)	Flying fox
(b)	*Pteropus*	(ii)	Angel fish
(c)	*Pterophyllum*	(iii)	Lamprey
(d)	*Petromyzon*	(iv)	Penguin

(1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

BAD208

NEET (UG) - 2021

32. Read the following statements.

(a) Metagenesis is observed in Helminths.

(b) Echinoderms are triploblastic and coelomate animals.

(c) Round worms have organ-system level of body organization.

(d) Comb plates present in ctenophores help in digestion.

(e) Water vascular system is characteristic of echinoderms.

Choose the **correct** answer from the options given below.

(1) (c), (d) and (e) are correct

(2) (a), (b) and (c) are correct

(3) (a), (d) and (e) are correct

(4) (b), (c) and (e) are correct

BAD209

33. Match **List - I** with **List - II.**

List-I		**List-II**	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(i)	(ii)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(iv)	(i)	(ii)	(iii)

BAD210

34. Match the following:

List-I		**List-II**	
(a)	*Physalia*	(i)	Pearl oyster
(b)	*Limulus*	(ii)	Portuguese Man of War
(c)	*Ancylostoma*	(iii)	Living fossil
(d)	*Pinctada*	(iv)	Hookworm

Choose the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iii)	(i)	(iv)
(2)	(iv)	(i)	(iii)	(ii)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(i)	(iv)	(iii)	(ii)

BAD211

35. Which one of the following organisms bears hollow and pneumatic long bones?

(1) *Neophron* (2) *Hemidactylus*

(3) *Macropus* (4) *Ornithorhynchus*

BAD212

NEET (UG) - 2022

36. Nitrogenous waste is excreted in the form of pellet or paste by :

(1) *Salamandra* (2) *Hippocampus*

(3) *Pavo* (4) *Ornithorhynchus*

BAD213

37. **Assertion (A) :** All vertebrates are chordates but all chordates are not vertebrates.

Reason (R) : Notochord is replaced by vertebral column in the adult vertebrates.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is not correct but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

BAD214

38. In which of the following animals, digestive tract has additional chambers like crop and gizzard ?

(1) *Bufo, Balaenoptera, Bangarus*

(2) *Catla, Columba, Crocodilus*

(3) *Pavo, Psittacula, Corvus*

(4) *Corvus, Columba, Chameleon*

BAD215

Re-NEET (UG) - 2022

39. Select the **incorrect** statements with respect to cyclostomes:

(a) They lack scales and paired fins.

(b) They have circular mouth with jaws.

(c) They bear 6-15 pairs of gills

(d) They migrate to deep sea for spawning.

Choose **the most appropriate answer** from the options given below.

(1) (a) and (b) only (2) (b) and (c) only

(3) (b) and (d) only (4) (a) and (d) only

BAD216

40. Which of the following animals has three chambered heart?

(1) *Scoliodon* (2) *Hippocampus*

(3) *Chelone* (4) *Pteropus*

BAD217

NEET (UG) - 2023

41. Radial symmetry is NOT found in adults of phylum _____.

(1) Hemichordata (2) Coelenterata

(3) Echinodermata (4) Ctenophora

BAD268

42. Select the **correct** statements with reference to chordates.

A. Presence of mid-dorsal, solid and double nerve cord.

B. Presence of closed circulatory system

C. Presence of paired pharyngeal gill slits

D. Presence of dorsal heart

E. Triploblastic pseudocoelomate animals

Choose the **correct** answer from the options given below:

(1) B and C only (2) B, D and E only

(3) C, D and E only (4) A, C and D only

BAD269

43. The unique mammalian characteristics are :

(1) hairs, pinna and mammary glands

(2) hairs, pinna and indirect development

(3) pinna, monocondylic skull and mammary glands

(4) hairs, tympanic membrane and mammary glands

BAD270

NEET (UG) (Manipur) - 2023

44. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Contractile vacuole	(I)	*Asterias*
(B)	Water vascular system	(II)	*Amoeba*
(C)	Canal system	(III)	*Spongilla*
(D)	Flame cells	(IV)	*Taenia*

Choose the **correct** answer from the options given below :

(1) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(2) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(3) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(4) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

BAD271

45. Select the **correct** statements :

(A) Platyhelminthes are triploblastic pseudocoelomate and bilaterally symmetrical organisms.

(B) Ctenophores reproduce only sexually and fertilization is external.

(C) In tapeworm, fertilization is internal but sexes are not separate.

(D) Ctenophores are exclusively marine, diploblastic and bioluminescent organisms.

(E) In sponges, fertilization is external and development is direct.

Choose the **correct** answer from the options given below :

(1) (A), (C) and (D) only

(2) (B), (C) and (D) only

(3) (A) and (E) only

(4) (B) and (D) only

BAD272

NEET (UG) 2024

46. Match **List I** with **List II**.

List I		List II	
A.	*Pleurobrachia*	I.	Mollusca
B.	Radula	II.	Ctenophora
C.	Stomochord	III.	Osteichthyes
D.	Air bladder	IV.	Hemichordata

Choose the **correct** answer from the options given below :-

(1) A-IV, B-II, C-III, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-III, C-II, D-I

BAD275

47. Consider the following statements :-

A. Annelids are true coelomates

B. Poriferans are pseudocoelomates

C. Aschelminthes are acoelomates

D. Platyhelminthes are pseudocoelomates

Choose the **correct** answer from the options given below :-

(1) B only

(2) A only

(3) C only

(4) D only

BAD276

48. Match **List I** with **List II**.

List-I		List-II	
A.	*Pterophyllum*	I.	Hag fish
B.	*Myxine*	II.	Saw fish
C.	*Pristis*	III.	Angel fish
D.	*Exocoetus*	IV.	Flying fish

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-III, B-I, C-II, D-IV

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-II, C-I, D-IV

BAD277

49. The following are the statements about non-chordates :-

A. Pharynx is perforated by gill slits.

B. Notochord is absent.

C. Central nervous system is dorsal.

D. Heart is dorsal if present.

E. Post anal tail is absent.

(1) A & C only (2) A, B & D only

(3) B, D & E only (4) B, C & D only

BAD278

RE-NEET (UG) 2024

50. Match **List-I** with **List-II**.

List-I		List-II	
A.	Chondrichthyes	I.	*Clarias*
B.	Cyclostomata	II.	*Carcharodon*
C.	Osteichthyes	III.	*Myxine*
D.	Amphibia	IV.	*Ichthyophis*

Choose the **correct** answer from the options given below :-

(1) A-II, B-IV, C-I, D-III

(2) A-I, B-III, C-II, D -IV

(3) A-II, B-III, C-I, D-IV

(4) A-I, B-II, C-III, D-IV

BAD279

51. Open Circulatory system is present in:

(1) *Palaemon*, *Nereis*, *Balanoglossus*

(2) *Hirudinaria*, *Bombyx*, *Salpa*

(3) *Anopheles*, *Limax*, *Limulus*

(4) *Pheretima*, *Musca*, *Pila*

BAD280

52. Which of the following pairs is an **incorrect** match ?

(1) Annelids and arthropods – Bilateral symmetry

(2) Sponges - Acoelomates

(3) Coelenterates and Ctenophores – Radial symmetry

(4) Platyhelminthes – Diploblastic organisation

BAD281

53. One is labelled as are **Assertion A** and the other is labelled as **Reason R**.

Assertion (A) : Members of subphylum vertebrate possess notochord during the embryonic period. The notochord is replaced by a cartiliaginous or bony vertebral column in the adult.

Reason (R) : Thus all chordates are vertebrates but not all vertebrates are chordates.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) **A** is true but **R** is false

(2) **A** is false but **R** is true

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both A and R are true but **R** is NOT the correct explanation of **A**.

BAD282

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	2	2	1	3	4	3	1	1	3	4	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	3	2	2	1	2	4	2	1	3	3	2	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	3	3	1	3	4	3	3	3	1	1	1	4	2
Question	46	47	48	49	50	51	52	53							
Answer	2	2	2	3	3	3	4	1							

Exercise - III Master Your Understanding

1. Which one of the following is a radially symmetrical, blind sac body plan and diploblastic animal?
(1) *Spongilla* (2) *Euspongia*
(3) *Asterias* (4) *Physalia*
BAD218

2. Which one is not a platyhelminthes?
(1) Tapeworm (2) Liver fluke
(3) *Planaria* (4) Hookworm
BAD219

3. Consider the following statements :-
(A) Protochordates are exclusively marine
(B) In cephalochordates, Notochord extends from head to tail region.
(C) In urochordates, Notochord is present only in larval tail.
(D) Cranium and vertebral column are bony in cyclostomates.
Which of the above statement is/are correct?
(1) A alone (2) B, C and D
(3) A, B and C (4) All
BAD220

4. Consider the following characteristic of fishes:
(A) They have four pairs of gills which are covered by an operculum.
(B) They have air bladder which regulates buoyancy
(C) They are mostly viviparous and development is direct
(D) Their body is streamlined and covered with cycloid/ctenoid scales
Which of the above characteristics regarding bony fishes are correct?
(1) A, B and C (2) C and D
(3) A, B and D (4) B alone
BAD221

5. Which of the following animal is radially symmetrical but has a bilaterally symmetrical larva?
(1) *Ascaris* (Round worm)
(2) *Nereis* (Ring worm)
(3) *Pila* (Apple snail)
(4) *Asterias* (Star fish)
BAD222

6. Identified the correct statement from the following with reference to Amphibians :
(A) They have internal fertilisation and development is direct.
(B) Amphibian skin is moist and without scales
(C) Their body is divisible into head and trunk
(D) They have three chambered heart
Select the **correct** answer :
(1) A and D (2) A and B
(3) B, C and D (4) D alone
BAD223

7. Which of the following pairs are correctly matched?
(A) Poikilothemous - Birds and mammals
(B) Agnatha - *Petromyzon* and *Myxine*
(C) Tunicates - *Ascidia, Salpa* and *Doliolum*
(D) Cartilaginous fishes - *Betta* and *Labeo*
(1) A and B (2) A, B and C
(3) B and C (4) A, B and D
BAD224

8. Water vascular system is found in :

(1) Porifera (2) Echinodermata

(3) Mollusca (4) Elasmobranch

BAD225

9. Match the column :-

(a) *Dentalium*	(i) Brittle star
(b) *Ophiura*	(ii) Cuttle fish
(c) *Antedon*	(iii) Sea Urchin
(d) *Echinus*	(iv) Tusk shell
(e) *Sepia*	(v) Sea lily

(1) a-(iv), b-(i), c-(iv), d-(ii), e-(iii)

(2) a-(iv), b-(i), c-(v), d-(iii), e-(ii)

(3) a-(i), b-(iv), c-(v), d-(iii), e-(ii)

(4) a-(iv), b-(i), c-(iii), d-(v), e-(ii)

BAD226

10. True coelom and metameric segmentation in the body is first observed in which of the following phylum?

(1) Platyhelminthes (2) Aschelminthes

(3) Annelida (4) Arthropoda

BAD227

11. Read the following statements (a to e) carefully :

(a) Triploplastic

(b) Bilateral symmetric

(c) Organ system organisation

(d) Pseudocoelomate

(e) Metameric segmentation

Which of the above statement(s) is/are correct for phylum Aschelminthyes

(1) a and b only

(2) a, b, c and e

(3) a, b, c and d

(4) a, b, c, d and e

BAD228

12. Which of the following coelenterate exist in both forms polyp and medusa and exhibit metagenesis?

(1) *Hydra* (2) *Adamsia*

(3) *Obelia* (4) *Gorgonia*

BAD229

13. Mark the incorrectly matched pair

(1) *Petromyzon* – Body devoid of scales and paired fins

(2) *Scoliodon* – Operculum present but air bladder is absent

(3) *Exocoetus* – Heart two chamered with sinus venosus

(4) *Rana* – Cold blooded with 10 pairs of cranial nerves and mesonephric kidneys

BAD230

14. Match list 'A' and List 'B' correctly and find your answer from the code given below :

List-A (Animal)	**List-B (Common name)**
(i) *Physalia*	(A) Brain coral
(ii) *Pennatula*	(B) Sea fan
(iii) *Gorgonia*	(C) Sea pen
(iv) *Meandrina*	(D) Portuguese man of war

Code :

(1) (i)-A, (ii)-B, (iii)-C, (iv)-D

(2) (i)-D, (ii)-C, (iii)-B, (iv)-A

(3) (i)-C, (ii)-D, (iii)-B, (iv)-A

(4) (i)-D, (ii)-C, (iii)-A, (iv)-B

BAD231

15. Which of the following is gregarious pest?

(1) Bed bug (2) Rat

(3) Cockroach (4) *Locusta*

BAD232

16. Pick the odd pair out :
(1) Cellular level : Porifera
(2) Tissue level : Aschelminthes
(3) Organ level : Platyhelminthes
(4) Organ system level : Annelida

BAD233

17. Select the group of organisms given below those have diploblastic members only :
(1) *Ctenoplana, Taenia, Fasciola*
(2) *Aurelia, Physalia, Meandrina*
(3) *Wuchereria, Culex, Limulus*
(4) *Aedes, Ascaris, Hydra*

BAD234

18. Fresh water sponge is :
(1) *Spongilla* (2) Scypha
(3) *Euspongia* (4) *Leucosolenia*

BAD235

19. The symmetry found in an animal which can be divided into indentical left and right halves in only one plane is:
(1) Spherical (2) Bilateral
(3) Radial (4) Biradial

BAD236

20. Select the pseudocoelomates from the list of organisms given below :
(1) *Ascaris, Fasciola, Taenia*
(2) *Culex, Locusta, Limulus*
(3) *Wuchereria, Ascaris, Ancylostoma*
(4) *Nereis, Hirudinaria, Wuchereria*

BAD237

21. Which of the following pairs of animals comprise 'Comb jellies' ?
(1) *Balanoglossus* and *Saccoglossus*
(2) *Pleurobranchia* and *Ctenoplana*
(3) Sea anemone and sea pen
(4) Sea lily and brittle star

BAD238

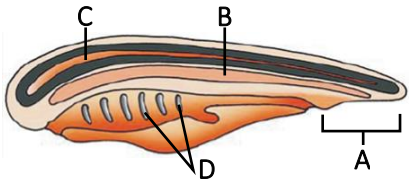
22. Which one of the following pairs of animals comprise 'Cartilaginous fishes'?
(1) *Labeo* and *Catla*
(2) *Pterophyllum* and *Scoliodon*
(3) *Pristis* and *Carcharodon*
(4) *Petromyzon* and *Myxine*

BAD239

23. Which of the following fishes is also known as "Great white shark'?
(1) *Pristis* (2) *Trygon*
(3) *Clarias* (4) *Carcharodon*

BAD240

24. A schematic representation of a chordate's embryo is shown in the above figure.



The structures marked **A, B, C** and **D** are respectively :
(1) Notochord, Nerve cord, Intestine and gill slits
(2) Post anal part, Gill slits, Nerve cord and Notochord
(3) Post anal part, nerve cord, Notochord and Gill
(4) Post anal part, Notochord, Nervecord and Gill slits.

BAD241

25. In which one of the following protochordates notochord present only in the tail of larva ?
(1) *Ascidia* (2) *Balanoglossus*
(3) *Myxine* (4) *Branchiostoma*

BAD242

26. In which of the following group notochord present in whole life?

(1) Hemichordata (2) Urochordata

(3) Cephalochordata (4) Vertebrata

BAD243

27. Match the column-I with the column-II and find out the correct answer :

Column-I	**Colum-II**
(A) Spongocoel	(i) Arthropoda
(B) Dorso-ventrally flattened body	(ii) Cnidaria
(C) Coelenteron cavity	(iii) Porifera
(D) Chitinous exoskeleton	(iv)Platyhelminthes

(1) A-iii, B-iv, C-i, D-ii

(2) A-iv, B-iii, C-ii, D-i

(3) A-iii, B-iv, C-ii, D-i

(4) A-ii, B-iv, C-iii, D-I

BAD244

28. Which of the following is a matching pair ?

(1) *Ophiura* – Sea lily

(2) *Octopus* – Cuttle fish

(3) *Torpedo* – Electric ray

(4) *Aptenodytes* – Ostrich

BAD245

29. Which one is not the character of Osteichthyes?

(1) Terminal mouth

(2) Gills with operculum

(3) Body covered by placoid scales

(4) Air bladder regulate buoyancy.

BAD246

30. Which one of the following pairs is not correctly matched?

(1) *Chaetopleura* - Chiton

(2) *Ascaris* - Round worm

(3) *Wuchereria* - Filaria worm

(4) *Enterobius* - Hook worm **BAD247**

31. The greatest evolutionary change that enabled the land vertebrates to be completely free from water, was the development of :

(1) four appendages

(2) lungs

(3) cleidoic eggs

(4) four chambered heart

BAD248

32. A mammal which completely lives in water is:

(1) Bear (2) Camel

(3) Whale (4) Rhinoceros

BAD249

33. How many in the given examples of animals are Coelentrates?

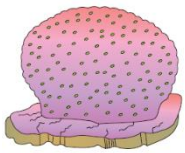
Physalia, Obelia, Planaria, Pennatula, Gorgonia, Pleurobrachia, Meandrina and *Nereis*

(1) Three (2) Four

(3) Five (4) Six

BAD251

34. Identify the following four animals (A, B, C and D) given below. Which one of these is not correctly identified in the option given along with its correct taxonomic group and their common name ?



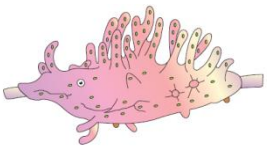
(A)



(B)



(C)



(D)

Options :-

	Fig.	Name of animal	Taxonomic group	Common Name
(1)	B	*Adamsia*	Ctenophora	Star coral
(2)	D	*Spongilla*	Porifera	Fresh water sponge
(3)	C	*Aurelia*	Coelenterata	Jelly fish
(4)	A	*Euspongia*	Porifera	Bath sponge

BAD252

35. How many organism are related with Echinodermata phylum?

Star fish, Sea-urchin, Sea-lily, Sea-mouse, Sea-hare, Sea-cucumber, Brittle star, Sea-squirt, Sea-fan, Sea-pen

(1) 4 (2) 5 (3) 6 (4) 7

BAD253

36. Which of the following shows the sexual dimorphism?

(1) *Hydra* & *Ascaris*

(2) *Hydra* & *Macaca*

(3) *Ascaris* & *Pheretima*

(4) *Ascaris* & *Macaca* **BAD255**

37. Which one of the following features is common in - silverfish, scorpion, dragonfly and prawn ?

(1) Three pairs of legs and segmented body

(2) Chitinous cuticle and two pairs of antennae

(3) Jointed appendages and chitinous exoskeleton

(4) Cephalothorax involves **BAD256**

38. In which animal nerve cell is present but brain is absent?

(1) Sponge (2) Earthworm

(3) Cockroach (4) *Hydra*

BAD257

39. Which one of the following is not a living fossil?

(1) *Peripatus* (2) King crab

(3) *Sphenodon* (4) *Archaeopteryx*

BAD260

40. Body is divisible into head and trunk in :

(1) Lizard (2) Frog

(3) Peacock (4) Platypus

BAD261

41. The evidence for the origin of birds from reptiles is the presence in them of :

(1) feathers (2) scales

(3) claws (4) hairs

BAD262

42. In which one of the following groups all animals are hermaphrodite?

(1) *Hydra, Ascaris, Pheretima*

(2) *Hydra, Homo sapiens*, Leech

(3) Tapeworm, Toad, Starfish

(4) *Hydra*, Leech, Tapeworm **BAD263**

43. Given below are two statements:

Statement-I : The most distinct feature of echinoderms is the presence of water canal system.

Statement-II : Echinodermata exhibits radial or bilateral symmetry depending on the stage.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BAD264

44. Given below are two statements:

Statement-I : In cephalochordate notochord is persistent throughout their life.

Statement-II : *Branchiostoma* is an example of cephalochordata

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BAD265

45. Given below are two statements:

Statement-I : All chordates are vertebrates but all vertebrates are not chordates.

Statement-II : Sub phylum urochordata and cephalochordate are often referred as protochordata.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BAD266

46. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : Water vascular system helps in locomotion, capture and transport of food and respiration in porifera.

Reason (R) : The presence of water vascular system is most unique feature of porifera.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) & (R) are correct but (R) is the not correct explanation of (A).

(2) (A) and (R) both are not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BAD267

47. Match the following:

(a) Operculum	(i) Ctenophora
(b) Parapodia	(ii) Mollusca
(c) Scales	(iii) Porifera
(d) Comb plates	(iv) Reptilia
(e) Radula	(v) Annelida
(f) Hairs	(vi) Cyclostomata and Chondrichthyes
(g) Choanocytes	(vii) Mammalia
(h) Gill slits	(viii) Osteichthyes

(1) a-vi, b-v, c-vii, d-i, e-ii, f-viii, g-iii, h-iv

(2) a-viii, b-v, c-iv, d-i, e-ii, f-vii, g-iii, h-vi

(3) a-viii, b-vii, c-vi, d-v, e-ii, f-iv, g-ii, h-i

(4) a-i, b-vi, c-v, d-viii, e-ii, f-iii, g-iv, h-vii

BAD273

48. Find incorrectly matched parasitic animal with its phylum.
(1) *Fasciola hepatica* – Platyhelminthes
(2) *Ascaris lumbricoids* – Aschelminthes
(3) *Taenia solium* – Flatworms
(4) *Wuchereria* – Platyhelminthes

BAD274

49. Which one of the following is a matching set of a phylum and its three examples ?
(1) Porifera – *Spongilla, Euplectella, Pennatula*
(2) Cnidaria – *Bonellia, Physalia, Aurelia*
(3) Platyhelminthes – *Planaria, Schistosoma, Enterobius*
(4) Mollusca – *Loligo, Teredo, Octopus*

BAD139

50. Metameric segmentation is the characteristic of :
(1) Mollusca and Choradata
(2) Platyhelminthes and Arthropoda
(3) Echinodermata and Annelida
(4) Annelida and Arthropoda

BAD140

51. Two common characters found in Centipede, Cockroach and Crab are **:**
(1) Book lungs and antennae
(2) Compound eyes and anal cerci
(3) Jointed legs and chitinous exoskeleton
(4) Green gland and tracheae

BAD141

52. Biradial symmetry and lack of cnidoblasts are the characteristic of :
(1) *Hydra* and *Starfish*
(2) *Starfish* and Sea Anemone
(3) *Ctenoplana* and *Beroe*
(4) *Aurelia* and *Paramoecium*

BAD142

53. In which one of the following sets of animals all the four give birth to young ones ?
(1) Kangaroo, Hedgehog, Dolphin, Loris
(2) Lion, Bat, Whale, Ostrich
(3) Platypus, Penguin, Bat, *Hippopotamus*
(4) Shrew, Bat, Cat, Kiwi

BAD144

54. Which one of the following is a matching pair of a body feature and the animal possessing it?
(1) Ventral heart – Scorpion
(2) Post-anal tail – *Octopus*
(3) Ventral central nervous system – Leech
(4) Pharyngeal gill slits absent in embryo – *Chamaeleon*

BAD145

55. What is true about *Nereis*, Scorpion, Cockroach and Silver fish ?
(1) They all belong to the same phylum
(2) They all have paired jointed appendages
(3) They all possess dorsal heart
(4) None of them is aquatic

BAD146

56. Which one of the following pairs is mismatched?
(1) *Bombyx mori* – Silk
(2) *Pila globosa* – Pearl
(3) *Apis indica* – Honey
(4) *Kenia lacca* – Lac

BAD147

57. Which of the following pairs are correctly matched ?

Animals	**Morphological features**
(A) Crocodile	4-chambered heart
(B) Sea Urchin	Parapodia
(C) *Obelia*	Metagenesis
(D) Lemur	Thecodont

(1) Only A and B (2) A, C and D
(3) B, C and D (4) Only A and D

BAD148

58. What is common between parrot, platypus and kangaroo ?
(1) Ovoparity
(2) Homoiothermy
(3) Toothless jaws
(4) Functional post-anal tail

BAD149

59. Which one of the following is not a characteristic of phylum Annelida?
(1) Closed circulatory system
(2) Segmentation
(3) Pseudocoelom
(4) Ventral nerve cord

BAD150

60. Which one of the following phyla is correctly matched with its two general characteristics?
(1) Arthropoda : Body divided into head, thorax and abdomen and respiration by tracheae
(2) Chordata : Notochord at some stage and separate anal and urinary openings to the outside
(3) Echinodermata : Pentamerous radial symmetry and mostly internal fertilisation
(4) Mollusca : Normally oviparous and development through a trochophore or veliger larva

BAD151

61. *Ascaris* is characterized by :
(1) absence of true coelom but presence of metamerism
(2) presence of neither true coelom nor metamerism
(3) presence of true coelom but absence of metamerism
(4) presence of true coelom and metamerism

BAD152

62. Which one of the following groups of the three animals each is correctly matched with their one characteristic morphological feature?

	Animals	***Morphological feature***
(1)	Liver fluke, Sea anemone, Sea cucumber	– Bilateral symmetry
(2)	Centipede, Prawn, Sea urchin	– Joined appendages
(3)	Scorpion, Spider, Cockroach	– Ventral solid central nervous system
(4)	Cockroach, Locust, *Taenia*	– Metameric segmentation

BAD153

63. Which one of the following groups of animals is bilaterally symmetrical and triploblastic?
(1) Sponges
(2) Coelenterates (Cnidarians)
(3) Aschelminthes (round worms)
(4) Ctenophores

BAD154

64. *Peripatus* is a connecting link between :
(1) Coelenterata and Porifera
(2) Ctenophora and Platyhelminthis
(3) Mollusca and Echinodermata
(4) Annelida and Arthropoda

BAD155

65. One example of animals having a single opening to the outside that serves both as mouth as well as anus is :
(1) *Fasciola* (2) *Octopus*
(3) *Asterias* (4) *Herdmania*

BAD157

66. Which one of the following statement about all the four of *Spongilla*, Leech, Dolphin and Penguin is **correct** ?

(1) All are bilaterally symmetrical

(2) Penguin is homoiothermic while the remaining three are poikilothermic

(3) Leech is a fresh water form while all others are marine

(4) *Spongilla* has special collared cells called choanocytes, not found in the remaining three.

BAD158

67. Which one of the following animals are triploblastic*?*

(1) Corals (2) Flat worms

(3) Sponges (4) Ctenophores

BAD159

68. Which one of the following statement about certain given animals is correct *?*

(1) Flatworms (Platyhelminthes) are coelomates

(2) Roundworms (Aschelminthes) are pseudocoelomates

(3) Molluscs are acoelomates

(4) Insects are pseudocoelomates

BAD160

69. Crocodile and Penguin are similar to Whale and Dogfish in which one of the following features ?

(1) Lay eggs and guard them till they hatch

(2) Possess bony skeleton

(3) Have gill slits at some stage

(4) Possess a solid single stranded central nervous system

BAD161

70. Which one of the following groups of animals is correctly matched with its one characteristic feature without even a single exception ?

(1) Reptilia : possess 3-chambered heart with one incompletely divided ventricle

(2) Chordata : possess a mouth provided with an upper and a lower jaw

(3) Chondrichthyes : possess cartilaginous endoskeleton

(4) Mammalia : give birth to young ones

BAD162

71. Which one of the following animals is correctly matched with its particular named taxonomic category ?

(1) Tiger – *tigris*, the species

(2) Cuttlefish – Mollusca, a class

(3) Humans – Primata, the family

(4) Housefly – *Musca*, an order

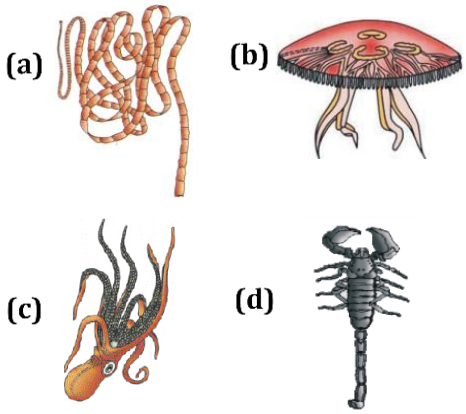
BAD163

72. In which one of the following the genus name, its two character and its class/phylum are correctly matched?

Genus name	Two characters	Class/ Phylum
(1) *Ascaris*	(a) Body segmented (b) Males and females distinct	Annelida
(2) *Salamandra*	(a) A tympanum represents ear (b) Fertilization is external	Amphibia
(3) *Pteropus*	(a) Skin possesses hair (b) Oviparous	Mammalia
(4) *Aurelia*	(a) Cnidoblasts (b) Organ level of organization	Coelenterata

BAD164

73. The figure shows four animals (a), (b), (c) and (d). Select the correct answer with respect to a common characteristics of two of these animals.



(1) (c) and (d) have a true coelom

(2) (a) and (d) respire mainly through body wall

(3) (b) and (c) show radial symmetry

(4) (a) and (b) have cnidoblasts for self-defence.

BAD165

74. In which one of the following, the genus name, its two characters and its phylum are **not correctly** matched, whereas the remaining three are correct ?

	Genus name	Two characters	Phylum
(1)	*Sycon*	(a) Pore bearing (b) Canal system	Porifera
(2)	*Periplaneta*	(a) Jointed appendages (b) Chitinuous exoskeleton	Arthropoda
(3)	*Pila*	(a) Body segmented (b) Mouth with Radula	Mollusca
(4)	*Asterias*	(a) Spiny skinned (b) Water vascular system	Echino-dermata

BAD166

75. Which one of the following categories of animals, is correctly described with no single exception in it?

(1) All sponges are marine and have collared cells

(2) All mammals are viviparous and possess diaphragm for breathing

(3) All reptiles possess scales, have a three chambered heart and are cold blooded (poikilothermal)

(4) All bony fishes have four pairs of gills and an operculum on each side

BAD168

76. Which one of the following pairs of animals are similar to each other pertaining to the feature stated against them?

(1) *Ascaris* and *Ancylostoma* – Metameric segmentation

(2) Sea horse and Flying fish – Cold blooded (poikilothermal)

(3) *Pteropus* and *Ornithorhyncus* – Viviparity

(4) Garden lizard and Crocodile – Three chambered heart

BAD169

77. One of the representatives of Phylum Arthropoda is :

(1) flying fish (2) cuttle fish

(3) silver fish (4) puffer fish

BAD170

78. Which group of animals belong to the same phylum?

(1) Sponge, Sea anemone, Starfish

(2) Malarial parasite, *Amoeba*, Mosquito

(3) Earthworm, Pinworm, Tapeworm

(4) Prawn, Scorpion, *Locusta* **BAD171**

79. Infection of *Ascaris* usually occurs by :

(1) mosquito bite

(2) drinking water containing eggs of *Ascaris*

(3) eating imperfectly cooked pork.

(4) Tse-tse fly

BAD172

80. Which of the following are correctly matched with respect to their taxonomic classification?

(1) Spiny anteater, sea urchin, sea cucumber –Echinodermata

(2) Flying fish, cuttlefish, silverfish – Pisces

(3) Centipede, millipede, spider, scorpion – Insecta

(4) House fly, butterfly, tsetsefly, silverfish – Insecta

BAD173

81. Match the name of the animal (column-I), with one characteristics (column-II), and the phylum/ class (column-III) to which it belongs :

	Column I	Column II	Column III
(1)	*Adamsia*	radially symmetrical	Porifera
(2)	*Petromyzon*	ectoparasite	Cyclostomata
(3)	*Ichthyophis*	terrestrial	Reptilia
(4)	Limulus	Body covered by chitinous exoskeleton	Pisces

BAD174

EXERCISE-III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	3	4	3	3	2	2	3	3	3	2	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	1	2	3	2	3	4	4	1	3	3	3	3	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	3	1	3	4	3	4	4	2	2	4	3	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	4	4	4	3	3	1	3	3	2	2	2	3	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	3	3	4	1	4	2	2	3	3	1	2	1	3	4
Question	76	77	78	79	80	81									
Answer	2	3	4	2	4	2									

Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

HOW TO ANALYSE CHEMICAL COMPOSITION

1. The most abundant intracellular cation is :

(1) K^+ (2) Na^+ (3) Ca^{++} (4) H^+

BBM075

PRIMARY AND SECONDARY METABOLITES

2. Codiene is :-

(1) A polymeric substance

(2) An alkaloid

(3) An essential oil

(4) A toxin

BBM197

CARBOHYDRATES

3. Which is odd –

(1) Chitin – Carbohydrate

(2) Pectin - Protein

(3) Steroid – Lipid

(4) Wax – Lipid

BBM001

4. Which is a disaccharide?

(1) Galactose (2) Fructose

(3) Maltose (4) Dextrin

BBM002

5. Sugar found in haemolymph of insects is called-

(1) Maltose (2) Lactose

(3) Trehalose (4) Galactose

BBM003

6. Carbohydrates are stored in mammals as:

(1) Glucose in liver

(2) Glycogen in muscles and spleen

(3) Lactic acid in muscles

(4) Glycogen in liver and muscles

BBM004

7. Translocation of sugars in flowering plants occur in the form of -

(1) Glucose (2) Sucrose

(3) Fructose (4) Maltose

BBM005

8. Sucrose is composed of -

(1) Glucose & Fructose

(2) Glucose & Glycogen

(3) Two molecules of Glucose

(4) Glycogen & Fructose

BBM006

9. Sugars with five membered ring are called:-

(1) Pyranose (2) Furanose

(3) Dextrorotatory (4) Levorotatory

BBM007

10. Amylose and Amylopectin chains occur in-

(1) Glycogen (2) Starch

(3) Cellulose (4) Chitin

BBM008

11. Which of the following disaccharide gives two molecules of glucose on hydrolysis?

(1) Maltose (2) Lactose

(3) Both (1) and (2) (4) Sucrose

BBM009

12. Which monosaccharide does not show optical isomerism :

(1) Dihydroxy acetone

(2) Glyceraldehyde

(3) Erythrose

(4) Ribose

BBM064

13. Which one of the following sets of monosaccharides forms sucrose?

(1) β-D-Glucopyranose and α-D-fructofuranose

(2) α-D-Glucopyranose and β-D-fructopyranose

(3) α-D-Galactopyranose and α-D-Glucopyranose

(4) α-D-Glucopyranose and β-D-fructofuranose

BBM072

14. Macro molecule chitin is :
(1) Simple polysaccharide
(2) Nitrogen containing polysaccharide
(3) Phosphorus containing polysaccharide
(4) Sulphur containing polysaccharide

BBM076

LIPIDS

15. All lipids are -
(1) Composed of fatty acids
(2) Triglycerides
(3) Insoluble in water
(4) All the above

BBM010

16. Which one of the following is not a constituent of cell membrane ?
(1) Phospholipids (2) Cholesterol
(3) Glycolipids (4) Proline

BBM065

PROTEINS

17. Proteins present in protoplasm are very important because-
(1) They provide definite shape to cell
(2) They function as biocatalyst
(3) They yield energy
(4) They are stored food

BBM011

18. Unit of protein is: -
(1) Amino acid (2) Monosaccharide
(3) NH_3 (4) Nucleotide

BBM012

19. Milk protein is-
(1) Lactogen (2) Myosin
(3) Casein (4) Pepsin

BBM013

20. The amino acids which are not synthesized in the body are called:
(1) Non-essential (2) Essential
(3) Deaminated (4) All of them

BBM014

21. For body growth and repair, one needs:
(1) Carbohydrates (2) Fats
(3) Proteins (4) Vitamins

BBM015

22. Proteins in the body are composed of: -
(1) Amino acids (2) Natural proteins
(3) Enzymes (4) nucleic acids

BBM016

23. Which is most important structural part of the body?
(1) Protein (2) Carbohydrates
(3) Lipid (4) Nucleic acid

BBM017

24. Histone is a basic protein due to -
(1) Alanine & glycine
(2) Methionine & serine
(3) Tryptophan & tyrosine
(4) Lysine & Arginine

BBM018

25. Which one is the most abundant protein in the animal world ?
(1) Collagen (2) Insulin
(3) Trypsin (4) Haemoglobin

BBM074

NUCLEIC ACID

26. Nucleic acids are made up of -
(1) Amino acids (2) Pentose sugars
(3) Glycerol (4) Nucleotides

BBM019

27. Circular and double stranded DNA occurs in
(1) Golgi body (2) Mitochondria
(3) Nucleus (4) ER

BBM020

28. Double helix model of DNA was proposed by-
(1) Watson and Crick
(2) Schleiden and Schwann
(3) Singer and Nicolson
(4) Kornberg and Khorana

BBM021

29. Back bone in structure of DNA molecule is made up of -
(1) Pentose Sugar and phosphate
(2) Hexose sugar and phosphate
(3) Purine and pyrimidine
(4) Sugar and phosphate

BBM022

30. Nucleotide is -
(1) N-base, pentose sugar and phosphoric acid
(2) Nitrogen, Hexose sugar and phosphoric acid
(3) Nitrogen base, pentose sugar
(4) Nitrogen base, trioses and phosphoric acid

BBM023

31. Which element is not found in nitrogenous base?
(1) Nitrogen (2) Hydrogen
(3) Carbon (4) Phosphorus

BBM024

32. A N-base together with pentose sugar and phosphate forms: -
(1) Nucleoside (2) Polypeptide
(3) Nucleotide (4) Amino acid

BBM025

33. The purine & pyrimidine pairs of complementary strands of DNA are held together by –
(1) H – bonds (2) O – bonds
(3) C – bonds (4) N – bonds

BBM026

34. On an average, how many purine N bases are present in single coil of DNA?
(1) Four (2) Five
(3) Ten (4) Uncertain

BBM027

35. ATP is-
(1) Nucleotide (2) Nucleoside
(3) Nucleic acid (4) Vitamin

BBM028

36. In DNA, purine nitrogen bases are: -
(1) Uracil and Guanine
(2) Guanine and Adenine
(3) Adenine and cytosine
(4) None

BBM029

37. Adenylic acid is: -
(1) Ribose + Adenine + phosphate
(2) Ribose + cytosine
(3) Nucleoside
(4) Deoxyribonucleotide

BBM030

38. The Watson-Crick model of DNA double helix was proposed in :-
(1) 1953 (2) 1963
(3) 1954 (4) 1952

BBM031

ENZYMES

39. The fastest acting enzyme, in the biological kingdom among the following is:
(1) Lipase
(2) Amylase
(3) Peptidase
(4) Carbonic anhydrase

BBM032

40. Apoenzyme is:
(1) Protein (2) Vitamin
(3) Carbohydrate (4) Amino acid

BBM033

41. Which of the following inactivates the enzyme by changing its shape?
(1) Competitive inhibitor
(2) Allosteric inhibitor
(3) Coenzyme
(4) All of the above

BBM034

42. Enzyme were discovered for the first time in:-
(1) Bacteria (2) Yeast
(3) Algae (4) Spinach

BBM035

43. What is NADP?
(1) An enzyme
(2) An ion carrier
(3) H acceptor coenzyme
(4) A part of t-RNA
BBM036

44. The cofactor of various respiratory enzymes is: -
(1) Ca (2) B
(3) Mg (4) Mo
BBM037

45. Most enzymes consist of two parts these are :-
(1) Enzyme & substrate
(2) Enzyme & coenzyme
(3) Apoenzyme and enzyme
(4) Apoenzyme & cofactor
BBM038

46. The first enzyme isolated in crystalline form was :-
(1) Catalase (2) Urease
(3) Peroxidase (4) Amylase
BBM039

47. In plants, enzymes are present: -
(1) Only in flowers
(2) Only in leaves
(3) In all the living cells
(4) Only in parenchyma
BBM040

48. Enzyme inhibition caused by a substrate analogue is :-
(1) Competitive
(2) Non-competitive
(3) In competitive
(4) Semi-competitive
BBM041

49. Allosteric enzymes have allosteric sites for :
(1) Inhibition only
(2) Activation only
(3) Reduction in activation energy
(4) Both activation and inhibition
BBM042

50. An example of competitive inhibition is :
(1) Succinic dehydrogenase by malonic acid
(2) Cytochrome oxidase by cyanide
(3) Hexokinase by glucose - 6 phosphate
(4) Carbonic anhydrase by carbon - dioxide
BBM043

51. First discovered Enzyme was :-
(1) Isomerase (2) Transaminase
(3) Zymase (4) Transferase
BBM044

52. Enzyme were discovered for the first time by:
(1) Sumner (2) Buchner
(3) Kuhne (4) None of the above
BBM045

53. Who coined the term enzyme ?
(1) Pasteur (2) Buchner
(3) Kuhne (4) Sumner
BBM046

54. Vitamin serves the function of :-
(1) An enzyme (2) A coenzyme
(3) A holoenzyme (4) A hormone
BBM047

55. Which of the following is a coenzyme ?
(1) NAD (2) NADP
(3) FAD (4) All the above
BBM048

56. The Coenzyme is -
(1) often a metal
(2) often a vitamin
(3) always a protein
(4) always an inorganic compound
BBM049

57. Which of the following is produced with the combination of apoenzyme and coenzyme-
(1) ES complex (2) Prosthetic gp
(3) Holoenzyme (4) EP complex's
BBM050

58. Which of the following coenzyme is a derivative of Niacin ?
(1) NAD (2) NADP
(3) FAD (4) Both (1) and (2)
BBM051

59. Which of the following is not consumed in a biochemical process ?
(1) Hormones (2) Enzymes
(3) Vitamins (4) Nucleotides
BBM052

60. How the presence of an enzyme affects the activation energy of a reaction ?
(1) It becomes increased
(2) It becomes decreased
(3) It is first increased and then decreased
(4) It is not affected at all
BBM053

61. Enzyme inhibition caused by a product of the same biochemical reaction is :-
(1) Competitive inhibition
(2) Feed-back inhibition
(3) In competitive inhibition
(4) Semi-competitive inhibition
BBM054

62. At boiling temperature of water, what will be effect on an enzyme :-
(1) become denatured
(2) remain unaffected
(3) become inactivated
(4) become more active
BBM055

63. Enzyme have very narrow optima for :-
(1) Light (2) Temperature
(3) pH (4) Humidity
BBM056

64. Which enzyme is not a protein ?
(1) Hexokinase (2) Synthetase
(3) Endonuclease (4) Ribozyme
BBM057

65. Which of the following statements regarding enzyme inhibition is correct ?
(1) Non-competitive inhibition of an enzyme can be overcome by adding large amount of substrate
(2) Competitive inhibition is seen, when a substrate competes with an enzyme for binding to an inhibitor protein
(3) Competitive inhibition is seen, when the substrate and the inhibitor compete for the active site on the enzyme
(4) Non-competitive inhibitors often bind to the enzyme irreversibly.
BBM060

66. The catalytic efficiency of two different enzyme can be compared by the :
(1) The K_m value
(2) The pH optimum value
(3) Formation of the product
(4) Molecular size of the enzyme
BBM061

67. Which of the following is/are structural analogue(s) of p-amino benzoic acid?
(1) Malonate
(2) Glucose-6-phosphate
(3) AMP
(4) Sulpha drugs
BBM062

68. An organic substance bound to an enzyme and essential for its activity is called
(1) Apoenzyme
(2) Isoenzyme
(3) Coenzyme
(4) Holoenzyme
BBM063

69. A competitive inhibitor of succinic dehydrogenase is :-
(1) α-ketoglutarate (2) Malate
(3) Malonate (4) Oxaloacetate
BBM066

70. For its activity, carboxypeptidase requires :-
(1) Niacin (2) Copper
(3) Zinc (4) Iron
BBM071

71. Transition state structure of the substrate formed during an enzymatic reaction is :
(1) permanent and stable
(2) transient but stable
(3) permanent but unstable
(4) transient and unstable
BBM077

72. The essential chemical components of many coenzymes are :
(1) Vitamins
(2) Proteins
(3) Nucleic acids
(4) Carbohydrates
BBM078

73. Enzyme catalysing the linking together of 2 compounds :-
(1) Lyases
(2) Transferases
(3) Hydrolases
(4) Ligases
BBM198

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	2	3	3	4	2	1	2	2	1	1	4	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	1	3	2	3	1	1	4	1	4	2	1	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	1	3	1	2	1	1	4	1	2	2	3	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	3	1	4	1	3	2	3	2	4	2	3	4	2	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73		
Answer	2	1	3	4	3	1	4	3	3	3	4	1	4		

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which one of the following is a non - reducing carbohydrate ?

(1) Maltose (2) Sucrose

(3) Lactose (4) Ribose

BBM079

Re-AIPMT 2015

2. The chitinous exoskeleton of arthropods is formed by the polymerisation of :

(1) lipoglycans

(2) keratin sulphate and chondroitin sulphate

(3) D-glucosamine

(4) N-acetyl glucosamine

BBM081

NEET-I 2016

3. One of the major components of cell wall of most fungi is :-

(1) Chitin

(2) Peptidoglycan

(3) Cellulose

(4) Hemicellulose

BBM082

4. A typical fat molecule is made up of :-

(1) Three glycerol molecules and one fatty acid molecule

(2) One glycerol and three fatty acid molecules

(3) One glycerol and one fatty acid molecule

(4) Three glycerol and three fatty acid molecules

BBM083

NEET-II 2016

5. Which of the following is the least likely to be involved in stabilizing the three–dimensional folding of most proteins?

(1) Hydrophobic interaction

(2) Ester bonds

(3) Hydrogen bonds

(4) Electrostatic interaction

BBM084

6. A non-proteinaceous enzyme is :-

(1) Ligase

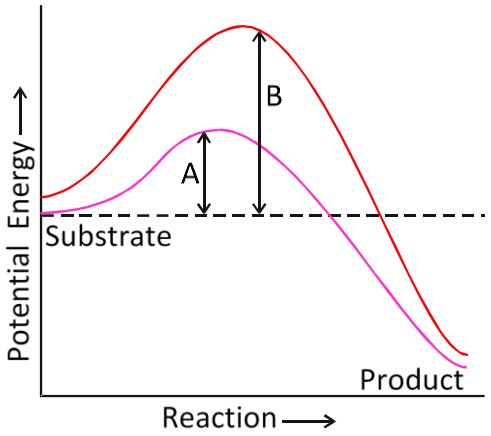
(2) Deoxyribonuclease

(3) Lysozyme

(4) Ribozyme

BBM085

7. Which of the following describes the given graph **correctly**?



(1) Endothermic reaction with energy A in absence of enzyme and B in presence of enzyme

(2) Exothermic reaction with energy A in absence of enzyme and B in presence of enzyme

(3) Endothermic reaction with energy A in presence of enzyme and B in absence of enzyme

(4) Exothermic reaction with energy A in presence of enzyme and B in absence of enzyme.

BBM086

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NEET (UG) 2017

8. Which of the following are not polymeric ?
(1) Proteins
(2) Polysaccharides
(3) Lipids
(4) Nucleic acids

BBM087

9. Which one of the following statements is **correct**, with reference to enzymes ?
(1) Holoenzyme = Apoenzyme + Coenzyme
(2) Coenzyme = Apoenzyme + Holoenzyme
(3) Holoenzyme = Coenzyme + Co-factor
(4) Apoenzyme = Holoenzyme + Coenzyme

BBM088

NEET (UG) 2018

10. The two functional groups characteristic of sugars are
(1) hydroxyl and methyl
(2) carbonyl and methyl
(3) carbonyl and phosphate
(4) carbonyl and hydroxyl

BBM089

11. Which of the following is an amino acid derived hormone ?
(1) Epinephrine (2) Ecdysone
(3) Estradiol (4) Estriol

BBM090

NEET (UG) 2019

12. Which of the following glucose transporters is insulin-dependent ?
(1) GLUT I (2) GLUT II
(3) GLUT III (4) GLUT IV

BBM091

13. Concanavalin A is :
(1) an alkaloid
(2) an essential oil
(3) a lectin
(4) a pigment

BBM092

14. Consider the following statements :
(A) Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group.
(B) A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.
Select the **correct** option.
(1) Both **(A)** and **(B)** are true.
(2) **(A)** is true and **(B)** is false.
(3) Both **(A)** and **(B)** are false.
(4) **(A)** is false and **(B)** is true.

BBM093

NEET (UG) 2019 (Odisha)

15. Which of the following organic compounds is the main constituent of Lecithin ?
(1) Arachidonic acid
(2) Phospholipid
(3) Cholesterol
(4) Phosphoprotein

BBM094

16. "Ramachandran plot" is used to confirm the structure of :-
(1) RNA
(2) Proteins
(3) Triacylglycerides
(4) DNA

BBM095

17. Prosthetic groups differ from co-enzymes in that :-
(1) they require metal ions for their activity.
(2) they (prosthetic groups) are tightly bound to apoenzymes.
(3) their association with apoenzymes is transient.
(4) they can serve as co-factors in a number of enzyme-catalyzed reactions.

BBM096

NEET (UG) 2020

18. Match the following :

(a) Inhibitor of catalytic activity	(i) Ricin
(b) Possess peptide bonds	(ii) Malonate
(c) Cell wall material in fungi	(iii) Chitin
(d) Secondary metabolite	(iv) Collagen

Choose the **correct** option from the following

	(a)	(b)	(c)	(d)
(1)	(ii)	(iii)	(i)	(iv)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(iii)	(i)	(iv)	(ii)
(4)	(iii)	(iv)	(i)	(ii)

BBM097

19. Identify the basic amino acid from the following.

(1) Valine (2) Tyrosine
(3) Glutamic Acid (4) Lysine

BBM098

20. Which of the following is the most abundant protein in the animals ?

(1) Insulin (2) Haemoglobin
(3) Collagen (4) Lectin

BBM099

21. Identify the substances having glycosidic bond and peptide bond, respectively in their structure :

(1) Inulin, insulin (2) Chitin, Cholesterol
(3) Glycerol, trypsin (4) Cellulose, lecithin

BBM100

NEET (UG) 2020 (COVID-19)

22. Identify the statement which is **incorrect**.

(1) Sulphur is an integral part of cysteine.
(2) Glycine is an example of lipids.
(3) Lecithin contains phosphorus atom in its structure.
(4) Tyrosine possesses aromatic ring in its structure.

BBM101

NEET (UG) 2021

23. Which of the following are **not** secondary metabolites in plants ?

(1) Morphine, codeine
(2) Amino acids, glucose
(3) Vinblastin, curcumin
(4) Rubber, gums

BBM102

24. Match **List - I** with **List - II.**

List -I		List – II	
(a)	Protein	(i)	C = C double bonds
(b)	Unsaturated fatty acid	(ii)	Phosphodiester bonds
(c)	Nucleic acid	(iii)	Glycosidic bonds
(d)	Polysaccharide	(iv)	Peptide bonds

Choose the **correct** answer from the options given below.

	(a)	(b)	(c)	(d)
(1)	(iv)	(i)	(ii)	(iii)
(2)	(i)	(iv)	(iii)	(ii)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(iv)	(iii)	(i)	(ii)

BBM103

25. Identify the **incorrect** pair.

(1) Alkaloids – Codeine
(2) Toxin – Abrin
(3) Lectins – Concanavalin A
(4) Drugs – Ricin

BBM104

26. Following are the statements with reference to 'lipids'.

(a) Lipids having only single bonds are called unsaturated fatty acids.
(b) Lecithin is a phospholipid.
(c) Trihydroxy propane is glycerol.
(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
(e) Arachidonic acid has 16 carbon atoms.

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Choose the **correct** answer from the options given below.

(1) (a) and (b) only

(2) (c) and (d) only

(3) (b) and (c) only

(4) (b) and (e) only

BBM105

NEET (UG) 2022

27. Exoskeleton of arthropods is composed of

(1) Cellulose (2) Chitin

(3) Glucosamine (4) Cutin

BBM106

28. Read the following statements on lipids and find out correct set of statements:

(a) Lecithin found in the plasma membrane is a glycolipid

(b) Saturated fatty acids possess one or more c = c bonds

(c) Gingelly oil has lower melting point, hence remains as oil in winter

(d) Lipids are generally insoluble in water but soluble in some organic solvents

(e) When fatty acid is esterified with glycerol, monoglycerides are formed

Choose the correct answer from the options given below:

(1) (a), (d) and (e) only

(2) (c), (d) and (e) only

(3) (a), (b) and (d) only

(4) (a), (b) and (c) only

BBM107

29. A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose ?

(1) $C_{12}H_{24}O_{12}$ (2) $C_{12}H_{22}O_{11}$

(3) $C_{12}H_{24}O_{11}$ (4) $C_{12}H_{20}O_{10}$

BBM108

RE-NEET (UG) 2022

30. Match List-I with List-II :

List - I	List - II
(a) Adenine	(i) Pigment
(b) Anthocyanin	(ii) Polysaccharide
(c) Chitin	(iii) Alkaloid
(d) Codeine	(iv) Purine

Choose the correct answer from the options given below :

(1) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

(2) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

(3) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

(4) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii)

BBM109

31. Primary proteins are also called as polypeptides because :

(1) They are linear chains

(2) They are polymers of peptide monomers

(3) Successive amino acids are joined by peptide bonds

(4) They can assume many conformations

BBM110

32. In the enzyme which catalyses the breakdown of:

$H_2O_2 \rightarrow H_2O + O_2$ the prosthetic group is :

(1) Nicotinamide adenine dinucleotide

(2) Haem

(3) Zinc

(4) Niacin

BBM111

33. Given below are two statements:

Statement-I : Amino acids have a property of ionizable nature of $-NH_2$ and $-COOH$ groups, hence have different structures at different pH.

Statement-II : Amino acids can exist as Zwitterionic form at acidic and basic pH.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct

(2) Both Statement-I and Statement-II are Incorrect

(3) Statement-I is correct but Statement-II is incorrect

(4) Statement-I is incorrect but Statement-II is correct

BBM112

NEET (UG) 2023

34. Cellulose does not form blue colour with Iodine because

(1) It is a helical molecule.

(2) It does not contain complex helices and hence cannot hold iodine molecules.

(3) It breaks down when iodine reacts with it.

(4) It is a disaccharide.

BBM180

35. Malonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

(1) Amylase

(2) Lipase

(3) Dinitrogenase

(4) Succinic dehydrogenase

BBM181

36. Given below are two statements :

Statement-I : A protein is imagined as a line, the left end represented by first amino acid (C-terminal) and the right end represented by last amino acid (N-terminal).

Statement-II : Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits β type.)

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement I and Statement II are false.

(2) Statement I is true but Statement II is false.

(3) Statement I is false but Statement II is true.

(4) Both Statement I and Statement II are true.

BBM182

37. Given below are two statements :

Statement-I : Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement-II : When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are false.

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is true.

(4) Both **Statement I** and **Statement II** are true.

BBM183

NEET (UG) 2023 (Manipur)

38. Match List-I with List-II.

	List-I		List-II
(A)	Protein	(I)	C=C double bonds
(B)	Unsaturated fatty acid	(II)	Phosphodiester bond
(C)	Nucleic acid	(III)	Glycosidic bonds
(D)	Polysaccharide	(IV)	Peptide bonds

Choose the **correct** answer from the options given below :

(1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(4) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

BBM184

39. Inulin is a polymer of :

(1) Fructose (2) Galactose

(3) Amino acids (4) Glucose

BBM185

40. Which of the following is not a secondary metabolite?

(1) Curcumin (2) Morphine

(3) Anthocyanin (4) Lecithin

BBM186

41. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Terpenoids	(I)	Codeine
(B)	Lectins	(II)	Diterpenes
(C)	Alkaloids	(III)	Ricin
(D)	Toxins	(IV)	Concanavalin A

Choose the **correct** answer from the options given below :

(1) A-(II), B-(IV), C-(III), D-(I)

(2) A-(II), B-(I), C-(IV), D-(III)

(3) A-(II), B-(III), C-(I), D-(IV)

(4) A-(II), B-(IV), C-(I), D-(III)

BBM187

NEET-UG 2024

42. Lecithin, a small molecular weight organic compound found in living tissues, is an example of :

(1) Amino acids (2) Phospholipids

(3) Glycerides (4) Carbohydrates

BBM188

43. Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of :

(1) Cofactor inhibition

(2) Feedback inhibition

(3) Competitive inhibition

(4) Enzyme activation

BBM189

44. Match List I with List II :-

	List I		List II
A.	GLUT-4	I.	Hormone
B.	Insulin	II.	Enzyme
C.	Trypsin	III.	Intercellular ground substance
D.	Collagen	IV.	Enables glucose transport into cells

Choose the correct answer from the options given below:

(1) A–IV, B–I, C–II, D–III

(2) A–I, B–II, C–III, D–IV

(3) A–II, B–III, C–IV, D–I

(4) A–III, B–IV, C–I, D–II

BBM190

45. Regarding catalytic cycle of an enzyme action, select the correct sequential steps :

A. Substrate enzyme complex formation.

B. Free enzyme ready to bind with another substrate.

C. Release of products.

D. Chemical bonds of the substrate broken

E. Substrate binding to active site.

Choose the correct answer from the options given below:

(1) E, A, D, C, B (2) A, E, B, D, C

(3) B, A, C, D, E (4) E, D, C, B, A

BBM191

46. Match List I with List-II :

List-I		List-II	
A.	Lipase	I.	Peptide bond
B.	Nuclease	II.	Ester bond
C.	Protease	III.	Glycosidic bond
D.	Amylase	IV.	Phosphodiester bond

Choose the correct answer from the options given below:

(1) A-IV, B-II, C-III, D-I

(2) A-III, B-II, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-I, C-III, D-II

BBM192

RE-NEET-UG 2024

47. Ligases is a class of enzymes responsible for catalysing the linking together of two compounds. Which of the following bonds is **not** catalysed by it ?

(1) C–C (2) P – O

(3) C – O (4) C – N

BBM193

48. Which of the following are **not** fatty acids ?

A. Glutamic acid B. Arachidonic acid

C. Palmitic acid D. Lecithin

E. Aspartic acid

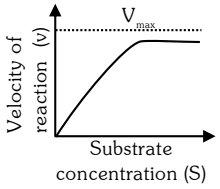
Choose the **correct** answer from the options given below :

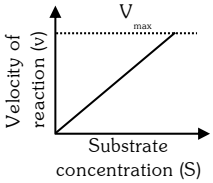
(1) C, D and E only (2) A and B only

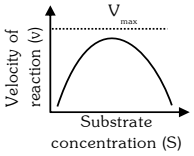
(3) A, D and E only (4) B and C only

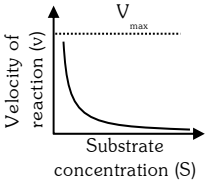
BBM194

49. Which of the following graphs depicts the effect of substrate concentration on velocity of enzyme catalysed reaction ?

(1) 

(2) 

(3) 

(4) 

BBM195

50. Match **List-I** with **List-II**.

List-I		List-II	
A.	Primary structure of protein	I.	Human haemoglobin
B.	Secondary structure of protein	II.	Disulphide bonds
C.	Tertiary structure of protein	III.	Polypeptide chain
D.	Quaternary structure of protein	IV.	Alpha helix and β sheet

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-II, C-I, D -IV

(3) A-I, B-III, C-II, D-IV

(4) A-IV, B-III, C-II, D-I

BBM196

ERERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	1	2	2	4	4	3	1	4	1	4	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	2	4	3	1	2	2	1	4	3	2	2	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	1	2	4	3	4	3	1	4	4	2	3	1	1
Question	46	47	48	49	50										
Answer	3	1	3	1	1										

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which of the following compound is present in acid soluble pool when we analyse chemical composition?
(1) Protein (2) Lipid
(3) Glucose (4) Carotenoid
BBM113

2. Which of the following elements present most abundantly on earth crust?
(1) Carbon (2) Hydrogen
(3) Oxygen (4) Silicon
BBM114

3. If $-NH_2$ group and –COOH group are attached on same carbon in any amino acid, then these types of amino acids are known as :
(1) α-AA (2) β-AA
(3) γ-AA (4) All
BBM115

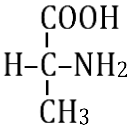
4. Variations in amino acids depend on :
(1) Side group
(2) Ester group
(3) Complexity of cell
(4) None of them
BBM116

5. How many carbon atoms are found in arachidonic acid?
(1) 16 (2) 18 (3) 20 (4) 22
BBM117

6. Which of the following AA is basic ?
(1) Valine (2) Lysine
(3) Glutamic acid (4) Glycine
BBM118

7. How many total carbons are found in Palmitic acid?
(1) 15 (2) 16 (3) 17 (4) 18
BBM119

8. Lipid may be :
(1) Monoglyceride (2) Diglyceride
(3) Triglyceride (4) All the above
BBM120

9.  This amino acid is :
(1) Serine (2) Alanine
(3) Glycine (4) Arginine
BBM121

10. Which functional group is common in fatty acid & amino acid ?
(1) –COOH (2) $-NH_2$
(3) –OH (4) All
BBM122

11. In proteins, amino acids are attached together by :
(1) Peptide bond (2) Amide bond
(3) Ester bond (4) Both (1) & (2)
BBM123

12. Proteins can act as :
(1) transporter of nutrients across cell membrane
(2) Hormones
(3) Enzymes
(4) All
BBM124

13. Which of the following is homopoly-saccharide?
(1) Cellulose (2) Starch
(3) Glycogen (4) All
BBM125

14. All the elements present in a sample of earth's crust are also present in a sample of living tissue, but which of the following element is higher in any living organism than in earth's crust?

(1) Hydrogen, Oxygen, Carbon and Silicon.
(2) Hydrogen, Oxygen, Carbon, Nitrogen and Silicon
(3) Hydrogen, Oxygen, Carbon, Nitrogen and Sulphur.
(4) Hydrogen, Oxygen, Carbon, Nitrogen, Sulphur and Sodium

BBM126

15. Pigments are considered as the secondary metabolites, having some particular functions. Which of the following is an example of pigments?
(1) Carotenoids (2) Codeine
(3) Concanavalin-A (4) Curcumin

BBM127

16. Which of the following can not be considered as the example of polymeric substances?
(1) Rubber (2) Gums
(3) Cellulose (4) Vinblastin

BBM128

17. Select the correctly matched.
(1) Anthocyanins - Alkaloids
(2) Carotenoids - Toxins
(3) Ricin - Drugs
(4) Lemon grass oil - Essential oils.

BBM129

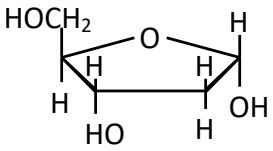
18. Which of the following amino acid having H(Hydrogen) as R group?
(1) Glycine (2) Alanine
(3) Serine (4) Leucine

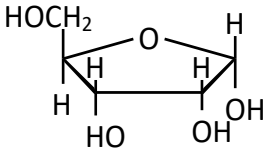
BBM130

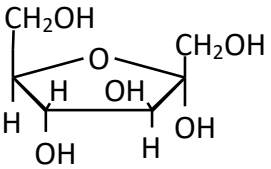
19. Physical and chemical properties of amino acids depend on
(1) Amino group (2) Carboxylic group
(3) R-group (4) All of the above

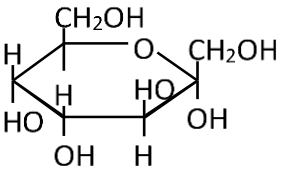
BBM131

20. Which of the following represents the structure of Ribose ?

(1) 

(2) 

(3) 

(4) 

BBM132

21. Enzymes, which catalyse the transfer of hydrogen (H) between two substrates, are called:-
(1) Transferases
(2) Oxidoreductases
(3) Isomerases
(4) Hydrolases

BBM133

22. Which of the following statements is **correct**?
(1) All the proteins are enzymes
(2) All enzymes are proteins
(3) Almost all enzymes are proteins
(4) All enzymes are nucleic acids

BBM134

23. Low temperature preserves the enzyme in a ...'A'... state whereas high temperature destroys enzymatic activity because proteins are ...'B'... by heat.
Choose the **correct** option:-

(1) A = temporarily inactive, B = denatured
(2) A = permanently inactive,
B = temporarily inactive
(3) A = permanently inactive, B = denatured
(4) A = denatured, B = temporarily inactive

BBM135

24. Which of the following is not correct?
(1) Catabolic pathways lead to the release of energy
(2) Anabolic pathways consume energy
(3) The living state and metabolism are synonymous
(4) CO_2 dissolving in water is an uncatalysed reaction in living systems

BBM136

25. A general rule of thumb is that rate of enzymatic reactions doubles or decreases by half for every ____ change in either direction.

Choose correct option for above blank.

(1) 50° (2) 5°C
(3) 100°C (4) 10°C

BBM138

26. Non proteinaceous part of holoenzyme can be:

(1) metal ion
(2) prosthetic group
(3) Coenzyme
(4) All of the above

BBM139

27. Which of the following can have catalytic power?

(1) Protein
(2) Nucleic acid
(3) Fat
(4) Both (1) and (2)

BBM140

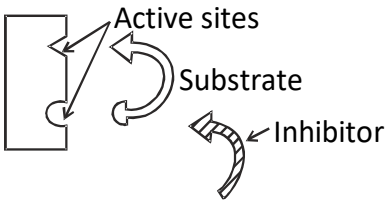
28. Catalytic activity is lost when the co-factor is removed from the enzyme. It testifies that :-
(a) Cofactors form one or more coordination bonds with enzymes.
(b) Cofactors play a crucial role in the catalytic activity of the enzyme
(c) Cofactors are not made up of proteins
(d) Cofactors show permanent association with enzymes

Options :

(1) (a) and (d) (2) Only (b)
(3) (b) and (c) (4) (a), (b) and (c)

BBM141

29.



Above diagram is the indicative diagram of a type of inhibition. Choose the correct match:-

	Type of inhibition	Substrate	Inhibitor
(1)	Non-competitive reversible	Malonate	Succinate
(2)	Competitive	Succinic dehydrogenase	Malonate
(3)	Non-competitive irreversible	Succinic dehydrogenase	Succinate
(4)	Competitive	Succinate	Malonate

BBM142

30. On the basis of type of reactions they catalyse, enzymes are divided into _____ classes each with ____ subclasses.

Choose the **correct** numbers for above blanks respectively from the following options :-

(1) 6, 6 (2) 6, 4–13
(3) 4–13, 6 (4) 4, 13

BBM143

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31. Inorganic catalysts work efficiently at high temperatures while enzymes get damaged at high temperatures (above 40°C). This feature of enzymes in not applicable to :-

(1) the enzymes which are made up of proteins

(2) The enzymes isolated from thermophobic organisms

(3) the enzymes which have active sites

(4) the enzymes of organisms who live in hot vents

BBM144

32. In the absence of any enzyme, reaction $CO_2 + H_2O \longrightarrow H_2CO_3$ is very slow, formation of 200 molecules of H_2CO_3 per hour. However by the use of carbonic anhydrase enzyme reaction speeds with about :-

(1) formation of 6×10^5 molecules of H_2CO_3 every second

(2) formation of 36×10^5 molecules of H_2CO_3 every minute

(3) formation of 36×10^6 molecules of H_2CO_3 every second

(4) formation of 6×10^6 molecules of H_2CO_3 every minute

BBM145

33. Stability is something related to energy status of the molecule. Transition state is high energy state as compare to substrate. So we can conclude that :-

(1) Transition state is more stable than substrate

(2) substrate is more stable than transition state

(3) Both substrate and transition state are very much unstable

(4) Stability can not be compared

BBM146

34. The activity of an enzyme can be affected by a change in the conditions which can alter the:-

(1) tertiary structure of nucleic acids

(2) primary structure of fats

(3) tertiary structure of proteins

(4) secondary structure of carbohydrates

BBM147

35. Which of the following statements is **not correct** regarding enzymes ?

(1) They are specific in nature

(2) They are used up in reaction

(3) They are proteinaceous in nature

(4) They speed up the rate of biochemical reactions.

BBM148

36. Which of the following statements regarding cofactors is **not correct** ?

(1) Every coenzyme is a cofactor but every cofactor is not a coenzyme

(2) Catalytic activity is lost when the cofactor is removed from the enzyme

(3) Every coenzyme is a cofactor and every cofactor is a coenzyme

(4) In the catalase enzyme, haem group work as a cofactor

BBM149

37. The prosthetic group can be distinguished from other cofactors of enzymes because prosthetic groups :-

(1) are organic in nature

(2) are tightly bound to the apoenzyme

(3) Both (1) and (2)

(4) are inorganic in nature and are loosely bound to the apoenzyme

BBM150

38. Ribozymes are :-

(1) Enzymes with tertiary structure of protein

(2) Fragments of endoplasmic reticulum those contain ribosomes

(3) Inorganic enzymes

(4) Nucleic acids that behave like enzymes

BBM151

39. Represented below is the effect of change in pH on enzyme activity. Which one of the following conclusions can be made by this diagram ?

(1) Enzyme show its highest activity at a particular pH, both below and above this pH, activity declines

(2) With increasing pH, activity increases upto a limit then it become constant

(3) At a particular pH, activity is minimum, both below and above this pH, activity increases

(4) Enzyme activity is affected by temperature not by pH so it is completely a hypothetical graph

BBM152

40. Match the column-I with column-II :-

Column-I (Example)		Column-II (Type of cofactor)	
(A)	NAD with dehydrogenase enzyme	(i)	Metal ion
(B)	Zn with carboxypeptidase enzyme	(ii)	Prosthetic group
(C)	Haem group with catalase enzyme	(iii)	Coenzyme

Choose the option from the following which denotes correct match of column-I with column-II :

(1) A–i, B–ii, C–iii (2) A–iii, B–ii, C–i

(3) A–iii, B–i, C–ii (4) A–ii, B–iii, C–i

BBM153

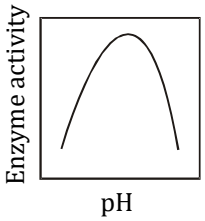
41. $CO_2 + H_2O \xrightarrow{B} A$

Choose the correct option for A and B in the given reaction :-

(1) B-Product of this reaction, A-Enzyme of this reaction

(2) A-Product of this reaction, B-Rate of formation of product of this enzyme is 360 lakh molecules/sec.

(3) A-Carbonic acid, B-Rate of formation of product of this enzyme is 6,00,000 molecules/sec.



(4) A-Transient state, B-Enzyme of this reaction

BBM154

42. Which of the following statements is not correct ?

(1) Enzymes bring down the activation energy of the processes

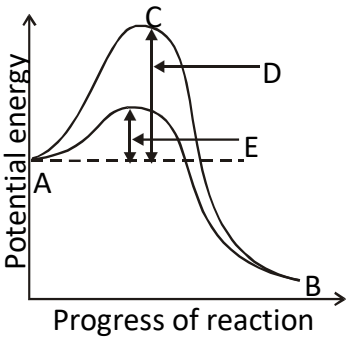
(2) Enzymes are preserved in a temporary inactive state at high temperature

(3) Enzymes isolated from thermophilic organisms have thermal stability

(4) Enzymes are differ from inorganic catalysts because enzymes get *damaged* at high temperature.

BBM155

43.



(1) Energy level of A is low as compared to B

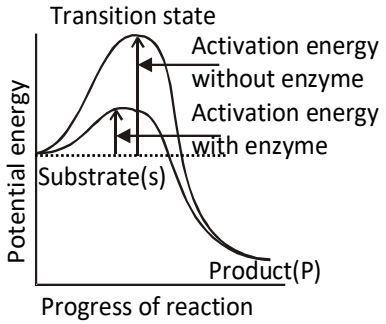
(2) Energy level of B is high as compared to C

(3) Energy level of D is high as compared to E

(4) Energy level of C is low as compared to A and B

BBM156

44.



Above diagram showing the concept of activation energy. If in this diagram energy level of 'P' become higher than the energy level of 'S' then which of the following will not apply for this diagram ?

(1) The process will be endothermic

(2) Activation energy with enzyme will be less than activation energy without enzyme

(3) Energy difference between transition state and product will be more than activation energy

(4) The substrate will go through the transition state

BBM157

45. (i) Transient association with the protein part of conjugated enzyme

(ii) Non protein organic substance

(iii) Usually made up of vitamins

All the above characters are applicable to :-

(1) Apoenzyme (2) Holoenzyme

(3) Coenzyme (4) Isoenzyme

BBM158

46. An enzyme catalysing a transfer of a group, G between a pair of substrate S and S' as follows:

$S - G + S' \longrightarrow S + S' - G$

G = phosphate or hydrogen or any other group, the enzyme is related with which of the following class?

(1) Transferases

(2) Dehydrogenases

(3) Both (1) and (2)

(4) Either (1) or (2)

BBM159

(B) ANALYTICAL QUESTIONS

47. Which element is present negligible in human body?

(1) O (2) C (3) H (4) Si

BBM160

48. Which of the following compound is found in acid soluble pool during analysis of a living tissue?

(1) Protein (2) Lipid

(3) Polysaccharide (4) Monosaccharide

BBM161

49. Which of the following is not a macromolecule?

(1) Amino acid (2) Nucleotide

(3) Monosaccharide (4) All the above

BBM162

50. α-amino acids are those, which –

(1) Participates in lipid synthesis

(2) –COOH group and $-NH_2$ group are attached on same carbon

(3) 21 types

(4) All the above are correct

BBM163

51. Which of the following is correct statement?

(1) Amino acids may be considered as substituted methane

(2) α-AA have difference only in R group

(3) In serine, R-group is hydroxy methyl

(4) All of the above

BBM164

52. What is the R group in Glycine?

(1) $-CH_2-OH$ (2) –H

(3) $-CH_3$ (4) None of them

BBM165

53. Which of the following is acidic amino acid?
(1) Valine (2) Glutamic acid
(3) Arginine (4) Lysine
BBM166

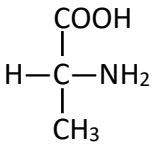
54. Which of the following is aromatic AA?
(1) Tyrosine (2) Tryptophan
(3) Phenyl alanine (4) All the above
BBM167

55. Which of the following is component of simple lipid?
(1) Glycerol (2) Glycol
(3) Fatty acid (4) Both (1) & (3)
BBM168

56. The lipid which is found in cell membrane is
(1) Phospholipid (2) Lecithin
(3) Both (1) & (2) (4) Palmitic acid
BBM169

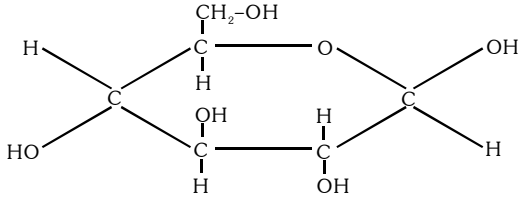
57. Heterocyclic ring is found in which compound?
(1) N-bases
(2) Ring structure of monosaccharides
(3) Protein
(4) Both (1) & (2)
BBM170

58. Following structure is related to which AA?



(1) Glycine (2) Alanine
(3) Serine (4) Tyrosine
BBM171

59. Following structure is –



(1) α-Glucose (2) β-Glucose
(3) Ribose (4) Deoxyribose
BBM172

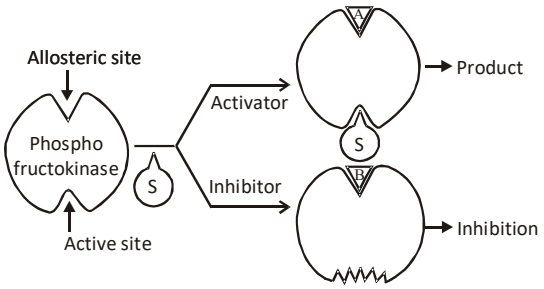
60. Which of the following is secondary metabolites?
(1) Rubber (2) Alkaloids
(3) Terpenoids (4) All the above
BBM173

61. Which of the following statement is correct?
(1) Lipid is biomacromolecule
(2) Lipid has <1000 daltons molecular weight
(3) Proteins are heteropolymer
(4) All the above are correct except (1)
BBM174

62. DNA is –
(1) Biomacromolecule
(2) Acidic in nature
(3) Polymer of nucleotide
(4) All of the above
BBM175

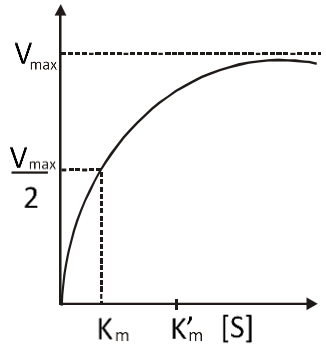
63. All the following statements about the structure of glycogen are true, except :-
(1) Branched chains occur, after every ten residues
(2) It is a copolymer of glucose and mannose
(3) It contains α–1, 4–glycosidic linkages
(4) It contains α–1, 6–glycosidic linkages
BBM176

64.



In the following diagrammatic representation A and B is :-
(1) A → G-6-P, B → ATP
(2) A → AMP, B → ATP
(3) A → Fructose-6-P, B → ATP
(4) A → ATP, B → ADP
BBM177

65.



Above graph is showing the effect of substrate concentration (S) on enzymatic activity. If due to change in enzymatic activity K_m value changes to K'_m then which of the following can be conclude on the basis of this change?

(1) There is a significant increase in enzymatic activity

(2) Entry of positive allosteric modulator in enzymatic activity

(3) Entry of non-competitive inhibitor in enzymatic activity

(4) In enzymatic activity, entry of inhibitor which closely resembles the substrate in its molecular structure

BBM178

66. One molecule of enzyme convert 5 molecules of substrate into product in 5 minutes.

20 molecules of substrate and two molecules of enzyme are mixed in a test tube. After 10 minutes the test tube will have :-

(1) products and 10 unreacted molecules of substrate

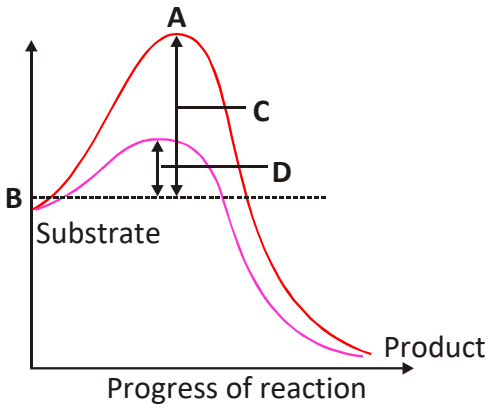
(2) only products

(3) 10 unreacted molecules of substrate, enzymes and products

(4) products and enzymes

BBM179

67. The figure given below shows the conversion of a substrate into product by an enzyme. In which one of the four options (1-4) the components of reaction labelled as A, B, C and D are identified correctly?



	A	B	C	D
(1)	Transition State	Potential energy	Activation energy without enzyme	Activation energy with enzyme
(2)	Potential Energy	Transition state	Activation energy with enzyme	Activation energy without enzyme
(3)	Activation energy with enzyme	Transition state	Activation energy without enzyme	Potential Energy
(4)	Potential Energy	Transition state	Activation energy with enzyme	Activation energy without enzyme

BBM067

68. Three of the following statements about enzymes are correct and one is wrong. Which one is wrong?

(1) Enzymes are denatured at high temperatures but in certain exceptional organisms they are effective even at temperatures 80°-90°C

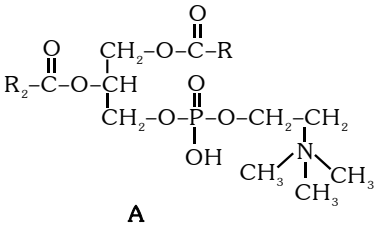
(2) Enzymes are highly specific

(3) Most enzymes are proteins but some are lipids

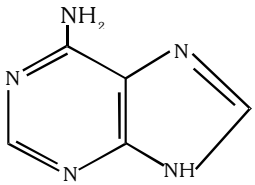
(4) Enzymes require optimum pH for maximal activity

BBM068

69. Which one of the following structural formulae of two organic compounds is correctly identified along with its related function ?



A



B

(1) B : adenine - a nucleotide that makes up nucleic acids

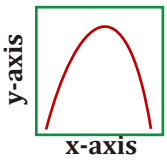
(2) A : Triglyceride - major source of energy

(3) B : Uracil - a component of DNA

(4) A : Lecithin - a component of cell membrane

BBM069

70. The curve given below shows enzymatic activity with relation to three conditions (pH, temperature and substrate concentration)

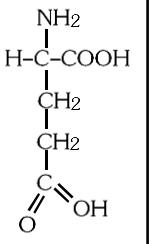
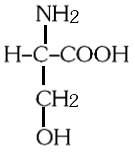
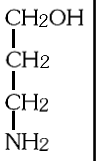
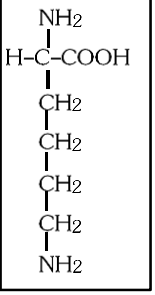


What do the two axises (x and y) represent ?

	x-axis	**y-axis**
(1)	Enzymatic activity	pH
(2)	Temperature	Enzyme activity
(3)	Substrate concentration	Enzymatic activity
(4)	Enzymatic activity,	Temperature

BBM070

71. Which one out of A - D given below correctly represents the structural formula of the basic amino acid ?

A	B	C	D
			

Options:

(1) A (2) B

(3) C (4) D

BBM073

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	1	1	3	2	2	4	2	1	4	4	4	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	1	4	2	2	3	1	4	4	4	4	2	4	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	2	3	2	3	2	4	1	3	3	2	3	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	4	4	2	4	2	2	4	4	3	4	2	2	4
Question	61	62	63	64	65	66	67	68	69	70	71				
Answer	4	4	2	2	4	4	1	3	4	2	4				

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

THE MALE REPRODUCTIVE SYSTEM

1. Which one of the following is a primary sex organ of human male ?
(1) Scrotum (2) Penis
(3) Testis (4) Prostate gland
BHR001

2. Secondary sex organ in human male is :-
(1) testis (2) ovary
(3) beard (4) vas deferens
BHR002

3. Vasa efferentia connect the :-
(1) testis with epididymis
(2) kidney with urinary bladder
(3) testis with scrotal wall
(4) None of the above
BHR003

4. In human, failure of testes to descend into scrotum is known as :-
(1) paedogenesis (2) castration
(3) cryptorchidism (4) impotency
BHR004

5. Common duct formed by union of vas deferens and duct from seminal vesicle is :-
(1) urethra
(2) tunica vasculosa
(3) ejaculatory duct
(4) spermatic duct
BHR005

6. Scrotum communicates with abdominal cavity through :-
(1) urethra (2) inguinal canal
(3) vas deferens (4) epididymis
BHR006

7. Tunica albuginea is the covering around :-
(1) oviduct (2) testis
(3) kidney (4) heart
BHR007

8. The functional unit of testis is :-
(1) uriniferous tubules
(2) malpighian tubules
(3) seminiferous tubules
(4) acini or lobules
BHR008

9. Penile urethra traverses through :-
(1) corpus cavernosum
(2) corpus spongiosum
(3) corpus callosum
(4) corpus striatum
BHR009

10. Seminiferous tubules are composed of :-
(1) spermatogonia
(2) glandular epithelium
(3) sensory epithelium
(4) germinal epithelium
BHR010

11. In human, the testes are located in :-
(1) abdominal cavity
(2) thoracic cavity
(3) extra-abdominal cavity
(4) pericardial cavity
BHR011

12. Bundles of erectile tissues in penis are :-
(1) corpora cavernosa
(2) corpus spongiosum
(3) Both (1) & (2)
(4) None of the above
BHR012

13. Glans penis is covered by :-
(1) areolar membrane (2) prepuce
(3) metrium (4) None of these
BHR013

14. In human, the testes occur in scrotal sacs, outside the viscera because of the :-
(1) presence of urinary bladder
(2) presence of rectum
(3) long vas-deferens
(4) requirement of low temperature for spermatogenesis
BHR014

15. Read the following statements.
(a) It is paired structure
(b) It is present on lateral side of male urethra
(c) It help in lubrication of penis
In above statements 'it' refers to :-
(1) seminal vesicle
(2) Bartholin's gland
(3) bulbourethral gland
(4) prostate

BHR015

16. Cells of Leydig occur in :-
(1) liver (2) ovary
(3) testis (4) spleen

BHR016

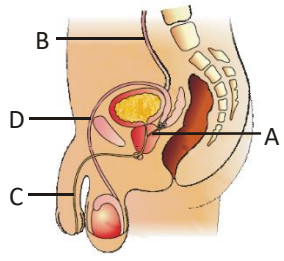
17. Seminiferous tubules occur in the :-
(1) liver (2) kidney
(3) ovary (4) testis

BHR017

18. Fructose is present in the secretion of:-
(1) seminal vesicle (2) perineal gland
(3) Cowper's gland (4) Bartholin's gland

BHR018

19. It is a diagrammatic sectional view of male reproductive system. Identify common duct which forms from the fusion of duct of seminal vesicle and vas deferens.



(1) A (2) B (3) D (4) C

BHR019

20. Temperature of scrotum as compared to abdominal cavity is less by :-
(1) 1^0C (2) 5^0C
(3) 3^0C (4) 10^0C

BHR020

21. Outer coat of seminiferous tubules is composed of fibrous connective tissue and is called as :-
(1) tunica propria (2) lamina propria
(3) plica semilunaris (4) tunica albuginea

BHR021

22. Secretions from which one of the following glands are rich in fructose, calcium and some enzymes ?
(1) Salivary glands
(2) Female sex accessory glands
(3) Male sex accessory glands
(4) Liver

BHR022

23. Which is an unpaired gland of male reproductive system ?
(1) Bartholin's gland (2) Seminal vesicle
(3) Prostate gland (4) Cowper's gland

BHR023

24. In human, maturation of sperms take place at a temperature :-
(1) equal to that of body
(2) higher than that of body
(3) lower than that of body
(4) at any temperature

BHR024

25. Circumcission is the procedure of :-
(1) cutting the glans penis
(2) removal of whole skin of penis
(3) removal of movable skin (prepuce) of glans penis
(4) reduction of the body part of penis

BHR025

26. Prostate gland produces a secretion for :-
(1) attracting sperms
(2) stimulating sperm activity
(3) attracting egg
(4) None of the above

BHR026

27. Testes descend into scrotum in human for:-
(1) spermatogenesis
(2) fertilization
(3) development of sex organs
(4) development of visceral organs
BHR027

28. Tunica vaginalis is found in :-
(1) ovary of female (2) testis of male
(3) vagina of female (4) None of the above
BHR028

29. Spermatozoa are nourished during their development by :-
(1) Sertoli cells
(2) interstitial cells
(3) connective tissue cells
(4) None of the above
BHR029

30. Release of sperms from testis is called as :-
(1) spermiation (2) semination
(3) insemination (4) ejaculation
BHR030

THE FEMALE REPRODUCTIVE SYSTEM

31. A secondary sexual character of human female is :-
(1) breasts (2) ovaries
(3) testes (4) thyroid gland
BHR031

32. Which is not a secondary sex organ in human?
(1) Vagina (2) Penis
(3) Prostate (4) Mammary gland
BHR032

33. At puberty, woman starts producing :-
(1) sperms (2) urine
(3) young ones (4) ova
BHR033

34. Eggs from ovary are released in :-
(1) oviduct (2) kidney
(3) ureter (4) coelom
BHR034

35. Lower narrow end of uterus is called as :-
(1) urethra (2) cervix
(3) clitoris (4) vulva
BHR035

36. Germinal epithelial cells are cuboidal and these are found in :-
(1) testes (2) ovaries
(3) Both (1) & (2) (4) None of the above
BHR036

37. Degenerative process of follicles or eggs in ovary is called :-
(1) metagenesis (2) atresia
(3) regression (4) None of the above
BHR037

38. Endometrium is lining of :-
(1) testis (2) urinary bladder
(3) uterus (4) ureter
BHR038

39. Clitoris is present at the upper junction of :-
(1) labia majora (2) mons pubis
(3) perineum (4) labia minora
BHR039

40. Parturition canal in female is :-
(1) uterus (2) oviduct
(3) vagina (4) urethra
BHR041

SPERMATOGENESIS AND STRUCTURE OF HUMAN SPERM

41. During differentiation, the spermatids remain associated with :-
(1) Leydig's cells (2) Kupffer's cells
(3) Spermatogonia (4) Sertoli cell
BHR042

42. In gametogenesis, reduction division takes place during :-
(1) multiplication phase
(2) growth phase
(3) first maturation division
(4) second maturation division
BHR043

43. Which type of division takes place during second maturation division ?
(1) Reduction division
(2) Equational division
(3) Amitosis
(4) None of the above

BHR044

44. Longest phase of spermatogenesis is :-
(1) multiplication phase
(2) growth phase
(3) maturation phase
(4) germinal phase

BHR045

45. During spermatogenesis, how many spermatozoa are formed from a single primary spermatocyte ?
(1) 1 (2) 2 (3) 4 (4) 8

BHR046

46. Correct order of spermatogenesis is :-
(1) Spermatids → Spermatogonium → Spermatocytes → Sperms
(2) Spermatogonium → Primary spermatocyte → Secondary Spermatocytes → Spermatids→ Sperms
(3) Primary spermatocyte → Spermatogonium → Secondary spermatocytes → Sperms → Spermatids
(4) Spermatogonium → Secondary spermatocytes → Primary spermatocyte → Spermatids → Sperms

BHR047

47. Which part of sperm is called power house ?
(1) Head (2) Neck
(3) Middle piece (4) Tail

BHR048

48. The acrosome plays a role in :-
(1) fusion of nuclei of gametes
(2) motality of sperm
(3) penetration of sperm into ovum
(4) All of the above

BHR049

49. The head of a mature sperm is mainly composed of :-
(1) elongated nucleus and acrosomal material
(2) mitochondria, cytoplasm and nucleus
(3) two centrioles and the axial filament
(4) All of the above

BHR050

50. A mature sperm has :-
(1) a pair of flagella
(2) a nucleus, an acrosome and a centriole
(3) a nucleus, an acrosome and a pair of centrioles
(4) a nucleus, an acrosome and a pair of centrioles and a tail

BHR051

51. Which part of the spermatid forms acrosome of sperm ?
(1) Mitochondria (2) Golgi body
(3) Nucleus (4) Lysosome

BHR052

52. Which of the following sets of vitamins is essential for gametogenesis ?
(1) Vitamin A and vitamin E
(2) Vitamin C and vitamin D
(3) Vitamin E and vitamin K
(4) Vitamin E and vitamin B complex

BHR053

OOGENESIS

53. At the time of birth, egg is arrested in the form of :-
(1) oogonia (2) primary oocyte
(3) secondary oocytes (4) ovum

BHR054

54. A human female has the maximum number of primary oocytes in her ovaries :-
(1) at menopause
(2) at puberty
(3) at birth
(4) early in her fertile years

BHR055

55. Egg is librated from ovary in human at :-
(1) secondary oocyte stage
(2) primary oocyte stage
(3) oogonial stage
(4) mature ovum stage
BHR056

56. During oogenesis, first meiotic division starts in :-
(1) first polar body
(2) second polar body
(3) primary oocyte
(4) secondary oocyte
BHR057

57. Which of the following are haploid cells ?
(1) Primary spermatocytes and primary oocytes
(2) Secondary spermatocytes and secondary oocytes
(3) Spermatogonia and oogonia
(4) Germinal cells.
BHR058

58. The process of spermatogenesis and oogenesis in human are under the influence of which hormone ?
(1) Oxytocin (2) FSH
(3) ACTH (4) ICSH
BHR059

59. During which stage of gametogenesis meiosis occurs?
(1) Growth phase
(2) Multiplication phase
(3) Maturation phase
(4) None of the above
BHR060

60. The primary egg membrane of human egg is:-
(1) chorion
(2) corona radiata
(3) zona pellucida
(4) None of the above
BHR061

61. Which of the following are diploid cells ?
(1) Secondary spermatocytes
(2) Spermatozoa and ova
(3) Spermatogonia, oogonia and primary spermatocytes
(4) Secondary oocytes
BHR062

62. Stored food of oocyte is :-
(1) nucleus (2) cytoplasm
(3) corticle granules (4) yolk
BHR063

63. Polar body is produced during the formation of :-
(1) sperm (2) secondary oocyte
(3) oogonium (4) spermatocytes
BHR064

INTERNAL STRUCTURE OF OVARY, OLLICULAR DEVELOPMENT AND MENSTRUAL CYCLE

64. Central stroma of ovary is made up of :-
(1) fibrous connective tissue
(2) reticular tissue
(3) adipose connective tissue
(4) None of the above
BHR065

65. Ovarian follicles are present in :-
(1) medulla
(2) germinal epithelium
(3) cortex
(4) mesovarium
BHR066

66. Graafian follicle contains :-
(1) many oocytes
(2) many sperms
(3) a single oocyte
(4) site for fertilisation of egg
BHR067

67. In human, corpus luteum is found in :-
(1) brain (2) ovary
(3) liver (4) eyes
BHR068

68. Antrum is filled with fluid and is found in :-
(1) bone marrow of bone
(2) cavity of brain
(3) Graffian follicle of ovary
(4) pericardium of heart

BHR069

69. Which one of the following is fibrous layer of follicle ?
(1) Theca externa
(2) Zona pellucida
(3) Membrana granulosa
(4) Vitelline membrane

BHR070

70. In case of non fertilization, corpus luteum :-
(1) stops secreting progesterone
(2) changes into corpus albicans
(3) starts producing progesterone
(4) None of the above

BHR071

71. Lutein cells are found in :-
(1) primary follicle (2) corpus albicans
(3) corpus luteum (4) All of the above

BHR072

72. Corpus luteum is a/an :-
(1) excretory structure
(2) endocrine structure
(3) digestive structure
(4) respiratory structure

BHR073

73. Luteal phase is the other name of :-
(1) follicular phase
(2) proliferative phase
(3) menstrual flow phase
(4) secretory phase

BHR074

74. Follicular phase of menstrual cycle is also known as :-
(1) proliferative phase
(2) secretory phase
(3) luteal phase
(4) menstruation phase

BHR075

75. Loss of reproductive capacity in women after age of approximately 45 years is known as :-
(1) menstruation (2) ageing
(3) menopause (4) menarche

BHR076

76. Which of the following hormones induces the development of corpus luteum ?
(1) LH (2) Oestrogen
(3) FSH (4) LTH

BHR077

77. The process of releasing the mature female gamete from the ovary is called as :-
(1) ovulation (2) parturition
(3) implantation (4) fertilisation

BHR078

78. Ovulation hormone is :-
(1) FSH (2) ICSH
(3) LH (4) testosterone

BHR079

79. Onset of pregnancy :-
(1) stimulates testosterone secretion
(2) inhibits further ovulation
(3) leads to degeneration of ovary
(4) inhibits fusion nuclei of egg and sperm

BHR080

80. Which of the following statements is correct ?
(1) Menstrual cycle is present in all mammals.
(2) Menstrual cycle is present in all primates.
(3) Estrous cycle occurs in all mammals.
(4) Most of the mammals are ovoviviparous.

BHR081

81. Yellow corpus luteum occurs in a human in :-
(1) heart to initiate heart beat
(2) skin to function as pain receptor
(3) brain and connects cerebral hemispheres
(4) ovary for the secretion of progesterone

BHR082

82. The fall in progesterone level leads to :-
(1) gestation (2) menopause
(3) lactation (4) mensturation
BHR083

83. In uterus, endometrium proliferates in response to :-
(1) relaxin (2) oxytocin
(3) progesterone (4) oestrogen
BHR084

84. Pregnancy hormone is :-
(1) estrogen (2) progesterone
(3) LH (4) FSH
BHR085

85. Pregnancy is detected with the help of presence of which hormones in urine of a pregnant female ?
(1) LH (2) Progesterone
(3) FSH (4) hCG
BHR086

STRUCTURE OF HUMAN EGG AND FERTILIZATION

86. Site of fertilization in human is :-
(1) ovary (2) uterus
(3) vagina (4) fallopian tube
BHR087

87. During fertilization, intermixing of chromosome of male and female gametes is called :-
(1) syngamy
(2) plasmogamy
(3) karyogamy
(4) amphimixis
BHR088

88. After cortical reaction, formation of fertilization membrane occurs :-
(1) outside of corona radiata
(2) inside of corona radiata
(3) in perivitelline space
(4) inside of membrane of oocyte
BHR089

89. Which of the following is responsible for division in fertilised egg ?
(1) Centriole of ovum
(2) Proximal centriole of sperm
(3) Distal centriole of sperm
(4) Mitochondria of sperm
BHR090

90. Capacitation of sperm occurs in :-
(1) urethra (2) vas deferens
(3) vagina (4) seminal vesicle
BHR091

91. Which part of sperm enters in egg in human?
(1) Complete sperm
(2) Only head
(3) Head and middle piece
(4) Head and acrosome
BHR092

EMBRYONIC DEVELOPMENT

92. Cleavage starts in :-
(1) fallopian tube (2) uterus
(3) vagina (4) None of these
BHR093

93. Cells formed as a result of cleavage are called as :-
(1) megameres (2) micromeres
(3) blastocyst (4) blastomeres
BHR094

94. Which of the following characteristics does not belong to cleavage ?
(1) Decrease in size of blastomeres
(2) Rapid mitotic cell division
(3) Interphase of very short duration
(4) Differention of blastomeres
BHR095

95. How many of the following structures may found in fallopian tube ?
Morula, Secondary oocyte, Primary oocyte, Secondary follicle, Ovum, Zygote, Blastocyst, Gastrula.
(1) 3 (2) 2 (3) 4 (4) 5
BHR096

96. What is true for cleavage ?
(1) Size of embryo is increased
(2) Size of cells is decreased
(3) Size of cells is increased
(4) Size of embryo is decreased

BHR097

97. Solid ball of cell produced by repeated cleavage is called as :-
(1) gastrula (2) blastocyst
(3) morula (4) None of the above

BHR098

98. In gastrula stage, which layer(s) is/are formed ?
(1) ectoderm (2) mesoderm
(3) endoderm (4) All of the above

BHR099

99. Morphogenetic movements of blastomeres result in:-
(1) morula stage (2) blastocyst stage
(3) gastrula stage (4) zygote

BHR100

100. Gastrulation is a process :-
(1) which starts the formation of germ layers
(2) that occurs just after morulation
(3) that occurs just after cleavage
(4) of rapid growth in blastomeres

BHR101

101. In which stage of development, the embryonic cells form the germinal layers?
(1) Morula (2) Blastocyst
(3) Gastrula (4) Zygote

BHR102

102. Sexually reproducing multicellular animals start their embryonic development from :-
(1) gastrula stage
(2) morula stage
(3) unicellular zygote
(4) ovum

BHR103

103. Placenta is a region where :-
(1) foetus is attached to mother by spermatic cord
(2) foetus is provided with mother's blood
(3) foetus receives nourishment from mother's blood
(4) foetus is covered by membranes

BHR104

104. In human embryo, the extra embryonic membranes are formed by :-
(1) inner cell mass (2) trophoblast
(3) formative cells (4) follicular cells

BHR105

PARTURITION

105. The expulsion of fully developed foetus from the uterus is known as :-
(1) ovulation (2) menstruation
(3) gestation (4) parturition

BHR106

106. Oxytocin helps in mainly :-
(1) milk production (2) child birth
(3) urine formation (4) gametogenesis

BHR107

107. Which of the following is not true about parturition process ?
(1) Secretion of oxytocin hormone by posterior pituitary
(2) Narrowing of pelvic cavity by relaxin hormone
(3) Secretion of progesterone hormone is stopped
(4) General position of foetus is occipito-anterior

BHR108

108. Foetal ejection reflex in human female is induced by :-
(1) differentiation of mammary glands
(2) pressure exerted by amniotic fluid
(3) release of oxytocin from pituitary gland
(4) fully developed foetus and placenta

BHR109

EXTRA POINTS

109. In the female, which structure is homologous to penis of male ?
(1) Cervix (2) Vagina
(3) Uterus (4) Clitoris

BHR111

110. Puberty occurs in females at the age of :-
(1) 8 - 10 years (2) 11-14 years
(3) 15-17 years (4) 18-20 years

BHR112

111. Which is not correct about secondary sexual characters of female ?
(1) Development of mammary glands
(2) Presence of pubic hair
(3) Voice of low pitch
(4) Menarche

BHR113

112. Hyaluronic acid which binds cells of corona radiata is a :-
(1) homopolysaccharide
(2) amino acid
(3) glycoprotein
(4) mucopolysaccharide

BHR115

113. Identical twins will be produced when :-
(1) one spermatozoon fertilises two ova
(2) one ovum is fertilised by two spermatozoa
(3) two eggs are fertilised
(4) one fertilised egg divides into two blastomeres and they become separate

BHR117

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	3	3	2	2	3	2	4	3	3	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	1	1	3	1	3	3	3	3	2	1	2	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	4	4	4	2	3	2	3	4	3	4	3	2	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	3	3	1	4	2	1	2	3	1	3	2	2	3	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	4	2	1	3	3	2	3	1	2	3	2	4	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	1	3	2	2	4	4	4	2	4	4	4	3	2	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	1	4	4	3	2	3	4	3	1	3	3	3	2	4
Question	106	107	108	109	110	111	112	113							
Answer	2	2	4	4	2	3	4	4							

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. The main function of mammalian corpus luteum is to produce :
(1) estrogen only
(2) progesterone
(3) human chorionic gonadotropin
(4) relaxin only

BHR143

2. Select the correct option describing gonadotropin activity in a normal pregnant female :-
(1) High level of FSH and LH stimulates the thickening of endometrium.
(2) High level of FSH and LH facilitate implantation of the embryo.
(3) High level of hCG stimulates the synthesis of estrogen and progesterone.
(4) High level of hCG stimulates the thickening of endometrium.

BHR144

AIPMT 2015

3. Capacitation refers to changes in the :-
(1) Ovum before fertilization
(2) Ovum after fertilization
(3) Sperm after fertilization
(4) Sperm before fertilization

BHR145

4. Hysterectomy is surgical removal of :-
(1) Prostate gland (2) Vas-deferense
(3) Mammary glands (4) Uterus

BHR146

5. Which of the following cells during gametogenesis is normally dipoid?
(1) Spermatid
(2) Spermatogonia
(3) Secondary polar body
(4) Primary polar body

BHR147

6. Which of these is not an important component of initiation of parturition in humans?
(1) Synthesis of prostaglandins
(2) Release of oxytocin
(3) Release of prolactin
(4) Increase in estrogen and progesterone ratio

BHR148

Re-AIPMT 2015

7. Ectopic pregnancies are referred to as :-
(1) Pregnancies terminated due to hormonal imbalance
(2) Pregnancies with genetic abnormality.
(3) Implantation of embryo at site other than uterus.
(4) Implantation of defective embryo in the uterus

BHR149

8. Which of the following events is not associated with ovulation in human female?
(1) LH surge
(2) Decrease in estradiol
(3) Full development of Graafian follicle
(4) Release of secondary oocyte

BHR150

9. In human females, meiosis-II is not completed until?
(1) birth
(2) puberty
(3) fertilization
(4) uterine implantation

BHR151

10. Which of the following layers in an antral follicle is acellular?
(1) Zona pellucida (2) Granulosa
(3) Theca internal (4) Stroma

BHR152

NEET-I 2016

11. Fertilization in humans is practically feasible only if:-
(1) the sperms are transported into vagina just after the release of ovum in fallopian tube
(2) the ovum and sperms are transported simultaneously to ampullary isthmic junction of the fallopian tube
(3) the ovum and sperms are transported simultaneously to ampullary - isthmic junction of the cervix
(4) the sperms are transported into cervix within 48 hrs of release of ovum in uterus

BHR153

12. Select the incorrect statement :-
(1) FSH stimulates the sertoli cells which help in spermiogenesis
(2) LH triggers ovulation in ovary
(3) LH and FSH decrease gradually during the follicular phase
(4) LH triggers secretion of androgens from the Leydig cells

BHR154

13. Changes in GnRH pulse frequency in females is controlled by circulating levels of :-
(1) estrogen and progesterone
(2) estrogen and inhibin
(3) progesterone only
(4) progesterone and inhibin

BHR155

14. Identify the correct statement on 'inhibin' :-
(1) Inhibits the secretion of LH, FSH and Prolactin.
(2) Is produced by granulose cells in ovary and inhibits the secretion of FSH.
(3) Is produced by granulose cells in ovary and inhibits the secretion of LH.
(4) Is produced by nurse cells in testes and inhibits the secretion of LH.

BHR156

NEET-II 2016

15. Several hormones like hCG, hPL, estrogen, progesterone are produced by :-
(1) Fallopian tube (2) Pituitary
(3) Ovary (4) Placenta

BHR157

16. Which of the following depicts the correct pathway of transport of sperms?
(1) Rete testis → Vas deferens → Efferent ductules → Epididymis
(2) Efferent ductules → Rete testis → Vas deferens → Epididymis
(3) Rete testis → Efferent ductules → Epididymis → Vas deferens
(4) Rete testis → Epididymis → Efferent ductules → Vas deferens

BHR158

17. Match Column-I with Column-II and select the correct option using the codes given below:

Column-I	**Column-II**
a. Mons pubis	i. Embryo formation
b. Antrum	ii. Sperm
c. Trophectoderm	iii. Female external genitalia
d. Nebenkern	iv. Graafian follicle

Codes:
(1) a–iii, b–i, c–iv, d–ii
(2) a–i, b–iv, c–iii, d–ii
(3) a–iii, b–iv, c–ii, d–i
(4) a–iii, b–iv, c–i, d–ii

BHR159

NEET-UG 2016

18. Capacitation occurs in :-
(1) Epididymis
(2) Vas deferens
(3) Female reproductive tract
(4) Rete testis

BHR160

19. A temporary endocrine gland in the human body is:-
(1) Corpus cardiacum (2) Corpus luteum
(3) Corpus allatum (4) Pineal gland

BHR161

20. GnRH, a hypothalamic hormone, needed in reproduction, acts on:-
(1) anterior pituitary gland and stimulates secretion of LH and FSH.
(2) posterior pituitary gland and stimulates secretion of oxytocin and FSH.
(3) posterior pituitary gland and stimulates secretion of LH and relaxin.
(4) anterior pituitary gland and stimulates secretion of LH and oxytocin.

BHR162

NEET-UG 2018

21. Hormones secreted by the placenta to maintain pregnancy are :
(1) hCG, hPL, progestogens, prolactin
(2) hCG, hPL, estrogens, relaxin, oxytocin
(3) hCG, hPL, progestogens, estrogens
(4) hCG, progestogens, estrogens, glucocorticoids

BHR163

22. The difference between spermiogenesis and spermiation is :-
(1) In spermiogenesis spermatids are formed, while in spermiation spermatozoa are formed.
(2) In spermiogenesis spermatozoa are formed, while in spermiation spermatids are formed.
(3) In spermiogenesis spermatozoa from Sertoli cells are released into the cavity of seminiferous tubules, while in spermiation spermatozoa are formed.
(4) In spermiogenesis spermatozoa are formed, while in spermiation spermatozoa are released from Sertoli cells into the cavity of seminiferous tubules.

BHR164

23. The amnion of mammalian embryo is derived from :-
(1) ectoderm and mesoderm
(2) endoderm and mesoderm
(3) mesoderm and trophoblast
(4) ectoderm and endoderm

BHR165

NEET-UG 2019

24. Select the correct sequence of transport of sperm cells in male reproductive system :-
(1) Testis → Epididymis → Vasa efferentia →Rete testis→Inguinal canal → Urethra
(2) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus
(3) Seminiferous tubules → Vasa efferentia →Epididymis→Inguinal canal → Urethra
(4) Testis → Epididymis → Vasa efferentia → Vas deferens → Ejaculatory duct → Inguinal canal→Urethra→Urethral meatus

BHR166

25. Extrusion of second polar body from egg nucleus occurs :
(1) after entry of sperm but before fertilization
(2) after fertilization
(3) before entry of sperm into ovum
(4) simultaneously with first cleavage

BHR167

NEET-UG 2019 (Odisha)

26. No new follicles develop in the luteal phase of the menstrual cycle because :
(1) Follicles do not remain in the ovary after ovulation
(2) FSH levels are high in the luteal phase
(3) LH levels are high in the luteal phase
(4) Both FSH and LH levels are low in the luteal phase

BHR168

NEET-UG 2020

27. Which of the following hormone levels will cause release of ovum (ovulation) from the Graafian follicle?

(1) Low concentration of FSH

(2) High concentration of Estrogen

(3) High concentration of Progesterone

(4) Low concentration of LH

BHR169

28. Match the following columns and select the correct option :-

	Column-I		**Column-II**
(a)	Placenta	(i)	Androgens
(b)	Zona pellucida	(ii)	Human Chorionic Gonadotropin (hCG)
(c)	Bulbo-urethral glands	(iii)	Layer of the ovum
(d)	Leydig cells	(iv)	Lubrication of the Penis

(1) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)

(2) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)

(3) (a)–(i), (b)–(iv), (c)–(ii), (d)–(iii)

(4) (a)–(iii), (b)–(ii), (c)–(iv), (d)–(i)

BHR170

29. Meiotic division of the secondary oocyte is completed :-

(1) At the time of fusion of a sperm with an ovum

(2) Prior to ovulation

(3) At the time of copulation

(4) After zygote formation

BHR171

NEET-UG 2021

30. Receptors for sperm binding in mammals are present on:-

(1) Corona radiata

(2) Vitelline membrane

(3) Perivitelline space

(4) Zona pellucida

BHR172

31. Which of these is not an important component of initiation of parturition in humans?

(1) Increase in estrogen and progesterone ratio

(2) Synthesis of prostaglandins

(3) Release of Oxytocin

(4) Release of Prolactin

BHR173

32. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy?

(1) Graafian follicle

(2) Corpus luteum

(3) Foetus

(4) Uterus

BHR174

NEET-UG 2022

33. Given below are two statements:

Statement I: The release of sperms into the seminiferous tubules is called spermiation.

Statement II: Spermiogenesis is the process of formation of sperms from spermatogonia.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both **Statement I** and **Statement II** are incorrect

(2) **Statement I** is correct but **Statement II** is incorrect

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct

BHR175

34. Which of the following statements are true for spermatogenesis but **do not** hold true for Oogenesis?

(a) It results in the formation of haploid gametes

(b) Differentiation of gamete occurs after the completion of meiosis

(c) Meiosis occurs continuously in a mitotically dividing stem cell population

(d) It is controlled by the Luteinising hormone (LH) and Follicle Stimulating Hormone (FSH) secreted by the anterior pituitary

(e) It is initiated at puberty

Choose the **most appropriate** answer from the options given below:

(1) (b) and (c) only

(2) (b), (d) and (e) only

(3) (b), (c) and (e) only

(4) (c) and (e) only

BHR176

Re-NEET-UG 2022

35. Arrange the components of mammary gland. (from proximal to distal)

(a) Mammary duct

(b) Lactiferous duct

(c) Alveoli

(d) Mammary ampulla

(e) Mammary tubules

Choose the most appropriate answer from the options given below :

(1) (c) → (a) → (d) → (e) → (b)

(2) (b) → (c) → (e) → (d) → (a)

(3) (c) → (e) → (a) → (d) → (b)

(4) (e) → (c) → (d) → (b) → (a)

BHR177

36. How many secondary spermatocytes are required to form 400 million spermatozoa ?

(1) 50 million (2) 100 million

(3) 200 million (4) 400 million

BHR178

37. Given below are two statements :

One is **labelled** as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : During pregnancy the level of thyroxine is increased in the maternal blood.

Reason (R) : Pregnancy is characterised by metabolic changes in the mother.

In the light of the above statements, choose **the most appropriate answer** from the options given below :

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct

(4) **(A)** is not correct but **(R)** is correct

BHR179

NEET-UG 2023

38. Which of the following statements are correct regarding female reproductive cycle?

A. In non-primate mammals cyclical changes during reproduction are called oestrus cycle.

B. First menstrual cycle begins at puberty and is called menopause.

C. Lack of menstruation may be indicative of pregnancy.

D. Cyclic menstruation extends between menarche and menopause.

Choose the **most appropriate** answer from the options given below :

(1) A and B only

(2) A, B and C only

(3) A, C and D only

(4) A and D only

BHR265

39. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Endometrium is necessary for implantation of blastocyst.

Reason R : In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

(2) **A** is true but **R** is false.

(3) **A** is false but **R** is true.

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

BHR266

40. Given below are two statements :

Statement I : Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Statement II : The cavity of the cervix is called cervical canal which along with vagina forms birth canal.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement I and Statement II are false.

(2) Statement I is correct but Statement II is false.

(3) Statement I incorrect but Statement II is true.

(4) Both Statement I and Statement II are true.

BHR267

NEET-UG 2023 (Manipur)

41. Arrange the sequence of different hormones for their role during gametogenesis.

(A) Gonadotropin LH stimulates synthesis and secretion of Androgen

(B) Gonadotropin releasing hormone from hypothalamus

(C) Androgen stimulates spermatogenesis

(D) Gonadotropin FSH helps in the process of spermiogenesis

(E) Gonadotropins from anterior pituitary gland.

Choose the **correct** answer from the options given below :

(1) (E), (A), (D), (B), (C)

(2) (C), (A), (D), (E), (B)

(3) (B), (E), (A), (C), (D)

(4) (D), (B), (A), (C), (E)

BHR268

42. Given below are two statements regarding oogenesis:

Statement I : The primary follicles get surrounded by more layers of granulosa cells, a theca and shows fluid filled cavity antrum. Now it is called secondary follicle.

Statement II : Graffian follicle ruptures to release the secondary oocyte from the ovary by the process called ovulation.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Statement I is correct but Statement II is false.

(2) Statement I is incorrect but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false.

BHR269

NEET-UG 2024

43. Which of the following is not a component of Fallopian tube?

(1) Uterine fundus (2) Isthmus

(3) Infundibulum (4) Ampulla

BHR270

44. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A : Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby.

Reason R : Colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is NOT the correct explanation of A.

(3) A is correct but R is not correct.

(4) A is not correct but R is correct.

BHR271

45. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: FSH acts upon ovarian follicles in female and Leydig cells in male.

Reason R: Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in male human being.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

BHR272

46. Given below are two statement :

Statement I : The presence or absence of hymen is not a reliable indicator of virginity.

Statement II : The hymen is torn during the first coitus only.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

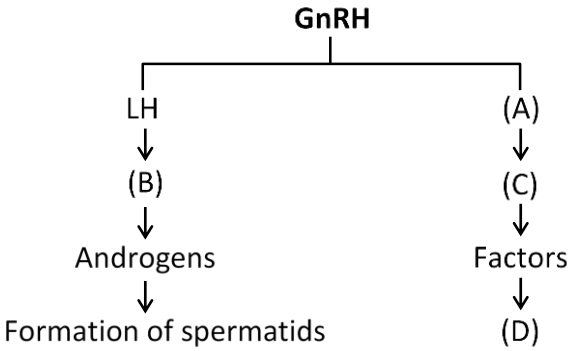
(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

BHR273

47. Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.



(1) FSH, Leydig cells, Sertoli cells, spermiogenesis

(2) ICSH, Interstitial cells, Leydig cells, spermiogensis.

(3) FSH, Sertoli cells, Leydig cells, spermatogenesis.

(4) ICSH, Leydig cells, Sertoli cells, spermatogenesis.

BHR274

RE-NEET-UG 2024

48. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion : During menstrual cycle, the ovulation takes place approximately on 14th day.

Reason : Rapid secretion of LH in the middle of menstrual cycle induces rupture of Graafian follicle and thereby the release of ovum.

In the light of the above statements, choose the **most appropriate** answer form the options given below.

(1) **A** is correct but **R** is not correct.

(2) **A** is not correct but **R** is correct.

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are correct and **R** is NOT the correct explanation of **A**.

BHR275

49. Arrange the following parts in human Mammary gland, traversing the route of milk ejection.

A. Mammary duct

B. Lactiferous duct

C. Mammary alveolus

D. Ampulla

E. Mammary tubule

Choose the **correct** answer form the options given below :

(1) D → C → E → A → B

(2) C → E → B → A → D

(3) C → E → A → D → B

(4) A → C → E → D → B

BHR276

50. Match **List-I** with **List-II** relating to human female external genitalia :

List-I (Structures)		**List-II (Features)**	
A.	Mons pubis	I.	A fleshy fold of tissue surrounding the vaginal opening
B.	Clitoris	II.	Fatty cushion of cells covered by skin and hair
C.	Hymen	III.	Tiny finger – like structure above labia minora
D.	Labia majora	IV.	A thin membrane-like structure covering vaginal opening

Choose the **correct** answer from the options given below :

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-IV, C-III, D-II

(4) A-II, B-III, C-I, D-IV

BHR277

51. Match **List-I** with **List-II**.

List-I		**List-II**	
A.	Parturition	I.	Several antibodies for new-born babies
B.	Placenta	II.	Collection of ovum after ovulation
C.	Colostrum	III.	Foetal ejection reflex
D.	Fimbriae	IV.	Secretion of the hormone hCG

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-I, D-II

(2) A-I, B-IV, C-II, D-III

(3) A-II, B-III, C-IV, D-I

(4) A-III, B-IV, C-II, D-I

BHR278

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	4	4	2	3	3	2	3	1	2	3	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	3	2	1	3	4	1	2	1	4	2	1	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	2	3	3	3	1	3	1	4	3	2	1	1	4
Question	46	47	48	49	50	51									
Answer	3	1	3	3	1	1									

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. How many testicular lobules are present in each testis ?
(1) About 100 (2) About 150
(3) About 250 (4) Infinite
BHR180

2. The male accessory glands include :-
(1) prostate gland
(2) bulbourethral glands
(3) seminal vesicles
(4) All of the above
BHR181

3. In male, penis is covered by a loose fold of skin called as :-
(1) foreskin (2) urethral meatus
(3) external genitalia (4) fimbriae
BHR182

4. The function of fimbriae is :-
(1) to collect ovum after ovulation
(2) to maintain the shape of ovary
(3) to provide the path to sperm during fertilization
(4) None of the above
BHR183

5. Which of the following statements is wrong ?
(1) Mammary lobes contain clusters called as alveoli.
(2) Uterus is also called as womb.
(3) The last part of the oviduct is called as ampulla.
(4) Stroma of ovary is divided into two zones.
BHR184

6. Secretion of which of the following hormones induces spermatogenesis ?
(1) GnRH (2) LTH
(3) Oxytocin (4) Relaxin
BHR185

7. Antrum is :-
(1) follicular cavity filled with fluid
(2) an inner theca layer
(3) the mature follicle
(4) follicular cavity with no fluid
BHR186

8. Ovulation is :-
(1) Release of secondary oocyte from ovary
(2) Release of primary oocyte from ovary
(3) Release of polar body
(4) Release of Graafian follicle
BHR187

9. For normal fertility, how much percentage of total sperms must have normal shape and size ?
(1) 50 % (2) 25 %
(3) 40 % (4) 60 %
BHR188

10. When do both LH & FSH attain a peak level in a menstrual cycle ?
(1) In last week of the cycle
(2) In mid of the cycle
(3) During Initial days of cycle
(4) On 4th day of cycle
BHR189

11. Which of the following statements is not correct ?
(1) In the absence of fertilization, the corpus luteum is degenerated.
(2) During pregnancy, all events of menstrual cycle are stopped.
(3) The secretion of LH and FSH decreases gradually during the follicular phase.
(4) The menstrual flow results due to breakdown of endometrial lining.
BHR190

12. In which phase of menstrual cycle Graafian follicle is transformed into corpus luteum ?
(1) Luteal phase
(2) Proliferative phase
(3) Follicular phase
(4) Growth phase

BHR191

13. Which of the hormones is essential for maintenance of the endometrium ?
(1) FSH (2) LH
(3) Progesterone (4) Testosterone

BHR192

14. Which of the following is an indicator of normal reproductive phase and extends between menarche and menopause ?
(1) Menstrual cycle
(2) Estrous cycle
(3) Ovulation
(4) Implantation

BHR193

15. In human, fertilization takes place in :-
(1) cervix (2) ampulla
(3) isthmus (4) vagina

BHR194

16. The embryo with 8 to 16 blastomeres is called as :-
(1) morula (2) blastocyst
(3) gastrula (4) foetus

BHR195

17. During Implantation, the blastocyst gets embedded in which layer of the uterus?
(1) Trophoblast (2) Endometrium
(3) Myometrium (4) Perimetrium

BHR196

18. The mitotic division starts as the zygote moves through the _____ of the oviduct called cleavage towards the uterus.
(1) ampulla (2) isthmus
(3) fimbriae (4) infundibulum

BHR197

19. During fertilization, a sperm comes in contact with which layer of the ovum ?
(1) Jelly coat
(2) Zona pellucida
(3) Vitelline membrane
(4) Perivitelline space

BHR198

20. Function of placenta is to :-
(1) supply O_2 to embryo
(2) removal CO_2 produced by the embryo
(3) produce several hormones
(4) All of the above

BHR199

21. Stem cells are found in :-
(1) inner cell mass (2) ectoderm
(3) endoderm (4) mesoderm

BHR200

22. The first sign of growing foetus may be noticed by :-
(1) listening to the heart sound carefully with the help of stethoscope
(2) appearance of hair
(3) appearance of head
(4) appearance of eye lids

BHR201

23. Placenta contains :-
(1) only chorionic villi
(2) only uterine tissue
(3) chorionic villi and uterine tissue
(4) trophoblast and chorionic villi

BHR202

24. The average duration of human pregnancy is about nine months which is known as :-
(1) gestation period (2) parturition
(3) lactation (4) implantation

BHR203

25. The signals for parturition originate from :-
(1) fully developed foetus
(2) placenta
(3) both fully developed foetus and placenta
(4) uterus

BHR204

26. Which hormone acts on uterine myometrium during parturition ?
(1) Oxytocin (2) LH
(3) Estrogen (4) Relaxin
BHR205

27. Which gland of female human undergoes differentiation during pregnancy ?
(1) Adrenal gland (2) Mammary gland
(3) Pituitary gland (4) Thymus gland
BHR206

28. Which of the following is correct for colostrum ?
(1) It contains severel antibodies.
(2) It is produced during the last few days of lactation.
(3) It is a pheromone.
(4) It is white in colour.
BHR207

29. Human beings are :-
(1) ovoviviparous (2) oviparous
(3) parthenogenetic (4) viviparous
BHR208

(B) ANALYTICAL QUESTIONS

30. What does happen if vasa deferentia are cut in a man ?
(1) Sperms become non nucleated.
(2) Spermatogenesis does not occur.
(3) Semen is ejaculated without sperms.
(4) Sperms have no motality.
BHR209

31. If epididymis is removed, then what does happen in a man ?
(1) Sperms have shorter life span.
(2) Sperms travel the pathway in lesser time.
(3) Functional maturation of sperms takes lesser time.
(4) Sperms are incapable for fertilization.
BHR210

32. Which of the following is not correct about sustentacular cells ?
(1) These are present in between the germinal epithelial cells.
(2) These are related with the nutrition of sperm.
(3) These form blood testis barrier.
(4) These form testosterone from oestrogen.
BHR211

33. After some time of ejaculation, semen liquefies due to presence of an enzyme which is found in the secretion of :-
(1) vagina (2) seminal vesicle
(3) prostate gland (4) Cowper's gland
BHR212

34. If Cowper's gland is removed, then which of the following would be affected ?
(1) Sexual attraction
(2) Capacitation of sperms
(3) Hardness of penis
(4) Copulation and fertilization
BHR213

35. Which of the following releases inhibin to control spermatogenesis ?
(1) Rete testis
(2) Follicular cells
(3) Leydig's cells
(4) Sustentacular cells
BHR214

36. Correct order of spermatogenesis is :-
(1) Spermatocytes, Spermatogonium, Spermatids, Sperms
(2) Spermatogonium, Spermatids, Spermatocytes, Sperms
(3) Spermatids, Spermatogonium, Spermatocytes, Sperms
(4) Spermatogonium, Primary Spermatocytes, Secondary Spermatocytes, Spermatids, Sperms
BHR215

37. Given below are two statements:

Statement-I : The edge of the infundibulum posses finger like projections called fimbriae, which helps in collection of ovum after ovulation.

Statement-II : Uterus is supported by tendons attached to the pelvic wall.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II both are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BHR216

38. How many secondary spermatocytes form 400 spermatozoa ?

(1) 100 (2) 400

(3) 40 (4) 200

BHR217

39. Match the column-A with column-B.

Column A		Column B	
A	Mons pubis	i	Fleshy folds of tissue
B	Labia majora	ii	Paired folds of tissue
C	Labia minora	iii	Finger-like structure
D	Clitoris	iv	Cushion of fatty tissue

(1) A-iv, B-i, C-ii, D-iii

(2) A-ii, B-iv, C-i, D-iii

(3) A-iv, B-ii, C-i, D-iii

(4) A-iii, B-iv, C-ii, D-i

BHR218

40. At which stage of spermatogenesis, sperms acquire their whole structural maturity and now these contain a haploid nucleus & other organelles ?

(1) Spermiogenesis

(2) Growth phase

(3) Multiplication phase

(4) Maturation phase

BHR219

41. How many sperms and ova are formed respectively from 50 secondary oocytes and 50 secondary spermatocytes in human ?

(1) 50 ova and 200 sperm

(2) 50 ova and 100 sperm

(3) 100 ova and 200 sperm

(4) 100 ova and 400 sperm

BHR220

42. Human egg has :-

(1) one Y-chromosome

(2) one X-chromosome

(3) two Y-chromosome

(4) one X-chromosome and one Y-chromosome.

BHR221

43. Given below are two statements:

Statement-I : LH act on leydig cells and stimulates synthesis and secretion of androgens.

Statement-II : FSH acts on the sertoli cells and stimulates secretion of some factors which help in the process of spermiogenesis .

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II both are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

BHR222

44. Oocyte is liberated from ovary under the influence of LH, after completing :
(1) meiosis and before liberating polar bodies
(2) meiosis I and before liberating second polar body
(3) meiosis
(4) meiosis II after release of first polar body

BHR223

45. Extrusion of second polar body from egg nucleus occurs :-
(1) after entry of sperm but before completion of fertilization
(2) after completion of fertilization
(3) before entry of sperm
(4) without any relation to sperm entry.

BHR224

46. If both ovaries are removed from human then level of which hormone is decreased in blood?
(1) Oxytocin
(2) Prolactin
(3) Estrogen
(4) Gonadotrophic releasing factor

BHR225

47. In a 30 years old lady, eggs are released in the form of :-
(1) oogonia
(2) primary oocyte
(3) secondary oocyte
(4) atretic follicle

BHR226

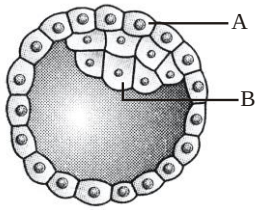
48. Phases in a menstrual cycle are :-
(1) Recovery phase and proliferative phase
(2) Proliferative phase and secretory phase
(3) Proliferative phase, secretory phase and menstrual phase
(4) Recovery phase, secretory phase and menstrual phase

BHR227

49. After ovulation, follicles are converted into:-
(1) corpus luteum (2) corpus albicans
(3) corpus cavernosa (4) corpus callosum

BHR228

50. In following structure, identify A & B and their respective functions.



Blastocyst

	A	B	Function of	
			A	B
(1)	Trophoblast	Inner cell mass	Attachment with endometrium	Differentiation as embryo
(2)	Trophoblast	Inner cell mass	Differentiation as embryo	Attachment with endometrium
(3)	Mesoderm	Inner cell mass	Differentiation as embryo	Attachment with endometrium
(4)	Ectoderm	Mesoderm	Attachment with endometrium	Differentiation as embryo

BHR229

51. In a menstrual cycle, ovulation normally takes place :-
(1) at the end of proliferative phase
(2) at the mid of secretory phase
(3) just before the end of secretory phase
(4) at the beginning of the proliferative phase

BHR230

52. In a menstrual cycle, hormones start increasing in which order from the beginning of menstruation ?
(1) FSH, progesterone, estrogen
(2) Estrogen, FSH, progesterone
(3) FSH, estrogen, progesterone
(4) Esterogen, progesterone, FSH

BHR231

53. At the time of ovulation, the human egg is covered by a membrane called as :-
(1) chorion (2) zona pellucida
(3) corona radiata (4) Both (2) & (3)

BHR232

54. How many cleavage divisions are required for the formation of 16 blastomeres ?
(1) 2 (2) 4
(3) 6 (4) 8

BHR233

55. The blastomeres in the blastocyst are arranged into an outer layer called ____**A**_____, and inner group of cells attached to trophoblast called the ____**B**___. What are 'A' and 'B' respectively?
(1) trophoblast, inner cell mass
(2) inner cell mass, trophoblast
(3) chorion, amnion
(4) amnion, chorion

BHR234

56. What is true about cleavage ?
(1) Nucleocytoplasmic ratio remains unchanged.
(2) Size of an embryo does not increase.
(3) There is less consumption of oxygen.
(4) The division is similar to meiosis.

BHR235

57. What is true about cleavage in a fertilized egg of human ?
(1) Cleavage require low O_2.
(2) It starts when egg reaches in uterus.
(3) It starts in fallopian tube.
(4) It is identical to normal mitosis.

BHR236

58. Which type of cell division occurs during cleavage?
(1) Amitosis (2) Mitosis
(3) Closed mitosis (4) Meiosis

BHR237

59. Fertilisation has following processes :-
(a) Plasmogamy (b) Karyogamy
(c) Syngamy (d) Amphimixis
Arrange these in correct sequence.
(1) a → b → c → d (2) b → a → d → c
(3) c → a → b → d (4) c → a → d → b

BHR238

60. Correct sequence of embryonic development is :-
(1) Fertilization ⟶ Zygote ⟶ Cleavage ⟶ Morula ⟶ Blastula ⟶ Gastrula
(2) Fertilization ⟶ Zygote ⟶ Blastula ⟶ Morula ⟶ Cleavage ⟶ Gastrula
(3) Fertilization ⟶ Cleavage ⟶ Morula ⟶ Zygote ⟶ Blastula ⟶ Gastrula
(4) Cleavage ⟶ Zygote ⟶ Fertilization ⟶ Morula ⟶ Blastula ⟶ Gastrula

BHR239

61. The lytic enzyme present in semen is :-
(1) ligase (2) estrogenase
(3) androgenase (4) hyaluronidase

BHR240

62. Given below are two statements:
Statement-I : hCG, hPL and relaxin are produced in women only during pregnancy.
Statement-II : By end of about 12 weeks, the body of foetus is covered with hairs, eye-lids separate, and eye lashes are formed.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement-I and II both are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

BHR241

63. Which one of following parts is present in male but not in female ?
(1) Urethra (2) Fallopian tube
(3) Vagina (4) Vas deferens
BHR242

64. Which hormone is not involved in the process of parturition ?
(1) Oxytocin (2) Prolactin
(3) Estrogen (4) Cortisol
BHR243

65. Accessory sexual characters in female is promoted by :-
(1) androgen (2) progesterone
(3) estrogen (4) testosterone
BHR244

66. The cellular layer that disintegrates and regenerates again and again in human is :-
(1) endometrium of uterus
(2) cornea of eye
(3) dermis of skin
(4) endothelium of blood vessels
BHR245

67. The functional maturation of sperms takes place in :-
(1) oviduct (2) epididymis
(3) vagina (4) All of these
BHR246

68. In human, the estrogen is secreted by the Graafian follicle from its :-
(1) external theca (2) internal theca
(3) zona pellucida (4) corona radiata
BHR248

69. Supporting cells found in the germinal epithelium of testis are called as :-
(1) Interstitial cells of Leydig
(2) Sertoli cells
(3) granular cells
(4) phagocytes
BHR249

70. Atretic follicles are found in the :-
(1) fallopian tube (2) uterus
(3) labia majora (4) ovary
BHR250

71. Which of the following hormones are produced in women only during pregnancy ?
(a) hPL (b) Relaxin
(c) Androgen (d) FSH
(e) hCG (f) LH
(1) a,b,e (2) b & e
(3) c,d,e,f (4) a,b,c,d,e,f
BHR251

72. Which of the following cells undergoes spermiogenesis ?
(1) Spematids
(2) Spermatogonia
(3) Primary spermatocytes
(4) Secondary spermatocytes
BHR252

73. Various changes in human sperm which prepare it to fertilise the ovum are called collectively as :-
(1) capacitation
(2) regeneration
(3) growth
(4) None of these
BHR253

74. During oogenesis, the small structure separated from egg is known as :-
(1) polar body
(2) secondary endosperm
(3) Herring body
(4) Hela cell
BHR254

75. In human foetus, the heart begins to beat at developmental age of :-
(1) 4 weeks (2) 3 weeks
(3) 6 weeks (4) 8 weeks
BHR255

76. Polar bodies are produced during the formation of :-
(1) sperms
(2) oogonia
(3) spermatocytes
(4) secondary oocytes

BHR256

77. Which of the following does not occur during natural menopause in a female ?
(1) Progesterone level in blood is decreased.
(2) FSH and LH levels in blood are decreased.
(3) Estrogen level in blood is decreased.
(4) Uterine changes are stopped.

BHR257

78. In a sperm, the mitochondria are occured :-
(1) in tail (2) in acrosome
(3) in middle piece (4) in head

BHR259

79. Which of the following sets of enzymes is found in the acrosome of human sperm ?
(1) Hyaluronidase, Collagenase
(2) Hyaluronidase, Zona lysin
(3) Hyaluronidase, Peptidase
(4) Only hyaluronidase

BHR260

80. Fixing up of the blastocyst in the wall of the uterus is known as :-
(1) fertilization
(2) implantation
(3) impreganation
(4) placentation

BHR261

81. Placenta in human beings is formed by :-
(1) amnion
(2) chorion
(3) allantois
(4) chorion and uterine wall

BHR262

82. The phenomenon of nuclear fusion of sperm and egg is known as :-
(1) karyogamy
(2) parthenogenesis
(3) vitellogenesis
(4) oogenesis

BHR263

83. Sertoli cells are involved in :-
(1) respiration
(2) nutrition of sperms
(3) excretion
(4) development of sex organs

BHR264

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	1	3	1	1	1	4	2	3	1	3	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	2	4	1	1	3	1	3	1	2	1	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	3	4	4	4	2	4	1	1	2	2	4	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	3	1	1	1	3	4	2	1	2	3	3	3	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	2	4	2	3	1	2	2	2	4	1	1	1	1	1
Question	76	77	78	79	80	81	82	83							
Answer	4	2	3	2	2	4	1	2							

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION, PROBLEMS AND STRATEGIES

1. Which one of the following was first country of world to adopt family planning programme?
(1) Japan (2) USA
(3) India (4) Bangladesh
BRH018

2. Family planning programmes were initiated in :-
(1) 1947 (2) 1951 (3) 1977 (4) 1955
BRH019

AMNIOCENTESIS

3. Amniocentesis is:-
(1) analysis of chemical composition of fluids of pregnant woman
(2) withdrawal of allantoic fluid from pregnant women
(3) done before pregnancy
(4) study of metaphase chromosomes from amniotic fluid to identify chromosomal abnormality
BRH001

4. Study of chromosomal abnormalities by taken out the amniotic fluid of embryo is called :-
(1) endoscopy (2) amniocentesis
(3) laproscopy (4) natal endoscopy
BRH003

5. Amniocentesis is used for determining :-
(1) length of embryo
(2) size of embryo
(3) hereditary disease of embryo
(4) All of the above
BRH013

6. The permissible use of the technique amniocentesis is for:
(1) Detecting any genetic abnormality
(2) Detecting sex of the unborn foetus
(3) Artificial insemination
(4) Transfer of embryo into the uterus of a surrogate mother
BRH026

POPULATION STABILISATION

7. Which one of the following is not legitimate for reducing birth rate?
(1) Ban on marriages
(2) Medical termination of pregnancy
(3) Use of contraceptives
(4) Late marriages
BRH004

BIRTH CONTROL – TEMPORARY CONTRACEPTIVE METHODS

8. An example of IUD is :-
(1) vasectomy (2) copper T
(3) condom (4) All of the above
BRH007

9. Example(s) of contraceptive is/are :-
(1) condom, cervical cap and diaphragm
(2) intrauterine device
(3) pill
(4) All of the above
BRH008

10. A contraceptive pill contains:-
(1) progesterone and estrogen
(2) spermicidal agents
(3) chemicals that cause abortion
(4) chemicals that prevent fertilization of ovum
BRH009

11. A contraceptive pill prevents ovulation by:-
(1) blocking fallopian tube
(2) inhibiting release of FSH and LH
(3) stimulating release of FSH and LH
(4) causing immediate degeneration of released ovum

BRH012

12. Most important component of oral contraceptive is :-
(1) thyroxine (2) LH
(3) progesterone (4) FSH

BRH014

13. The chances of fertilisation are very high during :-
(1) first five days of menstrual cycle
(2) last seven days of menstrual cycle
(3) day 10 to day 17 of menstrual cycle
(4) Both (1) & (2)

BRH017

14. 'Saheli', a female contraceptive pill is used:-
(1) daily (2) weekly
(3) quarterly (4) monthly

BRH020

15. What is the work of copper T?
(1) to inhibit ovulation
(2) to prevent fertilisation
(3) to inhibit implantation of blastocyst
(4) to inhibit gametogenesis

BRH022

16. Which one of the following is the most widely accepted method of contraception in India, as at present?
(1) Cervical caps
(2) Tubectomy
(3) Diaphragms
(4) IUDs' (Intra uterine devices)

BRH027

17. One of the legal methods of birth control is:
(1) by a premature ejaculation during coitus
(2) abortion by taking an appropriate medicine
(3) by abstaining from coitus from day 10 to 17 of menstrual cycle
(4) by having coitus at the time of day break

BRH031

BIRTH CONTROL – SURGICAL CONTRACEPTIVE METHODS

18. Purpose of tubectomy is to prevent:-
(1) egg formation
(2) embryonic development
(3) fertilisation
(4) coitus

BRH005

19. Vasectomy is :-
(1) Cutting of fallopian tube
(2) Cutting of vas deferens
(3) A factor of population growth
(4) None of the above

BRH006

20. Tubectomy, a method of population control is performed on :-
(1) both males and females
(2) only males
(3) only females
(4) only pregnant females

BRH015

21. Which is related to males?
(1) IUD
(2) tubectomy
(3) vasectomy
(4) None of the above

BRH016

MTP

22. Full form of MTP with respect to reproductive health is:-

(1) Maternal termination of pregnancy

(2) Mid-term pregnancy

(3) Medical try for pregnancy

(4) Medical termination of pregnancy

BRH011

23. Medical Termination of Pregnancy (MTP) is considered safe up to have many weeks of pregnancy?

(1) Eight weeks

(2) Twelve weeks

(3) Eighteen weeks

(4) Six weeks

BRH028

INFERTILITY (ART- ASSISTED REPRODUCTIVE TECHNOLOGIES) & STDS

24. What is correct about test tube baby?

(1) Fertilisation inside female genital tract and growth in test tube

(2) Rearing of prematurely born baby in incubator

(3) Fertilisation outside and gestation inside womb of mother

(4) Both fertilisation and development outside the female genital tract

BRH002

25. The partner(s) responsible for sex of the child is/are:-

(1) male

(2) female

(3) both male and female

(4) at times male & at times female

BRH010

26. Test tube baby means a baby born when :-

(1) it is developed in a test tube.

(2) it is developed through tissue culture method.

(3) the ovum is fertilised externally and there after implanted in the uterus.

(4) it develops from a non-fertilized egg.

BRH021

27. In vitro fertilisation is a technique that involves transfer of which one of the following into the fallopian tube?

(1) Zygote only

(2) Embryo only, up to 8 cell stage

(3) Either zygote or early embryo up to 8 cell stage

(4) Embryo of 32 cell stage

BRH025

28. The Test-tube Baby programme employs which one of the following techniques?

(1) Gamete intra fallopian transfer (GIFT)s

(2) Zygote intra fallopian transfer (ZIFT)

(3) Intra cytoplasmic sperm injection (ICSI)

(4) Intra uterine insemination (IUI)

BRH029

29. Artificial insemination means:

(1) introduction of sperms of a healthy donor Directly into the ovary

(2) transfer of sperms of a healthy donor to a test tube containing ova

(3) transfer of sperms of husband to a test tube containing ova

(4) artificial introduction of sperms of a healthy donor into the vagina

BRH030

MIXED

30. Given below are four methods (A–D) and their modes of action (a–d) in achieving contraception. Select their correct matching from the four options that follow:

	Method		Mode of Action
A.	The pill	(a)	Prevents sperms reaching cervix
B.	Condom	(b)	Prevents implantation
C.	Vasectomy	(c)	Prevents ovulation
D.	Copper T	(d)	Semen contains no sperms

Match the following:

(1) A – (c), B – (d), C – (a), D – (b)

(2) A – (b), B – (c), C – (a), D – (d)

(3) A – (c), B– (a), C – (d), D – (b)

(4) A – (d), B– (a), C – (b), D – (c)

BRH023

31. Consider the statements given below regarding contraception and answer as directed there after:

(A) Medical Termination of Pregnancy (MTP) during first trimester is generally safe

(B) Generally, chances of conception are nil until mother breast-feeds the infant upto two years

(C) Intrauterine devices like copper T are effective contraceptives

(D) Contraception pills may be taken up to one week after coitus to prevent conception

Which two of the above statements are correct?

(1) A, C (2) A, B

(3) B, C (4) C, D

BRH024

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	2	3	1	1	2	4	1	2	3	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	3	2	3	3	4	2	3	1	3	3	2	4	3
Question	31														
Answer	1														

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Tubectomy is a method of sterilization in which:
(1) small part of the fallopian tube is removed or tied up
(2) ovaries are removed surgically
(3) small part of vas deferens is removed or tied up
(4) uterus is removed surgically

BRH032

2. Which of the following is a hormone releasing Intra Uterine Device (IUD)?
(1) Multiload 375 (2) LNG – 20
(3) Cervical cap (4) Vault

BRH033

3. Assisted reproductive technology, IVF involves transfer of:
(1) Ovum into the Fallopian tube
(2) Zygote into the fallopian tube
(3) Zygote into the uterus
(4) Embryo with 16 blastomeres into Fallopian tube

BRH034

Re-AIPMT 2015

4. A childless couple can be assisted to have a child through a technique called GIFT. The full form of this technique is:
(1) Germ cell internal fallopian transfer
(2) Gamete inseminated fallopian transfer
(3) Gamete intra fallopian transfer
(4) Gamete internal fertilization and transfer

BRH035

NEET-I 2016

5. Which of the following approaches does not give the defined action of contraceptive?

(1)	Barrier Methods	Prevent fertilization
(2)	Intrauterine Devices	Increase phagocytosis of sperm, Suppress sperm mortality and fertilising Capacity of sperms
(3)	Hormonal contraceptives	Prevent/retard entry of sperms, prevents ovulation and Fertilization
(4)	Vasectomy	Prevents spermatogenesis

BRH036

NEET-II 2016

6. Which of the following is hormone releasing IUD?
(1) Lippes loop (2) Cu7
(3) LNG-20 (4) Multiload 375

BRH037

7. Which of the following is incorrect regarding vasectomy?
(1) Vasa deferentia is cut and tied
(2) Irreversible sterility
(3) No sperm occurs in seminal fluid
(4) No sperm occurs in epididymis

BRH038

8. Embryo with more than 16 blastomeres formed due to in vitro fertilization is transferred into:
(1) Fimbriae (2) Cervix
(3) Uterus (4) Fallopian tube

BRH039

NEET-UG 2017

9. In case of a couple where the male is having a very low sperm count, which technique will be suitable for fertilisation?
(1) Gamete intracytoplasmic fallopian transfer
(2) Artificial Insemination
(3) Intracytoplasmic sperm injection
(4) Intrauterine transfer

BRH040

10. The function of copper ions in copper releasing IUD's is:
(1) They inhibit gametogenesis
(2) They make uterus unsuitable for implantation
(3) They inhibit ovulation
(4) The suppress sperm motility and fertilising capacity of sperms

BRH041

NEET-UG 2018

11. The contraceptive 'SAHELI' :
(1) blocks estrogen receptors in the uterus, preventing eggs from getting implanted
(2) increases the concentration of estrogen and prevents ovulation in females
(3) is an IUD
(4) is a post-coital contraceptive

BRH042

NEET-UG 2019

12. Select the hormone-releasing Intra-Uterine Devices:
(1) Vaults, LNG-20
(2) Multiload 375, Progestasert
(3) Progestasert, LNG-20
(4) Lippes Loop, Multiload 375

BRH043

13. Which of the following contraceptive methods do involve a role of hormone?
(1) Lactational amenorrhea, Pills, Emergency contraceptives
(2) Barrier method, Lactational amenorrhea, Pills
(3) Cut, Pills, Emergency contraceptives
(4) Pills, Emergency contraceptives, Barrier methods

BRH044

NEET-UG 2019 (Odisha)

14. Which of the following is a correct statement?
(1) IUDs once inserted need not be replaced
(2) IUDs are generally inserted by the user herself
(3) IUDs increase phagocytosis of sperms in the uterus
(4) IUDs suppress gametogenesis

BRH045

NEET-UG 2020

15. In which of the following techniques, the embryos are transferred to assist those females who cannot conceive?
(1) GIFT and ICSI (2) ZIFT and IUT
(3) GIFT and ZIFT (4) ICSI and ZIFT

BRH046

NEET-UG 2021

16. Which one of the following is an example of Hormone releasing IUD?
(1) CuT (2) LNG 20
(3) Cu 7 (4) Multiload 375

BRH047

17. Veneral diseases can spread through :
(a) Using sterile needles
(b) Transfusion of blood from infected person
(c) Infected mother to foetus
(d) Kissing
(e) Inheritance
Choose the correct answer from the options given below.
(1) (a), (b) and (c) only
(2) (b), (c) and (d) only
(3) (b) and (c) only
(4) (a) and (c) only

BRH048

18. Match List-I with List-II.

	List-I		List-II
(a)	Vaults	(i)	Entry of sperm through Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of fallopian tube

Choose the correct answer from the options given below.
(1) (a)–(iv), (b)–(ii), (c)–(i), (d)–(iii)
(2) (a)–(i), (b)–(iii), (c)–(ii), (d)–(iv)
(3) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i)
(4) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

BRH049

NEET-UG 2022

19. Lippe's loop is a type of contraceptive used as:
(1) Vault barrier
(2) Non–Medicated IUD
(3) Copper releasing IUD
(4) Cervical barrier

BRH050

20. Match **List -I** with **List -II** with respect to methods of Contraception and their respective actions.

	List-I		List-II
(a)	Diaphragms	(i)	Inhibit ovulation and Implantation
(b)	Contraceptive Pills	(ii)	Increase phagocytosis of sperm within Uterus
(c)	Intra uterine Devices	(iii)	Absence of Menstrual cycle and ovulation following parturition
(d)	Lactational Amenorrhea	(iv)	They cover the cervix blocking the entry of sperms

Choose the **correct answer** from the options given below:
(1) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)
(2) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)
(3) (a)–(iii), (b)–(ii), (c)–(i), (d)–(iv)
(4) (a)–(iv), (b)–(i), (c)–(iii), (d)–(ii)

BRH051

Re-NEET-UG 2022

21. Which of the following is not an Intra Uterine Device?
(1) Progestogens (2) Multiload 375
(3) Lippes loop (4) Progestasert

BRH052

22. IUDs are small objects made up of plastic or copper that are inserted in the uterine cavity. Which of the following statements are **correct** about IUDs ?
(a) IUDs decrease phagocytosis of sperm within the uterus.
(b) The released copper ions suppress the sperm motility.
(c) IUDs do not make the cervix hostile to the sperm.
(d) IUDs suppress the fertilization capacity of sperm.
(e) The IUDs require surgical intervention for their insertion in the uterine cavity.
Choose **the most appropriate answer** from the options given below:
(1) (a), (d) and (e) only (2) (b) and (c) only
(3) (b) and (d) only (4) (d) only

BRH053

NEET-UG 2023

23. Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly ?
(1) Gonorrhoea (2) Hepatitis-B
(3) HIV Infection (4) Genital herpes

BRH081

24. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A : Amniocentesis for sex determination is one of the strategies of Reproductive and Child Health Care Programme.
Reason R : Ban on amniocentesis checks increasing menace of female foeticide.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is NOT the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

BRH082

25. Match **List I** with **List II.**

List I	List II
A. Vasectomy	I. Oral method
B. Coitus interruptus	II. Barrier method
C. Cervical caps	III. Surgical method
D. Saheli	IV. Natural method

Choose the **correct** answer from the options given below :
(1) A-III, B-IV, C-II, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

BRH083

NEET-UG 2023 (Manipur)

26. Which of the following sexually transmitted infections are completely curable ?
(1) HIV infection and Trichomoniasis
(2) Syphilis and trichomoniasis
(3) Hepatitis – B and Genital herpes
(4) Genital herpes and Genital warts

BRH084

27. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Non-medicated IUDs	(I)	Multiload 375
(B)	Copper releasing IUDs	(II)	Rubber barrier
(C)	Hormone releasing IUDs	(III)	Lippes loop
(D)	Vaults	(IV)	LNG-20

Choose the correct answer from the options given below :
(1) A-(IV), B-(III), C-(I), D-(II)
(2) A-(II), B-(IV), C-(III), D-(I)
(3) A-(III), B-(I), C-(IV), D-(II)
(4) A-(III), B-(IV), C-(II), D-(I)

BRH085

28. Given below are two statements :

Statement I :
Intra Cytoplasmic Sperm Injection (ICSI) is another specialised procedure of in-vivo fertilisation.

Statement II :
Infertility cases due to inability of the male partner to inseminate female can be corrected by artificial insemination (AI).

In the light of the above statements, choose the **correct** answer from the options given below:
(1) **Statement I** is correct but **statement II** is false
(2) **Statement I** is incorrect but **Statement II** is true
(3) Both **Statement I** and **Statement II** are true
(4) Both **Statement I** and **Statement II** are false.

BRH086

29. Which of the following statements are **correct** ?
(A) Reproductive health refers to total well-being in all aspects of reproduction.
(B) Amniocentesis is legally banned for sex determination in India.
(C) "Saheli" – a new oral contraceptive for females was developed in collaboration with ICMR (New Delhi).
(D) Amniocentesis is used to determine genetic disorders and survivability of foetus.

Choose the **most appropriate** answer from the options given below :

(1) (B) and (C) only

(2) (D) and (C) only

(3) (A), (B) and (D) only

(4) (A) and (C) only

BRH087

NEET-UG 2024

30. Which of the following is not a natural/traditional contraceptive method ?

(1) Coitus interruptus

(2) Periodic abstinence

(3) Lactational amenorrhea

(4) Vaults

BRH088

31. Match List-I with List-II :

List-I		List-II	
A.	Non-medicated IUD	I.	Multiload 375
B.	Copper releasing IUD	II.	Progestogens
C.	Hormone releasing IUD	III.	Lippes loop
D.	Implants	IV.	LNG-20

Choose the correct answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-I, B-III, C-IV, D-II

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-I, C-IV, D-II

BRH089

RE-NEET-UG 2024

32. Match **List-I** with **List-II** relating to examples of various kind of IUDs and barrier :

	List-I		List-II
(A)	Copper releasing IUD	(I)	Vaults
(B)	None-medicated IUD	(II)	Multiload 375
(C)	Contraceptive barrier	(III)	LNG-20
(D)	Hormone releasing IUD	(IV)	Lippes loop

Choose the **correct answer** from the options given below :

(1) A-II, B-IV, C-III, D-I

(2) A-IV, B-III, C-I, D-II

(3) A-II, B-I, C-III, D-IV

(4) A-II, B-IV, C-I, D-III

BRH090

33. Following is the list of STDs. Select the diseases which are not completely curable.

A. Genital warts B. Genital herpes

C. Syphilis D. Hepatitis-B

E. Trichomoniasis

Choose the **correct answer** form the options given below :

(1) A and D only

(2) B and D only

(3) A and C only

(4) D and E only

BRH091

Exercise - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	2	3	4	3	4	3	2	4	1	3	1	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	2	2	1	1	3	1	3	1	2	3	2	3	4
Question	31	32	33												
Answer	4	4	2												

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Which of the following is a nonsteroidal pill?
(1) 'Mala-D' (2) 'Mala-N'
(3) 'Saheli' (4) None of these
BRH054

2. Which of the following contraceptive methods make uterus unsuitable for implantation ?
(1) Diaphragm (2) Condom
(3) IUD (4) Natural method
BRH055

3. Which of the following is a natural method of contraception ?
(1) Sterilisation
(2) IUD
(3) Diaphragm
(4) Periodic abstinence
BRH056

4. Which method of contraception has high failure rate?
(1) Barrier method (2) IUD
(3) Sterilization (4) Natural method
BRH057

5. Which of the following can be used as an emergency contraceptive ?
(1) 'Mala-D' (2) 'Saheli'
(3) 'i-pill' (4) Condom
BRH058

6. MTP is relatively safe ?
(1) upto 12 weeks
(2) upto 18 weeks
(3) during first trimester
(4) Both (1) & (3)
BRH059

7. Which of the following methods of contraception has least side effects?
(1) IUD
(2) Pills
(3) Coitus interruptus
(4) Cervical cap
BRH060

8. Which of the following methods of contraception prevents ovulation ?
(1) Pills (2) LNG-20
(3) 'Norplant' (4) All of these
BRH061

9. Which of the following techniques is relative with in-vivo fertilisation?
(1) ZIFT (2) GIFT (3) ICSI (4) IUT
BRH062

10. In which of the following methods, zygote upto 32 blastomeres is transferred into the uterus?
(1) IUT (2) ZIFT (3) GIFT (4) ICSI
BRH063

11. A contraceptive pill developed by the scientists of CDRI, Lucknow is?
(1) 'Mala-D' (2) 'Mala-N'
(3) 'Saheli' (4) 'i-pill'
BRH064

12. Which of the following techniques is banned in India?
(1) USG (2) Sterilisation
(3) Amniocentesis (4) IVF
BRH065

13. IVF technique includes ?
(1) GIFT (2) AI
(3) IUT (4) All of these
BRH066

14. Which contraceptive method provides protection against HIV?
(1) IUD
(2) Pills
(3) Condom
(4) Periodic abstinence

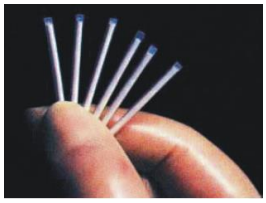
BRH067

(B) ANALYTICAL QUESTIONS

15. Progestasert makes the uterus unsuitable for implantation and cervix hostile to the sperms. It is a :-
(1) copper releasing IUD
(2) hormone releasing IUD
(3) non-medicated IUD
(4) Both (2) & (3)

BRH068

16. Identify the contraceptive device shown below as well as the related right site of its implantation into a woman. Then, select the option with correct matching.



	Contraceptive device	Site of implant
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Implant	Subcutaneous tissue
(4)	Multiload 375	Uterine wall

BRH069

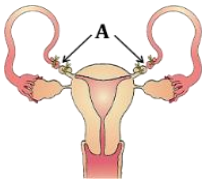
17. Which of the following is a Cu releasing IUD and its function?
(1) Multiload 375 : It suppresses sperm motility and fertilising capacity of sperms
(2) Progestasert : It makes uterus unsuitable for implantation and cervix hostile to sperms
(3) Lippes loop : It increases phagocytosis of sperms
(4) LNG-20 : It inhibits ovulation and implantation as well as alter the quality of cervical mucus to prevent/retard entry of sperms

BRH070

18. What percentage of total number of conceived pregnancies in a year all over the world are medically terminated?
(1) 30% (2) 45% (3) 20% (4) 10%

BRH071

19. In the given diagram, the technique depicted and 'A' are respectively :-



(1) vasectomy, vas deferens
(2) tubectomy, fallopian tube
(3) tubectomy, ovary
(4) vasectomy, epididymis

BRH072

20. Use of which of the following has/have been found to be very effective as emergency contraceptive?
(a) IUDs
(b) Progestogen-estrogen combinations
(c) Cervical cap
(d) Tubectomy
(1) (a), (b) and (c) (2) (b), (d) and (e)
(3) (a) and (b) (4) (c), (d) and (e)

BRH073

21. If vasectomy is done, then which condition would occur in a male?
(1) Teratospermia (2) Asthenospermia
(3) Azospermia (4) Both (1) & (2)

BRH074

22. Natural method of contraception includes:-
(1) coitus interruptus
(2) lactational amenorrhea
(3) periodic abstinence
(4) All of the above

BRH075

23. If male is impotent and female is normal then which of the following technique can be used to treat infertility?
(1) ICSI (2) ZIFT (3) GIFT (4) AI

BRH076

24. Diaphragms are contraceptive devices used by the females. Choose the correct option from the statements given below.
(i) They are introduced into the uterus.
(ii) They are placed to cover the cervical region.
(iii) They act as physical barriers to sperm entry.
(iv) They act as spermicidal agents.
(1) (i) and (ii) (2) (i) and (iii)
(3) (ii) and (iii) (4) (iii) and (iv)

BRH077

25. What happens after vasectomy?
(1) Absence of formation of semen
(2) Formed Sperms are dead or inactive
(3) Sperms immediately disappear in semen
(4) Sperms gradually disappear in semen

BRH078

26. If the duration of menstrual cycle is of 35 days, then what is risk period (cycle start on day 1) for this menstrual cycle?
(1) Day 9 to day 17
(2) Day 11 to day 18
(3) Day 16 to day 24
(4) Day 18 to day 35

BRH079

27. A small part of the fallopian tube is removed or tied up in :-
(1) tubectomy
(2) hysterectomy
(3) castration
(4) vasectomy

BRH080

Exercise - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	4	4	3	4	3	4	2	1	3	3	3	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	3	1	3	2	3	3	4	4	3	4	3	1			

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Exercise - I (Conceptual Questions) Build Up Your Understanding

1. *Agrobacterium tumefaciens* used in Genetic engineering for :-
(1) DNA-mapping
(2) DNA-modification
(3) Gene transfer
(4) DNA finger printing
BBP001

2. Who isolated the first restriction endonucleases?
(1) Temin & Baltimore (2) Sanger
(3) Smith (4) Paul berg
BBP002

3. Genetic engineering is :-
(1) Study of extra nuclear gene
(2) Manipulation of genotype by artificial method
(3) Manipulation of RNA
(4) Manipulation of enzymes
BBP003

4. Polymerase chain reaction technology (PCR-technique) is used for :-
(1) DNA identification
(2) DNA repair
(3) DNA amplification
(4) Cleave DNA
BBP004

5. Which structure is involved in genetic engineering?
(1) Plastid (2) Plasmid
(3) Codon (4) None
BBP005

6. Which of the following is the example of molecular scissors?
(1) EcoR I (2) Hind III
(3) BamH I (4) All the above
BBP006

7. Restriction endonucleases are used in genetic engineering because: -
(1) They can degrade harmful proteins
(2) They can join DNA fragments
(3) They can cut DNA at variable site
(4) They can cut DNA at specific base sequences
BBP007

8. When the genotype of an organism is improved by the addition of foreign gene, the process is called :-
(1) Tissue culture
(2) Genetic diversity
(3) Genetic engineering
(4) Plastic surgery
BBP008

9. Which vector is commonly used in the transfer of gene in a crop plant ?
(1) Plasmid of *Bacillus subtilis*
(2) Bacteriophages
(3) Ti-plasmid of *Agrobacterium*
(4) *E. coli* Phages
BBP009

10. The tumour inducing capacity of *Agrobacterium tumefaciens* is located in extrachromosomal plasmid called:-
(1) Ti – plasmid (2) Ri – plamid
(3) Lambda phage (4) Plasmid pBR322
BBP010

11. Genetic engineering aims at :-
(1) Destroying wild gene
(2) Preserving defective gene
(3) Curing human disease by introducing new gene
(4) All the above
BBP011

12. *Taq*-polymerase which is used for amplification of DNA related with: -
(1) Hybridoma technique
(2) PCR-technique
(3) Gene cloning
(4) r-DNA technology

BBP012

13. What is true for plasmid?
(1) Plasmids are widely used in gene transfer
(2) These are found in virus
(3) Plasmid contains gene for main vital activities
(4) These are main part of chromosome

BBP013

14. Which of the following cuts the DNA from specific places?
(1) Restriction endonuclease
(2) Ligase
(3) Exonuclease
(4) Alkaline phosphate

BBP014

15. Manipulation of DNA in genetic engineering became possible due to the discovery of: -
(1) Restriction endonuclease
(2) DNA polymerase
(3) Transcriptase
(4) Primase

BBP015

16. Which one of the following has found extensive use in genetic engineering work in plants?
(1) *Bacillus coagulens*
(2) *Agrobacterium tumefaciens.*
(3) *Clostridium septicum*
(4) *Xanthomonas citri*

BBP016

17. Restriction endonucleases: -
(1) Are enzymes which cleave DNA at specific sites
(2) Make DNA complementary to an existing DNA or RNA
(3) Cut or join DNA fragments
(4) Are required in vectorless direct gene transfer.

BBP017

18. Restriction endonucleases :-
(1) Are synthesized by bacteria as part of their defence mechanism
(2) Are present in mammalian cells for degradation of DNA when the cell dies
(3) Are used in genetic engineering for ligating two DNA molecules
(4) Are used for *in vitro* DNA synthesis

BBP018

19. The Ti plasmid, is often used for making transgenic plants. This plasmid is found in :-
(1) Yeast (2) *Azotobacter*
(3) *Rhizobium* (4) *Agrobacterium*

BBP019

20. Which of the following is the example of direct gene transfer?
(1) Microinjection (2) Electroporation
(3) Particle gun (4) All the above

BBP020

21. How many copies of DNA are produced in PCR technique after 6–cycle?
(1) 4 (2) 32 (3) 64 (4) 16

BBP021

22. Thermal cycle takes place in which technique :-
(1) Gel electrophoresis
(2) PCR-technique
(3) Centrifugation
(4) Southern blotting

BBP022

23. BAC and YAC are :-
(1) Natural DNA obtained from bacteria and yeast
(2) Useful vectors for eukaryotic gene transfer
(3) Artificial DNA obtained from bacteria and yeast
(4) Both (2) and (3)

BBP023

24. Restriction enzymes are :-
(1) Not always required in genetic engineering
(2) Essential tool in genetic engineering
(3) Nucleases that cleave DNA at specific sites
(4) Both (2) and (3)

BBP024

25. Electroporation procedure involves :-
(1) Fast passage of food through sieve pores in phloem elements with the help of electric stimulation.
(2) Opening of stomatal pores during night by artificial light
(3) Making transient pores in the cell membrane to introduce gene constructs
(4) Purification of saline water with the help of a membrane system.

BBP025

26. Which of the following restriction endonuclease enzyme produce blunt end in DNA :-

(1) Bam HI
```
      ↓
G G A T C C
C C T A G G
        ↑
```

(2) Eco RI
```
      ↓
G A A T T C
C T T A A G
        ↑
```

(3) Hae III
```
    ↓
G G C C
C C G G
  ↑
```

(4) All the above

BBP026

27. A bacterium modifies its DNA by adding methyl groups to the DNA, It does so to :-
(1) Clone its DNA
(2) Be able to transcribe many genes simultaneously
(3) Turn its gene on
(4) Protect its DNA from its own restriction enzyme

BBP027

28. The restriction enzyme EcoR I has the property of:-
(1) endonuclease activity
(2) exonuclease activity
(3) ligation activity
(4) correcting the topology of replicating DNA

BBP028

29. DNA ligase is an enzyme that catalyses the:-
(1) splitting of DNA threads into small bits
(2) joining of the fragments of DNA
(3) denaturation of DNA
(4) synthesis of DNA

BBP029

30. *Agrobacterium tumefaciens* contains a large plasmid, which induces tumour in the plants it is termed as :-
(1) Ti plasmid
(2) Ri plasmid
(3) Recombinant plasmid
(4) Shine Dalgarno sequence

BBP030

31. More advancement in genetic engineering is due to :-
(1) Restriction endonuclease
(2) Lipase
(3) Protease
(4) Zymase

BBP031

32. Which of the following is used as a best genetic vector in plants?
(1) *Bacillus thuringiensis*
(2) *Agrobacterium tumefaciens*
(3) *Pseudomonas putida*
(4) All of these

BBP032

33. Which of the following enzyme is used to join DNA fragments?
(1) Terminase (2) Endonuclease
(3) Ligase (4) DNA polymerase

BBP033

34. A kind of biotechnology involving manipulation of DNA is :-
(1) DNA replication
(2) Genetic engineering
(3) Denaturation
(4) Renaturation

BBP034

35. What is *true* of plasmid ?
(1) Found in viruses
(2) Contains genes for main vital activities
(3) Part of nuclear chromosome
(4) Widely used in gene transfer

BBP035

36. A suitable vector for gene cloning in higher organism is :-
(1) Baculovirus
(2) Retrovirus
(3) *Salmonella typhimurium*
(4) *Neurospora crassa*

BBP036

37. PCR proceeds in three distinct steps governed by temperature they are in order of:-
(1) Denaturation, Annealing, Synthesis
(2) Synthesis, Annealing, Denaturation
(3) Annealing, Synthesis, Denaturation
(4) Denaturation, Synthesis, Annealing

BBP037

38. What is the source of the Ti (Tumor inducing) plasmid which is modified and used as a cloning vector to deliver the desirable genes into plant cells ?
(1) *Agrobacterium tumefaciens*
(2) *Thermophilus aquaticus*
(3) *Pyrococcus furiosus*
(4) *Aedes aegypti*

BBP038

39. The thermostable enzymes, 'Taq' isolated from thermophilic bacteria is :-
(1) RNA polymerase
(2) DNA polymerase
(3) Restriction endonuclease
(4) DNA ligase

BBP039

40. The term "molecular scissors" generally refers to:-
(1) DNA polymerases
(2) RNA polymerases
(3) Restriction endonucleases
(4) DNA ligases

BBP040

41. In the PCR technology the DNA segment is replicated over a billion times. This repeated replications catalysed by the enzyme :-
(1) RNA polymerase
(2) Taq polymerase
(3) DNA dependent RNA polymerase
(4) Primase

BBP041

42. The restriction enzyme(s) used in recombinant DNA technology making staggered cuts in DNA leaving sticky ends is/are :-
(1) EcoR I (2) Hind III
(3) BamH I (4) All of the above
BBP042

43. Restriction enzyme EcoR I cuts the DNA between bases G and A only when the sequence in DNA is :-
(1) GATATC (2) GAATTC
(3) GATTCC (4) GAACTT
BBP043

44. For a DNA to function as a cloning vector the most essential requirement is :-
(1) multiple restriction sites
(2) several selectable markers
(3) circular nature
(4) 'ori' sequence
BBP044

45. According to EFB, "The integration of natural science and organisms, cells, parts thereof and molecular analogues for products and services," is known as:-
(1) Biochemistry (2) Bioinformatics
(3) Biotechnology (4) Biology
BBP045

46. The stickiness of DNA ends facilitates the action of which enzyme :-
(1) DNA polymerase
(2) DNA Ligase
(3) Restriction endonuclease
(4) Alkaline phosphatase
BBP046

47. Which technique is used to check the progression of restriction enzyme digestion?
(1) PCR
(2) Gel electrophoresis
(3) Southern Blotting
(4) Staining
BBP047

48. In gel electrophoresis, at which end of the gel the sample is loaded?
(1) In the wells
(2) Towards positive electrode
(3) Towards negative electrode
(4) Both (1) and (3)
BBP048

49. An antibiotic resistance gene of plasmid vector which get inactivated due to insertion of alien DNA, helps in the selection of:-
(1) Transformants
(2) Recombinants
(3) Non-Transformants
(4) Both (2) & (3)
BBP049

50. In which type of bioreactor air bubbles dramatically increases the oxygen transfer area?
(1) Simple stirred tank bioreactor
(2) Sparged stirred tank bioreactor
(3) Both (1) and (2)
(4) None of these
BBP050

51. In EcoR I, R stands for :-
(1) Strain (2) Species
(3) Genus (4) Order
BBP051

52. Restriction endonucleases are used in genetic engineering to form:-
(1) Recombinant molecule of protein
(2) Recombinant molecule of DNA
(3) Recombinant molecule of protein & DNA
(4) Recombinant cell
BBP052

53. Which technique is used for the separation of DNA fragments?
(1) PCR
(2) Gel electrophoresis
(3) Bioreactor
(4) Restriction endonuclease

BBP053

54. Which of following feature is not necessary for cloning vector?
(1) Origin of replication
(2) Low copy number
(3) Selectable marker
(4) Cloning sites

BBP054

55. Which of the following is not true for cloning vector?
(1) more than two origin sites of replication
(2) vector should have selectable marker gene
(3) single recognition site for the commonly used restriction enzyme
(4) pBR-322 have tetracycline resistance gene

BBP055

56. Transformation is a procedure through which:
(1) A piece of DNA is introduced in a host bacterium
(2) A piece of DNA is introduced in a vector
(3) A piece of DNA is introduced from protein
(4) All of these

BBP056

57. Second letter of the name of restriction endonuclease came from the :-
(1) Genus of organism
(2) Species of organism
(3) Family of organism
(4) Class of organism

BBP057

58. To isolate DNA from fungi we have to break the wall. This is done by:
(1) Lysozyme (2) Cellulase
(3) Invertase (4) Chitinase

BBP058

59. Which of the following enzyme will get inactivated in insertional inactivation?
(1) Transacetylase
(2) Permease
(3) β-galactosidase
(4) Taq-polymerase

BBP059

60. In presence of chromogenic substrate recombinant bacteria gives:-
(1) Red coloured colonies
(2) Colourless colonies
(3) Blue colonies
(4) Green colonies

BBP060

61. Which of the following enzyme is known as molecular scissors?
(1) Ligase
(2) DNA polymerase
(3) Restriction enzyme
(4) Helicase

BBP061

62. Which of the following is not required in PCR?
(1) DNA primer (2) DNA template
(3) RNA primer (4) *Taq* polymerase

BBP062

63. The substrate for restriction enzyme is :-
(1) Single stranded RNA
(2) Proteins
(3) Double stranded DNA
(4) Single stranded DNA

BBP063

64. In recombinant DNA technology, the term vectors refers to :-
(1) the enzyme that cuts DNA into restriction fragments
(2) the sticky end of a DNA fragment
(3) a plasmid used to transfer DNA into a living cell
(4) a DNA probe used to identify a particular gene

BBP064

65. pBR322 which is frequently used as a vector for cloning gene is:–
(1) an original bacterial plasmid
(2) a modified bacterial plasmid
(3) a viral genome
(4) a transposon

BBP065

66. Two microbes found to be very useful in genetic engineering are :–
(1) *Escherichia coli* and *Agrobacterium tumefaciens*
(2) *Vibrio cholerae* and a tailed bacteriophage
(3) *Diplococcus* sp. and *Pseudomonas* sp.
(4) Crown gall bacterium and *Caenorhabditis elegans*

BBP066

67. Restriction endonuclease :–
(1) Cuts the DNA molecule randomly
(2) Cuts the DNA molecule at specific sites
(3) Restricts the synthesis of DNA inside the nucleus
(4) Synthesizes DNA

BBP067

68. Which one of the following is commonly used in transfer of foreign DNA into crop plants ?
(1) *Penicillium expansum*
(2) *Trichoderma harzianum*
(3) *Meloidogyne incognita*
(4) *Agrobacterium tumefaciens*

BBP068

69. Which one of the following is used as vector for cloning genes into higher organisms?
(1) *Rhizopus nigricans*
(2) Retrovirus
(3) Baculovirus
(4) *Salmonella typhimurium*

BBP069

70. There is a restriction endonuclease called Eco RI. What does "co" part in it stand for ?
(1) Colon (2) Coelom
(3) Coenzyme (4) coli

BBP070

71. Agarose extracted from sea weeds finds use in:
(1) Spectrophotometry
(2) Tissue Culture
(3) PCR
(4) Gel electrophoresis

BBP071

72. Which one of the following techniques made it possible to genetically engineer living organisms?
(1) Hybridization
(2) Recombinant DNA techniques
(3) X-ray diffraction
(4) Heavier isotope labelling

BBP072

73. Which one is a true statement regarding DNA polymerase used in PCR ?
(1) It is isolated from a virus
(2) It remains active at high temperature
(3) It is used to ligate introduced DNA in recipient cells
(4) It serves as a selectable marker

BBP073

74. For transformation, micro-particles coated with DNA to be bombarded with gene gun are made up of :-

(1) Silicon or Platinum

(2) Gold or Tungsten

(3) Silver or platinum

(4) Platinum or zinc

BBP074

75. Which one of the following represents a palindromic sequence in DNA ?

(1) 5'-CATTAG-3' / 3'-GATAAC-5'

(2) 5'-GATACC-3' / 3'-CCTAAG-5'

(3) 5'-GAATTC-3' / 3'-CTTAAG-5'

(4) 5'-CCAATG-3' / 3'-GAATCC-5'

BBP075

76. In genetic engineering, the antibiotic resistant genes are used :-

(1) as sequences from where replication starts

(2) to keep the cultures free of infection

(3) as selectable markers

(4) to select healthy vectors

BBP076

77. Biolistic (gene-gun) is suitable for :-

(1) Constructing recombinant DNA by joining with vectors

(2) DNA finger printing

(3) Disarming pathogen vectors

(4) Transformation of plants cells

BBP077

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	3	2	4	4	3	3	1	3	2	1	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	1	4	4	3	2	4	4	3	3	4	1	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	3	2	4	2	1	1	2	3	2	4	2	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	4	2	2	1	2	2	2	1	1	2	4	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	3	3	3	2	1	2	4	2	4	4	2	2	2	3
Question	76	77													
Answer	3	4													

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which vector can clone only a small fragment of DNA?
(1) Bacterial artificial chromosome
(2) Yeast artificial chromosome
(3) Plasmid
(4) Cosmid

BBP078

Re-AIPMT 2015

2. The DNA molecules to which the gene of interest is integrated for cloning is called :-
(1) Carrier
(2) Transformer
(3) Vector
(4) Template

BBP079

3. The cutting of DNA at specific locations became possible with the discovery of :-
(1) Ligases
(2) Restriction enzymes
(3) Probes
(4) Selectable markers

BBP080

4. The introduction of T-DNA into plants involves: -
(1) Allowing the plant roots to stand in water
(2) Infection of the plant by *Agrobacterium tumefaciens*
(3) Altering the pH of the soil, then heat shocking the plants
(4) Exposing the plants to cold for a brief period

BBP081

NEET-I 2016

5. Which of the following is not a feature of the plasmids ?
(1) Independent replication
(2) Circular structure
(3) Transferable
(4) Single – stranded

BBP082

6. The *taq*-polymerase enzyme is obtained from:-
(1) *Thermus aquaticus*
(2) *Thiobacillus ferroxidans*
(3) *Bacillus subtilis*
(4) *Pseudomonas putida*

BBP083

7. Which of the following is a restriction endonuclease?
(1) Hind II (2) Protease
(3) DNase I (4) RNase

BBP084

NEET-II 2016

8. Stirred-tank bioreactors have been designed for :-
(1) availability of oxygen throughout the process
(2) ensuring anaerobic conditions in the culture vessel
(3) purification of product
(4) addition of preservatives to the product

BBP085

9. A foreign DNA and plasmid cut by the same restriction endonuclease can be joined to form a recombinant plasmid using :-
(1) Polymerase-III (2) Ligase
(3) *Eco* RI (4) *Taq* polymerase

BBP086

10. Which of the following is **not** a component of downstream processing ?
(1) Preservation
(2) Expression
(3) Separation
(4) Purification

BBP087

11. Which of the following restriction enzymes produces blunt ends ?
(1) *Xho* I (2) *Hind* III
(3) *Sal* I (4) *EcoR* V

BBP088

NEET (UG) 2017

12. The DNA fragments separated on an agarose gel can be visualised after staining with :-
(1) Acetocarmine
(2) Aniline blue
(3) Ethidium bromide
(4) Bromophenol blue

BBP089

13. The process of separation and purification of expressed protein before marketing is called:
(1) Downstream processing
(2) Bioprocessing
(3) Postproduction processing
(4) Upstream processing

BBP090

14. A gene whose expression helps to identify transformed cell is known as :-
(1) Vector
(2) Plasmid
(3) Structural gene
(4) Selectable marker

BBP091

NEET (UG) 2018

15. Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes ?
(1) Retrovirus (2) Ti plasmid
(3) λ phage (4) pBR322

BBP092

16. The correct order of steps in Polymerase Chain Reaction (PCR) is :-
(1) Extension, Denaturation, Annealing
(2) Annealing, Extension, Denaturation
(3) Denaturation, Extension, Annealing
(4) Denaturation, Annealing, Extension

BBP093

NEET (UG) 2019

17. DNA precipitation out of a mixture of biomolecules can be achieved by treatment with :-
(1) Isopropanol
(2) Chilled ethanol
(3) Methanol at room temperature
(4) Chilled chloroform

BBP094

18. Following statements describe the characteristics of the enzyme Restriction endonuclease. Identify the **incorrect** statement:
(1) The enzyme cuts DNA molecule at identified position within the DNA
(2) The enzyme binds DNA at specific sites and cuts only one of the two strands.
(3) The enzyme cuts the sugar-phosphate backbone at specific sites on each strand.
(4) The enzyme recognizes a specific palindromic nucleotide sequence in the DNA

BBP095

NEET (UG) 2019 (Odisha)

19. Match the following enzymes with their functions:-

(a) Restriction endonuclease	(i) Joins the DNA fragements
(b) Restriction exonuclease	(ii) Extends primers on genomic DNA template
(c) DNA ligase	(iii) Cuts DNA at specific position
(d) Taq polymerase	(iv) Removes nucleotides from the ends of DNA

Select the correct option from the following: -

(1) a-iii, b-i, c-iv d-ii

(2) a-iii, b-iv, c-i, d-ii

(3) a-iv, b-iii, c-i, d-ii

(4) a-ii, b-iv, c-i, d-iii

BBP096

20. The two antibiotic resistance genes on vector pBR322 are :-

(1) Ampicillin and Tetracycline

(2) Ampicillin and Chloramphenicol

(3) Chloramphenicol and Tetracycline

(4) Tetracycline and Kanamycin

BBP097

21. A selectable marker is used to: -

(1) help in eliminating the non-transformants, so that the transformants can be regenerated

(2) identify the gene for a desired trait in an alien organism

(3) select a suitable vector for transformation in a specific crop

(4) mark a gene on a chromosome for isolation using restriction enzyme

BBP098

22. Given below are four statements pertaining to separation of DNA fragments using gel electrophoresis. Identify the incorrect statements: -

(a) DNA is negatively charged molecule and so it is loaded on gel towards the Anode terminal

(b) DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.

(c) Smaller the size of DNA fragment larger is the distance it travels through it.

(d) Pure DNA can be visualized directly by exposing UV radiation.

Choose correct answer from the options given below

(1) (a), (c) and (d)

(2) (a), (b) and (c)

(3) (b), (c) and (d)

(4) (a), (b) and (d)

BBP099

23. An enzyme catalysing the removal of nucleotides from ends of DNA is:-

(1) DNA ligase

(2) Endonuclease

(3) Exonuclease

(4) Protease

BBP100

NEET (UG) 2020

24. The specific palindromic sequence which is recognized by EcoR I is :-

(1) 5' - GGATCC - 3' , 3' - CCTAGG - 5'

(2) 5' - GAATTC - 3' , 3' - CTTAAG - 5'

(3) 5' - GGAACC - 3' , 3' - CCTTGG - 5'

(4) 5' - CTTAAG - 3', 3' - GAATTC - 5'

BBP101

25. Choose the **correct** pair from the following:-
(1) **Exonucleases:** - Make cuts at specific positions within DNA
(2) **Ligases :-** Join the two DNA molecules
(3) **Polymerases:** Break the DNA into fragments
(4) **Nucleases :** Separate the two strands of DNA

BBP102

26. Identify the **wrong** statement with regard to Restriction Enzymes.
(1) Sticky ends can be joined by using DNA ligases.
(2) Each restriction enzyme functions by inspecting the length of a DNA sequence.
(3) They cut the strand of DNA at palindromic sites.
(4) They are useful in genetic engineering.

BBP103

27. The sequence that controls the copy number of the linked DNA in the vector, is termed :-
(1) Recognition site
(2) Selectable marker
(3) Ori site
(4) Palindromic sequence

BBP104

NEET (UG) 2020 (COVID-19)

28. First discovered restriction endonuclease that always cuts DNA molecule at a particular point by recognising a specific sequence of six base pairs is:-
(1) EcoRI
(2) Adenosine deaminase
(3) Thermostable DNA polymerase
(4) Hind II

BBP105

29. In Recombinant DNA technology antibiotics are used :-
(1) to keep medium bacteria-free
(2) to detect alien DNA
(3) to impart disease-resistance to the host plant
(4) as selectable markers

BBP106

30. Match the following techniques or instruments with their usage :-

(a) Bioreactor	(i) Separation of DNA fragments
(b) Electrophoresis	(ii) Production of large quantities of products
(c) PCR	(iii) Detection of pathogen, based on antigen – antibody reaction
(d) ELISA	(iv) Amplification of nucleic acids

Select the correct option from following:-
(1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

BBP107

31. In a mixture, DNA fragments are separated by:
(1) Bioprocess engineering
(2) Restriction digestion
(3) Electrophoresis
(4) Polymerase chain reaction

BBP108

32. Spooling is :-
(1) Amplification of DNA
(2) Cutting of separated DNA bands from the agarose gel
(3) Transfer of separated DNA fragments to synthetic membranes
(4) Collection of isolated DNA

BBP109

33. Select the correct statement from the following :-

(1) Gel electrophoresis is used for amplification of a DNA segment.

(2) The polymerase enzyme joins the gene of interest and the vector DNA.

(3) Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzymes of optimum conditions.

(4) PCR is used for isolation and separation of gene of interest.

BBP110

NEET (UG) 2021

34. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out:-

(1) RNA (2) DNA

(3) Histones (4) Polysaccharides

BBP111

35. Which of the following is a **correct** sequence of steps in a PCR (Polymerase Chain Reaction)?

(1) Denaturation, Annealing, Extension

(2) Denaturation, Extension, Annealing

(3) Extension, Denaturation, Annealing

(4) Annealing, Denaturation, Extension

BBP112

36. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as :-

(1) Yellow bands

(2) Bright orange bands

(3) Dark red bands

(4) Bright blue bands

BBP113

37. Plasmid pBR322 has PstI restriction enzyme site within gene amp^R that confers ampicillin resistance. If this enzyme is used for inserting a gene for β-galactoside production and the recombinant plasmid is inserted in an *E. coli* strain.

(1) it will not be able to confer ampicillin resistance to the host cell.

(2) the transformed cells will have the ability to resist ampicillin as well as produce β-galactoside.

(3) it will lead to lysis of host cell.

(4) it will be able to produce a novel protein with dual ability.

BBP114

38. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :-

(1) Degenerate primer sequence

(2) Okazaki sequences

(3) Palindromic Nucleotide sequences

(4) Poly(A) tail sequences

BBP115

39. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first?

(1) Annealing

(2) Extension

(3) Denaturation

(4) Ligation

BBP116

NEET (UG) 2022

40. Which one of the following statement is **not true** regarding gel electrophoresis technique?

(1) The separated DNA fragments are stained by using ethidium bromide.

(2) The presence of chromogenic substrate gives blue coloured DNA bands on the gel.

(3) Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.

(4) The process of extraction of separated DNA strands from gel is called elution.

BBP117

41. Given below are two statements :-

one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Polymerase chain reaction is used in DNA amplification

Reason (R) : The ampicillin resistant gene is used as a selectable marker to check transformation.

In the light of the above statements, choose the **correct** answer from the options given below:-

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is not correct but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

BBP118

42. In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme ?

(1) 5' G A A T T C 3'; 3' C T T A A G 5'

(2) 5' C T C A G T 3'; 3' G A G T C A 5'

(3) 5' G T A T T C 3'; 3' C A T A A G 5'

(4) 5' G A T A C T 3'; 3' C T A T G A 5'

BBP119

43. Given below are two statements:-

Statement I : Restriction endonucleases recognise specific sequence to cut DNA known as palindromic nucleotide sequence.

Statement II : Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.

In the light of the above statements, choose the **most appropriate** answer from the options given below:-

(1) Both **Statement I** and **Statement II** are incorrect

(2) **Statement I** is correct but **Statement II** is incorrect

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct

BBP120

44. Which of the following is **not** a desirable feature of a cloning vector ?

(1) Presence of a marker gene

(2) Presence of single restriction enzyme site

(3) Presence of two or more recognition sites

(4) Presence of origin of replication

BBP121

Re-NEET 2022

45. Separation of DNA, fragments is done by a technique known as :-

(1) Polymerase Chain Reaction

(2) Recombinant technology

(3) Southern blotting

(4) Gel electrophoresis

BBP122

46. Match **List-I** with **List-II** :-

	List-I		List-II
(a)	Gene gun	(i)	Replacement of a faulty gene by a normal healthy gene
(b)	Gene therapy	(ii)	Used for transfer of gene
(c)	Gene cloning	(iii)	Total DNA in the cells of an organism
(d)	Genome	(iv)	To obtain identical copies of a particular DNA molecule

Choose the **correct answer** from the options given below :-

(1) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

(2) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

(3) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

(4) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

BBP123

47. Pathogenic bacteria gain resistance to antibiotics due to changes in their :-

(1) Cosmids

(2) Plasmids

(3) Nucleus

(4) Nucleoid

BBP124

48. Which of the following methods is not commonly used for introducing foreign DNA into the plant cell?

(1) *Agrobacterium* mediated transformation

(2) Gene gun

(3) 'Disarmed pathogen' vectors

(4) Bacteriophages

BBP125

49. Refer to the following statements for agarose-gel electrophoresis :-

(a) Agarose is a natural polymer obtained from sea weed.

(b) The separation of DNA molecules in agarose-gel electrophoresis depends on the size of DNA.

(c) The DNA migrates from negatively-charged electrode to the positively-charged electrode

(d) The DNA migrates from positively-charged electrode to the negatively-charged electrode.

Choose **the most appropriate answer** from the options given below :-

(1) (a) and (b) only

(2) (a), (b) and (c) only

(3) (a), (b) and (d) only

(4) (b), (c) and (d) only

BBP126

NEET (UG) 2023

50. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

(1) DNA (2) Histones

(3) Polysaccharides (4) RNA

BBP127

51. In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.

(1) Zinc (2) Tungsten or gold

(3) Silver (4) Copper

BBP128

52. Upon exposure to UV radiation, DNA stained with ethidium bromide will show

(1) Bright blue colour

(2) Bright yellow colour

(3) Bright orange colour

(4) Bright red colour

BBP129

53. Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.

A. Insertion of recombinant DNA into the host cell.

B. Cutting of DNA at specific location by restriction enzyme.

C. Isolation of desired DNA fragment.

D. Amplification of gene of interest using PCR.

Choose the **correct** answer from the options given below:

(1) C, A, B, D (2) C, B, D, A

(3) B, D, A, C (4) B, C, D, A

BBP130

54. Which of the following is not a cloning vector?

(1) YAC (2) pBR322

(3) Probe (4) BAC

BBP131

NEET (UG) 2023 (Manipur)

55. Thermostable DNA polymerase used in PCR was isolated from :

(1) *Thermus aquaticus*

(2) *Escherichia coli*

(3) *Agrobacterium tumifaciens*

(4) *Bacillus thuringiensis*

BBP132

56. Ligation of foreign DNA at which of the following site will result in loss of tetracycline resistance of pBR322 :

(1) Pst I (2) Pvu I

(3) EcoR I (4) BamH I

BBP133

57. Which of the following statement is **incorrect** about *Agrobacterium tumifaciens*?

(1) It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.

(2) 'Ti' plasmid from *Agrobacterium tumifaciens* used for gene transfer is not pathogenic to plant cell.

(3) It transforms normal plant cells into tumor cells.

(4) It delivers 'T-DNA' into plant cell.

BBP134

58. Match **List-I** with **List-II.**

	List-I		**List-II**
(A)	Kanamycin	(I)	Delivers genes into animal cells
(B)	Cla I	(II)	Selectable marker
(C)	Disarmed retroviruses	(III)	Restriction site
(D)	Kanamycin^R gene	(IV)	Antibiotic resistance

Choose the **correct** answer from the options given below :

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

BBP135

59. Which of the following can act as molecular scissors?

(1) Restriction enzymes

(2) DNA ligase

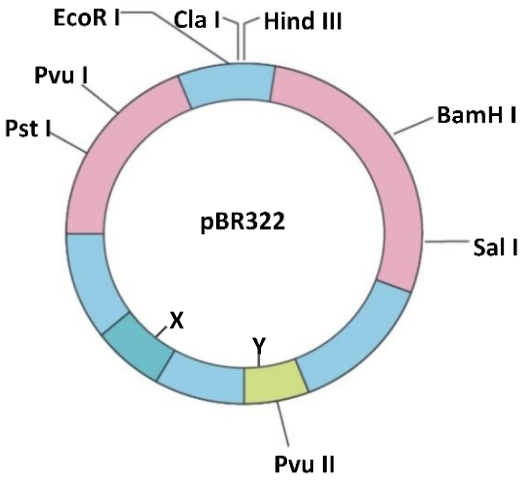
(3) RNA polymerase

(4) DNA polymerase

BBP136

NEET (UG) 2024

60. The following diagram showing restriction sites in *E.coli* cloning vector pBR322. Find the role of '*X*' and '*Y*' genes.



(1) The gene '*X*' is responsible for resistance to antibiotics and '*Y*' for protein involved in the replication of Plasmid.

(2) The gene '*X*' is responsible for controlling the copy number of the linked DNA and '*Y*' for protein involved in the replication of Plasmid.

(3) The gene '*X*' is for protein involved in replication of Plasmid and '*Y*' for resistance to antibiotics.

(4) Gene '*X*' is responsible for recognition sites and '*Y*' is responsible for antibiotic resistance.

BBP137

61. The "Ti plasmid" of *Agrobacterium tumefaciens* stands for :-

(1) Tumour inhibiting plasmid

(2) Tumor independent plasmid

(3) Tumor inducing plasmid

(4) Temperature independent plasmid

BBP138

62. Which of the following statements is incorrect?

(1) A bio-reactor provides optimal growth conditions for achieving the desired product.

(2) Most commonly used bio-reactors are of stirring type.

(3) Bio-reactors are used to produce small scale bacterial cultures.

(4) Bio- reactors have an agitator system, an oxygen delivery system and foam control system.

BBP139

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	2	4	1	1	1	2	2	4	3	1	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	2	2	1	1	4	3	2	2	1	3	4	4	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	3	2	1	2	1	3	3	2	1	1	4	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	2	4	2	1	2	3	4	3	1	4	1	3	1	2
Question	61	62													
Answer	3	3													

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. The science, which deals with techniques of using live organisms or enzymes from organism to produce products and processes useful to human is :-
(1) Genetics (2) Biotechnology
(3) Bioinformatics (4) None of these
BBP140

2. The first discovered restriction endonuclease which always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs is :-
(1) Hind II (2) Not I
(3) Hae III (4) All of these
BBP141

3. The first letter of the name of Restriction endonuclease came from the:-
(1) Genus of organism
(2) Species of organism
(3) Family of organism
(4) Class of organism
BBP142

4. Autonomously replicating circular extra chromosomal DNA of bacteria is :-
(1) Plastid (2) Nucleus
(3) Plasmid (4) None of these
BBP143

5. The specific DNA sequence in a chromosome which is responsible for initiation of replication is :-
(1) Cloning region
(2) Termination region
(3) Initiation region
(4) Origin of replication **BBP144**

6. Which of the following enzymes is known as 'genetic glue'?
(1) DNA polymerase
(2) Alkaline phosphatase
(3) DNA ligase
(4) All of the above
BBP145

7. pBR322 has two antibiotic resistance genes, they are :-
(1) Streptomycin and Ampicillin resistant gene
(2) Chloromycetin and tetracycline resistant gene
(3) Tetracycline and neomycin resistant genes
(4) Ampicillin and tetracycline resistant genes
BBP146

8. To isolate DNA from the plant cells we have to break the wall this is done by :-
(1) Lysozyme (2) Cellulase
(3) Chitinase (4) Invertase
BBP147

9. Restriction enzymes belong to a larger class of enzymes called :-
(1) Cellulases (2) Hydrolases
(3) Polymerases (4) Nucleases
BBP148

10. The construction of the first recombinant DNA was done by ?
(1) Stanley Cohen and Herbert Boyer
(2) Nathans and Smith
(3) Meselson and Stahl
(4) Alec Jeffreys
BBP149

11. EcoR I recognises palindromic sequence:-

(1) 5'-GGGCCC-3'
3'-CCCGGG-5'

(2) 5'-GAATTC-3'
3'-CTTAAG-5'

(3) 5'-AAGCTT-3'
3'-TTCGAA-5'

(4) None of the above

BBP150

12. The enzymes responsible for restricting the growth of bacteriophage in *E coli* were isolated in 1963, these enzyme are :-

(1) DNA ligases

(2) Alkaline phosphatases

(3) DNA polymerases

(4) Restriction endonuclease

BBP151

13. Vector which is commonly used to transfer foreign gene in a crop plant is :-

(1) Plasmids of *Salmonella*

(2) λ bacteriophage vector

(3) Ti plasmid of *Agrobacterium tumefaciens*

(4) None of the above

BBP152

14. Father of genetic engineering is :-

(1) Paul Berg

(2) Nathans

(3) Herbert Boyer

(4) Stanley Cohen

BBP153

15. A definition of biotechnology that encompasses both traditional view and modern view was given by :-

(1) European forum on Biotechnology

(2) European focus on Biotechnology

(3) European Federation of Biotechnology

(4) European Centre of Biotechnology

BBP154

16. Which one of the following is must in Genetic engineering?

(1) Restriction endonuclease + DNA ligase

(2) Restriction exonuclease + DNA polymerase

(3) Alkaline phosphate + DNA Ligase

(4) RNA polymerase + DNA polymerase

BBP155

17. Roman numbers following the names of restriction endonuclease indicate :-

(1) The order in which the enzymes were isolated from that strain of bacteria

(2) strain of bacteria

(3) the order in which genus is taken to isolate the enzyme

(4) None of the above

BBP156

18. Exonuclease removes nucleotides from:-

(1) Specific positions

(2) the ends of the DNA

(3) any where in DNA

(4) All the above

BBP157

19. In a chromosome there is a specific DNA sequence which is responsible for initiating replication is :-

(1) Ori

(2) Palindromic sequence

(2) Initiation sequence

(4) Promoter sequence

BBP158

20. First recombinant DNA was made by Stanley Cohen and Herbert Boyer in :-

(1) 1968 (2) 1970 (3) 1972 (4) 1974

BBP159

21. The first restriction endonuclease discovered was:-

(1) EcoR I (2) Sam I

(3) BamH I (4) Hind II

BBP160

22. In the vector pBR322 there is/are :-
(1) One selectable marker
(2) Two selectable markers
(3) Three selectable markers
(4) None of the above

BBP161

23. When the isolation of genetic material is done the RNA can be removed by treatment with :-
(1) Protease
(2) Chitinase
(3) Ribonuclease
(4) Deoxyribonuclease

BBP162

24. Knife of DNA :-
(1) DNA - ligase
(2) Restriction endonuclease
(3) Exonuclease
(4) Peptidase

BBP163

25. The enzymes, which remove nucleotides from the ends of the DNA are :-
(1) Exonuclease (2) Endonuclease
(3) Cellulase (4) Hydrolase

BBP164

26. Group of letters that form the same words when read both forward and backward is called:
(1) Palindrome (2) Same words
(3) Opposite words (4) None of the above

BBP165

27. Which type of ends are produced by EcoR I?
(1) Blunt ends
(2) Sticky ends
(3) Both (1) and (2)
(4) None of the above

BBP166

28. The sequence which is responsible for controlling the copy number of the linked DNA is :-
(1) Coding sequence
(2) Promoter sequence
(3) Terminator sequence
(4) Ori

BBP167

29. Apart from DNA in the bacterial nucleoid, there is a circular extrachromosomal DNA in a bacterial cell called :-
(1) Plasmid (2) Mesosome
(3) Chromosome (4) None of these

BBP168

30. The stickiness of the ends, facilitates the action of enzyme :-
(1) DNA ligase
(2) DNA polymerase
(3) Alkaline phosphatase
(4) All of the above

BBP169

31. Two enzymes responsible for restricting the growth of bacteriophage in *E. coli* were isolated in 1963, one of these cut DNA, while other :-
(1) Add propyl group to DNA
(2) Add ethyl group to DNA
(3) Add methyl group to DNA
(4) None of the above

BBP170

32. The linking of antibiotic resistant gene in the plasmid vector become possible with the enzyme :
(1) Reverse transcriptase
(2) DNA ligase
(3) DNA polymerase
(4) RNA polymerase

BBP171

33. If any protein encoding gene is expressed in a heterologous host then the protein is known as :-
(1) Recombinant gene
(2) Recombinant protein
(3) Selectable marker
(4) Homogenous protein

BBP172

34. The normal *E coli* cell carries resistance gene against:-
(1) Ampicillin
(2) Chloramphenicol
(3) Tetracycline
(4) None of the above

BBP173

35. Taq-polymerase is used in, polymerase chain reaction, because :-
(1) It becomes inactive at high temperature
(2) It makes other enzyme active at high temperature
(3) It remains active during high temperature
(4) It is obtained from thermostable virus.

BBP174

36. Small chemically synthesised oligonucleotides that are complementary to the regions of DNA at 3' end used in PCR are :-
(1) Primers (2) Dimers
(3) Small strands (4) Large fragments

BBP175

37. Bombardment of high velocity micro-particles of gold or tungsten coated with DNA on target cells is :-
(1) Biolistic (2) Micro-injection
(3) Electroporation (4) Bombing

BBP176

38. In micro injection :-
(1) DNA is bombarded on target cells
(2) DNA is placed through a vector
(3) DNA is directly injected into the nucleus of animal cell
(4) None of the above

BBP177

39. Most common matrix is agarose a natural polymer used in gel electrophoresis is extracted from :-
(1) an animal (2) a fungus
(3) Sea weeds (4) None of these

BBP178

40. *Agrobacterium tumefaciens* a pathogen transform normal plant cells into a tumor, similarly in animals the normal cells transformed into cancerous cells by:-
(1) Retroviruses (2) DNA viruses
(3) Ribo viruses (4) None of these

BBP179

41. Insertional inactivation results into inactivation of which enzyme ?
(1) Transacetylase (2) Permease
(3) Taq-polymerase (4) β-galactosidase

BBP180

42. If the bacterium does not have any insert, then in the presence of chromogenic substrate, it gives :-
(1) Red coloured colonies
(2) Colourless colonies
(3) Blue colonies
(4) Green colonies

BBP181

43. To make cell competent to take up DNA, heat shock is given to cells, the temperature of shock is :-
(1) 30°C (2) 42°C (3) 60°C (4) 90°C

BBP182

44. In gel electrophoresis technique the DNA fragments are forced to move through a medium towards :-
(1) Anode (2) Cathode
(3) Both (1) and (2) (4) None of the above

BBP183

45. Which one is not a basic step in genetically modifying an organism?
(1) Identification of DNA with desirable genes
(2) Introduction of the identified DNA into the host
(3) Introduction of unidentified DNA into the host
(4) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

BBP184

46. The most commonly used bioreactors are of:-
(1) Simple stirring type
(2) Sparged stirring type
(3) Both (1) and (2)
(4) None of the above

BBP185

47. Downstream processing is :-
(1) Process of separation of DNA fragments
(2) Process of joining the vector and the host DNA
(3) Process including separation and purification of the product
(4) Process of transferring DNA.

BBP186

48. Taq-polymerase is obtained from :-
(1) *Bacillus thuriengiensis*
(2) *Thermus aquaticus*
(3) *Salmonella typhimurium*
(4) *Escherichia coli*

BBP187

49. To denature the DNA template in PCR it is heated to
(1) 70°C (2) 54°C
(3) 80°C (4) 94°C

BBP188

50. Alternative selectable markers developed to differentiate non-recombinants from recombinants on the basis of :-
(1) Ability of separate them according to size.
(2) Ability to produce colour in the presence of a chromogenic substrate.
(3) Ability to kill the host.
(4) None of the above

BBP189

51. If DNA is inserted within the coding sequence of β-galactosidase enzyme then
(1) Non-recombinants will give blue coloured colonies in presence of chromogenic substrate
(2) Recombinant will give blue coloured colonies in presence of chromogenic substrate
(3) Both recombinants and non-recombinants give blue colour
(4) Non-recombinants do not produce colour due to insertional inactivation.

BBP190

52. In gel electrophoresis the DNA fragments separate according to size (smaller the fragment size, the faster it moves) this effect is called :-
(1) Sieving effect
(2) Movement effect
(3) Density effect
(4) Spooling

BBP191

53. Extraction, purification and packaging of products is collectively known as :-
(1) Upstream processing
(2) Distillation
(3) Downstream processing
(4) Genetic engineering

BBP192

54. You have three copies of a particular DNA molecule what technique would you use to make more copies of the molecule?
(1) Gel electrophoresis
(2) Sequencing
(3) PCR
(4) Restriction fragment analysis

BBP193

55. Ti plasmid is present in :-
(1) *E.coli*
(2) *Agrobacterium tumefaciens*
(3) *Pseudomonas*
(4) *Vibrio cholerae*

BBP194

56. DNA cannot pass through cell membrane as it is :-
(1) hydrophilic (2) hydrophobic
(3) lipophilic (4) All the above

BBP195

57. Which type of bioreactor is usually cylindrical or with a curved base to facilitate the mixing of the contents?
(1) Inverted tank bioreactor
(2) Stirred tank bioreactor
(3) Both (1) and (2)
(4) None of the above

BBP196

58. In PCR-technology primer is a :-
(1) Small chemically synthesized oligonucleotide that are complementary to region of DNA
(2) Large chemically synthesized polynucleotide that are identical to region of DNA
(3) Small segment of dsRNA
(4) None of these

BBP197

59. In gel electrophoresis the DNA fragment separate according to their size through sieving effect, which is provided by :-
(1) Agarose gel
(2) Nylon membrane
(3) Polyethylene glycol
(4) Ethidium Bromide

BBP198

60. Which of the following method of vector less gene transfer is suitable for plants ?
(1) Biolistic method
(2) Micro injection
(3) Liposome mediated
(4) Electroporation

BBP199

61. In gel electrophoresis, separated bands of DNA are cut out from the agarose gel and extracted from the gel pieces.
This step is known as :-
(1) Blotting (2) Elution
(3) Cloning (4) Tagging

BBP200

62. Which enzyme is used in PCR technique ?
(1) Thermostable DNA polymerase
(2) Thermostable RNA polymerase
(3) Thermostable ligase
(4) Thermostable vector

BBP201

63. Which of the following is used to deliver desirable gene in to animal cell?
(1) Disarmed Retrovirus
(2) Disarmed *Agrobacterium*
(3) Disarmed *E. coli*
(4) Disarmed plant pathogen

BBP202

64. *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA and it is known as :-
(1) R-DNA (2) S-DNA
(3) M-DNA (4) T-DNA

BBP203

65. When cut by the restriction enzyme, the DNA fragments can be joined together using :-

(1) DNA polymerase

(2) DNA ligase

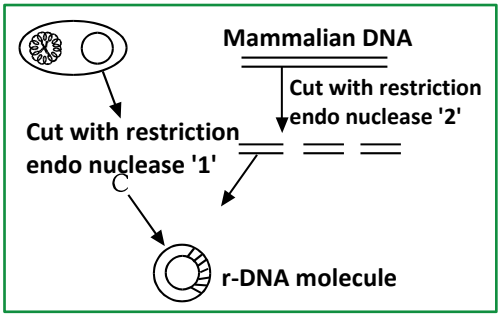
(3) Alkaline phosphatase

(4) DNA gyrase

BBP204

(B) ANALYTICAL QUESTIONS

66. The basic procedure involved in the synthesis of recombinant DNA molecule is depicted below. The mistake in the procedure is :-



(1) Enzyme polymerase is not included

(2) The mammalian DNA is shown double stranded

(3) Only one fragment is inserted

(4) Two different restriction enzymes are used

BBP205

67. If a recombinant DNA bearing gene for ampicillin resistance if transferred into *E. coli* cells and the host cells are spread on agar plates containing ampicillin, then :-

(1) both transformed and untransformed recipient cells will die.

(2) both transformed and untransformed recipient cell will grow.

(3) transformed recipient cells will grow and untransformed recipient cells will die.

(4) transformed recipient cells will die and untransformed recipient cells will grow.

BBP206

68. Which of the following tools of recombinant DNA technology is incorrectly paired with its use ?

(1) Restriction enzyme - Production of RFLPs

(2) DNA ligase-that cuts DNA, creating the sticky ends

(3) DNA polymerase - used in a polymerase chain reaction to amplify section of DNA

(4) reverse transcriptase - production of cDNA from mRNA

BBP207

69. Select the incorrect statement for continuous culture system:

(1) In this system, used medium is drained out from one side while fresh medium is added from other side.

(2) In this system, cells are maintained in their physiologically most active lag phase of growth.

(3) It produces larger biomass.

(4) It shows higher yields of desired product.

BBP208

70. In r-DNA technology or genetic engineering elution means :

(1) Remove the DNA from centrifuge tube after centrifugation

(2) The separated band of DNA are cut out from the gel and extracted from the gel piece

(3) Separation of the recombinant protein from recombinant cell

(4) Insertion of recombinant DNA into the host cell

BBP209

71. Restriction endonucleases are enzymes which :-

(1) restrict the action of the enzyme DNA polymerase

(2) remove nucleotides from the ends of the DNA molecule

(3) make cuts at specific positions within the DNA molecule

(4) recognize a specific nucleotide sequence for binding of DNA ligase

BBP210

72. Which one of the following palindromic base sequences in DNA can be easily cut at about the middle by some particular restriction enzyme?

(1) 5'_____GAATTC _____3'
3'_____ CTTAAG _____5'

(2) 5'_____CACGTA _____3'
3'_____ CTCAGT_____5'

(3) 5'_____CGTTCG _____3'
3'_____ ATGGTA _____5'

(4) 5'_____GATATG _____3'
3'_____ CTACTA _____5'

BBP211

73. Given below is a sample of a portion of DNA strand giving the base sequence on the opposite strands. What is so special shown in it?

5'_____GAATTC_____3'
3'_____CTTAAG_____5'

(1) Replication completed

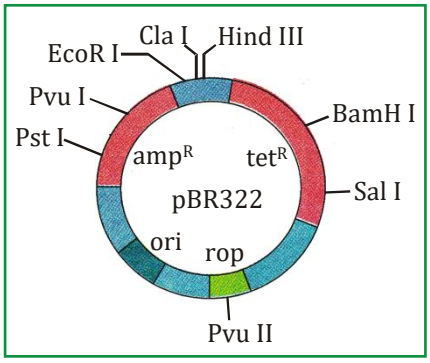
(2) Deletion mutation

(3) Start codon at the 5' end

(4) Palindromic sequence of base pairs

BBP212

74. The figure below is the diagrammatic representation of the *E. coli* vector pBR322. Which one of the given options correctly identifies its certain component(s)?



(1) Hind III, EcoR I-selectable markers

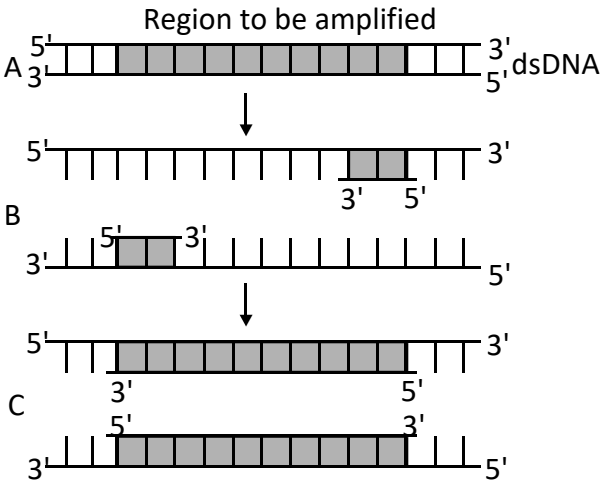
(2) amp^R, tet^R-antibiotic resistance genes

(3) ori-original restriction enzyme

(4) rop-reduced osmotic pressure

BBP213

75. The figure below shows three steps (A, B, C) of Polymerase Chain Reaction (PCR). Select the option giving correct identification together with what it represents?



(1) C-Extension in the presence of heat stable DNA polymerase

(2) A-Annealing with two sets of primers

(3) B-Denaturation at a temperature of about 98°C separating the two DNA strands

(4) A-Denaturation at a temperature of about 50°C

BBP214

76. The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of :-

(1) Inactivation of glycosidase enzyme in recombinant bacteria

(2) Non-recombinant bacteria containing beta-galactosidase

(3) Insertional inactivation of alpha-galactosidase in non-recombinant bacteria

(4) Insertional inactivation of beta-galactosidase in recombinant bacteria

BBP215

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	3	4	3	4	2	4	1	2	4	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	2	1	3	4	2	3	2	1	1	2	4	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	2	4	3	1	1	3	3	1	4	3	2	1	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	2	4	2	1	1	3	3	2	1	2	1	1	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	1	1	4	2	4	3	2	2	2	3	1	4	2	1
Question	76														
Answer	4														

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

1. A genetically manipulated organism containing in its genome one or more inserted gene of another species is called: -
(1) Transposon
(2) Gene expression
(3) Transgenic organism
(4) Retroposons

BBN001

2. A piece of nucleic acid using to find out a gene, by forming hybrid with it, is called as:-
(1) c – DNA (2) DNA probe
(3) Sticky end (4) Blunt end

BBN002

3. 'Transgenic' plants are produced by :-
(1) Inducing gene mutation
(2) Arresting spindle fibre formation
(3) Deleting sex chromosomes
(4) Introducing foreign genes

BBN003

4. Genetic modification (GM) has been used to:-
(1) Create tailor made plants
(2) Supply alternative resources to industries
(3) Enhanced nutritional value of food.
(4) All of the above

BBN004

5. The choice of Bt-gene for experiment depends upon:-
(1) The host plant/crop
(2) Targeted pest/insect
(3) *Bacillus* strain
(4) Both (1) & (2)

BBN005

6. In nematode resistance by RNA interference, some specific genes were introduced which form dsRNA. These were introduced in:-
(1) Nematode
(2) Host plant
(3) *Agrobacterium*
(4) All of these

BBN006

7. Select the incorrect match:-
(1) Transgenic mice - Polio vaccine
(2) Rosie cow - α lactalbumin gene
(3) ssDNA/RNA probe - Gene therapy
(4) PCR - Molecular diagnosis

BBN007

8. Genetically engineered bacteria have been used in commercial production of :-
(1) Thyroxine (2) Testosterone
(3) Human insulin (4) Melatonium

BBN008

9. Important objective of biotechnology in agriculture section is :-
(1) To produce pest resistant varieties of plants
(2) To increase the nitrogen content
(3) To decrease the seed number
(4) To increase the plant weight

BBN009

10. The name of drug used in cancer treatment produced by biotechnology is:-
(1) Interferon
(2) [HGH] Human growth hormone
(3) TSH
(4) Insulin

BBN010

11. The prerequisites for biotechnological production of antibiotics is :-
(1) To search an antibiotic producing microorganism
(2) To isolate the antibiotic gene
(3) To join antibiotic gene with *E. coli* plasmid
(4) All of the above

BBN011

12. Modern biotechnology consist :-
(1) Genetic engineering
(2) tissue culture
(3) Microbiology
(4) All the above

BBN012

13. First artificially synthesized hormone is :-
(1) Secretin (2) Insulin
(3) Glucagon (4) Renin

BBN013

14. The protein products of the following Bt toxin genes *cryIAc* and *cryIIAb* are responsible for controlling:-
(1) Bollworm (2) Roundworm
(3) Moth (4) Fruit fly

BBN014

15. A transgenic rice (Golden rice) has been developed for increased content of :-
(1) Vitamin A (2) Vitamin B_1
(3) Vitamin C (4) Vitamin D

BBN015

16. During the processing of the prohormone "proinsulin" into the mature "insulin" :-
(1) C-peptide is added to proinsulin
(2) C-peptide is removed from proinsulin
(3) B-peptide is added to proinsulin
(4) B-peptide is removed from proinsulin

BBN016

17. A genetically engineered bacteria used for clearing oil spills is :-
(1) *Escherischia coli*
(2) *Bacillus subtilis*
(3) *Agrobacterium tumifaciens*
(4) *Pseudomonas putida*

BBN017

18. First transgenic plant :-
(1) Potato (2) Tomato
(3) Tobacco (4) Maize

BBN018

19. *E. coli* are used in production of :-
(1) Rifampicin (2) LH
(3) Ecdyson (4) Interferon

BBN019

20. A giant rat is formed in the laboratory, what is the reason :-
(1) Gene mutation (2) Gene synthesis
(3) Gene manipulation (4) Gene replication

BBN020

21. Cultivation of Bt cotton has been much in the news. The prefix ‘‘Bt’’ means :-
(1) ‘‘Barium – treated’’ cotton seeds.
(2) ‘‘Bigger thread’’ variety of cotton with batter tensile strength.
(3) Produced by ‘‘biotechnology’’ using restriction enzymes and ligases.
(4) Carrying an endotoxin gene from *Bacillus thuringiensis*.

BBN021

22. *Cry*-gene which synthesize crystal protein isolated from :-
(1) *Bacillus thuriengensis*
(2) *Rhizobium*
(3) *Bacillus polymyxa*
(4) *Clostridium*

BBN022

23. Which of the following combination of risk are associated with genetically modified food?
(1) Toxicity
(2) Allergic reaction
(3) Antibiotic resistance in microorganism present in alimentary canal
(4) All the above
BBN023

24. Gene therapy first used in the treatment of:-
(1) Albinism (2) Haemophilia
(3) SCID (4) LIQID
BBN024

25. Bt–cotton is resistant for :-
(1) Round–worm (2) Fluke–worm
(3) Boll–worm (4) Pin–worm
BBN025

26. Genetically engineered human insulin is called :-
(1) Humulin (2) Haematin
(3) Hybridoma (4) Hybrid
BBN026

27. The C-peptide is :-
(1) not present in proinsulin
(2) present in mature insulin
(3) removed during maturation of insulin
(4) also present in artificial insulin
BBN027

28. GEAC makes decisions regarding :-
(1) the validity of GM research
(2) the safety of introducing GM organisms for public services
(3) the validity of biopatents
(4) more than one options are correct
BBN028

29. The Indian parliament has recently cleared which amendment of the Indian patents bill:-
(1) 1st (2) 2nd (3) 3rd (4) 4th
BBN029

30. The use of bio-resources by multinational companies & other organisations without proper authorisation from the countries & people concerned, is known as :–
(1) Biopatent (2) Biopiracy
(3) Biowar (4) Biodiversity
BBN030

31. Which of the following peptide chain in not present in mature insulin?
(1) A-peptide (2) B-peptide
(3) C-peptide (4) A & B peptide
BBN031

32. Golden rice is a promising transgenic crop. When released for cultivation, it will help in:-
(1) Alleviation of vitamin A deficiency
(2) Pest resistance
(3) Herbicide tolerance
(4) Producing a petrol-like fuel from rice
BBN032

33. A genetically engineered micro-organism used successfully in bioremediation of oil spills is a species of :-
(1) *Pseudomonas* (2) *Trichoderma*
(3) *Xanthomonas* (4) *Bacillus*
BBN033

34. Cry1 endotoxins obtained from *Bacillus thuringiensis* are effective against :-
(1) Flies (2) Nematodes
(3) Boll worms (4) Mosquitoes
BBN034

35. Human insulin is being commercially produced from a transgenic species of :-
(1) *Mycobacterium* (2) *Rhizobium*
(3) *Saccharomyces* (4) *Escherichia*
BBN035

36. A transgenic food crop which may help in solving the problem of night blindness in developing countries is :-
(1) Starlink maize
(2) Bt Soybean
(3) Golden rice
(4) Flavr Savr tomatoes
BBN036

37. The bacterium *Bacillus thuringiensis* is widely used in contemporary biology as :-
(1) Source of industrial enzyme
(2) Indicator of water pollution
(3) Insecticide
(4) Agent for production of dairy products
BBN037

38. The genetically-modified (GM) brinjal in India has been developed for :-
(1) Enhancing mineral content
(2) Drought-resistance
(3) Insect-resistance
(4) Enhancing shelf life
BBN038

39. Maximum number of existing transgenic animals is of :-
(1) Fish (2) Mice (3) Cow (4) Pig
BBN039

40. The process of RNA interference has been used in the development of plants resistant to:
(1) Nematodes (2) Fungi
(3) Viruses (4) Insects
BBN040

41. Silencing of mRNA has been used in producing transgenic plants resistant to:-
(1) Bacterial blights (2) Bollworms
(3) Nematodes (4) White rusts
BBN041

42. Consumption of which one of the following foods can prevent the kind of blindness associated with vitamin 'A' deficiency?
(1) Golden rice (2) Bt-Brinjal
(3) Flaver Savr 'tomato (4) Canolla
BBN042

43. About what percentage of all existing transgenic animals are mice?
(1) 75 (2) 50 (3) 100 (4) 95
BBN043

44. Tobacco plants resistant to a nematode have been developed by the introduction of DNA that produced (in the host cells) :-
(1) an antifeedant
(2) a toxic protein
(3) both sense and anti-sense RNA
(4) a particular hormone
BBN044

45. The first clinical gene therapy was given for treating :-
(1) Rheumatoid arthritis
(2) Adenosine deaminase deficiency
(3) Diabetes mellitus
(4) Chicken pox
BBN045

46. Somaclonal variations appears in :–
(1) Organism produced through somatic hybridization
(2) Plants growing in highly polluted conditions
(3) Apomictic plants
(4) Tissue culture raised plants
BBN046

47. Pomato is :–
(1) Somatic hybrid
(2) Allopolyploid
(3) Natural mutant
(4) Both (1) and (2)

BBN047

48. Plant part, used for culture is called :–
(1) Scion (2) Stock
(3) Explant (4) Callus

BBN048

49. Protoplast fusion causes :–
(1) Rapid growth of offspring
(2) Somatic hybridization
(3) Production of useful allopolyploid
(4) Both (2) and (3)

BBN049

50. Virus free plant can be obtained through :–
(1) Grafting
(2) Callus culture
(3) Shoot tip culture
(4) Suspension culture

BBN050

51. Which of the following hormone is used for shoot differentiation in callus ?
(1) 2, 4–D
(2) Benzyl amino purine (BAP)
(3) Deformylase
(4) Gibberellic acid

BBN051

52. Which one of the following is an example of somatic hybridisation?
(1) Bt cotton
(2) Pomato
(3) Golden rice
(4) All of these

BBN052

53. The process of transferring the cell culture from old medium to fresh culture medium is known as:-
(1) Sterilization
(2) Subculturing
(3) Introduction
(4) Suspension culture

BBN053

54. Tissue culture is beneficial for :-
(1) Micropropagation
(2) Production of disease free plants
(3) Androgenic haploid
(4) All the above

BBN054

55. In order to obtain virus-free plants through tissue culture the best method is :
(1) Protoplast culture
(2) Embryo rescue
(3) Anther culture
(4) Meristem culture

BBN055

56. Somaclones are obtained by :-
(1) Genetic engineering
(2) Tissue culture
(3) Plant breeding
(4) Irradiation

BBN056

57. Which one of the following is a case of wrong matching ?
(1) Micropropagation In-vitro production of plants in large numbers
(2) Callus-Unorganised mass of cells produced in tissue culture
(3) Somatic hybridization - Fusion of two diverse cells
(4) Vector DNA- Site for t-RNA synthesis

BBN057

58. Which part would be most suitable for raising virus-free plants for micropropagation?

(1) Meristem (2) Node

(3) Bark (4) Vascular tissue

BBN058

59. Polyethylene glycol method is used for :-

(1) Energy production from sewage

(2) Gene transfer without a vector

(3) Biodiesel production

(4) Seedless fruit production

BBN059

60. DNA or RNA segment tagged with a radioactive molecule is called:-

(1) Clone (2) Plasmid

(3) Vector (4) Probe

BBN060

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	4	4	2	3	3	1	1	4	4	2	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	3	4	3	4	1	4	3	3	1	3	4	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	1	3	4	3	3	3	2	1	3	1	4	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	3	4	3	2	2	2	4	4	2	4	1	2	4

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. To obtain virus - free healthy plants from a diseased one by tissue culture technique, which part/parts of the diseased plant will be taken :-
(1) Apical meristem only
(2) Palisade parenchyma
(3) Both apical and axillary meristems
(4) Epidermis only

BBN061

2. An analysis of chromosomal DNA using the Southern hybridization technique **does not** use:
(1) Electrophoresis
(2) Blotting
(3) Autoradiography
(4) PCR

BBN062

3. The first human hormone produced by recombinant DNA technology is :-
(1) Insulin (2) Estrogen
(3) Thyroxin (4) Progesterone

BBN063

AIPMT 2015

4. Which body of the Government of India regulates GM research and safety of introducing GM organisms for public services?
(1) Indian Council of Agricultural Research.
(2) Genetic Engineering Approval Committee.
(3) Research Committee on Genetic Manipulation.
(4) Bio-safety committee.

BBN064

5. The crops engineered for glyphosate are resistant/tolerant to :-
(1) Bacteria (2) Insects
(3) Herbicides (4) Fungi

BBN065

6. In Bt cotton, the Bt toxin present in plant tissue as pro-toxin is converted into active toxin due to:-
(1) Acidic pH of the insect gut
(2) Action of gut micro-organisms
(3) Presence of conversion factors in insect gut
(4) Alkaline pH of the insect gut

BBN066

7. A technique of micropropagation is :-
(1) Somatic embryogenesis
(2) Protoplast fusion
(3) Embryo rescue
(4) Somatic hybridization

BBN067

8. Which of the following enhances or induces fusion of protoplasts ?
(1) Polyethylene glycol and sodium nitrate
(2) IAA and kinetin
(3) IAA and gibberellins
(4) Sodium chloride and potassium chloride

BBN068

Re-AIPMT 2015

9. A protoplast is a cell :-
(1) without cell wall
(2) without plasma membrane
(3) without nucleus
(4) undergoing division

BBN069

10. Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of :-

(1) Vitamin A (2) Vitamin B

(3) Vitamin C (4) Omega 3

BBN070

NEET-I 2016

11. The two polypeptides of human insulin are linked together by :-

(1) Hydrogen bonds

(2) Phosphodiester bond

(3) Covalent bond

(4) Disulphide bridges

BBN071

NEET-II 2016

12. Which kind of therapy was given in 1990 to a four year old girl with adenosine deaminase (ADA) deficiency ?

(1) Immunotherapy

(2) Radiation therapy

(3) Gene therapy

(4) Chemotherapy

BBN072

NEET (UG) 2018

13. In India, the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is :-

(1) Indian Council of Medical Research (ICMR).

(2) Council for Scientific and Industrial Research (CSIR).

(3) Research Committee on Genetic Manipulation (RCGM).

(4) Genetic Engineering Approval Committee (GEAC).

BBN073

14. A 'new variety of rice was patented by a foreign company though such varieties have been present in India for a long time. This is related to :-

(1) Co-667 (2) Sharbati Sonora

(3) Lerma Rojo (4) Basmati

BBN074

15. Use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called :-

(1) Bio-infringement

(2) Biopiracy

(3) Biodegradation

(4) Bioexploitation

BBN075

NEET (UG) 2019

16. Which of the following is **true** for Golden rice?

(1) It is Vitamin A enriched, with a gene from daffodil.

(2) It is pest resistant, with a gene from *Bacillus thuringiensis*.

(3) It is drought tolerant, developed using *Agrobacterium* vector.

(4) It has yellow grains, because of a gene introduced from a primitive variety of rice

BBN076

17. Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology?

(1) Genetic code is not ambiguous.

(2) Genetic code is redundant.

(3) Genetic code is nearly universal.

(4) Genetic code is specific.

BBN077

18. What triggers activation of protoxin to active Bt toxin of *Bacillus thuringiensis* in boll worm?

(1) Body temperature

(2) Moist surface of midgut

(3) Alkaline pH of gut

(4) Acidic pH of stomach

BBN078

NEET (UG) 2019 (Odisha)

19. Exploitation of bioresources of a nation by multinational companies without authorization from the concerned country is referred to as: -

(1) Bioweapon

(2) Biopiracy

(3) Bioethics

(4) Biowar

BBN079

20. In RNAi, the genes are silenced using:-

(1) ds-RNA

(2) ss-DNA

(3) ss-RNA

(4) ds-DNA

BBN080

NEET (UG) 2020

21. Which of the following statements is **not correct**?

(1) Genetically engineered insulin is produced in *E coli*.

(2) In man insulin is synthesised as a proinsulin.

(3) The proinsulin has an extra peptide called C-peptide.

(4) The functional insulin has A and B chains linked together by hydrogen bonds.

BBN081

22. Match the organism with its use in biotechnology.

(a) *Bacillus thuringiensis*	(i) Cloning vector
(b) *Thermus aquaticus*	(ii) Construction of first rDNA molecule
(c) *Agrobacterium tumefaciens*	(iii) DNA polymerase
(d) *Salmonella typhimurium*	(iv) Cry proteins

Select the **correct** option from the following:

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(iv)	(i)	(ii)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(iii)	(ii)	(iv)	(i)

BBN082

23. Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to :-

(1) Insect predators

(2) Insect pests

(3) Fungal diseases

(4) Plant nematodes

BBN083

24. Match the following columns and select the **correct** option :-

Column-I		**Column-II**	
(a)	Bt cotton	(i)	Gene therapy
(b)	Adenosine deaminase	(ii)	Cellular defence deficiency
(c)	RNAi	(iii)	Detection of HIV infection
(d)	PCR	(iv)	*Bacillus thuringiensis*

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

BBN084

NEET (UG) 2020 (COVID-19)

25. RNA interference is used for which of the following purposes in the field of biotechnology ?

(1) to develop a plant tolerant to abiotic stresses

(2) to develop a pest resistant plant against infestation by nematode

(3) to enhance the mineral usage by the plant

(4) to reduce post-harvest losses

BBN085

26. The laws and rules to prevent unauthorised exploitation of bio-resources are termed as-

(1) Biopatenting (2) Bioethics

(3) Bioengineering (4) Biopiracy

BBN086

NEET (UG) 2021

27. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :-

(1) Biopiracy

(2) Gene therapy

(3) Molecular diagnosis

(4) Safety testing

BBN087

28. Which of the following is **not** an application of PCR (Polymerase Chain Reaction) ?

(1) Molecular diagnosis

(2) Gene amplification

(3) Purification of isolated protein

(4) Detection of gene mutation

BBN088

29. With regard to insulin choose correct options.

(a) C-peptide is not present in mature insulin.

(b) The insulin produced by rDNA technology has C-peptide.

(c) The pro-insulin has C-peptide.

(d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

(1) (b) and (d) only

(2) (b) and (c) only

(3) (a), (c) and (d) only

(4) (a) and (d) only

BBN089

30. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection?

(1) Western Blotting Technique

(2) Southern Blotting Technique

(3) ELISA Technique

(4) Hybridization Technique

BBN090

31. Match **List -I** with **List - II.**

List-I		**List-II**	
(a)	Protoplast fusion	(i)	Totipotency
(b)	Plant tissue culture	(ii)	Pomato
(c)	Meristem culture	(iii)	Somaclones
(d)	Micropropagation	(iv)	Virus free plants

Choose the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(iv)	(ii)	(i)
(2)	(ii)	(i)	(iv)	(iii)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iv)	(iii)	(ii)	(i)

BBN091

NEET (UG) 2022

32. Transposons can be used during which one of the following?

(1) Gene silencing

(2) Autoradiography

(3) Gene sequencing

(4) Polymerase Chain Reaction

BBN092

33. Statements related to human Insulin are given below :-

Which statement(s) is/ are **correct** about genetically engineered Insulin ?

(a) Pro-hormone insulin contain extra stretch of C-peptide

(b) A-peptide and B-peptide chains of insulin were produced separately in *E.coli*, extracted and combined by creating disulphide bond between them.

(c) Insulin used for treating Diabetes was extracted from Cattles and Pigs.

(d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.

(e) Some patients develop allergic reactions to the foreign insulin.

Choose the **most appropriate** answer from the options given below :-

(1) (b) only

(2) (c) and (d) only

(3) (c), (d) and (e) only

(4) (a), (b) and (d) only

BBN093

Re-NEET 2022

34. Milk of transgenic 'Cow Rosie' was nutritionally more balanced product for human babies than natural cow milk because it contained :-

(1) Human protein α–1–antitrypsin

(2) Human alpha–lactalbumin

(3) Human insulin–like growth factor

(4) Human enzyme Adenosine Deaminase (ADA)

BBN094

NEET (UG) 2023

35. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment ?

(1) Serum and Urine analysis

(2) Polymerase Chain Reaction (PCR) technique

(3) Enzyme Linked Immuno-Sorbent Assay (ELISA) technique

(4) Recombinant DNA Technology

BBN095

NEET (UG) 2023 (Manipur)

36. Select **incorrect** statement, regarding chemical structure of insulin.

(1) Mature insulin molecule consists of three polypeptide chains-A, B and C.

(2) Insulin is synthesized as prohormone which contains extra stretch of C-peptide.

(3) C-peptide is not present in mature insulin molecule.

(4) Polypeptide chains A and B are linked by disulphide bridges.

BBN096

37. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Gene therapy	(I)	Separation of DNA fragments
(B)	RNA interference	(II)	Diagnostic test for AIDS
(C)	ELISA	(III)	Cellular defence
(D)	Gel Electrophoresis	(IV)	Allows correction of a gene defect.

Choose the **correct** answer from the options given below :-

(1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(2) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

BBN097

NEET (UG) 2024

38. **Statement-I :** Bt toxins are insect group specific and coded by a gene *cry* IAc.

Statement-II : Bt toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below :-

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

BBN098

39. Which of the following are fused in somatic hybridization involving two varieties of plants ?

(1) Callus (2) Somatic embryos

(3) Protoplasts (4) Pollens

BBN099

40. Match **List-I** with **List-II**.

List-I		List-II	
A.	α-1 antitrypsin	I.	Cotton bollworm
B.	Cry IAb	II.	ADA deficiency
C.	Cry IAc	III.	Emphysema
D.	Enzyme replacement therapy	IV.	Corn borer

Choose the correct answer from the options given below :-

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-IV, C-I, D-III

BBN100

Re-NEET (UG) 2024

41. **Statement-I :** The Indian Government, has set up GEAC, which will make decisions regarding the validity of GM research.

Statement-II : Biopiracy is the term used to refer to the use of bio-resources by native people.

In the light of the above statements, choose the **correct** answer from the options given below :-

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

BBN101

42. Match **List-I** with **List-II**.

List-I		List-II	
A.	Genetically engineered Human Insulin	I.	Gene therapy
B.	GM Cotton	II.	*E. coli*

C.	ADA Deficiency	III.	Antigen-antibody interaction
D.	ELISA	IV.	*Bacillus thuringiensis*

Choose the **correct** answer from the options given below :

(1) A-III, B-II, C-IV, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-II, B-IV, C-I, D-III

BBN102

43. The Bt-toxin in genetically engineered Bt-cotton kills the pest by :-

(1) Creating pores in the midgut.

(2) Damaging the respiratory system.

(3) Degenerating the nervous system.

(4) Altering the pH of body fluids.

BBN103

44. Following are the steps involved in action of toxin in Bt. Cotton.

(A) The inactive toxin converted into active form due to alkaline pH of gut of insect.

(B) *Bacillus thuringiensis* produce crystals with toxic insecticidal proteins.

(C) The alkaline pH solubilises the crystals.

(D) The activated toxin binds to the surface of midgut cells, creates pores and causes death of the insect.

(E) The toxin proteins exist as inactive protoxins in bacteria.

(1) E → C → B → A → D

(2) B → C → A → E → D

(3) A → E → B → D → C

(4) B → E → C → A → D

BBN104

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	2	3	4	1	1	1	1	4	3	4	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	3	2	1	4	3	2	2	2	1	2	3	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Answer	2	1	1	2	1	1	3	3	3	3	1	4	1	4	

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. A transgenic food crop, which may help in solving the problem of night blindness in developing countries is :-
(1) Bt soyabean
(2) Golden rice
(3) Flavr savr tomatoes
(4) Starlink maize
BBN105

2. Bacterium *Bacillus thuringiensis* is widely used in contemporary biology as
(1) Source of industrial enzyme
(2) Indicator of water pollution
(3) Insecticide
(4) Agent for production of dairy products.
BBN106

3. *Bacillus thuringiensis* is a :-
(1) Bacterium (2) Protozoa
(3) Fungus (4) Virus
BBN107

4. GEAC stands for :-
(1) Gene evaluation approval committee
(2) Genetic engineering approval committee
(3) Genetic engineering applied committee
(4) Gene enhancement approval committee
BBN108

5. Conventional methods to diagnose a disease are :-
(1) Serum and urine analysis
(2) PCR
(3) ELISA
(4) All of the above
BBN109

6. The first transgenic cow, which produced human protein enriched milk was named :-
(1) Andy (2) Dolly
(3) Rosie (4) Dumpy
BBN110

7. Milk of transgenic cow 'Rosie' contains a substance that was nutritionally more balanced product for human babies is :-
(1) α-lactalbumin
(2) β-lactalbumin
(3) γ-lactalbumin
(4) δ-lactalbumin
BBN111

8. Which peptide is not present in the mature insulin and is removed during maturation into insulin?
(1) A-peptide (2) B-peptide
(3) C-peptide (4) Both (1) and (2)
BBN112

9. Two polypeptide chains of insulin are linked together by :-
(1) disulphide bonds
(2) hydrogen bonds
(3) Phosphodiester bonds
(4) Glycosidic bonds
BBN113

10. 'Flavr Savr' is a transgenic variety of :-
(1) Potato (2) Tomato
(3) Soyabean (4) Rice
BBN114

11. Golden rice is enriched in :-
(1) Vitamin C (2) Vitamin D
(3) Vitamin A (4) Vitamin E
BBN115

12. For the control of the cotton bollworms, which one of the genes is useful ?
(1) cry I Ac (2) cry I Ab
(3) cry I Ad (4) All of these
BBN116

13. Process involving silencing of a specific mRNA due to a complementary dsRNA molecule is called :-
(1) Transcription (2) RNA interference
(3) DNA interference (4) None of these

BBN117

14. *Meloidegyne incognitia* which infects the roots of tobacco plants causing a great reduction in yield is a :-
(1) Nematode (2) Bacterium
(3) Virus (4) Alga

BBN118

15. Toxin present in *Bacillus thuringiensis* does not kill the bacterium because it is inactive form what makes it active inside the insect?
(1) the alkaline pH of the gut, which solubilises the crystals
(2) the acid pH of the gut
(3) the neutral pH of the gut
(4) All of the above

BBN119

16. Crystals of Bt toxin produced by some bacteria do not kill the bacteria themselves because :-
(1) bacteria are resistant to the toxin
(2) toxin is immature
(3) toxin is inactive
(4) bacteria enclose toxin in a special sac

BBN120

17. Which one of the following statements about genetically engineered insulin is incorrect?
(1) *E. coli* is used for producing humulin
(2) Chains A, B were produced separately
(3) Eli lilly company prepared it for first time
(4) Genetically engineered insulin has C-peptide

BBN121

18. Infection by pathogen can be detected by the presence of antigens or by detecting the antibodies synthesised against the pathogen, on this principle a test is based which is ?
(1) PCR
(2) ELISA
(3) Both (1) and (2)
(4) None of the above

BBN122

19. Indian parliament recently cleared, which amendment of the Indian patents bill:-
(1) First amendment
(2) Second amendment
(3) Third amendment
(4) Fourth amendment

BBN123

20. How many documented varieties of basmati rice distinct for its unique aroma and flavour are grown in India?
(1) 27 varieties (2) 25 varieties
(3) 28 varieties (4) 26 varieties

BBN124

21. Over 95 percent of all existing transgenic animals are :-
(1) Pigs (2) Cows
(3) Fish (4) Mice

BBN125

22. The organisation set up for making decisions regarding the validity of GM research and the safety of introducing GM organism for public services is :-
(1) Genetic engineering approval committee
(2) Genetic engineering advanced company
(3) Genetic engineering applied committee
(4) None of these

BBN126

23. Use of bio-resources by multinational companies and other organisations without proper autorisation from the countries and people concerned without compensatory payment is called :-
(1) Biotheft (2) Biopatent
(3) Biopiracy (4) None of the above
BBN127

24. Which animal is being used to test the safety of polio vaccine?
(1) Transgenic mice (2) Transgenic pig
(3) Transgenic cow (4) Transgenic cat
BBN128

25. *CrylAb* gene produces proteins which control?
(1) Bollworms
(2) Corn borer
(3) Both (1) and (2)
(4) None of the above
BBN129

26. Nematode-specific genes were introduced into the host plant (tobacco plant) by using which vector ?
(1) Plasmid vector
(2) Cosmid vector
(3) Bacteriophage vector
(4) BAC
BBN130

27. Which one of the following statements are true regarding genetic modifications ?
(1) Genetic modifications reduced reliance on chemical pesticides
(2) Genetic modifications has enhanced nutritional value of food.
(3) Genetic modifications made crops more tolerant to abiotic stresses.
(4) All are correct
BBN131

28. Critical research areas of biotechnology are:-
(1) providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.
(2) Creating optimal conditions through engineering for a catalyst to act.
(3) Down stream processing technologies to purify the protein/organic compound.
(4) All the above
BBN132

29. Bacterium genetically engineered for cleaning oil spills is :-
(1) *Escherichia coli*
(2) *Pseudomonas putida*
(3) *Salmonella typhimurium*
(4) *Agrobacterium tumifaciens*
BBN133

30. Bacterium which is known as 'Super bug' is:-
(1) *Pseudomonas putida* (2) *Salmonella*
(3) *Escherichia* (4) *Agrobacterium*
BBN134

31. Animals those have had their DNA manipulated to possess and express an extra (foreign) gene are known as :-
(1) Transgenic animals
(2) Genetically modified animals
(3) Both (1) and (2)
(4) None of the above
BBN135

32. Genetically engineered human insulin is made in:-
(1) Fungus (2) Protista
(3) Plants (4) Bacterium
BBN136

33. Genetically engineered bacteria are being used in commercial production of :-
(1) melatonin (2) testosterone
(3) thyroxine (4) human insulin
BBN137

34. Insulin consists of two short polypeptide chains, which are linked together by?
(1) Sulphide bridges
(2) Peptide bridges
(3) Chloride bridges
(4) Disulphide bridges

BBN138

35. *Meloidegyne incognitia* infects the root of which plant ?
(1) Potato (2) Soyabean
(3) Tobacco (4) Tomato

BBN139

36. Genetics modified crops (GMC) are useful in agriculture because :-
(1) They are more tolerant to abiotic stresses
(2) They increase reliance on chemical pesticide
(3) They have reduced nutritional value
(4) All the above

BBN140

37. A nematode *Meloidegyne incognitia* infects the root of tobacco plant and causes a great reduction in yield. A novel strategy was adopted to prevent this infection which was based on the process of :-
(1) DNA interference (2) RNA interference
(3) PCR technique (4) DNA test

BBN141

38. In RNA interference (RNAi) :-
(1) The silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevent translation of the mRNA
(2) The silencing of a specific mRNA due to dsDNA
(3) The silencing of a specific mRNA due to tRNA
(4) All the above

BBN142

39. The protein encoded by the gene *cry*I*Ac* and *cry*II*Ab*, controls :-
(1) Cotton bollworm (2) Corn borer
(3) Cotton borer (4) All the above

BBN143

40. Transgenic animals produces biological product such as α-1-antitrypsin, which is used to treat :-
(1) Emphysema (2) Cystic fibrosis
(3) Phenyl ketonuria (4) Sickle cell anaemia

BBN144

41. The first transgenic cow was 'Rosie', produces :-
(1) Human protein-enriched milk (α-lactalbumin)
(2) Human protein α-1 antitrypsin riched milk
(3) Human protein enriched milk (insulin)
(4) All the above

BBN145

42. Transgenic mice are being developed for use in :-
(1) Testing the safety of polio vaccines before they are used on human
(2) Molecular diagnosis of diseases
(3) Production of human protein enriched milk
(4) Production of human insulin

BBN146

43. The technique that serves the purpose of early diagnosis of disease or pathogen :-
(1) Recombinant DNA technology
(2) Polymerase chain reaction (PCR)
(3) Enzyme linked immunosorbent assay (ELISA)
(4) All the above

BBN147

44. Transgenic tobacco which is developed through RNA interference, prevents the infection of :-
(1) A nematode – *Meloidegyne incognitia*
(2) A bacterium – *Pseudonomonas putida*
(3) A fungi – *Tricoderma*
(4) An insect

BBN148

45. The first clinical gene therapy was given to a 4-year old girl with ADA deficiency in :-
(1) 1984 (2) 1986 (3) 1992 (4) 1990

BBN149

(B) ANALYTICAL QUESTIONS

46. Production of a human protein in bacteria by genetic engineering is possible because
(1) Bacterial cell can carry out the RNA splicing reactions
(2) The mechanism of gene regulation is identical in humans and bacteria
(3) The human chromosome can replicate in bacterial cell
(4) The genetic code is universal

BBN150

47. If hemoglobin (Hb) of a normal individual and a sickle-cell patient are run in electrophoretic field, they will show :-
(1) same mobilities
(2) different mobilities
(3) Hb of patient will not move at all
(4) Hb^A are immobile

BBN151

48. Which one of the following is a correct statement ?
(1) "Bt" in "Bt-cotton" indicates that it is a genetically modified organism produced through biotechnology
(2) Somatic hybridization involves fusion of two complete plant cells carrying desired genes.
(3) The anticoagulant hirudin is being produced from transgenic ***Brassica napus*** seeds.
(4) "Flavr Savr" variety of tomato has enhanced the production of ethylene which improves its taste.

BBN152

49. An example of gene therapy is:-
(1) Production of injectable Hepatitis–B vaccine
(2) Production of vaccines in food crops like potatoes which can be eaten
(3) Introduction of gene for Adenosine deaminase in persons suffering from severe combined immuno-deficiency (SCID)
(4) Production of test tube babies by artificial insemination and implantation of fertilized eggs

BBN153

50. Transgenic plants are the ones :-
(1) Grown in artificial medium after hybridization in the field
(2) Produced by a somatic embryo in artificial medium
(3) Generated by introducing foreign DNA in to a cell and regenerating a plant from that cell
(4) Produced after protoplast fusion in artificial medium

BBN154

51. What is *true* about Bt toxin ?
(1) The concerned *Bacillus* has antitoxins
(2) The inactive protoxin gets converted into active form in the insect gut
(3) Bt protein exists as active toxin in the *Bacillus*
(4) The activated toxin enters the ovaries of the pest to sterilise it and thus prevent its multiplication

BBN155

52. The genetic defect adenosine deaminase (ADA) deficiency may be cured permanently by:
(1) Enzyme replacement therapy.
(2) Periodic infusion of genetically engineered lymphocytes having functional ADA cDNA.
(3) Administering adenosine deaminase activators.
(4) Introducing gene from bone marrow cells producing ADA into cells at early embryonic stages.

BBN156

53. An improved variety of transgenic basmati rice
(1) is completely resistant to all insect pests and diseases of paddy
(2) gives high yield but has no charactristic aroma
(3) does not require chemical fertilizers and growth hormones
(4) give high yield and is rich in vitamin A

BBN157

54. Genetic engineering has been successfuly used for producing :-
(1) transgenic Cow-Roise which produces high fat milk for making ghee.
(2) animals like bulls for farm work as they have super power.
(3) transgenic mice for testing safety of polio vaccine before use in humans.
(4) transgenic cow for obtaining α-1-antitrypsin.

BBN158

55. Some of the characteristics of Bt cotton are:
(1) High yield and production of toxic protein crystals which kill dipteran pests
(2) High yield and resistance to bollworms
(3) Long fibre and resistance to aphids
(4) Medium yield, long fibre and resistance to beetle pests

BBN159

56. Read the following four statements (A-D) about certain mistakes in two of them:-
(A) The first transgenic buffalo, Rosie produced milk which was human alpha-lactalbumin enriched.
(B) Restriction enzymes are used in isolation of DNA from other macro molecules.
(C) Isolation of DNA is one of the steps of R-DNA technology.
(D) Disarmed pathogen vectors are also used in transfer of R-DNA into the host.
Which are the two statements having mistakes ?
(1) Statements (A) and (B)
(2) Statements (B) and (C)
(3) Statements (C) and (D)
(4) Statements (A) and (C)

BBN160

57. *Bacillus thuringiensis* forms protein crystals which contain insecticidal protein.
This protein :-
(1) does not kill the carrier bacterium which is itself resistant to this toxin.
(2) binds with epithelial cells of midgut of the insect pest ultimately killing it.
(3) is coded by several genes including the gene lac z.
(4) is activated by acid pH of the foregut of the insect pest.

BBN161

58. Production of thousands of plants through tissue culture method is called :-
(1) Macropropagation
(2) Micropropagation
(3) Somatic embryo
(4) Totipotency

BBN162

59. Plants produced by tissue culture method are called :-
(1) Explant
(2) Somaclones
(3) Macropropagation
(4) SCP (Single cell protein)

BBN163

60. Which chemical is used in somatic hybridisation ?
(1) Polyethylene glycol (2) Acridine
(3) HNO_2 (4) Ethanol

BBN164

61. Each of the plant obtained through tissue culture are genetically identical to the original plant from which they were grown are known as:-
(1) Genocopies
(2) Somaclonal variants
(3) Somaclones
(4) Phenocopies

BBN165

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	1	2	1	3	1	3	1	2	3	1	2	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	2	2	1	4	1	3	1	2	1	4	4	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	4	4	3	1	2	1	1	1	1	1	4	1	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	2	3	3	3	2	4	4	3	2	1	2	2	2	1
Question	61														
Answer	3														

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

IMMUNITY

1. First line of defence of body is :-
(1) Skin and Mucous membrane
(2) Neutrophils and Monocytes
(3) Fever
(4) Interferon
BHD001

2. Antigen is :-
(1) Substances which stimulates the production of venom
(2) Vaccine
(3) Antibody production stimulating agent
(4) Part of the body defence system
BHD002

3. Which cell is not phagocytic in nature ?
(1) Monocyte (2) Macrophage
(3) Neutrophil (4) NK cell
BHD003

4. Cell mediated immunity is provided by :-
(1) B–lymphocytes (2) Plasma cells
(3) Mast cells (4) T-lymphocytes
BHD004

5. Which is a primary lymphoid organ ?
(1) Bone marrow and spleen
(2) Spleen and thymus
(3) Bone–marrow and tonsils
(4) thymus gland and bone marrow
BHD005

6. Which antibody is called secretory antibody?
(1) I_gE (2) I_gA
(3) I_gG (4) I_gM
BHD006

7. Neutrophils and monocytes are important cells participating in :-
(1) Phagocytosis
(2) Perforin production
(3) Passive immunity
(4) Antibody production
BHD007

8. Immune system retains the memory of which response in vaccination process :-
(1) Passive immunization response
(2) Primary immune response
(3) Secondary immune response
(4) All the above
BHD008

9. Number of polypeptide chains present in a molecule of antibody :-
(1) 2 (2) 4 (3) 6 (4) 8
BHD009

10. Colostrum, the first milk secretion of mammary gland is rich in immune-globulin:
(1) IgE (2) IgM (3) IgA (4) IgG
BHD010

11. An antibody is a :-
(1) Component of blood
(2) Secretion of mammalian erythrocyte
(3) Molecule that specifically reacts to an antigen
(4) White corpuscle which attack invading bacteria
BHD011

12. Surgical removal of thymus of a new born shall result in failure to mature :-
(1) Monocytes (2) B-lymphocytes
(3) T-lymphocytes (4) Basophils
BHD012

13. Substances that regulate or coordinate other leucocytes are called :-
(1) Interferon
(2) Cytokine
(3) Phagocytic chemical
(4) Enzyme

BHD013

GRAFTING AND VACCINES

14. Vaccination is a part of :-
(1) Treatment
(2) Passive immunisation
(3) Diagnosis
(4) Prophylaxis (Prevention)

BHD014

15. Vaccine is :-
(1) Type of antibody
(2) Inactivated antigen
(3) Inactivated pathogen
(4) Activated pathogen

BHD015

16. BCG vaccine is a preventive measure against :-
(1) Tuberculosis (2) Typhoid
(3) AIDS (4) Cholera

BHD016

17. D.P.T. vaccine in an example of :-
(1) Passive immunity
(2) Active immunity
(3) Both
(4) Interferon

BHD018

18. Immunisation is based on :-
(1) Memory of individuals
(2) Pathogenic power
(3) Phagocytosis
(4) Memory of immune system

BHD019

19. Congenital diseases are :
(1) Diseases present by birth
(2) Deficiency diseases
(3) Spread from one individual to another
(4) Occur during life

BHD020

20. Autoimmunity against the synovial membrane at joint is called :-
(1) Multiple sclerosis
(2) I.D.D.M.
(3) Hashimoto disease
(4) Rheumatoid arthritis

BHD024

COMMON HUMAN DISEASES

21. Widal test is employed for detecting :
(1) Pneumonia (2) Malaria
(3) Typhoid (4) Cholera

BHD025

22. Which is an infectious disease ?
(1) Coronary thrombosis
(2) Diphtheria
(3) Diabetes mellitus
(4) Hypertension

BHD026

23. Diphtheria is connected with :–
(1) Lungs (2) Throat
(3) Blood (4) Liver

BHD027

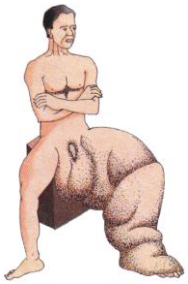
24. Which disease is caused by *Yersinia pestis*
(1) Plague
(2) Black fever
(3) *Xenopsylla cheopsis*
(4) All of these

BHD028

25. Dengue fever is transmitted by :-
(1) Aedes aegypti (Tiger mosquito)
(2) Culex fatigaus
(3) Anopheles
(4) *Ascaris*

BHD029

26. "Diagram" showing inflammation of lower limb is due to :-



(1) Filariasis
(2) Elephantiasis
(3) Ascariasis
(4) Both (1) and (2)

BHD030

27. *Plasmodium* enters the human body as :-
(1) Female *Anopheles* mosquito
(2) Sporozoite
(3) Trophozoite
(4) Haemozoin

BHD031

28. Toxin which is responsible for chill and high fever during malaria :-
(1) Haematin (2) Haemoglobin
(3) Haemozoin (4) Heam

BHD032

29. Incorrect about *Entamoeba histolytica* or amoebiasis is :-
(a) Parasite of small intestine
(b) Causes dysentery
(c) Houseflies are mechanical carriers
(d) Symptoms include constipation, abdominal pain and cramp.
(1) a (2) c
(3) a, c (4) All are correct

BHD033

30. Diagram showing ringworm affected area of skin. Which given pathogen is not related with disease?



(1) *Microsporum* (2) *Trichophyton*
(3) *Epidermophyton* (4) *Wuchereria*

BHD034

31. *Haemophilus influenzae* causes :-
(1) Typhoid (2) Plague
(3) Pneumonia (4) Influenza

BHD035

32. Match the columns with regards to Vector and Disease

	Column-I		Column-II
(p)	*Culex*	(i)	Dengue
(q)	*Anopheles*	(ii)	Filariasis
(r)	*Aedes*	(iii)	Malaria

(1) p-i, q-ii, r-iii (2) p-ii, q-iii, r-i
(3) p-ii, q-i, r-iii (4) p-i, q-iii, r-ii

BHD036

33. House flies are mechanical carriers of :-
(1) Amoebiasis
(2) Malaria
(3) Common cold
(4) Plague

BHD037

AIDS AND CANCER

34. AIDS is characterised by :-
(1) Reduction in number of helper T-cells
(2) Lack of interferon
(3) Reduction in number of killer T-cells
(4) Autoimmunity

BHD038

35. Carcinoma is a cancer of :-
(1) Lymphocytes
(2) Connective tissue
(3) Erythrocytes
(4) Ectoderm and endoderm

BHD039

36. Cancer cells are characterized by :-
(1) Uncontrolled growth
(2) Invasion of local tissue
(3) Spreading to other body parts
(4) All the above

BHD040

37. Carcinoma refers to :-
(1) Malignant tumours of the connective tissue
(2) Malignant tumours of the skin
(3) Benign tumours of the colon
(4) Benign tumours of the connective tissue
BHD042

38. AIDS is caused by HIV that principally infects :-
(1) all lymphocytes (2) activator B cells
(3) T_H lymphocytes (4) Cytotoxic T cells
BHD043

39. Treatment of cancer is :–
(1) Surgery (2) Radiation
(3) Chemotherapy (4) All
BHD044

40. Causative factor of cancer is called: –
(1) Oncogenes (2) Radiogen
(3) Estrogen (4) Carcinogen
BHD045

41. Nucleic acid in HIV: -
(1) ss RNA (2) ds RNA
(3) ss DNA (4) ds DNA
BHD046

42. Sarcoma is cancer of :–
(1) Epithelial tissue
(2) Mesodermal tissue
(3) Blood
(4) Endodermal tissues
BHD047

43. Which is not cancer :–
(1) Leukaemia (2) Glaucoma
(3) Carcinoma (4) Sarcoma
BHD048

44. Blood cancer is ?
(1) Leukemia (2) Thrombosis
(3) Haemophilia (4) Hemolysis
BHD049

DRUGS AND ALCOHOL ABUSE

45. Tobacco chewing results in :-
(1) Mouth cancer
(2) Lung cancer
(3) Bone cancer
(4) Leukaemia
BHD050

46. In drunk person, the part of brain to be affected first is :-
(1) Cerebellum
(2) Pons varolli
(3) Medulla oblongata
(4) Cerebrum
BHD051

47. Opium is obtained from :-
(1) *Thea sinensis*
(2) *Coffea arabica*
(3) *Oryza sativa*
(4) *Papaver somniferum*
BHD052

48. Marijuana, Ganja, and LSD are :-
(1) Narcotics (2) Hallucinogens
(3) Stimulants (4) Medicines
BHD053

49. In alcoholics, liver gets damaged as it :-
(1) Accumulates excess of fats
(2) Stores excess of glycogen
(3) Secretes more bile
(4) Has to detoxify alcohol
BHD054

50. LSD is obtained from :-
(1) *Cannabis* (2) *Claviceps*
(3) *Fusarium* (4) *Nostoc*
BHD055

51. *Cannabis sativa* (Hemp) yields:-
(1) Bhang (2) Charas
(3) Ganja (4) All the above
BHD056

52. In the liver, alcohol is converted into which toxic substance ?

(1) Formic acid (2) Acetaldehyde

(3) Nicotine (4) Urea

BHD057

53. (a) Smack

(b) Diacetylmorphine

(c) White

(d) Odourless

(e) Bitter crystalline compound

Above statements/information are **correct** for:-

(1) Morphine (2) Heroin

(3) Coccain (4) Barbiturates

BHD058

54. Cocaine is derived from :-

(1) *Erythroxylum coca*

(2) *Coffea arabica*

(3) *Thea sinensis*

(4) *Cannabis sativa*

BHD059

55. Which measure would be particularly useful for prevention and control of alcohol and drug abuse among adolescents ?

(a) Avoid undue peer pressure

(b) Seeking professional and medical help

(c) Looking for danger sign

(d) Education and counselling

(e) Seeking help from parents and peers

(1) a, b, d only

(2) a, c, d, e only

(3) c, e only

(4) a, b, c, d, e

BHD060

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	4	4	4	2	1	2	2	3	3	3	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	4	1	4	3	2	2	1	1	4	2	3	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	1	1	4	4	2	3	4	4	1	2	2	1	1
Question	46	47	48	49	50	51	52	53	54	55					
Answer	4	4	2	1	2	4	2	2	1	4					

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which is the particular type of drug that is obtained from the plant whose one flowering branch is shown below?



(1) Hallucinogen (2) Depressant
(3) Stimulant (4) Pain - killer

BHD095

2. At which stage of HIV infection does one usually show symptoms of AIDS ?
(1) Within 15 days of sexual contact with an infected person.
(2) When the infected retro virus enters host cells.
(3) When HIV damages large number of helper T-Lymphocytes.
(4) When the viral DNA is produced by reverse transcriptase.

BHD096

AIPMT 2015

3. HIV that causes AIDS, first starts destroying:
(1) Leucocytes
(2) Helper T-Lymphocytes
(3) Thrombocytes
(4) B-Lymphocytes

BHD097

4. Which of the following viruses is not transferred through semen of an infected male?
(1) Human immunodeficiency virus
(2) Chikungunya virus
(3) Ebola virus
(4) Hepatitis B virus

BHD098

5. Match each disease with its **correct** type of vaccine:

(a) tuberculosis	(i) harmless virus
(b) whooping cough	(ii) inactivated toxin
(c) diphtheria	(iii) killed bacteria
(d) polio	(iv) harmless bacteria

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(ii)	(iv)	(i)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(i)	(ii)	(iv)	(iii)
(4)	(ii)	(i)	(iii)	(iv)

BHD099

Re-AIPMT 2015

6. Grafted kidney may be rejected in a patient due to :-
(1) Innate immune response
(2) Humoral immune response
(3) Cell-mediated immune response
(4) Passive immune response

BHD100

7. If you suspect major deficiency of antibodies in a person, to which of the following would you look for confirmatory evidence?
(1) Serum globulins
(2) Fibrinogin in plasma
(3) Serum albumins
(4) Haemocytes

BHD101

8. Which of the following immunoglobulins does constitute the largest percentage in human milk?

(1) IgG (2) IgD (3) IgM (4) IgA

BHD103

NEET-I 2016

9. Asthma may be attributed to :

(1) bacterial infection of the lungs

(2) allergic reaction of the mast cells in the lungs

(3) inflammation of the trachea

(4) accumulation of fluid in the lungs

BHD105

10. In higher vertebrates, the immune system can distinguish self-cells and non-self. If this property is lost due to genetic abnormality and it attacks self-cells, then it leads to :-

(1) Allergic response

(2) Graft rejection

(3) Auto-immune disease

(4) Active immunity

BHD106

11. Antivenom injection contains preformed antibodies while polio drops that are administered into the body contain :-

(1) Activated pathogens

(2) Harvested antibodies

(3) Gamma globulin

(4) Attenuated pathogens

BHD107

12. Which of the following statements is **not true** for cancer cells in relation to mutations

(1) Mutations in proto-oncogenes accelerate the cell cycle.

(2) Mutations destroy telomerase inhibitor.

(3) Mutations inactive the cell control.

(4) Mutations inhibit production of telomerase.

BHD109

NEET-II 2016

13. Which kind of therapy was given in 1990 to a four year old girl with adenosine deaminase (ADA) deficiency ?

(1) Immunotherapy

(2) Radiation therapy

(3) Gene therapy

(4) Chemotherapy

BHD110

14. Which of the following is **correct** regarding AIDS causative agent HIV ?

(1) HIV is unenveloped retrovirus.

(2) HIV does not escape but attacks the aquired immune response.

(3) HIV is enveloped virus containing one molecule of single-stranded RNA and one molecule of reverse transcriptase.

(4) HIV is enveloped virus that contains two identical molecules of single-stranded RNA and two molecules of reverse transcriptase.

BHD111

15. Which of the following sets of diseases is caused by bacteria?

(1) Tetanus and mumps

(2) Herpes and influenza

(3) Cholera and tetanus

(4) Typhoid and smallpox

BHD112

NEET(UG) 2017

16. Transplantation of tissues/organs fails often due to non-acceptance by the patient's body. Which type of immune-response is responsible for such rejections?

(1) Cell - mediated immune response

(2) Hormonal immune response

(3) Physiological immune response

(4) Autoimmune response

BHD114

NEET(UG) 2018

17. Which of the following is ***not*** an autoimmune disease?

(1) Psoriasis

(2) Rheumatoid arthritis

(3) Alzheimer's disease

(4) Vitiligo

BHD115

18. In which disease does mosquito transmitted pathogen cause chronic inflammation of lymphatic vessels?

(1) Elephantiasis (2) Ascariasis

(3) Ringworm disease (4) Amoebiasis

BHD116

NEET (UG) 2019

19. Colostrum, the yellowish fluid, secreted by mother during the initial days of lactation is very essential to impart immunity to the newborn infants because it contains :-

(1) Natural killer cells

(2) Monocytes

(3) Macrophages

(4) Immunoglobulin A

BHD117

20. Which of the following immune responses is responsible for rejection of kidney graft?

(1) Auto- immune respones

(2) Humoral immune response

(3) Inflammatory immune response

(4) Cell-mediated immune response

BHD119

21. Drug called 'Heroin' is synthesized by :-

(1) Methylation of morphine

(2) Acetylation of morphine

(3) Glycosylation of morphine

(4) Nitration of morphine

BHD121

22. Identify the **correct** pair representing the causative agent of typhoid fever and the confirmatory test for typhoid.

(1) *Plasmodium vivax*/UTI test.

(2) *Streptococcus pneumoniae*/Widal test

(3) *Salmonella typhi*/Anthrone test

(4) *Salmonella typhi*/Widal test

BHD122

NEET (UG) 2019 (Odisha)

23. Humans have acquired immune system that produces antibodies to neutralize pathogens. Still innate immune system is present at the time of birth because it

(1) is very specific and uses different macrophages.

(2) produces memory cells for mounting fast secondary response.

(3) has natural killer cells which can phagocytose and destroy microbes.

(4) provides passive immunity.

BHD123

24. Which of the following diseases is an autoimmune disorder?

(1) Myasthenia gravis (2) Arthritis

(3) Osteoporosis (4) Gout

BHD124

25. Coca alkaloid or cocaine is obtained from:

(1) *Papaver somniferum*

(2) *Atropha belladona*

(3) *Erythroxylum coca*

(4) *Datura*

BHD126

NEET (UG) 2020

26. Match the following diseases with the causative organism and select the **correct** option.

Column-I	**Column-II**
(a) Typhoid	(i) *Wuchereria*
(b) Pneumonia	(ii) *Plasmodium*
(c) Filariasis	(iii) *Salmonella*
(d) Malaria	(iv) *Haemophilus*

(1) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

(2) (a)–(i), (b)–(iii), (c)–(ii), (d)–(iv)

(3) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

(4) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)

BHD127

NEET (UG) 2020 (COVID-19)

27. Match the following columns and select the correct option :

List-I		List-II	
(a)	Typhoid	(i)	*Haemophilus influenzae*
(b)	Malaria	(ii)	*Wuchereria bancrofti*
(c)	Pneumonia	(iii)	*Plasmodium vivax*
(d)	Filariasis	(iv)	*Salmonella typhi*

(1) (i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)

(2) (i)-(c), (ii)-(d), (iii)-(b), (iv)-(a)

(3) (i)-(a), (ii)-(c), (iii)-(b), (iv)-(d)

(4) (i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)

BHD129

28. The yellowish fluid "colostrum" secreted by mammary glands of mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. This type of immunity is called as :

(1) Passive immunity

(2) Active immunity

(3) Acquired immunity

(4) Autoimmunity

BHD131

NEET (UG) 2021

29. Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as:

(1) Arthritis

(2) Muscular dystrophy

(3) Myasthenia gravis

(4) Gout

BHD132

30. Match **List - I** with **List -II.**

List-I		List-II	
(a)	Filariasis	(i)	*Haemophilus influenzae*
(b)	Amoebiasis	(ii)	*Trichophyton*
(c)	Pneumonia	(iii)	*Wuchereria bancrofti*
(d)	Ringworm	(iv)	*Entamoeba histolytica*

Choose the **correct** answer from the options given below.

(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(3) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
(4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

BHD133

NEET (UG) 2022

31. Given below are two statements:

Statement I: Autoimmune disorder is a condition where body defense mechanism recognizes its own cells as foreign bodies.

Statement II: Rheumatoid arthritis is a condition where body does not attack self cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **II** are correct.

BHD134

32. Select the **incorrect** statement with respect to acquired immunity.

(1) Anamnestic response is elicited on subsequent encounters with the same pathogen.
(2) Anamnestic response is due to memory of first encounter.
(3) Acquired immunity is non-specific type of defense present at the time of birth.
(4) Primary response is produced when our body encounters a pathogen for the first time.

BHD135

Re-NEET (UG) 2022

33. Which of the following reasons in mainly responsible for graft rejection in transplantation of organs ?

(1) Inability of recipient to differentiate between 'self and 'non–self' tissues/ cells
(2) Humoral immune response only
(3) Auto–immune response
(4) Cell–mediated response

BHD136

34. Match **List-I** with **List-II.**

List-I		List-II	
(a)	Cellular barrier	(i)	Interferons
(b)	Cytokine barrier	(ii)	Mucus
(c)	Physical barrier	(iii)	Neutrophils
(d)	Physiological barrier	(iv)	HCl in gastric juice

Choose the correct answer from the options given below :

(1) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)
(2) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)
(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)
(4) (a) – (iii), (b) – (i), (c) – (ii), (d) – (iv)

BHD137

NEET (UG) 2023

35. Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly ?

(1) Gonorrhoea (2) Hepatitis-B
(3) HIV Infection (4) Genital herpes

BHD178

36. In which blood corpuscles, the HIV undergoes replication and produces progeny viruses ?

(1) B-lymphocytes (2) Basophils

(3) Eosinophils (4) T_H cells

BHD179

37. Match **List I** with **List II**.

List-I		List-II	
(A)	Heroin	(I)	Effect on cardiovascular system
(B)	Marijuana	(II)	Slow down body function
(C)	Cocaine	(III)	Painkiller
(D)	Morphine	(IV)	Interfere with transport of dopamine

Choose the **correct** answer from the options given below :

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

BHD180

38. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Ringworm	(I)	*Haemophilus influenzae*
(B)	Filariasis	(II)	*Trichophyton*
(C)	Malaria	(III)	*Wuchereria bancrofti*
(D)	Pneumonia	(IV)	*Plasmodium vivax*

Choose the **correct** answer from the options given below :-

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-II, C-IV, D-I

(4) A-II, B-III, C-IV, D-I

BHD181

NEET (UG) 2023 (Manipur)

39. Diacetyl morphine is also called as :

(1) Amphetamine

(2) Barbiturate

(3) Crack

(4) Smack

BHD257

40. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Typhoid	(I)	Protozoan
(B)	Elephantiasis	(II)	*Salmonella*
(C)	Ringworm	(III)	Aschelminthes
(D)	Malaria	(IV)	*Microsporum*

Choose the **correct** answer from the options given below :-

(1) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(2) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

BHD183

NEET (UG) 2024

41. Match **List-I** with **List-II**.

List-I		List-II	
A.	Common cold	I.	*Plasmodium*
B.	Haemozoin	II.	Typhoid
C.	Widal test	III.	Rhinoviruses
D.	Allergy	IV.	Dust mites

Choose the **correct** answer from the options given below :-

(1) A-II, B-IV, C-III, D-I

(2) A-I, B-III, C-II, D-IV

(3) A-III, B-I, C-II, D-IV

(4) A-IV, B-II, C-III, D-I

BHD185

42. Match **List-I** with **List-II**.

List-I		List-II	
A.	Cocaine	I.	Effective sedative in surgery
B.	Heroin	II.	*Cannabis sativa*
C.	Morphine	III.	*Erythroxylum*
D.	Marijuana	IV.	*Papaver somniferum*

Choose the **correct** answer from the options given below :-

(1) A-IV, B-III, C-I, D-II

(2) A-I, B-III, C-II, D-IV

(3) A-II, B-I, C-III, D-IV

(4) A-III, B-IV, C-I, D-II

BHD186

43. Which of the following are Autoimmune disorders?

(A) Myasthenia gravis

(B) Rheumatoid arthritis

(C) Gout

(D) Muscular dystrophy

(E) Systemic Lupus Erythematosus (SLE)

Choose the most appropriate answer from the options given below:

(1) A, B & D only

(2) A, B & E only

(3) B, C & E only

(4) C, D & E only

BHD187

44. Match **List I** with **List II**.

List-I		List-II	
A.	Typhoid	I.	Fungus
B.	Leishmaniasis	II.	Nematode
C.	Ringworm	III.	Protozoa
D.	Filariasis	IV.	Bacteria

Choose the **correct** answer from the options given below :-

(1) A-I, B-III, C-II, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-II, B-IV, C-III, D-I

BHD188

Re-NEET (UG) 2024

45. Match **List-I** with **List-II.**

List-I		List-II	
A.	Malignant tumors	I.	Destroy tumors
B.	MALT	II.	AIDS
C.	NACO	III.	Metastasis
D.	α-Interferons	IV.	Lymphoid tissue

Choose the **correct** answer from the options given below :-

(1) A-III, B-IV, C-II, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-I, C-IV, D-II

BHD189

46. Match **List-I** with **List-II**.

	List-I		List-II
A.	B-Lymphocytes	I.	Passive immunity
B.	Interferons	II.	Cell mediated immunity
C.	T-Lymphocytes	III.	Produce an army of proteins in response to pathogens
D.	Colostrum	IV.	Innate immunity

Choose the **correct** answer from the options given below :-

(1) A-I, B-IV, C-II, D-III

(2) A-IV, B-II, C-III, D-I

(3) A-III, B-IV, C-II, D-I

(4) A-II, B-IV, C-I, D-III

BHD190

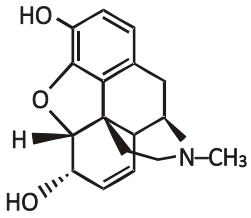
EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	2	2	2	3	1	4	2	3	4	4	3	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	1	4	4	2	4	3	1	3	3	4	1	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	4	4	1	4	4	4	4	3	3	4	2	2	1
Question	46														
Answer	3														

Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Diagram is showing chemical structure of Morphine, which is a natural opiate like codeine. Morphine is a strong analgesic, also has sedative & calming effect. It depresses respiratory centre, BP, heart beat. It is obtained from-



(1) *Cannabis sativa*

(2) *Datura*

(3) *Erythroxylum coca*

(4) *Papever sominiferum*

BHD138

2. Nicotine :-

(1) Stimulates adrenal gland

(2) Is an alkaloid

(3) Is present in tobacco

(4) All of these

BHD140

3. Opioids are the drugs which bind to specific opioid receptors present in our CNS and GIT. Heroin, commonly called smack is chemically...., which is white, odourless, bitter crystalline compound, obtained by.....of morphine.

(1) Diacetylmorphine, methylation

(2) Diacetylmorphine, acetylation

(3) Benzodiazepines, amination

(4) Amphetamines, acetylation

BHD141

4. Match the following :-

Column-I		Column-II	
I.	Allergy	(i)	Typhoid fever
II.	T-helper cells	(ii)	Single stranded RNA
III.	Hallucinogens	(iii)	Wuchereria
IV.	Liver test	(iv)	IgE
V.	Widal test	(v)	Cirrhosis
VI.	Filariasis	(vi)	Atropa belladona
VII.	ELISA test	(vii)	Activation of B-cells
VIII.	AIDS virus	(viii)	Carcinogens
IX.	Treatment of cancer	(ix)	AIDS
X.	X-rays	(x)	Immunotherapy

(1) I-(iv), II-(vi), III-(viii), IV-(v), V-(i), VI (iii), VII-(ix), VIII-(ii), IX-(x), X-(vii)

(2) I-(iv), II-(vii), III-(vi), IV-(v), V-(i), VI-(iii), VII-(ix), VIII-(ii), IX-(x), X-(viii)

(3) I-(iv), II-(vii), III-(v), IV-(ii), V-(i), VI-(iii), VII-(ix), VIII-(vi), IX-(x), X-(viii)

(4) I-(iv), II-(vii), III-(vi), IV-(v), V-(i), VI-(ix), VII-(x), VIII-(ii), IX-(iii), X-(viii)

BHD142

5. Match the Column I with Column II :-

Column-I		Column-II	
A.	Peyer's patches	(i)	*Aedes*
B.	Rheumatoid	(ii)	Neoplastic arthritis transformation
C.	IgA	(iii)	Cancer treatment

D.	Interferon	(iv)	Allergy
E.	*Gambusia*	(v)	Secondary lymphoid organ
F.	Chikungunya	(vi)	Metastasis
G.	Tetanus	(vii)	Colostrum
H.	IgE	(viii)	Autoimmunity
I.	Malignant tumor	(ix)	Antitoxin
J.	Carcinogen	(x)	Mosquito larvae

(1) A-(v), B-(viii), C-(vii), D-(iii), E-(x), F-(i), G-(ix), H-(iv), I-(vi), J-(ii)

(2) A-(vi), B-(viii), C-(vii), D-(v), E-(x), F-(ii), G-(ix), H-(iv), I-(i) J-(iii)

(3) A-(iv), B-(viii), C-(vii), D-(iii), E-(x), F-(i), G-(ix), H-(v), I-(vi), J-(ii)

(4) A-(x), B-(viii), C-(vii), D-(iii), E-(v), F-(i), G-(ix), H-(iv), I-(vi), J-(ii)

BHD143

6. Most cancers are treated by combination of:

(a) Surgery

(b) Radiotherapy

(c) Chemotherapy

(1) a and b only (2) a and c only

(3) b and c only (4) a, b and c

BHD144

7. Which radiations cause DNA damage leading to neoplastic transformation ?

(a) Ionising radiations

(b) X-rays

(c) Non-ionising radiations

(d) UV-rays

(1) a and c only

(2) b and d only

(3) a and b only

(4) a, b, c and d

BHD145

8. Below are some statements related to life cycle of plasmodium. Which of the following is correct?

(1) Gametocyte stage infects human body

(2) It multiplies by sexual method of reproduction in human liver

(3) Human RBC are ruptured and releases a toxin hemozoin

(4) Development of gametocyte possible only in female culex mosquito

BHD147

9. The factor which has effect on health is :-

(1) Genetic disorders (2) Infections

(3) Life style (4) All of the above

BHD148

10. Mark **correct** about a healthy person.

(i) More efficient

(ii) High productivity

(iii) Longevity of people

(iv) Brings economic prosperity

(1) Both (i) and (ii) only

(2) Only (iv)

(3) Both (iii) and (iv) only

(4) All are correct

BHD149

11. Mark the correct statement :-

(i) Health is state of complete physical, mental and social well being.

(ii) Common cold is infection of nose and upper respiratory tract.

(iii) Hepatitis is infectious diseases.

(iv) Cancer is non-infectious disease.

(v) Healthy person bring economic prosperity.

(1) (i), (ii) and (iii)

(2) (ii), (iii), (iv) and (v)

(3) (iii) and (iv)

(4) (i), (ii), (iii), (iv) and (v)

BHD150

12. Find out the incorrect matching with regards to innate immunity :-
(1) Physical barriers = Skin, mucosa
(2) Cytokine barriers = Interferons
(3) Cellular barriers = PMNL, neutrophils
(4) Physiological barriers = Epithelial lining
BHD151

13. H_2L_2 represents ..... molecule :-
(1) Antibody (2) Antigen
(3) Pathogen (4) Both (1) and (2)
BHD152

14. Which immune response / immunity is mainly responsible for graft rejection ?
(1) Cell-mediated
(2) Humoral
(3) Antibody mediated
(4) All of the above
BHD153

15. AIDS does not spread due to :-
(1) Mosquito bite (2) Physical touch
(3) Body fluids (4) Both (1) and (2)
BHD154

16. Incubation period for AIDS is usually :-
(1) 5–10 month (2) 5–10 years
(3) 2–12 month (4) 2–12 weeks
BHD155

17. Which cell acts as factory of HIV ?
(1) Macrophage (2) T_H cells
(3) T_K cells (4) N.K. cells
BHD156

18. In full blown AIDS, opportunistic infections are especially caused by :-
(1) *Mycobacterium*
(2) *Toxoplasma*
(3) Viruses and fungi
(4) All of the above
BHD157

19. Withdrawal syndrome is characterised by:-
(a) Anxiety (b) Shakiness
(c) Nausea (d) Sweating
(1) a, c only (2) b, c only
(3) a, c, d only (4) a, b, c and d
BHD158

20. (a) *Salmonella* is pathogenic bacterium
(b) Sustained high fever (39°C to 40°C)
(c) Intestinal perforation in severe cases
(d) Confirmed by "widal test"
(e) "Marry Mallon" was a carrier
Above statements are true for: -
(1) Plague (2) Diphtheria
(3) Typhoid (4) Dysentery
BHD159

21. Which of the following is not included in cellular barrier of innate immunity?
(1) Monocyte
(2) PMNL neutrophils
(3) T killer cell
(4) Macrophage
BHD160

22. Which of the following statement is true for interferons ?
(1) Secreted by non infected cells
(2) Protect virus infected cells
(3) Protect non-infected cells
(4) They are lipid in nature
BHD161

23. A certain patient is suspected to be suffering from Acquired Immuno Deficiency Syndrome. Which diagnostic technique will you recommend for its detection ?
(1) ELISA (2) MRI
(3) Ultrasound (4) WIDAL
BHD162

24. Treatment of which disease is only partially effective and can not prevent death, which is inevitable :-

(1) Cancer (2) AIDS

(3) SCID (4) Tuberculosis

BHD163

25. Cancer patients are given biological response modifiers such as _____ which activates their immune system and help in destroying the tumour:

(1) α-interferons (2) Cannabinoids

(3) Sedative (4) Opiate

BHD164

26. Which of the following STD is not caused by bacteria ?

(1) Trichomoniasis (2) Syphilis

(3) Gonorrhoea (4) Chlamydiasis

BHD165

27. Which disease is completely curable if detected early and treated properly ?

(1) Hepatitis-B (2) Gonorrhoea

(3) AIDS (4) Genital herpes

BHD166

28. Find the correct statement about tobacco smoking-

(A) Nicotine inhibits secretion of adrenal medulla.

(B) Smoking is associated with increased incidence of cancer of lungs and throat.

(C) Smoking increases carbon monoxide content in blood and decreases the concentration of haembound oxygen.

(D) Smoking decreases blood pressure.

(1) A and D (2) B and D

(3) A and B (4) B and C

BHD167

(B) ANALYTICAL QUESTIONS

29. BCG vaccine contains :-

(1) killed mycobacteria

(2) live attenuated mycobacteria

(3) recombinant protein of mycobacteria

(4) antibody against mycobacteria

BHD172

30. Primary response which is of ____(A)____ intensity. Subsequent encounter with the same pathogen elicit a ___(B)___ intensified ___(C)___ response, which is based on ___(D)___.

	(A)	(B)	(C)	(D)
(1)	High	Low	Anamnestic	Specificity
(2)	Low	High	Primary	Diversity
(3)	Low	High	Secondary	Memory
(4)	High	Low	Primary	Memory

BHD173

31. Read the following list carefully.

Neutrophils, erythrocytes, monocytes, macrophages, B-lymphocyte, NK cells, T-lymphocytes , thrombocytes

How many of the above cells constitute cellular barrier of innate immunity ?

(1) Three (2) Four

(3) Five (4) Six

BHD174

32. Which of the following option is true related to production and functions of a molecule that is a glycoprotein and represented as H_2L_2 ?

	Produced by	Functions
(1)	T-killer cell	cellular immunity
(2)	B cells	inflammation
(3)	NK cells	phagocytosis
(4)	Plasma cells	opsonization

BHD175

33. A plant, native of South America is a source of a drug of addiction which is CNS stimulant. This drug inhibits transport of

(1) Acetylcholine

(2) Dopamine

(3) Serotonin

(4) Adrenaline

BHD177

34. Antibodies in our body are complex :–

(1) Lipoproteins

(2) Steroids

(3) Prostaglandins

(4) Glycoproteins

BHD061

35. HIV that causes AIDS, first starts destroying:

(1) B–lymphocytes

(2) Leucocytes

(3) Thrombocytes

(4) Helper T–lymphocytes

BHD062

36. Increased asthmatic attacks in certain seasons are related to :-

(1) Low temperature

(2) Hot and humid environment

(3) Eating fruits preserved in tin containers

(4) Inhalation of seasonal pollen

BHD063

37. Lysozyme that is present in perspiration, saliva and tears, destroys :-

(1) Most virus - infected cells

(2) Certain fungi

(3) Certain types of bacteria

(4) All viruses

BHD064

38. Match the disease in **Column I** with the appropriate items (pathogen/prevention/ treatment) in **Column II**

Column-I		Column-II	
(a)	Amoebiasis	(i)	*Treponema pallidum*
(b)	Diphtheria	(ii)	Use only sterilized food and water
(c)	Cholera	(iii)	DPT Vaccine
(d)	Syphilis	(iv)	Use oral rehydration therapy

(1) a – (ii), b – (i), c – (iii), d – (iv)

(2) a – (ii), b – (iii), c – (iv), d – (i)

(3) a – (i), b – (ii), c – (iii), d – (iv)

(4) a – (ii), b – (iv), c – (i), d – (iii)

BHD065

39. To which type of barriers under innate immunity, do the saliva in the mouth and the tears from the eyes, belong?

(1) Physiological barriers

(2) Physical barriers

(3) Cytokine barriers

(4) Cellular barriers

BHD066

40. Which one of the following is the ***correct*** statement regarding the particular psychotropic drug specified ?

(1) *Morphine* leads to delusions and disturbed emotions

(2) *Barbiturates* cause relaxation and temporary euphoria

(3) *Hashish* alters thought perceptions and causes hallucinations

(4) *Opium* stimulates nervous system and causes hallucinations

BHD067

41. Which one of the following statements is correct?

(1) Malignant tumours may exhibit metastasis

(2) Patients who have undergone surgery are given cannabinoids to relieve pain

(3) Benign tumours show the property of metastasis

(4) Heroin accelerates body functions

BHD068

42. Which of the following is a pair of viral diseases?

(1) Typhoid, Tuberculosis

(2) Ringworm, AIDS

(3) Common Cold, AIDS

(4) Dysentery, Common Cold

BHD069

43. A person likely to develop tetanus is immunised by administering :-

(1) Weakened germs

(2) Dead germs

(3) Preformed antibodies

(4) Wide spectrum antibiotics

BHD070

44. Use of anti-histamines and steroids give a quick relief from :-

(1) headache (2) Allergy

(3) Nausea (4) Cough

BHD071

45. The genetic defect - adenosine deaminase (ADA) deficiency may be cured permanently by :-

(1) Enzyme replacement therapy

(2) Periodic infusion of genetically engineered lymphocytes having functional ADA cDNA

(3) Administering adenosine deaminase activators

(4) Introducing bone marrow cells producing ADA into cells at early embryonic stages

BHD072

46. The letter "T" in T-lymphocyte refers to :

(1) Thymus (2) Thyroid

(3) Thalamus (4) Tonsil

BHD073

47. Widal test is used for the diagnosis of :-

(1) Typhoid

(2) Malaria

(3) Pneumonia

(4) Tuberculosis

BHD074

48. Ringworm in humans is caused by :

(1) Viruses (2) Bacteria

(3) Fungi (4) Nematodes

BHD075

49. Consider the following four statements (a-d) regarding kidney transplant and select the two **correct** ones out of these.

(a) Even if a kidney transplant is proper the recipient may need to take immune suppressants for a long time.

(b) The cell-mediated immune response is responsible for the graft rejection.

(c) The B-lymphocytes are responsible for rejection of the graft.

(d) The acceptance or rejection of a kidney transplant depends on specific interferons.

The two **correct** statements are :-

(1) (a) and (b) (2) (b) and (c)

(3) (c) and (d) (4) (a) and (c)

BHD076

50. Which one of the following statements is **correct** with respect to AIDS ?

(1) The causative HIV retrovirus enters helper T-lymphocytes thus reducing their numbers

(2) The HIV can be transmitted through eating food together with an infected person

(3) Drug addicts are least susceptible to HIV infection

(4) AIDS patients are being fully cured cent per cent with proper care and nutrition

BHD077

51. Select the **correct** statement from the ones given below : -

(1) Cocaine is given to patients after surgery as it stimulates recovery.

(2) Barbiturates when given to criminals make them tell the truth.

(3) Morphine is often given to persons who have undergone surgery as a pain killer.

(4) Chewing tobacco lowers blood pressure and heart rate.

BHD078

52. Which one of the following techniques is safest for the detection of cancers ?

(1) Radiography (X–ray)

(2) Computed tomography (CT)

(3) Histopathological studies

(4) Magnetic resonance imaging (MRI)

BHD079

53. At which stage of HIV infection does one usually show symptoms of AIDS ?

(1) When the infecting retrovirus enters host cells.

(2) When viral DNA is produced by reverse transcriptase.

(3) When HIV replicates rapidly in helper T-lymphocytes and damages large number of these.

(4) Within 15 days of sexual contact with an infected person.

BHD080

54. A certain patient is suspected to be suffering from Acquired Immuno Deficiency Syndrome. Which diagnostic technique will you recommend for its detection?

(1) ELISA

(2) MRI

(3) Ultra sound

(4) WIDAL

BHD081

55. Common cold is not cured by antibiotics because it is:

(1) not an infectious disease

(2) caused by a virus

(3) caused by a Gram-positive bacterium

(4) caused by a Gram-negative bacterium

BHD082

56. Which one of the following options gives the correct matching of a disease with its causative organism and mode of infection?

	Disease	Causative Organisms	Mode of Infection
(1)	Malaria	Plasmodium vivax	Bite of male anopheles mosquito
(2)	Typhoid	Salmonella typhi	With inspired air
(3)	Pneumonia	Streptococcus pneumonia	Droplet infection
(4)	Elephantiasis	Streptococcus pneumonia	With infected water and food

BHD083

57. Select the **correct** statement with respect to diseases and immunisation :

(1) Injection of snake antivenom against snake bite is an example of active immunisation.

(2) If due to some reason B-and T-lymphocytes are damaged, the body will not produce antibodies against a pathogen.

(3) Injection of dead/inactivated pathogens causes passive immunity

(4) Certain protozoans have been used to mass produce Hepatitis-B vaccine

BHD084

58. Common cold differs from pneumonia in, that

(1) Pneumonia is caused by a virus while the common cold is caused by the bacterium *Haemophilus influenzae*

(2) Pneumonia pathogen infects alveoli whereas the common cold affects nose and respiratory passage but not the lungs.

(3) Pneumonia is a communicable disease whereas the common cold is a nutritional deficiency disease.

(4) Pneumonia can be prevented by a live attenuated bacterial vaccine whereas the common cold has no effective vaccine.

BHD085

59. Widal Test is carried out to test :

(1) HIV/AIDS (2) Typhoid fever

(3) Malaria (4) Diabetes mellitus

BHD086

60. Cirrhosis of liver is caused by the chronic intake of:

(1) Tobacco (Chewing)

(2) Cocaine

(3) Opium

(4) Alcohol

BHD087

61. Motile zygote of *Plasmodium* occurs in :-

(1) Human RBCs

(2) Human liver

(3) Gut of female *Anopheles*

(4) Salivary glands of *Anopheles*

BHD088

62. In which one of the following options the two examples are correctly matched with their particular type of immunity?

	Examples	Type of immunity
(1)	Saliva in mouth and Tears in eyes	Physical barriers
(2)	Mucus coating of epithelium lining the urinogenital tract and the HCl in stomach	Physiological barriers
(3)	Polymorpho nuclear leukocytes and monocytes	Cellular barriers
(4)	Anti-tetanus and anti-snake bite injections	Active immunity

BHD089

63. The first clinical gene therapy was given for treating :-

(1) Rheumatoid arthritis

(2) Adenosine deaminase deficiency

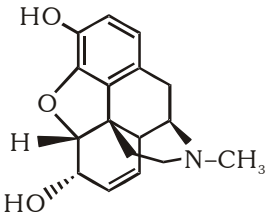
(3) Diabetes mellitus

(4) Chicken pox

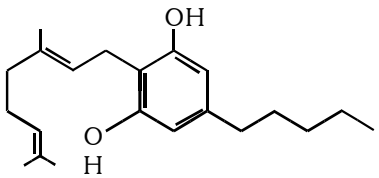
BHD090

64. Identify the molecules (a) and (b) shown below and select the right option giving their source and use.

(a)



(b)



	Molecule	Source	Use
(1)	(b) Cannabinoid	*Atropa belladona*	Produces hallucinations
(2)	(a) Morphine	*Papaver somniferum*	Sedative and pain killer
(3)	(a) Cocaine	*Erythroxylum coca*	Accelerates the transport of dopamine
(4)	(b) Heroin	*Cannabis sativa*	Depressant and slows down body functions

BHD091

65. Which one of the following statements is correct with respect to immunity?

(1) Antibodies are protein molecules, each of which has four light chains

(2) Rejection of a kidney graft is the function of B-lymphocytes.

(3) Preformed antibodies need to be injected to treat the bite by a viper snake.

(4) The antibodies against small pox pathogen are produced by T-lymphocytes

BHD092

66. The cell-mediated immunity inside the human body is carried out by :

(1) Erythrocytes

(2) T-lymphocytes

(3) B-lymphocytes

(4) Thrombocytes

BHD093

67. Infection of *Ascaris* usually occurs by :

(1) mosquito bite

(2) drinking water containing eggs of *Ascaris*

(3) eating imperfectly cooked pork.

(4) Tse-tse fly

BHD094

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	2	2	1	4	4	3	4	4	4	4	1	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	4	4	3	3	3	1	2	1	1	2	4	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	2	4	4	4	3	2	1	2	1	3	3	2	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	3	1	1	3	4	3	1	2	3	2	2	2	4
Question	61	62	63	64	65	66	67								
Answer	3	3	2	2	3	2	2								

Exercise - I (Conceptual Questions) Build Up Your Understanding

ORIGIN OF LIFE

1. Primitive atmosphere was made up of the mixture of :
(1) Oxygen, ammonia, methane, water
(2) Hydrogen, ammonia, methane, oxygen
(3) Hydrogen, steam, methane, ammonia
(4) Oxygen, methane, water, nickle

BEV001

2. Which compounds were formed in the direction of the origin of life ?
(1) Urea, nucleic-acid
(2) Urea, amino-acid
(3) Proteins, nucleic-acid
(4) Protein, amino-acid

BEV002

3. What is most important for origin of life ?
(1) Carbon
(2) Oxygen
(3) Water
(4) Nitrogen

BEV003

4. Pasteur succeeded in disproving the theory of spontaneous generation because :
(1) The laboratory was clean
(2) He pulled out the neck of flask into a tube
(3) He was lucky
(4) Yeast used in flask were dead

BEV004

5. Now the basis of origin of life is :
(1) Spontaneous generation
(2) God's desire
(3) Sunlight on mud
(4) None of them

BEV005

6. Oxygen in atmosphere has been formed by:
(1) Evaporation of water
(2) Photosynthesis of blue green algae
(3) Metabolism of microorganisms
(4) Decaying organisms

BEV006

7. Primitive atmosphere was reducing because :
(1) Hydrogen atoms were few
(2) Hydrogen atoms were active and in greater number
(3) Nitrogen atoms were more
(4) Oxygen atoms were more

BEV007

8. Who called water of primitive sea as pre biotic soup ?
(1) Haldane (2) Oparin
(3) Fox (4) Huxley

BEV009

9. Oparin's theory is based on :
(1) Artificial synthesis
(2) Spontaneous generation
(3) God's will
(4) All of these

BEV010

10. Which biologist gave most logical biochemical theory of origin of life ?
(1) Urey (2) Oparin
(3) Stanley Miller (4) Haeckel

BEV011

11. It is believed that the first organisms which inhabitated earth's surface were :
(1) Autotrophs (2) Mixotrophs
(3) Heterotrophs (4) Chromatotrophs

BEV012

12. A compound having very important role in prebiotic evolution was :

(1) SO_2 (2) CH_4 (3) SO_3 (4) NO

BEV013

13. Under certain conditions scientists have obtained non-cellular structures. These are known as :

(1) Microbes (2) Protists

(3) Coacervates (4) Prebiotic soup

BEV014

14. Who did an experiment to prove that "The organic compounds were the basis of life"?

(1) Darwin

(2) Stanley Miller and Harold C. Urey

(3) Melvin

(4) Fox

BEV015

15. First life on earth was :-

(1) Cyanobacteria

(2) Chemoheterotrophs

(3) Autotrophs

(4) Photoautotrophs

BEV017

16. Which of the following is favourite idea of astronomers ?

(1) Special creation theory

(2) Cosmic panspermia

(3) Biogenesis

(4) Abiogenesis

BEV018

17. During chemical evolution, key biological compounds were synthesised :-

(1) in the atmosphere

(2) along the ocean shore

(3) in the ocean

(4) none of the above

BEV019

18. Miller and Urey performed an experiment to prove the origin of life. They took gases NH_3 and H_2 along with :

(1) N_2 and H_2O (2) H_2O and CH_4

(3) CH_4 and N_2 (4) CO_2 and NH_3

BEV021

19. Abiogenesis is the :

(1) Origin of life from non-living material

(2) Origin of life from living organism

(3) Origin of viruses and microbes

(4) None

BEV022

20. Most modern hypothesis regarding origin of life was given by :

(1) Wallace (2) Hugo de Vries

(3) Oparin (4) Charles Darwin

BEV023

EVIDENCES OF ORGANIC EVOLUTION

21. According to recapitulation theory which statement is correct :-

(1) Development of individual metazon shown embryonic characters of ancestors.

(2) Ontogeny repeats phylogeny.

(3) Germplasm is immortal.

(4) Every organisms is produced by its parents.

BEV025

22. Which of the following set has homologous organs ?

(1) Hands of man, monkey and kangaroo and trunk of elephant.

(2) Wings of insects, birds and bats.

(3) Hind limbs of grasshopper, horse and bat

(4) Mouthparts of cockroach, mosquito and honey bee.

BEV026

23. Who was the first to explain recapitulation theory ?
(1) Weismann (2) Haeckel
(3) Darwin (4) Malthus
BEV027

24. Which of the following sets do not have homologous organs ?
(1) Wings of mosquito and butterfly
(2) Wings of butterfly and bat
(3) Mouth parts of cockroach and butter fly
(4) None of them
BEV030

25. Wings of locust, pigeon, and bat are example of :
(1) Vestigial organs
(2) Analogous organs
(3) Homologous organs
(4) Exoskeleton
BEV031

26. Homology is exhibited by :
(1) Wings of butterfly, birds and bat
(2) Paddle of whale, forearm of horse and forelimbs of man
(3) Tail of monkey and bird
(4) Sting of scorpion and honey bee
BEV032

27. Homologous organs are :
(1) Dissimilar origin and dissimilar structures
(2) Dissimilar origin but similar functions
(3) Similar origin with similar or dissimilar functions
(4) Similar origin with dissimilar functions
BEV039

28. Human hand, wing of bat and flipper of whale represent
(1) Analogous organs
(2) Vestigial organs
(3) Homologous organs
(4) Evolutionary organs
BEV040

29. Dinosaurs disappeared during :
(1) Jurassic (2) Triassic
(3) Cretaceous (4) Permian
BEV041

30. A connecting link between reptiles and birds is:
(1) Archaeopteryx (2) Platypus
(3) Java ape man (4) Whale
BEV042

31. Evolution of heart from one to two, three and four chambered proves :-
(1) Biogenetic law of Haeckel
(2) Lamarckism
(3) Hardy weinberg's law
(4) Neo Darwinism
BEV043

32. Which evidence of evolution is related to Darwin's finches ?
(1) Evidences from biogeographical distribution
(2) Evidences from vestigeal organs
(3) Evidences from embryology
(4) Evidences from palaeontology
BEV045

33. Similarities in organisms with different genotype indicate :-
(1) Microevolution
(2) Macroevolution
(3) Convergent evolution
(4) Divergent evolution
BEV046

34. Potato and sweet potato :–
(1) have edible parts which are homologous organs.
(2) have edible parts which are analogous organs.
(3) have been introduced in India from the same place.
(4) are two species of the same genus.
BEV048

35. Fossils are :
(1) animals living in burrows
(2) remnants of extinct animals and plants
(3) floating organisms
(4) fast runners

BEV050

36. The age of fossils is determined by :
(1) analysis of bones
(2) radioactive C^{14} dating
(3) electron microscopy
(4) weighing the fossils

BEV051

37. Missing link in evolution is :
(1) *Peripatus* (2) *Limulus*
(3) *Pheretima* (4) *Archaeopteryx*

BEV052

38. Convergent evolution of two species is associated with:
(1) analogous organs
(2) recent common ancestor
(3) homologous organs
(4) different habitat

BEV053

39. Organs which have the same fundamental structure but are different in function, are called ?
(1) vestigial organs
(2) homologous organs
(3) analogous organs
(4) homoplastic organs

BEV054

40. Wings of insects and wings of birds are the examples of :
(1) Analogy (2) Homology
(3) Serology (4) Mimicry

BEV055

41. Flippers of seal are modified :
(1) fins (2) hindlimb
(3) forelimb (4) gills

BEV057

42. Darwin's finches are an example of :
(1) Divergent evolution
(2) Adaptive radiation
(3) Allopatric speciation
(4) All of these

BEV058

THEORIES OF ORGANIC EVOLUTION

43. Theory of evolution is mainly concerned with:
(1) Spontaneous generation
(2) Theory of special creation
(3) Gradual change
(4) Conditions of environment

BEV059

44. Change with descent is the basis of which theory ?
(1) Recapitulation theory
(2) Oparin's theory
(3) Theory of organic evolution
(4) Cell theory

BEV060

45. Name of the scientist who gave Mutation Theory:
(1) Wallace (2) Malthus
(3) Darwin (4) De Vries

BEV061

46. Darwin's Theory of Natural Selection was based on:
(1) Inheritance of acquired characters
(2) Mutation
(3) Enormous rate of reproduction in organisms, struggle for existence and survival of the fittest
(4) Changes due to the use and disuse of organs

BEV062

47. Snakes do not have legs because :
(1) Legs are lost during their entry in tunnels
(2) Legs are lost during evolution
(3) The ancestors of reptiles did not have legs
(4) There are no legs in lizards

BEV063

48. One of the revolutionary concepts in biology was Charles Darwin's 'Origin of Species'. It deals with
(1) Gene mutation
(2) Use and disue of organs
(3) Germplasm Theory
(4) Natural selection leading to the survival of the fittest

BEV064

49. Which of the following is responsible for evolution according to Neo-Darwinism ?
(1) Mutation
(2) Natural selection
(3) Mutation and Natural selection
(4) Either (1) or (2)

BEV065

50. A Scientist kept 69 generations of Drosophila in darkness even after that the flies had normal eyes, this disproves the law of :
(1) Synthetic theory
(2) Natural selection
(3) Germplasm theory
(4) Acquired characters are inherited

BEV066

51. The ultimate source of variation is :
(1) Mutation
(2) Sexual reproduction
(3) Natural selection
(4) Hormonal action

BEV067

52. Which is the most important factor for evolution of new species ?
(1) Geographic isolation
(2) Extensive in-breeding
(3) Reproductive isolation
(4) Adaptation

BEV068

53. Gene pool is :
(1) Genotype of an individual of a population
(2) Different genes of all individuals of a species found in an area
(3) Pool of artificially synthesised genes
(4) Genes of a genus

BEV069

54. Which of the following evidences does not favour the Lamarckian concept ?
(1) Absence of limbs in snakes
(2) Presence of webbed toes in aquatic birds
(3) Melanization in peppered moth in industrial area
(4) Lack of pigment in cave dwelling animals

BEV070

55. Match the following columns and find correct combination :

	Column-I		Column-II
a.	Darwin	p.	Mutation theory
b.	De Vries	q.	Protobiont
c.	Pasteur	r.	Origin of species
d.	Fox	s.	Special Creation
		t.	Swan-Necked Flask Experiment

(1) a = r, b, = p, c = t, d = q
(2) a = p, b = q, c = r, d = s
(3) a = t, b = r, c = q, d = p
(4) a = r, b = t, c = p, d = q

BEV072

56. Frequency of an allele in an isolated population may change due to :-
(1) Genetic drift (2) Gene flow
(3) Mutation (4) Natural selection
BEV073

57. There are two alleles (A_1 & A_2) out of which one (A_1) has nil abundance in a population then the abundance of second allele (A_2) is
(1) 0.25 (2) 1.00 (3) 0.40 (4) 0.50
BEV388

58. The migration of gene in to a population from other population by interbreeding is called :-
(1) Gene pool (2) Gene flow
(3) Genetic drift (4) Gene erosion
BEV389

59. Equilibrium of gene frequencies is :-
(1) $P^2 \times 2Pq \times q^2 = 1$
(2) $\sigma = \sqrt{\frac{Pq}{N}}$
(3) Hardy Weinberg law
(4) Mutation
BEV390

60. In a random mating population of 28,800 individuals, percentage of dominant homozygous individuals is 49% find out the percentage of heterozygous individuals :
(1) 21% (2) 42% (3) 32% (4) 9%
BEV391

61. In a population that is in Hardy Weinberg equilibrium, the frequency of a recessive allele for a certain hereditary trait is 0.20. What percentage of the individual in the next generation would be expected to show the dominant trait :-
(1) 16% (2) 32% (3) 64% (4) 96%
BEV392

62. In a random mating population frequency of disease-causing recessive allele is 80%. What would be the frequency of carrier individuals in population :-
(1) 64% (2) 32%
(3) 16% (4) 100%
BEV393

63. In a random mating population frequency of dominant allele is 0.7. What will be the frequency of recessive phenotype :–
(1) 0.49 (2) 0.09 (3) 0.3 (4) 0.21
BEV394

64. At a particular locus, frequency of 'A' allele is 0·6 and that of 'a' is 0·4. What would be the frequency of heterozygotes in a random mating population at equilibrium :-
(1) 0·24 (2) 0·16 (3) 0·48 (4) 0·36
BEV395

65. The existence within a population of non-beneficial alleles in heterozygous genotype is:
(1) genetic load (2) genetic drift
(3) genetic flow (4) selection
BEV396

66. Some bacteria are able to grow in Streptomycin containing medium due to -
(1) Natural selection
(2) Induced mutation
(3) Reproductive isolation
(4) Genetic drift
BEV074

67. Which of the following is important for speciation ?
(1) Seasonal isolation
(2) Reproductive isolation
(3) Behavioural isolation
(4) Tropical isolation
BEV075

68. Genetic drift rapidly operates in :-
(1) Small isolated population
(2) Large isolated population
(3) Fast reproductive population
(4) Slow reproductive population
BEV076

69. De Vries gave his mutation theory on organic evolution while working on –
(1) *Oenothera lamarckiana*
(2) *Drosophila melanogaster*
(3) *Pisum sativum*
(4) *Althea rosea*
BEV077

70. Which of the following factors help in evolution but is not considered as the basic factor for evolution ?
(1) Isolation (2) Adaptation
(3) Variation (4) Mutation
BEV078

71. Factors helps in the formation of new species are:
(1) competition and variation
(2) isolation and competition
(3) competition and mutation
(4) isolation and mutation
BEV079

72. The idea not related to the Darwinian evolutionary theory is :
(1) survival of the best
(2) struggle for existence
(3) inheritance of acquired characters
(4) origin of species by natural selection
BEV080

73. Inheritance of acquired characters comes from :
(1) Lamarckism (2) Darwinism
(3) Neo-Lemarckism (4) Neo-Darwinism
BEV081

74. Coverstone of theory of Darwin was :
(1) natural selection
(2) inheritance of acquired characters
(3) omnis cellulae e cellulae
(4) higher productivity
BEV082

75. The chance of elimination of genes from a small population is an example of :
(1) selection pressure (2) speciation
(3) adaptation (4) genetic drift
BEV083

76. The classical example of adaptive radiation is:
(1) Darwin finches
(2) Marsupials of Australia
(3) Giant turtle
(4) All of these
BEV084

77. Struggle for existence and survival of the fittest theories were given by :
(1) Wallace (2) Darwin
(3) Lamarck (4) None of these
BEV085

78. Which of the following is not a concept of Lamarck?
(1) Environmental pressure causes variation
(2) Rate and survival of organism is different due to variation
(3) Inheritance of acquired character
(4) If an organ is used constantly it will continuously increase its size.
BEV086

79. Initiating force of evolution is :
(1) Variation
(2) Natural selection
(3) Adaptation
(4) Competition
BEV087

80. According to the Neo-Darwinian theory which of the following is responsible for the origin of new species?
(1) Mutations only
(2) Useful variations and natural selection
(3) Mutations together with natural selection
(4) Hybridization only

BEV088

81. Which of the following was not given by Darwin's theory of evolution ?
(1) Struggle for existence
(2) Over production
(3) Natural selection
(4) Genetic drift

BEV089

HUMAN EVOLUTION

82. Which primate is closest to man regarding organic evolution ?
(1) Gibbon
(2) Gorilla
(3) Sinanthropus
(4) Orangutan

BEV090

83. Which character applies to *Homo sapiens* ?
(1) Opposable toe
(2) Large canine
(3) Cranial capacity 1450 cc
(4) Chin prominence absent

BEV091

84. *Homo erectus* differed from Cro-magnon man in having :
(1) Jaws protruding out
(2) Tool making
(3) Sloping jaws
(4) Arts and paintings

BEV092

85. Which of the following statement is correct?
(1) Proconsul was ancestor of man and ape
(2) Proconsul was ancestor of man and not of ape
(3) Apes were ancestor of man anatomically
(4) None of them

BEV093

86. Most recent man found as fossil was :
(1) Java man
(2) Peking man
(3) Cro-magnon man
(4) Hiedelberg man

BEV094

87. What was the cranial capactiy of java man?
(1) 400 cc (2) 650 cc
(3) 900 cc (4) 1450 cc

BEV095

88. Evolution of man was possible because our ape-like ancestors :
(1) Showed bipedal movement on open land
(2) Used fire
(3) Felt difficulty in nutrition
(4) Developed community hunting

BEV096

89. Characteristics of primitive monkey which was in the direction of evolution of man :
(1) Thumb parallel to fingers
(2) 32 teeth
(3) Prehensile tail
(4) Flat nose

BEV099

90. Greatest advantage of bipedal movement :
(1) Fore arms becoming free for carrying out order of brain
(2) Greater speed
(3) Support the body properly
(4) Loss of weight

BEV100

91. Which fossil man had cranial capacity almost equal to modern man ?
(1) *Australopithecus*
(2) Java ape man
(3) Neanderthal man
(4) Peking man
BEV101

92. Largest cranial capacity was found in :
(1) Peking man
(2) Neanderthal man
(3) Java man
(4) Cro-magnon man
BEV102

93. Cro-magnon man was :
(1) herbivorous (2) frugivorous
(3) carnivorous (4) omnivorous
BEV103

94. Which of the following is the most primitive ancestor of man ?
(1) *Homo habilis*
(2) *Ramapithecus*
(3) *Australopithecus*
(4) *Homo neanderthalensis*
BEV104

95. *Homo habilis* refers to :
(1) Wandering species
(2) Ancient man
(3) Modern man
(4) Tool-maker
BEV105

96. Which of the following statement is true ?
(1) *Homo erectus* is direct ancestor of *Homo sapiens*
(2) Neanderthal man is direct ancestor of modern man
(3) *Australopithecus* is direct ancestor of modern man
(4) Fossils of Cromagnon man first found in Ethopia
BEV106

97. Character which is closely related to human evolution-
(1) Disappearance of tail
(2) Reduction in size of jaws
(3) Binocular vision
(4) Flat nails
BEV107

98. Who is directly related to man ?
(1) Gorilla (2) Rhesus
(3) Gibbon (4) Orangutan
BEV108

99. Which of the following is the closer relative of man ?
(1) Chimpanzee (2) Gorilla
(3) Oranguttan (4) Gibbon
BEV109

100. Which of the following is correct order of the evolutionary history of man ?
(1) Peking man, *Homo sapiens*, Neanderthel man, Cro-magnon man
(2) Peking man, Neanderthal man, *Homo sapiens*, Cro-magnon man
(3) Peking man, Neanderthal man, Cro-magnon man, Modern man
(4) Peking man, Neanderthal man, *Homo sapiens*, Heidelberg man
BEV110

101. According to fossils which are discovered up to present time, origin and evolution of man was started from which country ?
(1) France (2) Java
(3) Africa (4) China
BEV111

102. The banding pattern of chromosomes of 3 and 6 of human beings and chimpanzee shows that they had:
(1) common origin
(2) different origin
(3) same number of chromosomes
(4) similar blood groups
BEV113

103. Which of the following ancestor of man was fond of painting and weapons making?

(1) Neanderthal man

(2) Cro-magnon man

(3) Java man

(4) Peking man

BEV114

104. The scientific name of *Homo erectus erectus* has been given to :

(1) Cro-magnon man

(2) Neanderthal man

(3) Java man

(4) Peking man

BEV115

105. Closest ancestor to modern man was :

(1) Neanderthal man

(2) *Homo habilis*

(3) Cro-magnon man

(4) *Australopithecus*

BEV116

106. The cranial capacity of Peking man was about:

(1) 900 cc (2) 1660 cc

(3) 1075 cc (4) 1450 cc

BEV117

107. Ancestor of man who first stood erect was?

(1) *Australopithecus*

(2) Cro-magnon man

(3) Java man

(4) Peking man

BEV118

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	2	4	2	2	1	1	2	3	2	3	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	2	1	3	2	4	2	2	2	2	3	3	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	1	3	2	2	2	4	1	2	1	3	4	3	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	4	3	4	1	3	2	3	1	1	2	2	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	2	2	3	1	1	2	1	1	2	4	3	1	1	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	2	2	1	3	4	3	3	1	1	3	3	1	2	1
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	4	4	2	4	1	2	1	1	3	3	1	2	3	3
Question	106	107													
Answer	3	1													

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Forelimbs of cat, lizard used in walking; forelimbs of whale used in swimming and forelimbs of bats used in flying are an example of:
(1) Analogous organs
(2) Adaptive radiation
(3) Homologous organs
(4) Convergent evolution

BEV153

2. Which one of the discarded are analogous structures: -
(1) Wings of Bat and Wings of Pigeon.
(2) Gills of Prawn and Lungs of Man.
(3) Thorns of Bougainvillea and Tendrils of Cucurbita
(4) Flippers of Dolphin and Legs of Horse

BEV154

3. In a population of 1000 individuals 360 belong to genotype AA, 480 to Aa and the remaining 160 to aa. Based on this data, the frequency of allele A in the population is :-
(1) 0.4 (2) 0.5 (3) 0.6 (4) 0.7

BEV397

AIPMT 2015

4. Which of the following had the smallest brain capacity?
(1) *Homo sapiens*
(2) *Homo neanderthalensis*
(3) *Homo habilis*
(4) *Homo erectus*

BEV155

5. Which is the most common mechanism of genetic variation in the population of sexually reproducing organism?
(1) Chromosomal aberrations
(2) Genetic drift
(3) Recombination
(4) Transduction

BEV156

6. A population will not exist in Hardy - Weinberg equilibrium if :-
(1) There are no mutations
(2) There is no migration
(3) The population is large
(4) Individuals mate selectively

BEV398

Re-AIPMT 2015

7. The wings of a bird and the wings of an insect are:
(1) homologous structures and represent convergent evolution
(2) homologous structures and represent divergent evolution
(3) analogous structures and represent convergent evolution
(4) phylogenetic structures and represent divergent evolution

BEV157

8. Industrial melanism is an example of:
(1) Neo Lamarckism
(2) Neo Darwinism
(3) Natural selection
(4) Mutation

BEV158

9. Which the following are most suitable indicators of SO_2 pollution in the environment?

(1) Fungi (2) Lichens

(3) Conifers (4) Algae

BEV159

NEET-I 2016

10. Which of the following structure is homologous to the wing of a bird?

(1) Dorsal fin of a Shark

(2) Wing of a Moth

(3) Hind limb of Rabbit

(4) Flipper of Whale

BEV160

11. Analogous structures are a result of: -

(1) Divergent evolution

(2) Convergent evolution

(3) Shared ancestry

(4) Stabilizing selection

BEV161

12. Following are the two statements regarding the origin of life: -

(a) The earliest organisms that appeared on the earth were non-green and presumably anaerobes.

(b) The first autotrophic organisms were the chemoautotrophs that never released oxygen.

Of the above statements which one of the following options is correct?

(1) (a) is correct but (b) is false.

(2) (b) is correct but (a) is false.

(3) Both (a) and (b) are correct.

(4) Both (a) and (b) are false.

BEV162

NEET-II 2016

13. Genetic drift operates in:

(1) Non-reproductive population

(2) Slow reproductive population

(3) Small isolated population

(4) Large isolated population

BEV163

14. In Hardy-Weinberg equation, the frequency of heterozygous individual is represented by:-

(1) pq (2) q^2

(3) p^2 (4) 2pq

BEV164

15. The chronological order of human evolution from early to the recent is: -

(1) Ramapithecus → *Homo habilis* → Australopithecus → Homo erectus

(2) Australopithecus → *Homo habilis* → Ramapithecus → Homo erectus

(3) Australopithecus → Ramapithecus → *Homo habilis* → Homo erectus

(4) Ramapithecus → Australopithecus → *Homo habilis* → Homo erectus

BEV165

16. Which of the following is the correct sequence of events in the origin of life?

I. Formation of protobionts

II. Synthesis of organic monomers

III. Synthesis of organic polymers

IV. Formation of DNA-based genetic systems

(1) II, III, I, IV

(2) II, III, IV, I

(3) I, II, III, IV

(4) I, III, II, IV

BEV166

17. Genetic drift operates in :-

(1) Non-reproductive population

(2) Slow reproductive population

(3) Small isolated population

(4) Large isolated population

BEV399

18. In Hardy-Weinberg equation, the frequency of heterozygous individual is represented by:-

(1) pq (2) q^2

(3) p^2 (4) 2pq

BEV400

NEET-UG 2017

19. Artificial selection to obtain cows yielding higher milk output represents:

(1) Directional as it pushes the mean of the character in one direction

(2) Disruptive as it splits the population into two, one yielding higher output and the other lower output

(3) Stabilizing followed by disruptive as it stabilizes the population to produce higher yielding cows

(4) Stabilizing selection as it stabilizes this character in the population

BEV167

NEET-UG 2018

20. The similarity of bone structure in the forelimbs of many vertebrates is an example of:

(1) Homology

(2) Analogy

(3) Convergent evolution

(4) Adaptive radiation

BEV168

21. Among the following sets of examples for divergent evolution, select the incorrect option:-

(1) Forelimbs of man, bat and cheetah

(2) Heart of bat, man and cheetah

(3) Brain of bat, man and cheetah

(4) Eye of octopus, bat and man

BEV169

22. According to Hugo de Vries, the mechanism of evolution is:

(1) Multiple step mutations

(2) Saltation

(3) Phenotypic variations

(4) Minor mutations

BEV170

NEET-UG 2019

23. Variations caused by mutation, as proposed by Hugo de Vries, are :

(1) Random and directional

(2) Random and directionless

(3) Small and directional

(4) Small and directionless

BEV171

24. In a species, the weight of new-born ranges from 2 to 5 kg. 97% of the new-born with an average weight between 3 to 3.3 kg survive whereas 99% of the infants born with weights from 2 to 2.5 kg or 4.5 to 5 kg die. Which type of selection process is taking place?

(1) Directional Selection

(2) Stabilizing Selection

(3) Disruptive Selection

(4) Cyclical Selection

BEV172

25. Match the hominids with their correct brain size:

(a) *Homo habilis*	(i) 900 cc
(b) *Homo neanderthalensis*	(ii) 1350 cc
(c) *Homo erectus*	(iii) 650-800 cc
(d) *Homo sapiens*	(iv) 1400 cc

Select the correct option.

(1) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

(2) (a)–(iii), (b)–(ii), (c)–(i), (d)–(iv)

(3) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

(4) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)

BEV173

26. A gene locus has two alleles A, a. If the frequency of dominant allele A is 0.4 then what will be the frequency of homozygous dominant, heterozygous and homozygous recessive individuals in the population ?

(1) 0.36 (AA); 0.48(Aa); 0.16 (aa)

(2) 0.16 (AA); 0.24 (Aa); 0.36 (aa)

(3) 0.16 (AA); 0.48 (Aa); 0.36 (aa)

(4) 0.16 (AA); 0.36 (Aa); 0.48 (aa)

BEV401

NEET-UG 2019 (Odisha)

27. In Australia, marsupials and placental mammals have evolved to share many similar characteristics. This type of evolution may be referred to as:

(1) Adaptive Radiation

(2) Divergent Evolution

(3) Cyclical Evolution

(4) Convergent Evolution

BEV174

28. Which of the following statements is correct about the origin and evolution of men?

(1) Agriculture came around 50,000 years back.

(2) The *Dryopithecus* and *Ramapithecus* primates existing 15 million years ago, walked like men.

(3) Homo habilis probably ate meat.

(4) Neanderthal men lived in Asia between 1,00,000 and 40,000 years back.

BEV175

NEET-UG 2020

29. Embryological support for evolution was disapproved by :

(1) Oparin

(2) Karl Ernst von Baer

(3) Alfred Wallace

(4) Charles Darwin

BEV176

30. Flippers of Penguins and Dolphins are examples of:

(1) Natural selection

(2) Adaptive radiation

(3) Convergent evolution

(4) Industrial melanism

BEV177

31. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask :

(1) CH_3, H_2, NH_3 and water vapour at 600°C

(2) CH_4, H_2, NH_3 and water vapour at 800°C

(3) CH_3, H_2, NH_4 and water vapour at 800°C

(4) CH_4, H_2, NH_3 and water vapour at 600°C

BEV178

32. Which of the following refer to correct example(s) of organisms which have evolved due to changes in environment brought about by anthropogenic action?

(a) Darwin's Finches of Galapagos islands.

(b) Herbicide resistant weeds.

(c) Drug resistant eukaryotes.

(d) Man-created breeds of domesticated animals like dogs.

(1) Only (d) (2) Only (a)

(3) (a) and (c) (4) (b), (c) and (d)

BEV179

NEET (UG) 2020 (COVID-19)

33. Embryological support for evolution was proposed by :

(1) Ernst Heckel

(2) Karl Ernst von Baer

(3) Charles Darwin

(4) Alfred Wallace

BEV180

34. After about how many years of formation of earth, life appeared on this planet ?

(1) 500 billion years

(2) 50 million years

(3) 500 million years

(4) 50 billion years

BEV181

35. The phenomenon of evolution of different species in a given geographical area starting from a point and spreading to other habitats is called :-

(1) Saltation

(2) Co-evolution

(3) Natural selection

(4) Adaptive radiation

BEV182

36. A Hominid fossil discovered in Java in 1891, now extinct, having cranial capacity of about 900 cc was:

(1) *Homo erectus* (2)Neanderthal man

(3) *Homo sapiens* (4) *Australopithecus*

BEV183

NEET (UG) 2021

37. The factor that leads to Founder effect in a population is:

(1) Natural selection

(2) Genetic recombination

(3) Mutation

(4) Genetic drift

BEV184

38. Match **List-I** with **List-II**.

	List-I		**List-II**
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the correct answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(ii)	(i)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(i)	(iv)	(iii)	(ii)

BEV185

NEET (UG) 2022

39. Natural selection where more individuals acquire specific character value other than the mean character value, leads to :-

(1) Directional change

(2) Disruptive change

(3) Random change

(4) Stabilising change

BEV186

40. Which of the following statements is not true?

(1) Sweet potato and potato is an example of analogy

(2) Homology indicates common ancestry

(3) Flippers of penguins and dolphins are a pair of homologous organs

(4) Analogous structures are a result of convergent evolution

BEV187

Re-NEET (UG) 2022

41. Panspermia, an idea that is still a favourite for some astronomers, means :

(1) Creation of life from dead and decaying matter.

(2) Creation of life from chemicals.

(3) Origin of sperm in human testes.

(4) Transfer of spores as unit of life from other planets of Earth.

BEV188

42. Select the correct statement regarding mutation theory of evolution.

(1) This theory was proposed by Alfred Wallace.

(2) Variations are small directional changes.

(3) Single step large mutation is a cause of speciation.

(4) Large differences due to mutations arise gradually in a population.

BEV189

NEET (UG) 2023

43. Select the correct group/set of Australian Marsupials exhibiting adaptive radiation.

(1) Numbat, Spotted cuscus, Flying phalanger

(2) Mole, Flying squirrel, Tasmanian tiger cat

(3) Lemur, Anteater, Wolf

(4) Tasmanian wolf, Bobcat, Marsupial mole

BEV291

NEET (UG) 2023 (Manipur)

44. Identify the fossil of man who showed the following characteristics.

(A) Brain capacity of 1400 cc

(B) Used hides to protect their body

(C) Buried their dead bodies

In the light of above statements, choose the **correct** answer from the options given below:

(1) *Homo erectus*

(2) Neanderthal man

(3) *Homo habilits*

(4) *Australopithecus*

BEV292

NEET (UG) 2024

45. The flippers of the Penguins and Dolphins are the example of the

(1) Adaptive radiation

(2) Natural selection

(3) Convergent evolution

(4) Divergent evolution

BEV293

46. Given below are some stages of human evolution. Arrange them in correct sequence (Past to Recent).

A. *Homo habilis*

B. *Homo sapiens*

C. *Homo neanderthalensis*

D. *Homo erectus*

Choose the **correct** sequence of human evolution from the options given below :-

(1) D-A-C-B (2) B-A-D-C

(3) C-B-D-A (4) A-D-C-B

BEV294

47. Match **List I** with **List II.**

List-I		List-II	
A.	Mesozoic Era	I.	Lower invertebrates
B.	Proterozoic Era	II.	Fish & Amphibia
C.	Cenozoic Era	III.	Birds & Reptiles
D.	Paleozoic Era	IV.	Mammals

Choose the **correct** answer from the options given below :-

(1) A-II, B-I, C-III, D-IV

(2) A-III, B-I, C-II, D-IV

(3) A-I, B-II, C-IV, D-III

(4) A-III, B-I, C-IV, D-II

BEV295

RE-NEET (UG) 2024

48. Match **List-I** with **List-II**.

List-I		List-II	
A.	Gene pool	I.	Stable within a generation
B.	Genetic drift	II.	Change in gene frequency by chance
C.	Gene flow	III.	Transfer of genes into or out of population.
D.	Gene frequency	IV.	Total number of genes and their alleles

Choose the **correct** answer from the options given below :-

(1) A-III, B-II, C-I, D-IV

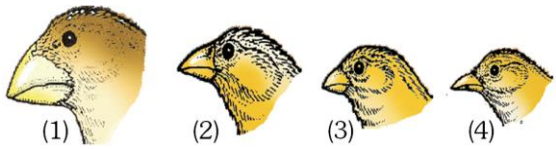
(2) A-IV, B-II, C-III, D -I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

BEV296

49. Which evolutionary phenomenon is depicted by the sketch given figure ?



(1) Artificial selection

(2) Genetic drift

(3) Convergent evolution

(4) Adaptive radiation

BEV297

50. Match **List-I** with **List-II** with respect to convergent evolution.

List-I		List-II	
A.	Lemur	I.	Flying phalanger
B.	Bobcat	II.	Numbat
C.	Ant eater	III.	Spotted cuscus
D.	Flying squirrels	IV.	Tasmanian tiger cat

Choose the **correct** answer from the options given below :-

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-II, C-IV, D -I

(3) A-IV, B-III, C-II, D-I

(4) A-IV, B-II, C-III, D-I

BEV298

51. Match **List-I** with **List-II.**

List-I		List-II	
A.	Living Fossil	I.	Elongated canine teeth
B.	Connecting Link	II.	Vermiform appendix
C.	Vestigial Organ	III.	*Echidna*
D.	Atavism	IV.	*Latimeria*

Choose the **correct** answer from the options given below :-

(1) A-IV, B-III, C-II, D-I

(2) A-IV, B-II, C-III, D-I

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-IV, C-I, D-II

BEV299

52. What is the correct order (old to recent) of periods in Paleozoic era ?

(1) Silurian, Devonian, Permian, Carboniferous

(2) Silurian, Devonian, Carboniferous, Permian

(3) Permian, Devonian, Silurian, Carboniferous

(4) Silurian, Carboniferous, Permian, Devonian

BEV300

EXERCISE - II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	3	3	3	4	3	3	2	4	2	3	3	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	4	1	1	4	2	2	2	3	3	4	4	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	1	3	4	1	4	1	1	3	4	3	1	2	3
Question	46	47	48	49	50	51	52								
Answer	4	4	2	4	1	1	2								

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Exercise - III Master Your Understanding

(A) NCERT BASED QUESTIONS

1. Time gap in between formation of earth & origin of life :
(1) 0.5 Billion years (2) 1 Billion years
(3) 4.5 Billion years (4) 4.0 Billion years
BEV190

2. Who completely discarded the theory of spontaneous generation ?
(1) Louis Pasteur
(2) Van Helmont
(3) Lazzaro spllanzani
(4) Fransisco Redi
BEV191

3. Stanley miller observed the formation of _____ during his experiment :
(1) Nucleic Acid (2) Nucleotides
(3) Amino Acid (4) Polypeptide
BEV192

4. First formation of life occured in :
(1) Water (2) Air
(3) At 800°C temp. (4) In Vapour state
BEV193

5. Ship used by Darwin :
(1) HSM Beagle
(2) His Majesty service
(3) HMS Beagle
(4) Her Major ship
BEV194

6. Marsupial mole, Koala, Bandicoot & Wombat are example of :
(1) Divergent evolution
(2) Convergent evolution
(3) Adaptive radiation
(4) Both (1) and (3)
BEV195

7. Darwin's finches represent the phenomenon of:
(1) Adaptive radiation
(2) Parallel evolution
(3) Convergent evolution
(4) Both (1) and (2)
BEV196

8. Flying squirrel & flying phalanger represent the phenomenon of :
(1) Divergent evolution
(2) Convergent evolution
(3) Adaptive radiation
(4) Divergent radiation
BEV197

9. According to Hugo de vries speciation due to mutation is also known as 'Saltation' which means :
(1) Single step variation
(2) Variations at regular intervals
(3) Single step large mutation
(4) Huge change due to Natural selection
BEV198

10. In some animals of different groups different structure developed along same direction due to Adaptation to same needs this is called as:
(1) Divergent evolution
(2) Convergent evolution
(3) Parallel evolution
(4) None of these
BEV199

11. Presence of coelacanth fish was observed in
(1) South America (2) South Africa
(3) North America (4) North Africa
BEV200

12. Which reptile went back into water to evolve into fish like reptile ?
(1) *Ichthyosaurs* (2) *Calotes*
(3) *Hemidactylus* (4) *Alligator*

BEV201

13. The biggest land reptile was :
(1) *Palycosaurs* (2) *Calotes*
(3) *Tyrannosaurus* (4) *Bangarus*

BEV202

14. Select the true statements :
(1) *Ramapithecus* and *Dryopithecus* were existing about 50 million years ago
(2) *Ramapithecus* was man like while *Dryopithecus* was more ape like
(3) *Ramapithecus* was more ape like while *Dryopithecus* was more man-like
(4) Both (1) and (2)

BEV203

15. Brain capacity of *Homo habilis* was :
(1) 650-800 cc (2) 800-900 cc
(3) 600-1000 cc (4) 900-1100 cc

BEV204

16. Fossils of *Homo erectus* was discovered in :
(1) Java (2) Bangladesh
(3) Ethiopia (4) Tanzania

BEV205

17. Neanderthal man lived near :
(1) East & West Africa (2) South Africa
(3) North Africa (4) East & Central Asia

BEV206

18. Prehistoric cave art developed about :
(1) 10000 years ago
(2) 15000 years ago
(3) 18000 years ago
(4) 20000 years ago

BEV207

19. The story of evolution of modern man in respect to brain & language appears as :
(1) Convergent evolution
(2) Divergent evolution
(3) Parallel evolution
(4) All of above

BEV208

20. Branching descent is accounted for :
(1) Homology (2) Analogy
(3) Vestigial organs (4) Atavism

BEV209

21. Highest brain capacity present in :
(1) *Australopithecus*
(2) *Homo habilis*
(3) *Homo erectus*
(4) Neanderthal

BEV210

22. Agriculture & Human settlement started about :
(1) 10000 years back
(2) 20000 years back
(3) 30000 years back
(4) 40000 years back

BEV211

23. Who proposed the first form of life from pre-existing non-living organic molecules ?
(1) Oparin & Haldane
(2) Stanley Miller & Harold Urey
(3) Lamark & Darwin
(4) Hugo De Vries & Dobzhansky

BEV212

24. In the experiment of Stanley Miller the temp. of water vapour was :
(1) 1200°C (2) 1000°C
(3) 800°C (4) 1500°C

BEV213

25. Darwin finches differ with each other on the basis of :

(1) Feather (2) Beak

(3) Colour of eye (4) Body colour

BEV214

26. Evolution is :

(1) Determinism (2) Chance

(3) Both (1) and (2) (4) None

BEV215

27. Darwin's finches represent :

(1) Adaptive radiation

(2) Divergent evolution

(3) Both (1) and (2)

(4) Parallel evolution

BEV216

28. Hugo de Vries worked on :

(1) *Oenothera lamarckiana*

(2) *Lathyrus odoratus*

(3) *Pisum sativum*

(4) *Arachis hypogea*

BEV217

29. Select the incorrect statements :

(1) Natural selection is a heritable variation & by reproduction leave greater number of progemy.

(2) During stabilisation of natural selection more individuals acquire value other than mean character value.

(3) By the time of 500 million years ago invertebrates were formed and were active.

(4) Reptiles lay thick shelled eggs which do not dry up in sun unlike those of Amphibians.

BEV218

30. Match the columns :

(a) *Australopithecus*	(i) First human like
(b) *Homo habilis*	(ii) Used hides to protect the body
(c) *Homo erectus*	(iii) Arose in Africa
(d) Neanderthal man	(iv) Probably ate meat
(e) *Homo sapiens*	(v) Hunted with stone weapons and ate fruits

(1) a-v, b-ii, c-iv, d-i, e-iii

(2) a-i, b-ii, c-iv, d-iii, e-v

(3) a-v, b-i, c-iv, d-ii, e-iii

(4) a-v, b-ii, c-iii, d-i, e-iv

BEV219

31. Which of the following is used as an atmospheric pollution indicator?

(1) Lepidoptera (2) Lichens

(3) Lycopersicon (4) Lycopodium

BEV220

32. The theory of spontaneous generation stated that:

(1) Life arose from living forms only.

(2) Life can arise from both living and non-living.

(3) Life can arise from non-living things only

(4) Life arises spontaneously, neither from living nor from the non-living.

BEV221

33. Animal husbandry and plant breeding programmes are the examples of :

(1) Reverse evolution

(2) Artificial selection

(3) Mutation

(4) Natural selection

BEV222

34. Palaentological evidences for evolution refers to the :
(1) Development of embryo
(2) Homologous organs
(3) Fossils
(4) Analogous organs.

BEV223

35. The bones of forelimbs of whale, bat, cheetah and man are similar in structure, because :
(1) One organism has given rise to another
(2) They share a common ancestor
(3) They perform the same function
(4) They have biochemical similarities

BEV224

36. Analogous organs arise due to :
(1) Divergent evolution
(2) Artificial selection
(3) Genetic drift
(4) Convergent evolution

BEV225

37. $(p+q)^2 = p^2 + 2pq + q^2 = 1$ represents an equation used in :
(1) Population genetics
(2) Mendelian genetics
(3) Biometrics
(4) Molecular genetics

BEV226

38. Appearance of antibiotic-resistant bacteria is an example of:
(1) Adaptive radiation
(2) Transduction
(3) Pre-existing variation in the population
(4) Divergent evolution

BEV227

39. Evolution of life shows that life forms had a trend of moving from:
(1) Land to water
(2) Dry land to wet land
(3) Fresh water to sea water
(4) Water to land

BEV228

40. Viviparity is considered to be more evolved because :
(1) The young ones are left on their own.
(2) The young ones are protected by a thick shell.
(3) The young ones are protected inside the mother's body and are looked after they are born leading to more chances of survival.
(4) The embryo takes a long time to develop

BEV229

41. Fossils are generally found in :
(1) Sedimentary rocks
(2) Igneous rocks
(3) Metamorphic rocks
(4) Any type of rock

BEV230

42. Which type of selection is industrial melanism observed in moth, *Biston betularia*?
(1) Stabilising
(2) Directional
(3) Disruptive
(4) Artificial

BEV231

43. The most accepted line of descent in human evolution is :

(1) *Australopithecus* → *Ramapithecus* → *Homo sapiens* → *homo habilis*

(2) *Homo erectus* → *Homo habilis* → *Homo sapiens*

(3) *Ramapithecus* → *Homo habilis* → *Homo erectus* → *Homo sapiens*

(4) *Australopithecus* → *Ramapithecus* → *Homo erectus* → *Homo habilis* → *Homo sapiens*

BEV232

44. Match the scientists listed under column 'A' with ideas listed column 'B'.

Column A		**Column B**	
i.	Darwin	M.	Abiogenesis
ii.	Oparin	N.	Use and disuse of organs
iii.	Lamarck	O.	Mutation theory
iv.	De-Vries	P.	Evolution by natural selection

Options :

(1) i-M; ii-P; iii-N; iv-O

(2) i-P; ii-M; iii-N; iv-O

(3) i-N; ii-P; iii-O; iv-M

(4) i-P; ii-O; iii-N; iv-M

BEV234

45. In 1953 S. L. Miller created primitive earth conditions in the laboratory and gave experimental evidence for origin of first form of life from preexisting non-living organic molecules. The primitive earth conditions created include :

(1) Low temperature, volcanic storms, atmosphere rich in oxygen

(2) Low temperature, volcanic storms, reducing atmosphere

(3) High temperature, volcanic storms, non-reducing atmosphere

(4) High temperature, volcanic storms, reducing atmosphere containing CH_4, NH_3 etc.

BEV235

46. Variations during mutations of meiotic recombination are :

(1) Random and directionless

(2) Random and directional

(3) Random and small

(4) Random, small and directional

BEV236

47. The naturalist who came about with the same ideas that of the Darwin :

(1) Alfred Nobel (2) Alfred Wallace

(3) Lamarck (4) H. Khorana

BEV237

48. According to Darwin, the term fitness refers to:

(1) Social fitness

(2) Mental fitness

(3) Physical fitness

(4) Reproductive fitness

BEV238

49. Dead remains of organisms found in the earth crust are the :

(1) Palaentological evidences

(2) Morphological evidences

(3) Anatomical evidences

(4) Physiological evidences

BEV239

50. When more than one adaptive radiations appear in an isolated geographical area, than it is called ?
(1) Natural selection
(2) Convergent evolution
(3) Divergent evolution
(4) Retrogressive evolution

BEV240

51. Who gave the theory of Genetic equilibrium
(1) Lamarck (2) Hugo DeVries
(3) Hardy - Weinberg (4) Darwin

BEV241

52. Giraffe's neck and forelimbs get elongated during the course of evolution, due to :
(1) Inheritance of acquired characters
(2) Natural selection
(3) Geographical isolation
(4) Convergent evolution

BEV242

53. First mammals were :
(1) Human (2) Apes
(3) Monkeys (4) Shrews

BEV243

54. The first human being like prehistoric man was:
(1) *Homo sapiens*
(2) *Homo erectus*
(3) *Homo habilis*
(4) Neanderthal man

BEV244

55. In which type of natural selection two peaks are formed ?
(1) Stabilizing selection
(2) Directional selection
(3) Disruptive selection
(4) Both (1) & (3)

BEV245

56. Which of the following statement is correct?
(1) The skull of adult chimpanzee is like modern adult human
(2) The skull of baby chimpanzee is like modern adult human
(3) Skull of baby chimpanzee is exactly similar to adult chimpanzee
(4) Skull of baby chimpanzee and adult chimpanzee has no resemblance to skull of human

BEV246

57. Which of the following is not amongst the evolutionary line of mammals?
(1) Therapsids (2) Pelycosaurs
(3) Synapsids (4) Sauropsids

BEV247

58. Before industrialization set in England :
(1) Only white winged moths were present on trees
(2) White winged moths were less in number on trees
(3) Dark winged moths were less in number on trees
(4) White winged moths were absent

BEV248

59. Proper burial of dead bodies for the first time started by which pre historic man ?
(1) Java man
(2) *Homo habilis*
(3) Neanderthal man
(4) *Australopithecines*

BEV249

60. *Homo sapiens* arose in _____ and moved across continents and developed into distinct races :
(1) America (2) Australia
(3) China (4) Africa

BEV250

61. Who lived in near east and central Asia between 1,00,000 - 40,000 years back?
(1) *Homo erectus*
(2) *Homo habilis*
(3) Neanderthal man
(4) *Australopithecines*

BEV251

62. How many of the following are examples of homologous structures in plants :
(a) Eyes of octopus and mammal
(b) Potato and sweet potato
(c) Forelimbs of whale, bat, cheetah and human
(d) Vertebrate heart
(e) Thorn of *bougainvillea* and tendril of *cucurbita*
(f) Potato and carrot
(g) Potato and radish
Options:
(1) Four (2) Three
(3) Two (4) One

BEV252

63. The most accepted view for origin of life is :
(1) Chemical evolution
(2) Spontaneous generation
(3) Religious literature
(4) Panspermia

BEV253

64. The study of fossils in different aged rocks shows all of the following except :
(1) They probably died during the formation of that particular sediment
(2) Some of them appear similar to modern organisms
(3) Life forms varied over time and certain forms were restricted to certain geological time spans
(4) They do not represent extinct organisms

BEV254

65. Which vertebrates evolved into the first amphibians?
(1) Jawless fishes
(2) Lobefins
(3) Salamanders
(4) *Ichthyosaurus*

BEV255

66. When reptiles came down which animals took over the earth ?
(1) Birds (2) Dinosaurs
(3) Mammals (4) Amphibians

BEV256

67. Forelimbs of whale, bat, cheetah and human do not show-
a. Similarities in the pattern of bones
b. Adaptive radiation
c. Analogous organs
d. Convergent evolution
e. Divergent evolution
f. Different structures evolving for same functions
(1) a, b and c (2) c, d and f
(3) a and d (4) a, d and c

BEV257

68. Therapsids were evolved into -
(1) Dinosaurs (2) Reptiles
(3) Birds (4) Mammals

BEV258

(B) ANALYTICAL QUESTIONS

69. Believers of spontaneous generation theory believed that :

(1) Life originated from other similar organisms or spontaneously

(2) Life originated only spontaneously

(3) Life originated from similar organisms

(4) Life originated from air

BEV259

70. Fossilization can occur where :

(1) Animals are buried and preserved by natural process

(2) Animals are destroyed by scavangers

(3) Animals are eaten by predators

(4) Animals are destroyed by environmental conditions

BEV261

71. What is sequence in the evolution of mammals?

(1) Fish-amphibian-bird-mammals

(2) Insect-fish bird-mammals

(3) Fish-amphibian-reptile-mammals

(4) Fish-amphibian-reptile-bird-mammals

BEV262

72. There would be no evolution if :

(1) The inheritance of acquired characters did not take place

(2) Somatic variations were not inheritable

(3) Gene variations were not found among members of population

(4) Somatic variations would not transform into germinal variations

BEV263

73. If population of a species is transferred to more suitable environment then it will show:

(1) Protection against enemies

(2) More individuals would survive

(3) Rate of reproduction increases

(4) Unlimited food would be available

BEV264

74. Which of the following facts develop suspicions in Lamarckism ?

(1) Human females are not born with bored ear pinna although they have been bored for thousands of years.

(2) Giraffe has long neck to eat leaves of tall trees

(3) A stag can run fast to protect against the enemies

(4) None of them

BEV265

75. Chances of inheritable and evolutionary changes are more in such species which reproduce by :

(1) Parthenogenesis

(2) Fission

(3) Sexual reproduction

(4) Asexual reproduction

BEV266

76. One major criticism of Darwin's theory is that:

(1) It presumes that environment upon earth has been changing through ages

(2) It does not explain variations with heredity

(3) It overestimates reproductive capacity of animals and plants

(4) It does not explain vestigeal organs

BEV267

77. Use of atomic bombs may lead to abnormalities even in upcoming generations because of :
(1) Body changes
(2) Air pollution
(3) Changed atoms in atmosphere
(4) Genetic mutation

BEV268

78. On Galapagos island Darwin observed variation in beaks of birds (Darwin's finches) and he concluded :
(1) Interspecies variation
(2) Intraspecies variation
(3) Natural selection according to food
(4) Inheritance of acquired characters

BEV269

79. Homologous organs are :-
(1) Wings of cockroach and wings of bats
(2) Wings of insects and wings of birds.
(3) Eye of the octopus and of mammals
(4) Pectoral fins of fishes and forelimbs of horse.

BEV270

80. Original idea of survival of fittest was of :-
(1) Herbert Spencer (2) Darwin
(3) Wallace (4) None of these

BEV271

81. Fore coming generations are less adaptive than their parental generation due to :-
(1) Natural selection
(2) Mutation
(3) Genetic drift
(4) Adaptation

BEV272

82. Which of the following are homologous organs?
(1) Wings of birds & Locust
(2) Wings of birds (Sparrow) & Pectoral fins of fish
(3) Wings of bat & Butterfly
(4) Legs of frog & Cockroach

BEV273

83. Convergent evolution is illustrated by :-
(1) Rat and dog
(2) Bacterium and protozoan
(3) Starfish and cuttle fish
(4) Dogfish and whale

BEV274

84. Which one of the following sequences was proposed by Darwin and Wallace for organic evolution ?
(1) Overproduction, variations, constancy of population size, natural selection
(2) Variations, constancy of population size, overproduction, natural selection
(3) Overproduction, constancy of population size, variations, natural selection
(4) Variations, natural selection, overproduction, constancy of population size

BEV275

85. Industrial melanism is an example of :
(1) Drug resistance
(2) Darkening of skin due to smoke from industries
(3) Protective resemblance with the surroundings
(4) Defensive adaptation of skin against ultraviolet radiations

BEV276

86. Darwin in his 'Natural Selection Theory' did not believe in any role of which one of the following in organic evolution :-

(1) Parasites and predators as natural enemies

(2) Survival of the fittest

(3) Struggle for existence

(4) Discontinuous variations

BEV277

87. Which one of the following describes correctly the homologous structures ?

(1) Organs with anatomical similarities, but performing different functions

(2) Organs with anatomical dissimilarities, but performing same function

(3) Organs that have no function now, but had an important function in ancestors

(4) Organs appearing only in embryonic stage and disappearing later in the adult

BEV278

88. What kind of evidence suggested that man is more closely related with chimpanzee than with other hominoid apes ?

(1) Comparison of chromosomes morphology only

(2) Evidence from fossil remains and the fossil mitochondrial DNA alone

(3) Evidence from DNA extracted from sex chromosomes, autosomes & mitochondria

(4) Evidence from DNA from sex chromosomes only

BEV279

89. Age of fossils in the past was generally determined by radio-carbon method and other methods involving radioactive elements found in the rocks. More precise methods, which were used recently and led to the revision of the evolutionary periods for different groups of organisms include :

(1) Study of the conditions of fossilization

(2) Electron spin resonance (ESR) & fossil DNA

(3) Study of carbohydrates/proteins in rocks

(4) Study of carbohydrates/proteins in fossils

BEV280

90. Presence of gills in the tadpole of frog indicates that :-

(1) Fishes evolved from frog like ancestors

(2) Frogs will have gills in future

(3) Frogs evolved from gilled ancestors

(4) Fishes were amphibious in the past

BEV281

91. Being all mammals, whales, dolphins, bat, monkey and horse have some common characters but they also show conspicuous differences. This is due to phenomenon of :-

(1) Normalisation (2) Genetic drift

(3) Divergence (4) Convergence

BEV282

92. Which one of the following phenomenon supports Darwin's concept of natural selection in organic evolution ?

(1) Production of 'Dolly', the sheep by cloning

(2) Development of organs from 'stem cells' for organ transplantation

(3) Development of transgenic animals

(4) Prevalence of pesticide resistant insects

BEV284

93. Man who firstly believed in immortality of soul was ?
(1) Neanderthal man (2) Cro-Magnon
(3) Java man (4) Peking man
BEV285

94. Which of the following is not true for a species?
(1) Members of a species can interbreed.
(2) Variations occur among members of a species.
(3) Gene flow does not occur between the populations of a species.
(4) Each species is reproductively isolated from every other species.
BEV286

95. In evolution the studies can be made at molecular level. For example the protein present in the blood of man and ape are similar. The base sequence in nucleic acids and amino acids sequence in protein in related organism is alike. These are the examples which one specifically referred to in:
(1) Convergent evolution
(2) Molecular analogy
(3) Molecular homology
(4) Homoplastic appearance
BEV287

96. Which of the following statement is incorrect?
(1) Life processes are consequences of reactions that occur in an organism
(2) Living organism are made of inorganic and organic compounds.
(3) Life comes from pre-existing life
(4) Genes are not responsible for the stability or changeability of species
BEV288

97. Select the correct statement from the following :-
(1) Darwinian variations are small and directionless.
(2) Fitness is the end result of the ability to adapt and gets selected by nature.
(3) All mammals except whales and camels have seven cervical vertebrae.
(4) Mutations are random and directional.
BEV289

98. Which of the following are not analogous organs ?
(1) Fins of fishes and flippers of whales
(2) Stings of honey bee and scorpion
(3) Thorn of *bougainvillea* and tendril of *Cucurbita*
(4) Wings of insect and wings of bird
BEV290

99. Which one of the following amino acid was not found to be synthesized in Miller's experiment–
(1) Glycine (2) Aspartic acid
(3) Glutamic acid (4) Alanine
BEV120

100. An important evidence in favour of organic evolution is the occurrence of –
(1) Homologous and vestigial organs
(2) Analogous and vestigial organs
(3) Homologous organs only
(4) Homologous and analogous organs
BEV121

101. Evolutionary history of an organism is known as:
(1) Phylogeny (2) Ancestry
(3) Paleontology (4) Ontogeny
BEV123

102. Sickle cell anemia has not been eliminated from the African population because –
(1) it is controlled by recessive genes
(2) it is not a fatal disease
(3) it provides immunity against malaria
(4) it is controlled by dominant genes

BEV124

103. One of the important consequences of geographical isolation is :-
(1) no change in the isolated fauna
(2) preventing Speciation
(3) speciation through reproductive isolation
(4) random creation of new species

BEV125

104. Among the human ancestors the brain size was more than 1000 CC in: -
(1) *Homo neanderthalensis*
(2) *Homo erectus*
(3) *Ramapithecus*
(4) *Homo habilis*

BEV126

105. Two plants can be conclusively said to belong to the same species if they: -
(1) can reproduce freely with each other and form seeds.
(2) have more than 90 percent similar genes.
(3) look similar and possess identical secondary metabolites.
(4) have same number of chromosomes.

BEV127

106. The Finches of Galapagos islands provide an evidence in favour of: -
(1) Special Creation
(2) Evolution due to Mutation
(3) Retrogressive Evolution
(4) Biogeographical Evolution

BEV128

107. What is common to whale, seal and shark :-
(1) Seasonal migration
(2) Thick subcutaneous fat
(3) Convergent evolution
(4) Homoiothermy

BEV129

108. Adaptive radiation refers to: -
(1) Adaptations due to Geographical isolation.
(2) Evolution of different species from a common ancestor.
(3) Migration of members of a species to different geographical areas.
(4) Power of adaptation in an individual to a variety of environments.

BEV130

109. When two species of different genealogy come to resemble each other as a result of adaptation, the phenomenon is termed :-
(1) Divergent evolution
(2) Microevolution
(3) Co-evolution
(4) Convergent evolution

BEV131

110. The concept of chemical evolution is based on:
(1) Crystallization of chemicals
(2) Interaction of water, air and clay under intense heat
(3) Effect of solar radiation on chemicals
(4) Possible origin of life by combination of chemicals under suitable environmental conditions

BEV132

111. Industrial melanism as observed in peppered moth proves that: -

(1) The true black melanic forms arise by a recurring random mutation.

(2) The melanic form of the moth has no selective advantage over lighter form in industrial area.

(3) The lighter-form moth has no selective advantage either in polluted industrial area or non-polluted area.

(4) Melanism is pollution-generated feature.

BEV133

112. Which one of the following statements is correct: -

(1) Stem cells are specialized cells

(2) There is no evidence of the existence of gills during embryogenesis of mammals

(3) All plant and animal cells are totipotent

(4) Ontogeny repeats phylogeny

BEV134

113. Which one of the following is incorrect about the characteristics of protobionts (coacervates and microspheres) as envisaged in the abiogenic origin of life?

(1) They were partially isolated from the surroundings.

(2) They could maintain an internal environment.

(3) They were able to reproduce.

(4) They could separate combinations of molecules from the surroundings.

BEV135

114. Which one of the following scientist's name is correctly matched with the theory put forth by him?

(1) de Vries – Natural selection

(2) Mendel – Theory of pangenesis

(3) Weismann – Theory of continuity of Germplasm

(4) Pasteur – Inheritance of acquired characters

BEV136

115. In the case of peppered moth (*Biston betularia*) the black-coloured form became dominant over the light-coloured form in England during industrial revolution. This is an example of:-

(1) Inheritance of darker colour character acquired due to the darker environment

(2) Natural selection whereby the darker forms were selected.

(3) Appearance of the darker coloured individuals due to very poor sunlight.

(4) Protective mimicry.

BEV138

116. Darwin's finches are a good example of

(1) Convergent evolution

(2) Industrial melanism

(3) Connecting link

(4) Adaptive radiation

BEV139

117. The most apparent change during the evolutionary history of Homo sapiens is traced in: -

(1) Walking upright

(2) Shortening of jaws

(3) Remarkable increase in the brain size

(4) Loss of body hair

BEV140

118. Given below are four statements (A-D) each with one or two blanks. Select the option which correctly fills up the blanks in two statements:

Statements:

(A) Wings of butterfly and birds look alike and are the results of ___(i)___ evolution

(B) Miller showed that CH_4, H_2, NH_3 and ___(i)__, when exposed to electric discharge in flask resulted in formation of ___(ii)__

(C) Vermiform appendix is a ___(i)___ organ and an ___(ii)___ evidence of evolution.

(D) According to Darwin evolution took place due to __(i)__ and __(ii)__ of the fittest.

Options:

(1) (A) – (i) convergent
(B) – (i) oxygen, (ii) nucleosides

(2) (B) – (i) water vapour, (ii) amino acids,
(C) – (i) rudimentary (ii) chemical

(3) (C) – (i) vestigial, (ii) anatomical,
(D) – (i) mutations, (ii) multiplication

(4) (D) – (i) small variations, (ii) survival,
(A) – (i) convergent

BEV141

119. What was the most significant trend in the evolution of modern man (*Homo sapiens*) from his ancestors?

(1) Upright posture

(2) Shortening of jaws

(3) Binocular vision

(4) Increasing brain capacity

BEV142

120. Evolution of different species in a given area starting from a point and spreading to other geographical areas is known as: -

(1) Migration

(2) Divergent evolution

(3) Adaptive radiation

(4) Natural selection

BEV143

121. What was the most significant trend in the evolution of modern man (*Homo sapiens*) from his ancestors?

(1) Increasing cranial capacity

(2) Upright posture

(3) Shortening of jaws

(4) Binocular vision

BEV144

122. The extinct human who lived 1,00,000 to 40,000 years ago, in Europe, Asia and parts of Africa, with short stature, heavy eye brows, retreating fore heads, large jaws with heavy teeth, stocky bodies, a lumbering gait and stooped posture was :-

(1) Cro-Magnons humans

(2) Ramapithecus

(3) Homo habilis

(4) Neanderthal human

BEV145

123. Which one of the following options gives one correct example each of convergent evolution and divergent evolution?

	Convergent evolution	Divergent evolution
(1)	Bones of forelimbs of vertebrates	Wings of butterfly and birds
(2)	Thorns of Bougainvillea and tendrils of Cucurbita	Eyes of Octopus and Mammals
(3)	Eyes of octopus and mammals	Bones of forelimbs of vertebrates
(4)	Thorns of Bougainvillea and tendrils of Cucurbita	Wings of butterflies and birds

BEV146

124. The idea of mutations was brought forth by:

(1) Hardy Weinberg, who worked on allele frequencies in a population

(2) Charles Darwin, who observed a wide variety of organisms during sea voyage

(3) Hugo do Vries, who worked on evening primrose

(4) Gregor Mendel, who worked on *Pisum sativum*

BEV147

125. The eye of octopus and eye of cat show different patterns of structure, yet they perform similar function. This is an example of:

(1) Analogous organs that have evolved due to divergent evolution.

(2) Homologous organs that have evolved due to convergent evolution.

(3) Homologous organs that have evolved due to divergent evolution.

(4) Analogous organs that have evolved due to convergent evolution.

BEV148

126. The tendency of population to remain in genetic equilibrium may be disturbed by:

(1) lack of random mating

(2) random mating

(3) lack of migration

(4) lack of mutations

BEV149

127. The process by which organisms with different evolutionary history evolve similar phenotypic adaptation in response to a common environmental challenge, is called:

(1) Adaptive radiation

(2) Natural selection

(3) Convergent evolution

(4) Non-random evolution

BEV150

128. According to Darwin, the organic evolution is due to:

(1) Reduced feeding efficiency in one species due to the presence of interfering species

(2) Intraspecific competition

(3) Interspecific competition

(4) Competition within closely related species

BEV151

129. Variation in gene frequencies within populations can occur by chance rather than by natural selection. This is referred to as:

(1) Genetic load

(2) Genetic flow

(3) Genetic drift

(4) Random mating

BEV152

EXERCISE - III ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	1	3	4	1	2	3	2	2	1	3	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	3	3	1	4	1	1	3	2	2	3	1	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	3	2	3	2	4	1	3	4	3	1	2	3	2	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	2	4	1	2	3	2	4	3	3	2	4	3	3	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	4	1	4	2	3	2	4	2	1	3	3	2	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	4	3	4	1	2	2	4	3	3	4	1	3	2	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	4	1	3	3	4	2	3	3	1	1	3	3	1	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	4	3	2	4	4	1	4	3	3	2	4	3	4	4	3
Question	121	122	123	124	125	126	127	128	129						
Answer	1	4	3	3	4	1	3	3	3						